Synergetics

Cooperative Phenomena in Multi-Component Systems

Edited by

H. Haken

Springer Fachmedien Wiesbaden GmbH

SYNERGETICS

Cooperative Phenomena in Multi-Component Systems

Proceedings of the Symposium on Synergetics from April 30 to May 6, 1972, Schloß Elmau

Edited by H. Haken

1973. With 151 Figures

Springer Fachmedien Wiesbaden GmbH

ISBN 978-3-519-03011-9 ISBN 978-3-663-01511-6 (eBook)
DOI 10.1007/978-3-663-01511-6

All rights reserved. This publication, or parts thereof,
may not be produced in any form, or by any means, electronic,
mechanical, photocopying, recording or otherwise, or trans-
mitted without written permission of the publisher.
© Springer Fachmedien Wiesbaden 1973
Originally published by B.G. Teubner, Stuttgart 1973
Softcover reprint of the hardcover 1st edition 1973
Cover design: W. Koch, Stuttgart

Editor

Haken, Prof. Dr. H., Institut für theoretische Physik der Universität, 7 Stuttgart 1, Azenbergstr. 12, Germany

With Contributions of

Adam, Dr. Gerold, Fachbereich Chemie der Universität, 775 Konstanz, Postfach 733, Germany

Chinitz, Dr. Benjamin, Brown University, Providence, USA

Crampton, Dr. Graham, Brown University, Providence, USA

Fröhlich, Prof. H., Chadwick Laboratory, University of Liverpool, Liverpool, England

Graham, Dr. Robert, Dept. of Physics, New York University, 4 Washington Place, New York, N. Y. 10003, USA

Grossmann, Prof. Dr. S., Fachbereich Physik, Philipps-Universität, 355 Marburg, Renthof 6, Germany

Haken, Prof. Dr. H., Institut für theoretische Physik der Universität, 7 Stuttgart 1, Azenbergstr. 12, Germany

Julesz, Dr. Bela, Head, Sensory and Perceptual Processes Dept., Bell Telephone Laboratories Inc., Murray Hill, N. J., USA

Kadanoff, Prof. Leo P., Dept. of Physics, Brown University, Providence, 02912, USA

Kawasaki, Prof. Kyozi, Dept. of Physics, Temple University, Philadelphia, Pa. 19122, USA

Kerner, Prof. Edward H., Physics Dept., University of Delaware, Newark, Del. 19711, USA

Kubo, Prof. Ryogo, Dept. of Physics, University of Tokyo, 3–1 Hongo 7–Chome, Bunkyo-ku, Tokyo, Japan

Kuhn, Prof. Hans, Karl-Friedrich-Bonhoeffer-Institut, MPI für biophys. Chemie, 34 Göttingen-Nikolausberg, Germany

Landauer, Dr. Rolf, IBM Thomas J. Watson Res. Center, P. O. Box 218, Yorktown Heights, N. Y. 10598, USA

Lefever, Dr. R., Chimie Physique II, Université Libre de Bruxelles, 50 Av. F. D. Roosevelt, 1050 Bruxelles, Belgium

Matsubara, Prof. Takeo, Dept. of Physics, Kyoto University, Kyoto, Japan

Matsuda, Prof. Hirotsugu, Res. Institute for Fundamental Physics, Kyoto University, Kyoto, Japan

Mesarovic, Prof. M., Systems Res. Center, Case Western Reserve University, 10900 Euclid Avenue, Cleveland, 44106, USA

Montroll, Prof. Elliot W., River Campus Station, University of Rochester, Rochester, N. Y. N. Y. 14627, USA

Mori, Prof. Hazime, Dept. of Physics, Kyushu University, Fukuoka, Japan

Ogita, Dr. Naotumi, The Institute of Physical and Chemical Research, Yamato-machi, Saitama, Japan

Prigogine, Prof. I., Chimie Physique II, Université Libre de Bruxelles, 50 Av. F. D. Roosevelt, 1050 Bruxelles, Belgium

Reichardt, Prof. Werner, MPI für biologische Kybernetik, 74 Tübingen, Spemannstr. 38, Germany

Schlögl, Prof. F. C., Insitut für theoretische Physik, Lehrstuhl B, TH Aachen, Templergraben 55, Germany

Schwartz, Dr. Susan R. C., Brown University, Providence, USA

Thomas, Prof. H., Institut für theoretische Physik der Universität, 6 Frankfurt/M., Robert-Mayer-Straße 8–10, Germany

Tomita, Prof. Kazuhisa, Dept. of Physics, Faculty of Science, University of Kyoto, Kyoto, Japan

Tucker, Dr. John R., Brown University, Providence, USA

Ueda, Dr. Akira, Dept. of Appl. Mathematics and Physics, University of Kyoto, Kyoto, Japan

Wagner, Prof. Dr. M., Institut für theoretische Physik der Universität, 7 Stuttgart 1, Herdweg 77, Germany

Weidlich, Prof. W., Institut für theoretische Physik der Universität, 7 Stuttgart 1, Herdweg 77, Germany

Weinblatt, Dr. Herbert, Brown University, Providence, USA

West, Dr. Bruce J., Institute for Fundamental Studies, Dept. of Physics and Astronomy, The University of Rochester, Rochester, N. Y. 14627, USA

Wilson, Dr. Hugh R., Dept. of Theoret. Biology, University of Chicago, 939 E. 57th Street, Chicago, Ill. 60637, USA

Woo, Dr. James W. F., IBM Thomas J. Watson Res. Center, P. O. Box 218, Yorktown Heights, N. Y. 10598, USA

Yonezawa, Dr. Fumiko, Res. Institute for Fundamental Physics, University of Kyoto, Kyoto, Japan Japan

Preface

At a first glance the reader of this book might be puzzled by the variety of its topics which range from phase-transition-like phenomena of chemical reactions, lasers and electrical currents to biological systems, like neuron networks and membranes, to population dynamics and sociology. When looking more closely at the different subjects the reader will recognize, however, that this book deals with one main problem: the behaviour of systems which are composed of many elements of one or a few kinds. We are sure the reader will be surprised in the same way as the participants of a recent symposium on synergetics, who recognized that such systems have amazingly common features. Though the subsystems (e. g. electrons, cells, human beings) are quite different in nature, their joint action is governed by only a few principles which lead to strikingly similar phenomena. It hardly needs to be mentioned that once such common principles are established, they are of an enormous stimulus and help for future research.

Though the articles of this book are based on invited papers given at the first International Symposium on Synergetics at Schloß Elmau from April 30 to May 6, 1972, it differs from usual conference proceedings in a distinct way. The authors and subjects were chosen from the very beginning so that finally a well organized total book arises. We hope that the reader will feel the same pleasure and enthusiasm the participants at the symposium had.

I use this occasion to express my deep gratidude to my secretary, Mrs. U. Funke, who helped me in organizing the symposium and in preparing this book in an extremely efficient way. Finally I wish to thank the Teubner Verlag, Stuttgart, whose management spontaneously agreed to edit this book on an new field.

Stuttgart, July 1972 H. Haken

Contents

H. Haken
Introduction to Synergetics 9

Mathematical and Physical Concepts for Cooperative Systems

F. C. Schlögl
Stability Criteria in Non-Equilibrium Thermodynamics 21

R. Kubo
Relaxation and Fluctuation of Macrovariables 28

K. Kawasaki
New Method in Non-Equilibrium Statistical Mechanics
of Cooperative Systems 35

H. Mori
Theory of Linear Non-Markoffian Processes 45

S. Grossmann
Fluctuations near Phase Transitions
in Restricted Geometries 54

Instabilities and Phase-Transition-Like Phenomena in Physical Systems far from Thermal Equilibrium

R. Graham
Phase-Transition-Like Phenomena in Lasers
and Nonlinear Optics 71

H. Thomas
Dynamics of Current Instabilities 87

R. Landauer, J. W. F. Woo
Cooperative Phenomena in Data Processing 97

I. Prigogine and R. Lefever
Theory of Dissipative Structures 124

Biochemical Kinetics and Population Dynamics

E. H. Kerner
A Gibbs Ensemble Approach to Biochemical Kinetics 137

E. W. Montroll, B. J. West
Models of Population Growth, Diffusion,
Competition and Rearrangement 143

H. Kuhn
Selforganization of Nucleic Acids
and the Evolution of the Genetic Apparatus 157

N. Ogita, A. Ueda, T. Matsubara, F. Yonezawa, H. Matsuda
Computer Simulation of Helix-Coil Transition 177

Biological Structures

W. Reichardt
Mechanism of Pattern Recognition
by the Visual System of Insects 187

B. Julesz
A Cooperative Model of Stereoscopic Depth Perception 194

H. R. Wilson
Cooperative Phenomena in a Homogeneous Cortical Tissue Model 207

G. Adam
Cooperative Transitions in Biological Membranes 220

K. Tomita
A Model for Muscle Contraction 232

H. Fröhlich
Organisation and Long Range Selective Interaction
in Biological and Other Pumped Systems 241

M. Wagner
Nonlinear Transport as a Possible Key to
Physical Understanding in Biology 246

General Structures

M. D. Mesarovic
Theory of Hierarchical Structures 255

B. Chinitz, G. Crampton, L. P. Kadanoff, S. C. Schwartz, J. R. Tucker, H. Weinblatt
Metropolitan Models and Public Policy 263

W. Weidlich
Fokker-Planck Equation Treatment of Interacting Social Groups 269

Introduction to Synergetics

H. Haken

In many disciplines of science we deal with systems composed of many subsystems. A few examples, mainly taken from topics in this book, are listed in Fig. 1. Very often the properties of the large system can not be explained by a mear random superposition of actions of the subsystems. Quite on the contrary the subsystems behave in a well organized manner, so that the total system is in an ordered state or shows actions which one might even call purposeful. Furthermore one often observes more or less abrupt changes between disorder and order or transitions between different states of order. Thus the question arises, who are the mysterious demons who tell the subsystems in which way to behave so to create order, or, in a more scientific language, which are the principles by which order is created.

Fig. 1

Examples of multi-component systems

Science	System	Subsystem
Physics	ferromagnet	elementary magnets (spins)
	laser	atoms
Chemistry	chemical ensemble	molecules
Biology	biological clocks	molecules
	neural network	neurons
Ecology	groups of animals	individual animals
	forest	" plants
Sociology	society	human beings

1. The order parameter

To answer this question let us consider a simple example from physics, the ferromagnet. We start from the fact that a magnet looses its magnetic field when the temperature T is increased above a certain critical temperature T_c and vice versa. In an atomic picture this effect is "explained" as follows (compare Fig. 2). The ferromagnet consists of many elementary magnets ("spins"), which are in the disordered state for $T > T_c$ and which are in an ordered state for $T < T_c$. At $T = T_c$ a so-called phase transition [1] occurs.

The next question to be answered is, of course, why get the spins aligned. This ordering is explained as follows. There is a certain elementary force (it is the Coulomb exchange interaction), which tries to bring 2 spins into parallel alignment. So what one has to deal with is finally the interaction of each spin with a number of other spins. This interaction is counteracted by the random temperature motion of the single elementary magnets. If we imagine that there are about 10^{22} spins per ccm, it becomes at once evident, that this many-body problem is extremely difficult and especially that it is absolutely hopeless to investigate the motion of each individual spin. One has rather to throw away all unnecessary information about the individual spins and replace it by a new quantity, which refers to the total system. This problem to which a great deal of effort has been devoted in theoretical physics has been treated using the concept of the o r d e r p a r a m e t e r . It should be noted here that the modern theory of phase transitions goes beyond this concept using scaling laws [2] or still more advanced methods (e. g. "3,99 dimensions" [3]).

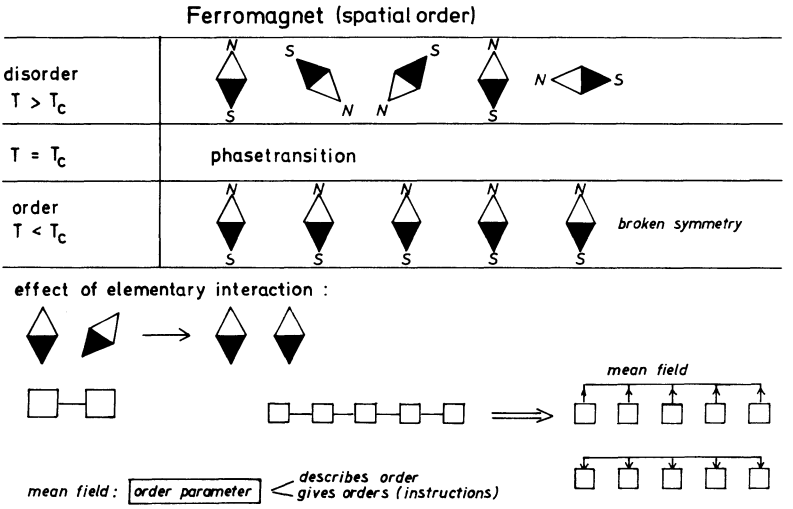


Fig. 2 Microscopic model of ferromagnet

For the field of synergetics the order parameter concept seems to be well suited, however. Its principle is indicated by the lower part of Fig. 2. By it the direct interaction between the elementary magnets, symbolized by the connected squares, is replaced by a 2 step procedure. First a fictitious mean field is assumed to be generated by all the individual spins. Then in the next step this mean field is assumed to react on the individual spins. We do not want to represent here the detailed theory but just stress that the order parameter obeys a rather simple equation.

The order parameter has at least two functions. It describes order because it is zero in the disordered state and assumes a maximum value in the completely ordered state. On the other hand it gives orders to the individual elementary magnets by acting with a certain force on them. In a more modern language one would say the order parameter gives i n s t r u c t i o n s to the

subsystems. Order parameters need not necessarily be fictitious quantities. For instance in the laser it is the lightfield which is generated by all the excited atoms. The concept of the order parameter is not limited to systems which are in thermal equilibrium like the ferromagnet. It

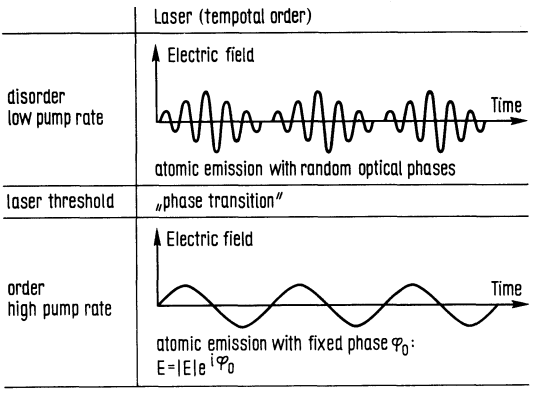


Fig. 3
Incoherent and coherent light of the laser as example of disordered and ordered states

applies equally well to physical systems far from thermal equilibrium. We just mentioned the laser [4], but it also applies to chemical ensembles and other systems. Examples are listed in Fig. 4.

Finding the right order parameter is not always a simple task even in physics. One may remind the reader that in superconductors it is certainly not quite obvious from the very beginning that

System	Orderparameter
ferromagnet	mean field
superconductor	pair wavefunction
laser	lightfield or photon number
chemical ensemble	numbers of molecules n_j
biological clocks	" "
neural network	pulserate
animals	numbers of animals
forest	density of plants
society	number of people of given opinion

Fig. 4 Examples of order parameters

the order parameter is here the pair wave function. Finding the adequate order parameters will be an important task for further research, in particular in the biological discipline.

The order parameter and the subsystems (in our example the spins) form the first step of a hierarchical system. In hierarchical systems there exists also a h i e r a r c h y o f t i m e c o n s t a n t s ("relaxation times", "response times") as has been observed e. g. by Mesarovic [5].

The subsystems usually have time constants much smaller than those of the higher system [1]), i. e. the subsystems follow immediately the changes of the order parameter. We observe a very important consequence: If the behaviour of the subsystems can be described by differential equations, the coordinates of the subsystems may be eliminated adiabatically.

2. Equations for order parameters

In most of the following articles the reader will be confronted with such equations. We present here some typical examples which stem also from other fields rather than physics.

2.1. Population dynamics (one kind of species). We consider an equation which describes the increase of a population, for instance a human population or an animal population. The following equation, however, holds equally well for a single mode laser or for an autocatalytic reaction. The dynamics is described by the equation

$$\frac{dn}{dt} = (\alpha - \beta n)\, n \quad (+ F(t)) \tag{1}$$

n is the number of individuals (human beings, photons, molecules). The growth rate $\frac{dn}{dt}$ is proportional to the number of individuals n with the rate factor α. If $\alpha < 0$ the "death" rate is bigger than the "birth" rate and the population dies out. The solution for the stationary state reads $n_0 = 0$. If $\alpha > 0$ an instability occurs. First the number of individuals increases exponentially. If the resources, who are ever limited (food, excited laser atoms, other molecules) get depleted the growth saturates, described by the factor $-\beta n$ in eq. (1). The total behaviour is shown schematically in Fig. 5. The term in brackets, F (t), describes creation of photons by spontaneous emission or random generation of molecules by reactions other than autocatalytic. In most theories it is assumed that F (t) has a very short memory, or in mathematical language, that F (t) is Markoffian. More recently considerable progress has been achieved e. g. by Mori who also treated in detail Non-Markoffian equations (compare his article).

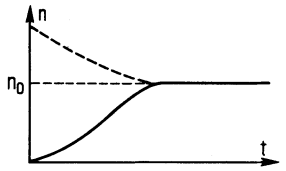


Fig. 5 Schematic representation of the solution of eq. (1)

2.2. Population dynamics (two kinds of species). An important example is the famous Lotka-Volterra problem [6]. We consider a biological system with two sorts of fishes. One sort of fishes, the predator, lives only on small fishes, the prey. The small fishes are assumed to live on unlimited food supply. Because without the presence of the prey the predators would die out, the rate of change of that population is given by $-\gamma_1 n_1$. On the other hand the number increases if there is food supply due to the presence of the prey. The growth rate is proportional

1) A counterexample is treated in the article by Kawasaki.

to the number of predator fishes present and to the number of the prey. Thus we obtain the equation

$$\frac{dn_1}{dt} = -\gamma_1 n_1 + \beta_{12} n_1 n_2 \tag{2}$$

and by similar considerations

$$\frac{dn_2}{dt} = \gamma_2 n_2 - \beta_{21} n_1 n_2 \tag{3}$$

for the prey.

The above mentioned authors found that equations (2) and (3) have an oscillatory solution (Fig. 6) which can be understood as follows. First only few predators are present, so the number of prey fishes increases. Then the number of predators increase eating more and more prey fishes. The number of prey fishes finally decreases and due to the lack of food finally the number of predator fishes decreases. So the circle may start again. Equations of the same type (2), (3) hold also for certain autocatalytic chemical reactions. This leads immediately to the conclusion that chemical clocks must exist, which has been indeed demonstrated in beautiful experiments. A somewhat more complicated system found by Zhabotinsky [7] shows an oscillatory change of colour from red to blue (Fig. 7). Chemical reactions being the foundation of biological events, this model offers also an understanding of the action of biological clocks (perhaps eventually of the control-mechanism of the heart).

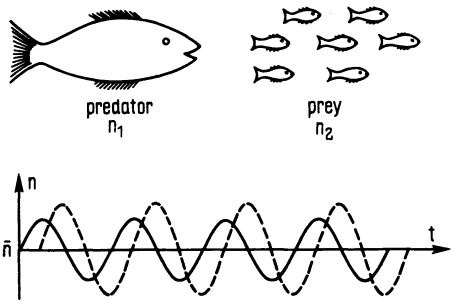


Fig. 6 The predator – prey – cycle (schematic)

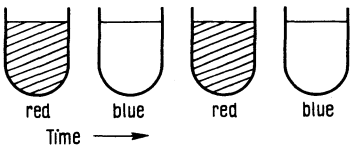


Fig. 7
Temporal oscillation of a chemical reaction

2.3. More complicated systems. The equations (2) and (3) can be generalized in a straight forward manner to systems with several constituents. They read[1])

$$\frac{dn_j}{dt} = E_j n_j - \sum_i \beta_{ij} n_i n_j + F_j(t) \tag{4}$$

---

1) In eq. (4) the random "forces" F (t) again describe random generation of individuals, e. g. in the laser process or in chemical reactions.

For special choices of the coefficients β_{ij} they have been already treated by Volterra. They describe population dynamics, chemical reactions, biological cells (see e. g. the contributions by Montroll, Schlögl, Lefever and Prigogine, Kerner). They apply to nonlinear transport (see the contribution by Wagner who considers a slightly more complicated case), and to the multimode laser [8]. In these examples the n's are used as adequate numbers or densities. If the β's are slightly generalized and if the n's are interpreted as pulse rates, equations (4) have been used by Cowan [9] to describe neuron networks. A further developed theory is presented in this book by Wilson. The Volterra case has been treated more recently in a detailed fashion by Montroll and coworkers.

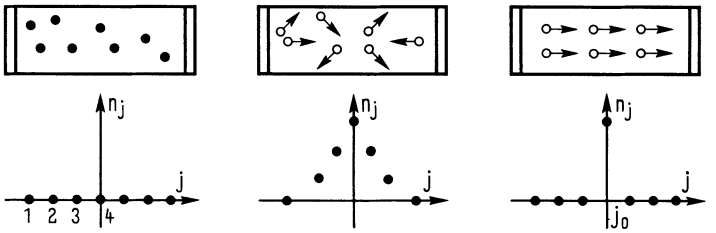


Fig. 8 Photon selection in a laser.
The above row sketches a laser with 2 end-mirrors. The dots symbolize excited atoms, the arrows photons emitted in the corresponding directions. The lower row represents the photon number n_j of the photon kind j. From left to right:
Before laser action starts the atoms are excited, no photons are present.
Laser action starts, photons are emitted in random positions etc. and are selected according to the different gain. The dots in the middle lower part of Fig. 8 are thus representative of the gain curve.
When a stationary state is reached all kinds of photons have died out but one (right hand side of Fig. 8).

Because the multimode laser is not treated in this way in the present book we present this example in slightly more detail. We consider in a cavity excited atoms which start to emit photons in random directions (and also randomly distributed in a certain frequence range). A discussion of eqs. (4) for the laser in which all β's have positive sign then shows the following. First the photon numbers for photons of different direction and energy ist built up according to Fig. 8, but then the nonlinear term becomes dominant, which leads to the selection of a single mode which has the highest gain and lowest loss [8]. All other modes disappear. Our laser equations thus describe selection. They apply equally well to autocatalytic reactions [10]. Indeed these equations were more recently used by Eigen [11] to explain selection within the process of evolution, which finally led to the genetic code. Contact may be also made with extremely interesting ideas by Fröhlich. He suggested at the first Versailles conference ("From theoretical physics to biology") that there may be highly excited longitudinal electric oscillations in biological systems which play a decisive role in the organization of living objects. His mathematical model is represented in this book.

We want to stress here a number of extensions if the number of n's becomes very large. About 15 years ago Kerner has applied the concept of thermodynamics to the equations of the type (4) by assuming that one has to deal with many j. An article in which he extends his work further to treat biological cells is included in the present book.

Another elegant treatment was given by Montroll and his coworkers, who treated the total sum over i in (4) as a random function. Another possibility which to our knowledge has not been

exploited so far, is to introduce new order parameters for the total system (4), so that one now obtains a sequence or a hierarchy of order parameters. There seems to be still a vaste field open to attack.

3. Systems far from thermal equilibrium. Stability, instability, broken symmetry, fluctuations, critical slowing down

So far we hardly touched the question whether thermal equilibrium is vital or not for the above considerations. Clearly enough the equations (4) apply to systems which have nothing to do with thermal equilibrium. Nevertheless it has turned out more recently that the analogy between systems in thermal equilibrium and far away from it can be quite close.

Because these questions will be treated in detail in the framework of physics by the articles of Graham, Grossmann, Landauer, and Thomas, and in the context with physics and biology by Prigogine and Lefever and by Fröhlich, I want to give here only a few introductory remarks. I take the following example mainly out of a former field of my research, that of laser physics.

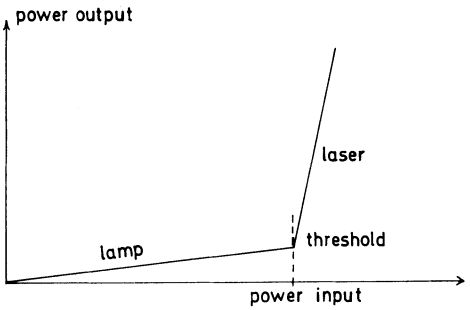


Fig. 9 Power output versus power input of a laser. Schematic diagram.

In Fig. 9 the emitted light intensity of a laser ("power output") is plotted against the pump rate ("power input"). For small pump rate the light intensity increases very slowly (small slope of the curve) or in other words, the lightsource efficiency, which is just the slope, is very small. This is the region of completely incoherent radiation (disordered state). When a certain threshold of the pump rate is reached the light intensity now increases very rapidly. The efficiency is very high. This is brought about by the coherent emission (ordered state). The analogy to the phase transition in thermal equilibrium is striking if we set the specific heat in correspondence to the efficiency.

To follow this analogy up more closely we discuss the order parameter and the equation it obeys. Instead of the photon number n, which occurred e. g. in eq. (1), we now choose the electric field strength E of the laser light[1]).

---

[1]) More exactly: If E_{tot} is the electric field strength of the laser light, we put $E_{tot}(x, t) = (E(t)\, e^{i\omega_0 t} + E^*(t)\, e^{-i\omega_0 t}) \sin k_0 x$, where ω_0 is the cavity frequency, k_0 the wave number and E (t) the slowly varying amplitude).

E is connected with the photon number n by

$$E = \sqrt{n}\, e^{i\varphi} \quad \text{or} \quad n = |E|^2 \tag{5}$$

(besides a constant factor).

Because E depends on the phase φ, it contains more information than n. As we will see below, this has very important consequences. As can be derived from a detailed microscopic theory [12], E obeys the following equation:

$$\dot{E} = \alpha' E - \beta' |E|^2 E \quad (+ F(t)) \tag{6}$$

for a single mode laser. α' is the socalled unsaturated gain, β' the saturation constant. The meaning of α', β' becomes obvious, when we use eq. (5). Then eq. (1) follows with $\alpha = 2\alpha', \beta = 2\beta'$. F (t) is again a fluctuating force (different from that in eq. (1)), describing spontaneous emission. Eq. (6) is one of the simplest, but very instructive examples for the study of the interplay between fluctuating forces (here: F (t)) and systematic forces (here: $\alpha' E - \beta' |E|^2 E$) or, in Monod's words [13]: of the interplay between "chance and necessity". In case of an infinitely extended laser with a continuous number of modes the expression $\gamma \Delta E$ must be added to the right hand side of eq. (6). In this case E is a function of time and space.

If we interpret eq. (6) as equation of a heavily overdamped motion [1]) of a particle with coordinate E we can interpret the first two terms on the right hand side of (6) as a force with a potential

$$V(E) = -\frac{\alpha'}{2} |E|^2 + \frac{\beta'}{4} |E|^4 \tag{7}$$

which is plotted in Fig. 8. For $\alpha' < 0$ the point E = 0 is stable: When the "particle" is pushed away from E = 0 by the random force F (t), it falls down the potential well and returns to E = 0.

When we let the parameter α' approach zero, the restoring force decreases, |E| comes down more and more slowly ("critical slowing down" near an instability). At the same time the fluctuations of E caused by F (t) are growing bigger and bigger ("critical fluctuations").

For $\alpha' \lesssim 0$ the position E = 0 becomes unstable. This instability is connected with a broken symmetry. The system has now the choice between all states for which $|E| = E_0$ and an arbitrary phase $\varphi = \varphi_0$. (If E had to be a real number, there are two stable values possible: $E = \pm E_0$. This bistable case will be treated below in several articles). At $\alpha' = 0$, where E changes from E = 0 to $E = |E_0| e^{i\varphi_0}$, a "phase-transition" occurs, which is very similar to a second order phase transition of systems in thermal equilibrium (see below).

A very similar problem occurs for the parametric oscillators and the tunnel diode which has been treated by Landauer in the connection with the theory of the computing process (see his article).

---

[1]) To see this explicitly, just add an acceleration term $m\ddot{E}$ with a very small "mass" to the left hand side of eq. (6).

Because "symmetry breaking instabilities" play an important role in synergetics, we mention a few further examples: Molecules which differ only in one property, e. g. optical dichroism, but having identical α, β, and F, or DNA or RNA, which differ by the arrangement of their constituents, but again having the same production factors α etc. Another example is provided by molecules, whose concentration is spacedependent. In the broken symmetry case e. g. s p a t i a l l y p e r i o d i c molecule concentrations may occur. Such systems and related ones have been studied in detail by Prigogine and coworkers and were called "dissipative structures" (see also the article of Prigogine and Lefever in this book). Analogies with respect to instabilities include a large variety of systems, e. g. plasmas [14], or current instabilities (see Thomas' article). A further example is provided by different states of interstellar matter [15]. Instabilities (or multi-stability) of exactly the same type as treated here are also found in visual perception (see the article by Reichardt) and in sociological models (see Weidlich's contribution). The ingenious experiments by Julesz are a challenge for a similar interpretation and considerable progress has been achieved in the model of neural networks by Wilson and Cowan.

Our laser example allows also to make contact with two important methods of stability theory:

a) Linearization. When we start with the stable point E = 0 (for $\alpha < 0$) we may neglect the nonlinear term $\sim |E|^2 E$ in eq. (6) ("linearization around a stable point"). The solution of (6) then reads

$$E = \text{const} \cdot e^{-|\alpha'|t}$$

E evidently, for $|\alpha'| \to 0$.. E becomes undamped with a frequency $\omega = 0$. This is the socalled "soft mode".

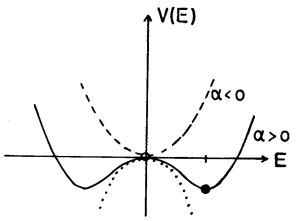


Fig. 10 The potential V(E), two-dimensional plot.
dashed curve: disordered state, incoherent emission
dotted curve: instability for $\alpha' > 0$.
solid line: situation with broken symmetry

b) Ljapunov's stability theory. V (E), eq. (7) has allowed us to discuss the stability and the stable points in a very simple manner. Indeed, V is just an example of a Ljapunov function (see also Schlögl's article).

We finally come back to the question how close the analogy can be between a system in and far from thermal equilibrium. Landauer has argued that there is in particular such an analogy if the principle of detailed balance is obeyed. Indeed it has been shown [16] that in this case the distribution function f can be written by means of the generalized thermodynamic potential G in the form

$$f = N e^{-G} \tag{8}$$

A consequence of this is demonstrated in the last figure 11 (which applies to the physicists among the readers) which gives a comparison between the distribution function of laser light and that of the pair wave function of the Ginzburg-Landau theory. The analogy is obviously perfect. If the detailed balance principle holds, the fluctuation and disssipation theorem of Kubo

[17] can be extended to systems far from thermal equilibrium as has been demonstrated in different ways by Agarwal [18] and Weidlich [19].

The distribution function of the laser

$$f = N \exp\left(-\frac{B}{Q}\right)$$

$$B = \int \{\tilde{\alpha} \mid E(x) \mid^2 + \tilde{\beta} \mid E(x) \mid^4 + \tilde{\gamma} \mid \left(\frac{d}{dx} - i\frac{\omega_0}{c}\right) E(x) \mid^2\} dx$$

E : electric field strength
$\tilde{\alpha} = \tilde{a}\,(d_c - d)\,; \tilde{a}, \tilde{\beta}\,\tilde{\gamma} > 0$ are laser constants
c : velocity of light
ω_0: atomic line frequency
N : normalization factor
d : atomic inversion
d_c : critical atomic inversion
B : is a generalized thermodynamic potential
Q : is the thrength of the fluctuations

The distribution function of the Ginzburg-Landau theory of superconductivity

$$f = N \exp\left(-\frac{F}{kT}\right)$$

$$F = \int \{\alpha \mid \psi(\underset{\sim}{x}) \mid^2 + \beta \mid \psi(\underset{\sim}{x}) \mid^4 + \frac{1}{2m} \mid \left(\underset{\sim}{\nabla} - \frac{2ei}{c}\underset{\sim}{A}\right) \psi(\underset{\sim}{x}) \mid^2\} d^3x$$

ψ : pair wave function
$\alpha = \alpha'(T - T_c)$
α', b': superconductor constants
k : Boltzmann's constant
T : absolute Temperature
T_c : critical Temperature
F : the free energy
m, e: mass and charge of electrons respectively

Fig. 11
Comparison of the statistical distribution function of the laser light amplitude with that of the Ginzburg-Landau theory of superconductivity.

4. How to obtain equations for order parameters

Our above considerations show clearly that the concept of the order parameter is extremely useful to describe multicomponent systems. So far we hardly said anything, however, how to obtain the adequate equations. We list here some possibilities:

a) derivation from a microscopic theory (example: ferromagnet, laser, nonlinear optics) by looking for "soft modes" or constructing suitable macroscopic variables or collective coordinates.

b) use of general principles, e. g. symmetry principles. This is the basis of the Landau theory of

phase transitions. As was shown by Graham (see his article), it is also applicable to nonlinear optics, i. e. systems far from thermal equilibrium.

c) by plausibility arguments (e. g. population dynamics, Volterra-Lotka problem).

5. Conclusion

In this introduction I have tried to "warm up" the reader for the following more detailed contributions. I hope he has got the impression that in spite of the completely different nature of the systems, their behaviour is governed, at a well defined level of consideration, by few very general principles which offers an explanation of the often amazingly similar performance of such systems.

References

[1] For a review on phase transitions see e. g. Brout, R.: Phase transitions. New York 1965.
[2] See e. g. Kadanoff, L. P. et al. Rev. of mod. Physics Vol. **33**, 2, 395 (1967).
[3] K. G. Wilson, Phys. Rev. **B 4**, 3174, (1971); K. G. Wilson and M. E. Fisher, Phys. Rev. Lett. **28**, 4 (1972)
[4] R. Graham and H. Haken, Z. Physik **213**, 420 (1968), and references in Graham's article.
[5] See his article in this book, where further references are given.
In the following we refer the reader to the detailed references contained in the articles of the individual authors of the present book.
[6] A. J. Lotka, Elements of Math. Biology, 1924 (Dover Reprint, 1956).
V. Volterra, Lécon sur la Théorie Mathematique de la lutte pour la vie (Paris 1931).
[7] A. M. Zhaboutinsky, Biofizika **2**, 306 (1964); Russ. J. Phys. Chem. **42**, 1649 (1968).
[8] Haken, H. and H. Sauermann, Z. Physik **176**, 47 (1963).
[9] See references in Wilson's article in this book.
[10] The application of laser type equations to ordering processes in biology has been proposed by Haken, H., in: Theoretical Physics and Biology, Ed. M. Marois, Amsterdam 1969.
[11] M. Eigen, Die Naturwissenschaften, Bd. **58**, 10, 465 (1971).
[12] H. Haken, Z. Physik **181**, 96 (1964), and Handbuch der Physik, Bd. XXV/2c, Heidelberg – Berlin – New York 1970, and in connection with phase transitions in: Festkörperprobleme Bd. X, Ed. O. Madelung, Marburg (1970). For further references see Graham's article.
[13] J. Monod: Le hasard et la nécessity, Paris 1970
[14] Klimontovich, Yu. L., The statistical Theory of non-equilibrium Processes in a Plasma, 1967.
[15] Graham et al., private communication. Kippenhahn et al., private communication.
[16] R. Graham and H. Haken, Z. Physik **243**, 289 (1971).
[17] R. Kubo, J. Phys. Soc. Japan **12**, 570 (1954); Rep. Progr. Physics **29**, 255 (1966).
[18] G. S. Agarwal, Z. Physik **252**, 25 (1972).
[19] W. Weidlich, Z. Physik **248**, 234 (1971).

Stability Criteria in Non-Equilibrium Thermodynamics

F. C. Schlögl

1. Chemical models for non-equilibrium phase transitions

The following considerations are concerned with transport states, which can be far away from thermal equilibrium. The considerations shall be restricted to transport states which can be described by relatively few macroscopic variables. If these variables are constant in time, they are called steady states.

The central question of this paper will be the stability of such steady states. At first examples shall be given for illustration and a special feature of some steady state systems shall be mentioned.

Thermodynamics of non-equilibrium steady states shows interesting features, as for instance phase transitions. This was stressed by Haken e. a. [1–4] in LASER theory, or by Landauer [5, 6] in the theory of electrical circuits. Here models of chemical reactions shall be given as examples of non-equilibrium phase transitions [7].

The first example is the case of two isothermic reactions

$$A + X \rightarrow 2X \tag{1.1}$$

$$B + X \rightarrow C \tag{1.2}$$

between species A, B, C, X of rarified gases in a stirring vessel. The concentration n of X may be variable, of all other species, however, hold fixed by feeding. Then the rate change of X is in convenient units given by

$$\dot{n} = -n^2 + (1 - \beta)\, n + \gamma. \tag{1.3}$$

β, γ are proportional to the concentrations of B, C. Vanishing n defines steady states in dependence on β, γ. The diagram of the steady states is given in Figure 1. The plot gives n in dependence on β for different values γ. For vanishing γ a phase transition of second order arises at the critical value 1 of β. The diagram can be compared with that of equilibrium states of a ferro-magneticum. n, β, γ have to be replaced by M, T, H.

The concentration n is proportional to the forward reaction rates and is a "parameter of order" with respect to the phase transition. Phase transitions of second order are connected with breaking of a symmetry. Here the symmetry operation is time inversion.

Another analogy to equilibrium phase transitions is the fact that the regression time of fluctuations becomes extremely large at the critical point.

The second example is similar to the first one. Only the first reaction (1.1) is replaced by

$$A + 2X \rightarrow 3X. \tag{1.4}$$

This gives by adequate choice of units

$$\dot{n} = -n^3 + 3n^2 - \beta n + \gamma \equiv -\psi(n) + \gamma. \tag{1.5}$$

In general three steady states can occur. Two states n_1, n_2 are stable. The third one with a concentration n_3 lying between n_1, n_2 is unstable.

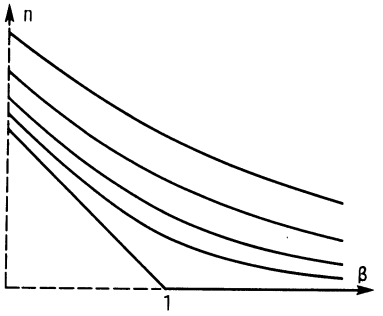


Fig. 1

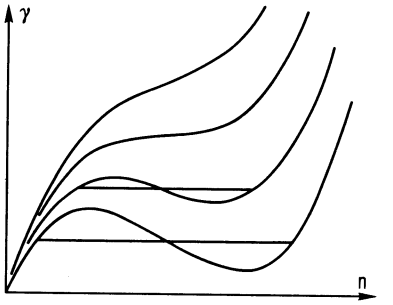


Fig. 2

In Figure 2 the diagram of the steady states is given. γ is plotted over n for different values β. The analogy to the equilibrium state diagram of a Van der Waals gas is clear when n, β, γ are replaced by v^{-1}, p, T. To a given γ three steady states exist in case only that β is less than a critical value β_c. The unstable states lie on the decreasing part of the curves ψ (n).

The analogy to a Van der Waals gas becomes more closed if the reactor is not a stirring vessel but diffusion of the component X takes place. The concentrations of the other species may be homogeneous in space, what is realized by sufficient high mobility of these species. In this case two "phases" n_1, n_2 can be in coexistence in different parts of the space. In the stable states the boundary layer between these phases accepts minimal surface. For plane layer the coexistence value γ for given β is found by the Maxwellian construction which connects the states n_1, n_2 in the diagram by a straight horizontal line cutting off equal areas below and above the curve ψ (n). For spherical phase domains the coexistence value γ depends on the radius of the sphere in the same way as vapor pressure of droplets or bubbles.

2. Liapounoff criterion

The following is concerned with stability of steady states of a physical system which is subject to fixed boundary conditions given by fixed values of the thermodynamic variables of the environment. In the examples given in the preceding chapter the boundary conditions fixed the concentration of A, B, C. In general the boundary conditions don't allow an equilibrium. They allow, however, one or more steady states. Of course they allow a multitude of other transport states which in the following shall be compared with a steady state.

The transport states which are compared with the considered steady state may be parametrized by parameters ξ_k. They may be chosen such that vanishing ξ_k correspond to the steady state. A region round the origin is called a stability region of the steady state if the states in this region during development with time do not move out of the region.

Now L (ξ) may be a convex function which vanishes in the origin only and increases monotonously such that any surface in the ξ-space with constant L-value encloses smaller values L only. We call L a Liapounoff-function [8]. Any region

$$L(\xi) \leqslant C \tag{2.1}$$

is a stability region if everywhere in its interior at any time is valid

$$\dot{L} \leqslant 0. \tag{2.2}$$

3. Gain of information

In the following a quantity will be used as Liapounoff-function L which in statistics is called "gain of information" [9].

If i is an index denumerating the microstates of a physical system and p_i is a probability distribution over the microstates describing a macrostate, then the negative of the entropy S

$$I(p) = -S = \sum_i p_i \ln p_i \tag{3.1}$$

is the information measure. It is the mean value of the negative of the bit number which is necessary to fix a certain microstate, if only the probability distribution was known [10].

The quantity

$$K(p, p') = \sum_i p_i \ln \frac{p_i}{p_i'}, \tag{3.2}$$

which is attached to two distributions p, p′ is called "gain of information" leading from p′ to p. It can be introduced as the mean value of the bit number [11] which leads from p′ to p. It is a measure of the knowledge which is gained by the information that the probabilities of the microstates are p_i if originally the state of knowledge gave expectations p_i'. K is positive definite and vanishing only if p and p′ are identical. It is quite different from the difference of the entropies in the states p, p′.

If p′ is a thermal state in which the system is in equilibrium with its environment, p′ is given by the knowledge of the environment state. p be another thermal state of the system. Its knowledge requires additional observations in the system itself. This knowledge is measured by K. One can show that K is precisely the entropy produced in the interior of the system by the change from p to p′. It's decrease, that is it's negative time derivative, is entropy production per second. Positivity of K gives the stability relations of equilibrium thermodynamics [12].

4. Stability criterion for steady states [13, 14]

The quantity K is a convex function of the p_i. The p_i are not independent, but it is always possible to characterize all possible distributions p by a set ξ of independent parameters such that K is a convex function of ξ. The ξ can moreover be chosen vanishing for p′.

If now p' is a steady state, we can apply the Liapounoff criterion by using K as Liapounoff function in ξ-space, p being another transport state which is possible for the given boundary conditions of the system. Thus we get the following result: Any region round the steady state p' defined by

$$K(p, p') \leqslant C. \tag{4.1}$$

is a stability region if everywhere in it and always is valid

$$\dot{K}(p, p') \leqslant 0. \tag{4.2}$$

So far this is a statistical criterion. It is the basis of a criterion in macroscopic thermodynamics. This is the criterion of Glansdorff and Prigogine in a generalized form.

To get it, certain assumption of a very general kind must be made. The first one is that there exists a statistical theory for the thermodynamic description. Thermodynamic description works with macroscopic variables M^ν which are mean values of observables M_i^ν in the microstates. This kind of variables which are functions in phase space may be called "direct" extensive variables. The assumption can be formulated by the requirement that there exists a unique mapping between the values M^ν and the distribution p in statistics. An example of such a mapping are the canonical distributions where M^ν are thermostatic variables. Another example are the hydrodynamic distributions of Mori [15], Zubarev [16] and Mc Lennan [17].

The second assumption is that the "direct" extensive variables M^ν are additive for independent systems if composed to a new one.

On account of these two assumptions one can show that K is equal to the negative of the non-linear part of the Taylor-expansion of the entropy difference

$$\delta S \equiv S - S' \tag{4.3}$$

with respect to powers of

$$\delta M^\nu \equiv M^\nu - M'^\nu. \tag{4.4}$$

It may be written in the form

$$K = -\delta_{NL} S. \tag{4.5}$$

Here δ symbolizes finite deviations from the steady state and δ_{NL} symbolizes the non-linear part in its expansion. It is essential that M^ν are "direct" extensive variables. The criterion states that any region

$$\delta_{NL} S \geqslant C \tag{4.6}$$

is a stability region if in it's interior always is fulfilled

$$\delta_{NL} \dot{S} \geqslant 0. \tag{4.7}$$

By restriction to infinitesimal deviations from the steady state the criterion gets the form

$$\delta^{(2)} \dot{S} \geqslant 0 \tag{4.8}$$

which was given by Glansdorff and Prigogine [18] by macroscopic thermodynamic arguments.

Afterwards one can interpret the inequality (4.2) as follows. If the observer knows only that the system with given boundary conditions is in a stability region round p', an unbiased estimate for the momentary state would lead to p' itself because all other states p have restricted life time. If the observer, however, knows e. g. by observation the momentary state p, then his excess knowledge, which is measured by K, can not increase after the last observation.

5. An example

The given chemical reaction models were one-dimensional in this sense that only one variable n was free. In this case stability can be seen directly and the criterion can be checked. In the given example and in other one-dimensional systems the criterion gives correct results.

The advantage of the criterion becomes conspicuous for more-dimensional systems. As the interesting systems are nonlinear, they in general are quite complicated for direct solution of the equations of motions. In the following an example is given which is so simple that it is possible to see the stability features directly [14]. Therefore it can be used as check for the criterion and, on the other hand, it shows the advantage of the criterion. The example is the system of two rarified gases X, Y which react one with another in the following way

$$X + Y \rightarrow 2X + Y \tag{5.1}$$

$$X + Y \rightarrow X + 2Y \tag{5.2}$$

$$X \rightarrow C \tag{5.3}$$

$$Y \rightarrow C. \tag{5.4}$$

These reactions may be irreversible, occuring only in one direction. The connection to a model of biological increase becomes obvious by calling X, Y "females" and "males". If X, Y enter symmetrically into the kinetic equations of the reaction process, the trajectories of the motion in the concentration diagram X, Y is given by Figure 3 which shows an unstable steady state in point (a, a). The sign of (4.7) is much easier found than the trajectories. It is given in Figure 4 and shows immediately the instability of the steady state.

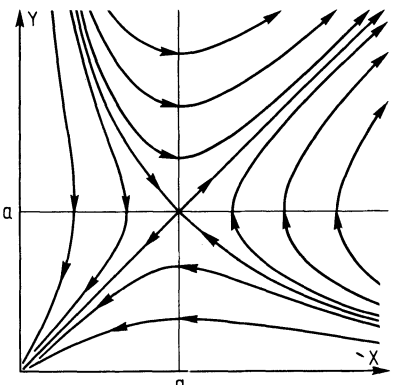


Fig. 3

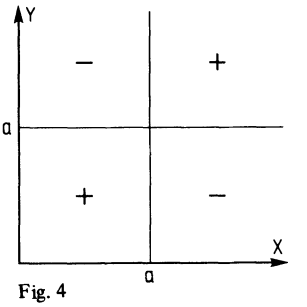


Fig. 4

6. Fluctuations from a steady state [14]

A given fluctuation from a steady state p' corresponds to a distribution p. The probability of the fluctuation can be introduced by the relative frequencies

$$N_i/N = p_i \tag{6.1}$$

of the results i if a large number N of independent measurements of the microstates is carried out in an ensemble representing the distribution p'. It gives the probability

$$W_N = w^N \tag{6.2}$$

where

$$w \sim \exp(-K(p, p')). \tag{6.3}$$

The last quantity w is the probability to find the fluctuation p in the state p'. (6.3) is a generalization of Einstein's formula which results if p' is a microcanonical equilibrium distribution.

For infinitesimal deviations from a stable steady state the equations of motion of regression can be assumed to be linear:

$$\dot{\alpha}^\mu = \sum_\nu L^{\mu\nu} X_\nu. \tag{6.4}$$

α^μ are the deviations of the macroscopic state variables from the steady state. The X_ν are defined by

$$X_\nu = -\frac{\partial K}{\partial \alpha^\nu} \tag{6.5}$$

and are linear in the deviations α^μ. The coefficient matrix $L^{\mu\nu}$ in general is non-symmetric. On account of (6.3) one can show that a kind of generalized Kubo-relations

$$L^{\mu\nu} = -\langle \dot{\alpha}^\mu \alpha^\nu \rangle \tag{6.6}$$

are valid in which the bracket signifies the mean value in the steady state distribution.

7. Quantum theory

In quantum theory a macroscopic thermal state is described by a statistical operator ρ which attaches to any complete measurement a probability distribution over the possible results of the measurement. To each of these distribution we could apply the notions information measure and gain of information. As well known, we can, however, in a more concise manner define a measure of information

$$I(\rho) = \mathrm{Tr}(\rho \ln \rho) \tag{7.1}$$

directly attached to ρ.

By introduction of a hermitian operator "bit number" one can define a measure of gain of information [7] which corresponds to the change from ρ' to ρ by the expectation value of the

difference of the bit numbers of ρ and ρ'. One gets

$$K(\rho, \rho') = \mathrm{Tr}(\rho\ (\ln \rho - \ln \rho')). \tag{7.2}$$

It is possible to show that this quantity is positive definite vanishing only if ρ, ρ' are identical, and that it is a convex function of an adequate set of parameters characterizing ρ. The stability criterion, the fluctuation probability, and the given generalized Kubo relation can be gained with the quantity $K(\rho, \rho')$ as in classical physics.

References

[1] Haken, H.: Laserlicht – ein neues Beispiel für eine Phasenumwandlung? Verhandl. DPG (VI) 5 (1970). Stuttgart 1970.

[2] Haken, H.: Festkörperprobleme X (ed. O. Madelung). Braunschweig 1970.

[3] Graham, R., Haken, H.: Z. Physik **243**, 289 (1971)

[4] Grossmann, S., Richter, P. H.: Z. Physik **242**, 458 (1971)

[5] Landauer, R., Woo, J. W. F.: Proceedings of the Sixth IUPAP Conference on Statistical Mechanics (eds. S. A. Rice, K. F. Freed, and J. C. Light). Chicago.

[6] Landauer, R.: Stability and Instability in Information Processing and in Steady State Dissipative Systems. Plenarvorträge der 36. Physikertagung 1971 Essen. Stuttgart 1971.

[7] Schlögl, F. C.: To be published in Z. Physik.

[8] Lasalle, J., Lefschetz, S.: Stability by Liapunov's Direct Method. New York 1961. German translation: Die Stabilitätstheorie von Ljapunoff. Mannheim 1967.

[9] Rényi, A.: Wahrscheinlichkeitsrechnung. Berlin 1966.

[10] Shannon, C., Weaver, W.: The mathematical theory of communication. Urbana 1949.

[11] Schlögl, R.: Z. Physik **249**, 1 (1971).

[12] Schlögl, F.: Z. Physik **191**, 81 (1966).

[13] Schlögl, F.: Z. Physik **243**, 303 (1971).

[14] Schlögl, F.: Z. Physik **248**, 446 (1971).

[15] Mori, H.: Phys. Rev. **115**, 298 (1959).

[16] Zubarev, D. N.: Soviet Phys. Doklady **10**, 526 (1965).

[17] Mc Lennan, I. A.: Advances in Chemical Physics, (ed. J. Prigogine). New York 1963.

[18] Glansdorff, P., Prigogine, I.: Physica **46**, 344 (1970).

Relaxation and Fluctuation of Macrovariables

Ryogo Kubo

1. We consider here a system composed of a great number of elementary units which are interacting among themselves and possibly with the environment [1]. As a typical example we shall discuss in some details an ideal Weiss-Ising spin system, but the general consideration is not limited to this particular system. For such a system a set of macrovariables will be defined. For a physical system in thermal equilibrium they are thermodynamic variables, and for a fluid they are the variables of velocity field. In these cases, the set of macrovariables is complete in the sense that a macroscopic description is possible in terms of those variables. This will not be at all obvious in other cases, for example in a biological or a sociological system, or even in a physical system far from equilibrium. We do not intend here to discuss this point further but will assume that such macroscopic variables are found in the system we consider.

Let q be such a macrovariable or a set of macrovariables. The evolution of q makes in general a stochastic process, rather than deterministic, which we assume to be Markoffian. The Chapman-Kolmogorov equation (or the master equation) is written as

$$\frac{\partial}{\partial t}P(q, t) = -\int dr\, W(q, r)P(q, t) + \int dr\, W(q - r, r)P(q - r, t) \tag{1}$$

where $P(q, t)$ is the probability density and $W(q, r)$ is the transition rate for a jump by r from the state q. If q is a set of variables, say $q_1, \ldots, q_n$, q and r are considered as vectors. If their values are discrete, integrations are understood as summations.

We consider extensive macrovariables. An extensive macrovariable consists of a large number of small contributions from the elementary units of the system. It could be the total magnetization of a magnetic system, or the population in a city. If the size of the system is denoted by Ω, it is normalized by the size as

$$x = q/\Omega, \tag{2}$$

which then characterizes the internal state of the system. The transition probability $W(q, r)$ in Eq. (1) may be considered to be determined by the internal state rather than q itself. We thus assume that

$$W(q, r) = \Omega w(x, r) \tag{3}$$

In statistical thermodynamics, the probability for an equilibrium state is given in the form,

$$P_e(q) = C \exp[-\Phi(q)] = C \exp[-\Omega\phi(x)], \tag{4}$$

where $\Phi(q)$ is a characteristic function or a thermodynamic potential multiplied by $1/kT$ and $\phi(x)$ is the corresponding function of x normalized by the size Ω. The probability (4) has a strong peak at $x = x_0$ which minimizes the function $\phi(x)$. Fluctuation of x around this most

probable value x_0 is practically of the order of $\Omega^{-1/2}$ very small if the size Ω is very large. This may be regarded as a manifestation of the central limit theorem which applies to an extensive random variable under some general mathematical conditions. It should be noted, however, that the distribution (4) is valid not only for fluctuations in the order of $\Omega^{-1/2}$ but also for deviations of the order of Ω^0.

Non-equilibrium evolution of the system is described by the probability P(q, t) which is a solution of Eq. (1) for a given initial distribution. One would expect that there may exist some general theorems which come out of the fact that the system is large, reflected on the asymptotic form of the probability P(q, t). This question must have been asked by many people, but here we refer to the work of van Kampen [2] to which the present work is closely related. Choosing the variable x, Eq. (2), and making the assumption (3), we write Eq. (1) as

$$\frac{1}{\Omega}\frac{\partial}{\partial t}P(x,t) = -\int dr\left\{1-\exp\left(-\frac{r}{\Omega}\frac{\partial}{\partial x}\right)\right\}w(x,r)P(x,t). \tag{5}$$

The problem is then to find an asymptotic form of P(x, t) for large values of Ω. If the existence of such an asymptotic form is proved, this will be an extension of the central limit theorem to non-stationary stochastic processes. We do not attempt here to prove it but shall be content with trying an Ansatz to the solution of (5). The asymptotic form (4) of equilibrium distribution suggests the Ansatz,

$$P(x,t) = C\exp\{\Omega f(x,t) + \cdots\}, \tag{6}$$

for a non-equilibrium probability.

Inserting (6) into Eq. (5) and taking the leading terms of the order of Ω, we obtain the equation

$$\frac{\partial}{\partial t}f(x,t) = -\int dr\, w(x,r)\left\{1-\exp\left(-r\frac{\partial}{\partial x}f(x,t)\right)\right\}. \tag{7}$$

To solve this equation we try a solution of the form

$$f(x,t) = g(z,t),\ x = y(t) + z \tag{8}$$

for which Eq. (7) is written as

$$\frac{\partial}{\partial t}g(z,t) - \dot{y}\frac{\partial g}{\partial z} = -\int dr\, w(y+z,r)\left\{1-\exp\left(-r\frac{\partial g}{\partial z}\right)\right\}. \tag{9}$$

Following van Kampen, we determine the function y(t) by the equation,

$$\frac{dy(t)}{dt} = c_1(y), \tag{10}$$

where
$$c_1(y) = \int dr\, w(y,r)r \tag{11}$$

is the average velocity of y when the system is in the state y. This choice of y erases the terms linear in $\partial g/\partial z$ from Eq. (9) and enables us to assume g(z, t) in the form

$$g(z,t) = a_1(t)z^2 + a_2(t)z^3 + \cdots$$

$$= -\frac{1}{2\chi(t)}z^2 + \cdots, \tag{12}$$

for which Eq. (9) then gives a set of ordinary differential equations to determine the coefficients $a_1(t), a_2(t), \ldots$ as functions of t. In particular the equation for the variance χ is given by

$$\frac{d}{dt}\chi(t) = 2\frac{dc_1(y)}{dy}\chi(t) + c_2(y) \tag{13}$$

where $$c_2(y) = \int dr\, w(y, r) r^2 \tag{14}$$

is the second moment of the transition rate. Eq. (10) determines the most probable path y(t) of the macrovariable x, and Eq. (13) the fluctuation around this path. The variance of x or q is given in terms of χ as

$$\langle (x - y(t))^2 \rangle = \Omega^{-1}\chi(t), \text{ or } \langle \{q - \Omega y(t)\}^2 \rangle = \Omega\chi(t). \tag{15}$$

The above treatment is closely related to van Kampen's analysis which uses the Ansatz,

$$P(x, t) = P'(\xi, t), \qquad \xi = \Omega^{1/2}\{x - y(t)\},$$

for the solution of Eq. (5). If y(t) is chosen as to satisfy Eq. (10) and ξ is considered to be of the order of Ω^0, the Kramers-Moyal expansion of Eq. (5),

$$\frac{\partial}{\partial t}P(x, t) = \left[\sum_{n=1}^{\infty} \frac{(-1)^n}{n!\Omega^{n-1}}\left(\frac{\partial}{\partial x}\right)^n \int dr \cdot r^n w(x, r)\right]P(x, t) \tag{16}$$

can be transformed into an equation for $P'(\xi, t)$ which is systematically ordered in the powers of $\Omega^{-1/2}$. In the asymptotic limit $\Omega \to \infty$, the equation becomes

$$\frac{\partial}{\partial t}P'(\xi, t) = \frac{\partial}{\partial \xi}\left\{-c_1'(y)\xi + \frac{1}{2}c_2(y)\frac{\partial}{\partial \xi}\right\}P'(\xi, t) \tag{17}$$

which looks like an ordinary Fokker-Planck equation but with time-dependent coefficients on the right hand side determined by the most probable motion y(t). In fact it is seen easily that

$$P'(\xi, t) = [2\pi\chi(t)]^{-1/2}\exp\{-\xi^2/2\chi(t)\} \tag{18}$$

is a solution of Eq. (17) if the variance $\chi(t)$ is determined by Eq. (13). The Ansatz (6) corresponds to an extension of (18) to a more general form in which contributions from higher order terms in the Kramers-Moyal expansion are summed up to the function g(z, t) on the exponent of the probability function P. It should be also mentioned that Eqs. (10) and (13) have been obtained by van Kampen [3] simply by calculating the first and second moments of transition from Eq. (5) and by using Eq. (15).

2. The analysis made in the above seems to have a wide application. All kinds of birth and death problems may be cast in this frame as long as the population is a macroscopic size. If A(n, r, t) is the rate of death of r individuals and B(n, r, t) is that of birth of r individuals in a population n, and if Ω is a large number to normalize the population, we may reasonably assume that

$$A(n, r, t) = \Omega a(x, r, t) \tag{19}$$

$$B(n, r, t) = \Omega b(x, r, t)$$

where $x = n/\Omega$

is the population density. Then the normalized average population, or the most probable population density, $y = \bar{n}/\Omega$ and the variance, $\chi = \Omega\langle(x - y)^2\rangle$, follow Eq. (10) and (13), in which the moments of the transition rate are

$$\begin{aligned} c_1(y, t) &= \sum_r r\{b(y, r, t) - a(y, r, t)\} \\ c_2(y, t) &= \sum_r r^2\{b(y, r, t) + a(y, r, t)\}. \end{aligned} \tag{20}$$

An interesting example of birth and death problem with a critical property is given by an ideal Weiss magnet consisting of N Ising spins, each of which flips over from one direction to another with some probability determined by the temperature and the local field. This is the so-called Glauber model or the kinetic Ising model, which is idealized to the Weiss limit, each spin interacting with all other spins in the system, so that the molecular field is dertermined by the total magnetization. Eq. (1) may be assumed to

$$\begin{aligned} \frac{\partial}{\partial t} P(N_+, N_-, t) = &-\left\{\exp\left[-\mu - \frac{\alpha}{N}(N_+ - N_-)\right] N_+ + \right. \\ &\left. + \exp\left[\mu + \frac{\alpha}{N}(N_+ - N_-)\right] N_-\right\} P(N_+, N_-, t) + \\ &+ \exp\left[\mu + \frac{\alpha}{N}(N_+ - N_- - 2)\right](N_- + 1)P(N_+ - 1, N_- + 1, t) + \\ &+ \exp\left[-\mu - \frac{\alpha}{N}(N_+ - N_- + 2)\right](N_+ + 1)P(N_+ + 1, N_- - 1, t) \end{aligned} \tag{21}$$

where $\mu = \mu_B H/kT$ and $\alpha = J/kT$ represent the external magnetic field and the exchange coupling respectively. Time is scaled by a suitable unit which generally depends on temperature. We put

$$(N_+ - N_-)/N = x.$$

The normalized transition rate w in Eq. (3) is found to be

$$\begin{aligned} w(x, r) &= \frac{1}{2}(1 - x)\, e^{\mu + \alpha x} \qquad \text{for } r = 2 \\ &= \frac{1}{2}(1 + x)\, e^{-\mu - \alpha x} \qquad \text{for } r = -2 \end{aligned} \tag{22}$$

so that Eqs. (10) and (13) now become

$$\dot{y} = \sinh(\mu + \alpha y) - y\cosh(\mu + \alpha y) \tag{23}$$

$$\begin{aligned} \dot{\chi} = &2\{(\alpha - 1)\cosh(\mu + \alpha y) - \alpha y \sinh(\mu + \alpha y)\}\chi + \\ &+ 2\{\cosh(\mu + \alpha y) - y \sinh(\mu + \alpha y)\}. \end{aligned} \tag{24}$$

The paramagnetic states above the critical temperature T_c are characterized by $\alpha < 1$, the ferromagnetic states below T_c by $\alpha > 1$, and the critical point by $\alpha = 1$. Eq. (23) $\dot{y} = 0$ gives the well-known equation of Weiss approximation for the magnetization.

In the absence of the external magnetic field, Eq. (23) may be expanded in y as

$$\dot{y} = (\alpha - 1)y + \left(\frac{1}{6}\alpha^3 - \frac{1}{2}\alpha^2\right) y^3 + \cdots. \tag{25}$$

Small deviations from the equilibrium decay in time exponentially with relaxation times which show up the critical slowing down as the system approaches the critical point. At the critical point Eq. (25) becomes

$$\dot{y} = -\frac{1}{3}y^3 + \cdots \tag{26}$$

so that the decay cannot be simply exponential or the relaxation cannot be linear. Larger deviations from equilibrium are generally non-linear and the nonlinearity is pronounced as the critical point is approached and may be accompanied by anomalous fluctuations.

Equations (10) and (13) have some interesting consequences when the macrovariable is just one-dimensional. In this case, Eq. (13) can be written as

$$c_1(y)\frac{d\chi}{dy} = -2c_1'(y)\chi + c_2(y),$$

which gives

$$\chi(y) = \chi_0\left\{\frac{c_1(y)}{c_1(y_0)}\right\}^2 + c_1(y)^2 \int_{y_0}^{y} \frac{c_2(y')\,dy'}{\{c_1(y')\}^3} \tag{27}$$

with the initial condition

$$y = y_0, \quad \chi = \chi_0, \quad \text{at } t = t_0.$$

If Δy is a small deviation (which needs not be small as $\Omega^{-1/2}$) from an equilibrium y_0, we may assume

$$c_1(y) = -\gamma\Delta y, \quad \text{and } c_2(y) \simeq c_2(y_0) = \text{const}$$

Then the relaxation is linear,

$$\frac{d}{dt}\Delta y = -\gamma\Delta y,$$

so that it decays exponentially

$$\Delta y = (\Delta y)_0\, e^{-\gamma t}$$

The variance is given by

$$\chi = \frac{c_2}{2\gamma} + \left(\chi_0 - \frac{c_2}{2\gamma}\right)(\Delta y/\Delta y_0)^2 \tag{31}$$

which attains its equilibrium value

$$\chi_e = c_2/2\gamma$$

as y relaxes to y_0. This type of relaxation and fluctuation is a common feature of near equilibrium states, a well-known example of which is the relaxation of velocity of a Brownian particle [5]. Eq. (17) is then the standard Fokker-Planck equation of Brownian motion. This is in accordance with Doob's theorem which asserts that a Gaussian-Markoffian process is necessarily associated with a linear relaxation process, or in other words, with a simple exponential decay of the correlation function [6].

3. In a more general case of non-linear relaxation, fluctuation is necessarily more complex, but the variance of fluctuation can be found from Eq. (27). A particularly interesting example of this

is an anomalous growth of fluctuation when the system relaxes to a stable equilibrium starting from a state close to an unstable equilibrium. Suppose that the system has more than one equilibrium positions as is illustrated by Fig. 1, where the point B is a stable equilibrium while A is an

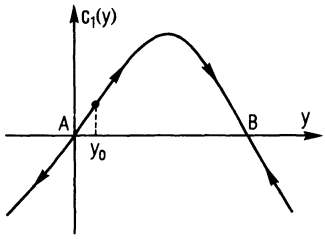


Fig. 1

unstable equilibrium. The arrows indicate how the system changes in relaxation process. Since $c_1(y_0)$ is initially very small, the first term on the right hand side of Eq. (27) grows very large as $c_1(y)$ attains its maximum value and also the second term will grow very large because the integral is almost divergent at the lower bound. An explicit expression can be easily worked out for example in the case that

$$c_1(y) = c\,y(a - y), \quad c_2(y) = \text{const}, \tag{32}$$

for which the expression (27) can easily be calculated.

Figure 2 shows some examples of such anomalous growth of fluctuation calculated for the example of a Weiss-Ising magnet [5]. The magnet is supposed to be initially almost demagnetized but with slight preference to one of the possible directions of magnetization. It is suddenly brought to a temperature somewhat below the Curie point. The magnetization will approach the

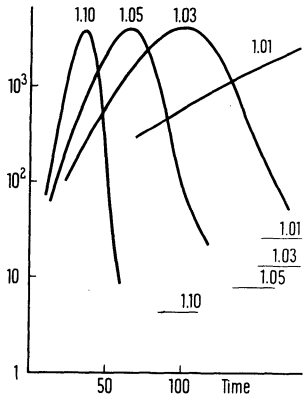


Fig. 2. Anomalous growth of the variance χ when a Weiss-Ising magnet relaxes to a finite magnetization below the Curie point. The initial value of magnetization is assumed to be $y_0 = 0.01$ with $\chi_0 = 0$. Horizontal lines indicate the final values of χ for different temperatures $\alpha = T_c/T$.

final equilibrium value. During this process of relaxation, fluctuation will increase anomalously, some thousand times as large as the final equilibrium values, in the intermediate stages. Intuitively this seems natural. When a society changes from an unstable state to a stable state it will experience enormous confusion and disorder. This sort of anomalous fluctuation has been observed by experiments for laser phenomena and has been given theoretical analysis [8]. Equation (27) seems to give a very simple account of this fluctuation.

If the initial state is exactly zero magnetization without any external field, the magnet does not know in which direction of magnetization it should relax when it is placed at some temperature

below the critical point. Correspondingly, Eq. (23) tells us only it stays at the unstable point. One may try Eq. (7) with a solution of the form

$$f(x, t) = b_1(t)x^2 + b_2(t)x^4 + \cdots \tag{33}$$

expecting it to approach a two peaked probability. Inserting (33) into Eq. (7), we obtain

$$\dot{b}_1(t) = 4(\alpha - 1)b_1 + 4b_1^2$$

for $b_1(t)$. In the ferromagnetic region, where $\alpha > 1$, $b_1(t)$ relaxes only to zero, not to a positive equilibrium value. Other coefficients are also shown to approach to zero, leading only to an unphysical solution. It seems thus that Eq. (7) has lost the guarantee of attaining equilibrium, which the original equation (1) may have possessed. It is not clear at this moment whether this is a fault of our Ansatz (6) or it bears something essential as a property of a macrovariable with critical behaviour.

4. In conclusion, we remark that real critical phenomena are far more delicate and complex than to be described by simple equations like (10) and (13). We would need a greater number of macrovariables, an infinite number of them finally, as the critical condition is approached. There the central limit theorem will not hold and any sort of asymptotic evaluation would become very much delicate for equilibrium as well as non-equilibrium distributions. However, our simple treatment serves for qualitative understanding of the phenomena and seems as itself to deserve more thorough studies.

References

[1] A part of this work was contributed by the author to the IUPAP conference on Statistical Mechanics at the University of Chicago, March 29-April 2, 1971.

[2] van Kampen, N. G.: Can. J. Phys. **39**, 551 (1961).

[3] van Kampen, N. G. in: Fluctuation Phenomena in Solids (ed. R. E. Burgess) New York-London, 1965.

[4] van Kampen, N. G. in: Fundamental Problems in Statistical Mechanics (Proceedings of the NUFFIC International Summer Course in Science The Netherland, Aug. 1961, ed. E. G. D. Cohen, Amsterdam, 1962, p. 199.

[5] Ming Chen Wang and Uhlenbeck, G. E.: Rev. Mod. Phys. **17**, 323 (1945).

[6] Doob, J. L.: Annals of Math. **43**, 351 (1942). See also Ref. [5].

[7] K. Kitahara, Master thesis submitted to Department of Physics, University of Tokyo, 1971.

[8] Arecchi, F. T., Degiorgio, V. and Querzola, B.: Phys. Rev. Letters **19**, 1168 (1967); Arecchi, F. T., and Degiorgio, V.: Phys. Rev. A3 1108 (1971); Risken, H. in: Progress in Optics (ed. E. Wolf), Amsterdam, 1970, p. 241.

New Method in Non-Equilibrium Statistical Mechanics of Cooperative Systems[1])

Kyozi Kawasaki

1. Introduction

It is a common every-day experience that a system of macroscopic size composed of a great number of interacting particles exhibits well-organized dynamical behavior which is qualitatively different from the behavior exhibited by individual constituent particles. The aim of the non-equilibrium statistical mechanics has been to understand such macroscopic behavior on the molecular level. The key ideas in this endeavor are (i) the identification of variables which describe such well-organized motions (also referred to as gross behavior) and are referred to variously as the collective coordinates, the macroscopic variables or the gross variables (in this article we adopt the last-mentioned terminology) and (ii) the dissipation of these well-organized motions into heat by the action of rapid random molecular motions (the so-called random forces). Here the underlying assumption is that the time scale characterizing the motion of gross variables is always well separated from the time scale for random forces, which appears to be justified under normal circumstances by the very existence of macroscopic laws such as classic hydrodynamic equations. Theoretically, however, the picture is not so clear cut. Consider, as an example, a gas described by the Boltzmann equation. If the above-mentioned assumption were correct, this implies that the Boltzmann collision operator has an eigenvalue spectrum which consists of five-fold zeros for five collision invariants corresponding to the five gross variables and the background spectrum for the remaining "random force" variables which is well separated from the zeros. This, however, seems to be true only for the gases with intermolecular forces which are Maxwellian or "harder." For "softer" intermolecular forces, there exists a continuous background spectrum extending to zero [1]. This means that the time scale of the gross behavior of the system can mingle with the time scale of "random force" motions and thus one may doubt the validity of classic hydrodynamic laws for these systems although to the author's knowledge the problem seems to have not been thoroughly investigated except of ionized gases where the classic hydrodynamics is drastically modified by the presence of long range Coulomb forces.

It has been realized recently that also in dense systems the situation is again not so clear cut. Modern derivations of classical hydrodynamic equations [2] assume that time correlation functions of "random force" entering transport coefficients should decay very fast compared to the macroscopic time scale. Contrary to the assumption, these time correlation functions were found to possess parts that decay as $t^{-d/2}$ at long times where d is the dimensionality of space [3,4].

[1]) The work supported by the National Science Foundation.

The influence of these long time tails is strong enough to invalidate the classical hydrodynamics in two-dimensions and to give singular corrections to the classical hydrodynamics in three-dimensions as well.

Other examples of this sort are provided by transport anomalies near the critical points where various transport coefficients either vanish or diverge, and strong non-hydrodynamical behavior is observed, which is a manifestation of the existence of large long time tails in time correlation functions near the critical point [5].

These circumstances call for a new approach to study long time behavior of time-correlation functions. The method to be described here emerged from attempts to understand transport anomalies near the critical point started by Fixman[1]). It is found that nonlinear coupling among gross variables is responsible for such a behavior in dense systems, and the theory is now known as the mode-mode coupling theory.

Although the mode-mode coupling theory as we now know was originally developed for the study of critical phenomena, we feel that the approach should be useful in other areas. Therefore, the theory will be developed in Sec. 2 in a somewhat general way, and a few possible areas of applications will be discussed in Sec. 4.

2. Formalism

Once the gross variables $(a_1, a_2, \ldots) \equiv \{a\}$ are identified, our new theory can be best illuminated by comparing it with closely related Mori's microscopic theory of the generalized Brownian motion for the gross variables [6]. From now on we extensively use the matrix notation where the molecular expression $A_1, A_2, \ldots$ for the gross variables $a_1, a_2, \ldots$ are arranged in a column matrix A where equilibrium averages of A are chosen to be zero. We also limit our presentation to the case of classical mechanics only since the extension to quantum mechanics can be done rather easily at the expense of complicating the notation a bit [7]. Mori derived an exact generalized Langevin equation for A of the following form:

$$\frac{d}{dt} A(t) = i\omega \cdot A(t) - \int_0^t ds \Phi(s) \cdot A(t-s) + F(t) \tag{2.1}$$

with

$$\omega \equiv -i \langle \dot{A} A^+ \rangle \cdot \langle A A^+ \rangle^{-1} \tag{2.2}$$

$$\Phi(t) \equiv \langle F(t) F(0)^+ \rangle \cdot \langle A A^+ \rangle^{-1} \tag{2.3}$$

$$F \equiv \dot{A} - i\omega \cdot A \tag{2.4}$$

where A^+ and F^+ are Hermitian adjoints of the column matrices A and F, respectively, and dots denote time derivatives.

The first, second, and third terms in (2.1) represent organized motions, their dissipation, and the random forces, respectively. The time evolution of F(t) is modified so that the organized motion represented by the first term of (2.1) does not influence its time dependence. Mori's formula beautifully embodies in its simplicity the results of correlation function theories of transport phenomena [2]. However, the real problem starts right here: how to evaluate the important memory function $\Phi(t)$. If the basic assumption mentioned in Sec. 1. were correct, F(t) would be genuinely random and $\Phi(t)$ decays quickly to zero with very little structure. We have already

[1]) Review in this area is given in References [7, 9, 10, 17] and [18];

seen that this is not always the case implying that the "random" force F(t) as well as $\Phi(t)$ should have rich structures buried in. This leads us to seek an alternative starting point which illuminates those structures and isolates genuinely the random force. Such an attempt led us to the following exact equation [7] for A(t):

$$\frac{d}{d}A(t) = v(\{A(t)\}) + \sum_j \int_0^t ds\, P_e^{-1}(\{A(t)\}) \frac{\partial}{\partial A_j^*(t-s)} [\hat{\phi}_j^0(s;\{A(t-s)\}) \times$$

$$\times P_e(\{A(t-s)\})] + f(t) \tag{2.5}$$

where P_e is the equilibrium probability distribution function and

$$v(\{a\}) \equiv \langle \dot{A}; \{a\} \rangle \tag{2.6}$$

$$f \equiv \dot{A} - \langle \dot{A}; \{A\} \rangle \tag{2.7}$$

Here $\langle X; \{a\} \rangle$ for any phase space function X denotes an average of X where $\{A\}$ assume fixed values $\{a\}$;

$$\langle X; \{a\} \rangle \equiv \langle X\delta(\{A\} - \{a\}) \rangle^0 / \langle \delta(\{A\} - \{a\}) \rangle^0 \tag{2.8}$$

Where $\langle \cdots \rangle^0$ is the equilibrium average and $\delta(\{A\} - \{a\})$ is the product of delta functions $\delta(A_j - a_j)$ over all the gross variables. $\hat{\phi}_j^0$ is a column matrix whose i-th element ϕ_{ij}^0 is given by

$$\phi_{ij}^0(t;\{a\}) \equiv \langle f_i(t) f_j^*(0); \{a\} \rangle. \tag{2.9}$$

The time evolution of f(t) is again modified so that the organized motion represented by the first term of (2.5) does not enter its dynamics. Each term of (2.5) is interpreted as follows. $v(\{A(t)\})$ describes slow organized time variation in which the state of the system instantly follows changes in A(t). The second term represents the dissipation of the organized motion due to the random force f(t) represented by the third term of (2.5).

Usefulness of (2.5) becomes apparent if the choice of the gross variables is made judiciously so that these variables exhaust slowly varying degrees of freedom. Then, any non-equilibrium state will quickly reduce into the local equilibrium state with fixed values of the gross variables, which then changes very slowly in the time scale of the gross variables. This means that a non-equilibrium average of f(t) quickly vanishes and also $\phi_{ij}^0(t; \{a\})$ decays to zero in a short time compared to the time scale of the gross variables. Therefore, (2.5) reduces to

$$\frac{d}{dt}A(t) = v(\{A(t)\}) + \sum_j P_e^{-1}(\{A(t)\}) \frac{\partial}{\partial A_j^*(t)} \times$$

$$\times [L_j^0(\{A(t)\}) P_e(\{A(t)\})] + f(t) \tag{2.10}$$

with $$L_j^0(\{a\}) \equiv \int_0^\infty dt \langle f(t) f_j^*(0); \{a\} \rangle. \tag{2.11}$$

Hence no memory effects remain.

In order to compare the equations (2.1) and (2.5), let us introduce a complete set of orthonormal functions ψ_n, n = 1, 2, . . ., of $\{a\}$ satisfying

$$\langle \psi_m^*(\{A\}) \psi_n(\{A\}) \rangle^0 = \delta_{mn} \tag{2.12}$$

$$\sum_n \psi_n(\{a\}) \psi_n^*(\{a'\}) = P_e^{-1}(\{a\}) \delta(\{a\} - \{a'\}). \tag{2.13}$$

When $P_e(\{a\})$ is Gaussian, the set $\{\psi_n\}$ reduces to that mentioned by Zwanzig [8]. In particular we can choose $\psi_0 = 1$ and the next members are

$$\psi_j(\{a\}) = a_j/\chi_j^{1/2} \tag{2.14}$$

with $$\chi_j = \langle |A_j|^2 \rangle^0 \tag{2.15}$$

provided that

$$\langle A_j^* A_l \rangle^0 = 0 \quad \text{for } j \neq l. \tag{2.16}$$

With the use of (2.8) and (2.13), we readily find

$$v(\{A\}) = i\omega \cdot A + v'(\{A\}) \tag{2.17}$$

$$v'(\{A\}) \equiv \sum_n{}' \langle \dot{A}\psi_n^*(\{A\})\rangle^0 \psi_n(\{A\}) \tag{2.18}$$

where the sum $\sum_n{}'$ starts with ψ_n quadratic in $\{A\}$. Equating (2.5) and (2.1) at $t = 0$, we then find

$$F = f + v'(\{A\}). \tag{2.19}$$

This result clearly demonstrates that even if our random force f changes rapidly[1]), Mori's "random force" F contains components which vary as slowly as A and gives (2.1) a non-Markovian character even if (2.10) is Markovian. The mixing of two time scales described in the preceding section may then be ascribed to slowly varying components of F. Even if we do not make the Markovian approximation (2.10), our new memory function ϕ_{ij}^0, (2.9), is expected to have far less structure than Mori's memory function $\Phi(t)$, (2.3), and hence should be easier to evaluate if necessary. Therefore, (2.5) or (2.10) provides us with a better starting point than (2.1) wherever mixing of time scales is important.

Having obtained our starting equation which still is a stochastic equation, we need to develop a method to obtain from it quantities of physical interest. We here describe a method to obtain time correlation functions or propagators of the form

$$G_{ij}(t) = \langle A_i(t) A_j^*(0) \rangle^0 \tag{2.20}$$

where A's are chosen to satisfy (2.16). We simplify the starting equation (2.5) by supposing that $\phi_{ij}^0(t; \{a\})$ does not depend upon $\{a\}$ and $P_e(\{a\})$ is Gaussian, which leads to

$$\frac{d}{dt} A_i(t) = \sum_j \left\{ i\omega_{ij} A_j(t) - \int_0^t \phi_{ij}^0(s) \chi_j^{-1} A_j(t-s) \right\} + v_i'(\{A(t)\}) + f_i(t) \tag{2.21}$$

where we have included the memory effects associated with ϕ_{ij}^0. At this point it is convenient to have an auxiliary set of variables $\{A_i^0(t)\}$ which satisfies

$$\frac{d}{dt} A_i^0(t) = \sum_j \left\{ i\omega_{ij} A_j^0(t) - \int_0^t \phi_{ij}^0(s) \chi_j^{-1} A_j^0(t-s) \right\} + f_i(t) \tag{2.22}$$

and $$A_i^0(0) = A_i(0), \tag{2.23}$$

[1]) Note that f is statistically independent of any functions of A: $\langle f; \{a\} \rangle = 0$.

and define a propagator $G^0_{ij}(t)$ for them which is analogous to (2.20). We then introduce fictitious free Boson creation and annihilation operators $\alpha_i^{0+}(t)$ and $\alpha_i^0(t)$ satisfying the commutation relations,

$$[\alpha_i^0(t), \alpha_j^0(t')] = [\alpha_i^{0+}(t), \alpha_j^{0+}(t')] = 0 \tag{2.24}$$

$$\begin{aligned}[\alpha_i^0(t), \alpha_j^{0+}(t')] &= -[\alpha_{i*}^{0+}(t), \alpha_{j*}^0(t')] \\ &= (\chi_i\chi_j)^{-1/2}\, G^0_{ij}(t-t'),\ t \geqq t'\end{aligned} \tag{2.25}$$

where i* is defined by $A_{i*} = (A_i)^*$ etc. Then it can be shown that $G_{ij}(t)$ is obtained as the following "vacuum expectation value" in the fictitious Boson space:

$$G_{ij}(t) = \langle 0 \mid a_i(t)a_j^+(0) \mid 0\rangle, \quad t \geqslant 0. \tag{2.26}$$

$a_i(t)$ is an operator in the Boson space corresponding to $A_i(t)$ and satisfies the integral form of the equation (2.21) for $A_i(t)$;

$$a_i(t) = a_i^0(t) + \sum_j \int_0^t ds\, G^0_{ij}(t-s)\chi_j^{-1} v_j'(\{a(s)\}) \tag{2.27}$$

where $a_i^0(t)$ corresponds to $A_i^0(t)$ and is given by

$$a_i^0(t) = \sqrt{\chi_i}\,[\alpha_i^0(t) + \alpha_{i*}^{0+}(t)]. \tag{2.28}$$

Note that $a_i(t)$ and $a_i^0(t)$ commute among themselves even for different times. $v_i'(\{a\})$ describes non-linear interaction among modes a's and determining its form is a problem of equilibrium statistical mechanics. Solving (2.27) and using (2.24) and (2.25), $G_{ij}(t)$ can be expressed in terms of $G^0_{lm}(t)$. The latter can be known if the $\phi^0(t)$'s are known. In this way our theory can be cast in the formalism of quantum field theory. The present formalism indeed is a somewhat different version of the one due to Zwanzig [8], which allows us to include the memory effects associated with $\phi^0_{ij}(t)$ as well.

3. Examples

Let us first illustrate the general formalism of the preceding section for the problem of decay of concentration fluctuations in binary fluid critical mixtures. Here the relevant gross variables are Fourier components of the local concentration fluctuation c_k and of the local transverse velocity v_k^α, (α = x, y, z) where k is restricted to be much smaller than the inverse of the average intermolecular distance [7, 9, 10]. After some simplifications the starting equation (2.21) reduces to

$$\frac{d}{dt}v_q^\alpha = -q^2\frac{\eta_0}{\rho}v_q^\alpha - i\frac{k_BT}{2\rho V}\sum_k\left(\frac{1}{\chi_k} - \frac{1}{\chi_{q-k}}\right)\left(k^\alpha - q^\alpha\frac{q\cdot k}{q^2}\right)c_k c_{q-k} + f_q^\alpha \tag{3.1}$$

$$\frac{d}{dt}c_q = -q^2\frac{L_0}{\chi_q}c_q - \frac{i}{V}\sum_k q\cdot v_k c_{q-k} + f_q^c \tag{3.2}$$

where $$\chi_k \equiv <|c_k|^2>/V, \tag{3.3}$$

ρ, T and V being the mass density, the temperature and the system volume, respectively, and η_0 and L_0 are the non-anomalous parts of the shear viscosity and concentration conductivity, respec-

tively. The shear viscosity can be shown to be finite at the critical point [9, 10], and as a first approximation we replace (3.1) by

$$\frac{d}{dt} v_{\mathbf{q}}^{\alpha} = -q^2 \frac{\eta^*}{\rho} v_{\mathbf{q}}^{\alpha} + f_{\mathbf{q}}^{\alpha} \tag{3.4}$$

where η^* is the so-called high frequency shear viscosity. Then the second order perturbation theory with respect to interaction in (3.2) allows us to eliminate $\mathbf{v}_{\mathbf{k}}$ and leads to the following result for the propagator of concentration fluctuation[1]):

$$\frac{d}{dt} G_{\mathbf{q}c}(t) = -\Gamma_q G_{\mathbf{q}c}(t) \tag{3.5}$$

with

$$\Gamma_q \equiv \frac{k_B T}{6\pi\eta^*} \xi^{-3} K_0(q\xi) \tag{3.6}$$

$$K_0(x) \equiv \frac{3}{4} [1 + x^2 + (x^3 - x^{-1}) \tan^{-1} x] \tag{3.7}$$

where ξ is the range of concentration fluctuation and the Ornstein-Zernike form for $\chi_{\mathbf{k}}$ has been assumed [9 10, 11]. The unimportant contribution from the first term of (3.2) has been dropped. For the binary mixture of 3-methylpentane-nitroethane, very detailed experimental data are available for ξ and Γ_q from the Rayleigh linewidth measurements [12], and also for the macroscopic shear viscosity $\eta(T)$ near the critical consolute point [13]. Tentatively identifying η^* with $\eta(T)$ the theoretical expression (3.6) (dashed line) is compared with data in Fig. 1. Although

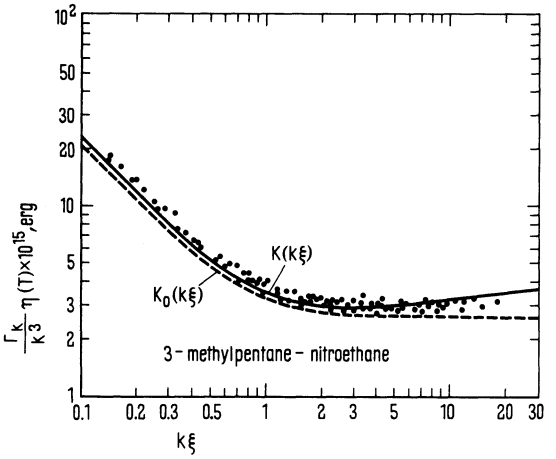


Fig. 1 Rayleigh linewidth in 3-methylpentane-nitroethane critical mixture. After ref. [14].

the theory fairly well describes the observed linewidth, there is a systematic discrepancy of 10 to 20%. On the other hand, the experimentally determined $\eta(T)$ exhibits a rather sharp cusp-like peak at the critical point. This suggests that η^* in fact should depend appreciably on q as well as on the frequency of the disturbance, which can be studied using both (3.1) and (3.2). This problem has been investigated recently by the author and Lo [14] as well as by Perl and

[1]) More precisely, this the second order perturbation theory for the "self-energy" that gives the decay rate in the more symmetrized scheme. For technical details, see Ref. [7].

Ferrell [15]. We considered the k-dependence of both η^* which appears in (3.6) and the real non-local shear viscosity η_q by treating the interaction terms in (3.1) and (3.2) self-consistently. The result is shown by the solid line Fig. 1, which improves agreement with experiments. As indicated by the work of Perl and Ferrel [15], consideration of frequency dependence would tend to bring down the theoretical curve somewhat especially at the large values of $q\xi$. However, the complete treatment of the frequency dependence appears to be quite formidable and is left for the future study.

Another problem which we briefly mention and in which the coupling among hydrodynamic modes plays a role is concerned with the non-linear dependence of stress tensor in incompressible fluids $T^{\alpha\beta}$ on the rate of shear, say, $D \equiv \partial v^y/\partial x$. This work was done in collaboration with J. D. Gunton. The starting equation would be

$$\frac{d}{dt} v_q^\alpha = -q^2 \frac{\eta}{\rho} v_q^\alpha - \frac{i}{V} \sum_k q' \cdot v_k \left[v_{q-k}^\alpha - \frac{q_\alpha}{q^2} q \cdot v_{q-k} \right] + f_q^\alpha \tag{3.8}$$

However, we have used a time-correlation function expression for $T^{\alpha\beta}$[3]. One might expect that for small D, $T^{\alpha\beta}$ would be a power series in D. However, mixing for time scales alters the situation completely, and we now obtain the following non-analytic expansion in $d \equiv | D\rho/\eta |$;

$$T^{xy} = -\eta D + 0.064 \frac{k_B T}{4\pi} d^{3/2} \tag{3.9}$$

$$T^{xx} - T^{yy} = 0.091 \frac{k_B T}{4\pi} d^{3/2} \tag{3.10}$$

$$T^{xx} - T^{zz} = 0.19 \frac{k_B T}{4\pi} d^{3/2}. \tag{3.11}$$

(3.9) represents dependence of shear viscosity on the rate of shear and (3.10) and (3.11) represent the normal stress effects.

4. Future Outlook

The method described in Sec. 2 and illustrated in Sec. 3 has been most successfully applied to transport anomalies near critical points where large fluctuations make it imperative to consider interactions among fluctuating modes. In a relatively short span of time most of the mysteries in this field have been fairly well elucidated. The most recent example is the finding that SF_6 whose strange behavior had mystified theorists for some time has now joined other well understood fluids [16]. Since progresses in this area have been reviewed occasionally [7, 9, 17, 18] we would like to discuss possible applications of the method in other areas.

Long time tails appearing in time correlation functions have already been mentioned and can be understood using the present theory or its variations [19]. From the point of view of molecular dynamics in liquids, it is interesting to know not only the asymptotic long time behavior of time correlation functions but also the behavior at shorter time. Since the present formalism keeps a room for the memory effects associated with $\phi_{ij}^0(t)$ which should become important for shorter

times, one may develop a kind of new expansion which works from long times back to shorter times although no such attempt has been made yet.

So far, we have been concerned with processes in which fluctuations decay in time. However, our approach can be used to study processes in which fluctuations grow in time such as nucleation phenomena, spinodal decomposition, etc. where we are no longer near an equilibrium state. One of the problems one encounters in trying to develop a general theory is to identify gross variables to describe a given physical process except possibly in a superconductor [20] where the order parameter provides such a variable. If we restrict ourselves to the vicinity of critical points, our approach seems to circumvent this difficulty. Again consider a binary fluid critical mixture. Since we may be dealing with spatially inhomogeneous states, we choose, instead of $c_{\mathbf{k}}$ and $v_{\mathbf{k}}^{\alpha}$, the coarse grained local gross variables $c(\mathbf{r})$ and $v^{\alpha}(\mathbf{r})$. Then we may start with the following Fokker-Planck type equation for the probability functional $P(\{c\},\{v\};t)$ which generalizes (3.1) and (3.2) to the situation where gross variables may largely deviate from their equilibrium values [8, 10].

$$\left(\frac{\partial}{\partial t}+\mathscr{H}\right) P(\{c\},\{v\};t)=0. \tag{4.1}$$

If we assume that η_0, ρ, L_0 and T remain constant independent of the gross variables[1]), the operator $\mathscr{H}$ is written as

$$\mathscr{H}=\mathscr{H}_c+\mathscr{H}_v+\mathscr{H}' \tag{4.2}$$

with

$$\mathscr{H}_c = L^0 \int d\mathbf{r} \frac{\delta}{\delta c(\mathbf{r})} \nabla^2 \left[k_B T \frac{\delta}{\delta c(\mathbf{r})}+\frac{\delta \Psi(\{c\})}{\delta c(\mathbf{r})}\right] \tag{4.3}$$

$$\mathscr{H}_v = \frac{\eta_0}{\rho^2} \sum_{\alpha\beta} \int d\mathbf{r} \frac{\delta}{\delta v^{\alpha}(\mathbf{r})}\left[\delta_{\alpha\beta}\nabla^2-\frac{\partial^2}{\partial r_{\alpha}\partial r_{\beta}}\right] \times \left[k_B T \frac{\delta}{\delta v^{\beta}(\mathbf{r})}+\rho v^{\beta}(\mathbf{r})\right] \tag{4.5}$$

$$\mathscr{H}' = -\int d\mathbf{r} \frac{\delta}{\delta c(\mathbf{r})} \nabla\cdot[\mathbf{v}(\mathbf{r})c(\mathbf{r})]$$

$$-\frac{1}{\rho} \sum_{\alpha} \int d\mathbf{r} \frac{\delta}{\delta v^{\alpha}(\mathbf{r})} c(\mathbf{r}) \left[\frac{\partial}{\partial r_{\alpha}} \frac{\delta \Psi(\{c\})}{\delta c(\mathbf{r})}\right], \tag{4.5}$$

where $\Psi(\{c\})$ is the extra free energy required to create a fluctuation $c(\mathbf{r})$ and is equal to $-(k_b T)^{-1}\ln P_e(\{c\})$ apart from a constant.

Near the critical point the local velocity changes very rapidly compared to the local concentration fluctuation. Thus, during slow variation of $c(\mathbf{r})$, the local velocity quickly reaches thermal equilibrium. This allows us to eliminate v by treating $\mathscr{H}'$ by the second order perturbation theory, and results in an additional $\mathscr{H}'_c$ involving only $c(\mathbf{r})$:

$$\mathscr{H}'_c = -\langle \mathscr{H}' \mathscr{H}_v^{-1} \mathscr{H}' \rangle_v^0 \tag{4.6}$$

---

[1]) This involves the assumption that fluctuations in ρ and T are statistically independent of concentration fluctuations. Although this is not strictly correct, the assumption considerably simplifies the problem.

where the average is over the thermal equilibrium state of velocity field. The final outcome is the following stochastic equation for the slowly varying probability distribution function of concentration fluctuation $P(\{c\};t)$:

$$\frac{\partial}{\partial t}P(\{c\};t) = -L^0 \int d\mathbf{r}\frac{\delta}{\delta c(\mathbf{r})}\nabla^2\left[k_BT\frac{\delta}{\delta c(\mathbf{r})}+\frac{\delta\Psi(\{c\})}{\delta c(\mathbf{r})}\right]P(\{c\};t)$$

$$+\int d\mathbf{r}\int d\mathbf{r}'\sum_{\alpha\beta}\frac{\delta}{\delta c(\mathbf{r})}\frac{\partial}{\partial r_\alpha}c(\mathbf{r})\left\{k_BT\frac{\delta}{\delta c(\mathbf{r}')}\frac{\partial}{\partial r'_\beta}c(\mathbf{r}')-\right.$$

$$\left.-c(\mathbf{r}')\left[\frac{\partial}{\partial r'_\beta}\frac{\delta\Psi(\{c\})}{\delta c(\mathbf{r}')}\right]\right\}\times\mathcal{T}^{\alpha\beta}(\mathbf{r}-\mathbf{r}')P(\{c\};t) \tag{4.7}$$

where $\mathcal{T}^{\alpha\beta}(\mathbf{r})$ is the Oseen tensor defined by

$$\mathcal{T}^{\alpha\beta}(\mathbf{r}) = \frac{1}{k_BT}\langle v^\alpha(\mathbf{r})\mathcal{H}_v^{-1}v^\beta(\mathbf{r}')\rangle_v^0 = \frac{1}{8\pi\eta}\left[\frac{1}{r}\delta_{\alpha\beta}+\frac{r_\alpha r_\beta}{r^3}\right], \tag{4.8}$$

the non-locality correction to the shear viscosity η being ignored here. As in Sec. 3, we expect that the second term of (4.7) dominates the first term near the critical point. The equation (4.7) should provide a good starting point to study dynamical behavior of concentration fluctuations near the critical point where the system may not be near the thermal equilibrium such as the decay of metastable states. The important point to note in this example is that the mode coupling not only results in the renormalization of transport coefficients [8] but also alters the form of the stochastic equation for slowly varying variables.

References

[1] Grad, H.: Accuracy and limits of applicability of solutions of equations of transport; dilute monatomic gases. Proceedings of the International Seminar on the Transport Properties of Gases (Eds. J. Kestin and J. Ross) (1964).

[2] Zwanzig, R.: The time-correlation functions and transport coefficients in statistical mechanics. Ann. Rev. Phys. Chem. **16**, 67 (1965).

[3] Yamada, T., and Kawasaki, K.: Non-linear effects in the shear viscosity of critical mixtures. Progr. Theor. Phys. (Kyoto) **38**, 1031 (1962).

[4] Alder, B. J. and Wainwright, T. E.: Decay of the velocity autocorrelation function. Phys. Rev. **A1**, 18 (1970).

[5] Sengers, J. V.: Transport Properties of Fluids near critical points. Varenna Lecture (to be published). Also see Refs. [12] and [16].

[6] Mori, H.: Transport, collective motion and Brownian motion. Progr. Theor. Phys. (Kyoto) **33**, 423 (1965).

[7] Kawasaki, K.: Mode-mode coupling theories with applications to dynamical critical phenomena (tentative) in: Phase transition and critical phenomena (eds. C. Domb and M. S. Green). New York – London (to be published).

[8] Zwanzig, R.: Nonlinear dynamics of collective modes. Presented at International Conference on Statistical Mechanics at Chicago, March-April 1971 (to be published).

[9] Kawasaki, K.: Dynamical behavior near critical point. Presented at International Conference on Statistical Mechanics at Chicago, March-April 1971 (to be published), and the references cited therein.

[10] Kawasaki, K.: Dynamical theory of fluctuations near the critical points. Varenna Summer Lectures, 1970 (to be published).

[11] Ferrell, R. A.: Decouple-mode dynamical scaling theory of the binary-liquid phase transition. Phys. Rev. Letters **24**, 1169 (1970).

[12] Chang, R. F., Keyes, P. H., Sengers, J. V. and Alley, C. O.: Dynamics of concentration fluctuations near the critical mixing point of a binary fluid. Phys. Rev. Letters, **27**, 1706 (1971), and also Ber. Bunsenges. phys. Chemie (to be published).

[13] Stein, A., Allegra, J. C., and Allen, G.: Shear viscosity of 3-methylpentane-nitroethane near the critical point. J. Chem. Phys. **55**, 4265 (1971). Tsai, B. C.: Master's Thesis. University of Akron 1970 (unpublished).

[14] Kawasaki, K, and Lo, S. M.: Nonlocal shear viscosity and order parameter dynamics in fluids. Phys. Rev. Letters (to be published). For further more detailed tests of the theory, see the following references:
S. P. Lee, W. Tscharnuter, and B. Chu: Kawasaki-Stokes-Einstein formula and dynamical scaling in the critical region of a binary liquid mixture: isobutyric acid in water. Phys. Rev. Letters **28**, 1509 (1972).
C. C. Lai and S. H. Chen: Evidence of mode-mode coupling and nonlocal shear viscosity in a binary mixture near the consolute point (preprint).
Chu, B. Lee, S. P. and Tscharnuter W.: Comparison of experiments with the modified mode-mode coupling theory of Kawasaki in binary fluids (preprint).

[15] Perl, R. and Ferrell, R. A.: Critical viscosity and diffusion in the binary liquid phase transition (preprint).

[16] Lim, T. K., Swinney, H. L., Langley, K. H., and Kachnowski, T. A.: The Rayleigh linewidth in SF_6 near the critical point. Phys. Rev. Letters **27**, Dec. (1971).
Feke, G. T., Hawkins, G. A., Lastovka, J. B., and Benedek, G. B.: Spectrum and intensity of light scattered from sulfur hexafluoride along the critical isochore. Phys. Rev. Letters **27**, 1780 (1971).

[17] Papers contained in: Dynamical Aspects of Critical Phenomena (eds. Budnick, J. I. and Kawatra, M. P.). New York, 1972.

[18] Kadanoff, L.: Mode-mode coupling near the critical point. J. Phys. Soc. (Japan) **26**, Supplement 122 (1969).

[19] Kawasaki, K. : Application of extended mode coupling theory to long time behavior of correlation functions. Progr. Theor. Phys. (Kyoto) **45**, 1691 (1971); **46**, Oct. (1971).
Ernst, M. H, Hauge, E. H., and van Leeuwen, J. M. J.: Asymptotic time behavior of correlation functions. Phys. Rev. Letters **25**, 1254 (1970).
Ernst, M. H. (preprint).

[20] Langer, J.: Statistical theory of the decay of metastable states. Ann. Phys. (N.Y.) **54**, 258 (1969); also a lecture delivered at International Workshop on Nucleation Theory and its Applications, Atlanta, April 1972.

Theory of Linear Non-Markoffian Processes

Hazime Mori

1. Introduction

Recently non-Markoffian properties have been studied in various linear dissipative phenomena. Markoffian theories assume that the time scale of the process of our interest is distinctly larger than the time scales involved in underlying processes [1]. If the process is not Markoffian, then we usually try to find Markoffian equations by choosing a new set of variables. Very often we can succeed in doing this just by adding new variables. Thus Markoffian equations have been constructed and used successfully in many problems. Recently, however, there begins to appear interesting phenomena which cannot be described by linear Markoffian equations. For example, we now have interesting high-frequency phenomena with short wavelengths where no simple hydrodynamic description holds. Another example is provided by anomalous transport phenomena near critical points where a nonlinear coupling among macroscopic modes seems to play a crucial role.

In this paper let us confine ourselves to linear dissipative processes. Linear responses to external perturbations such as light and neutron scattering are well-known to be described by the time-correlation functions of the type of [2, 3]

$$\xi_k(t) = \langle A_k(t) A_k^* \rangle, \tag{1}$$

Where $A_k(t)$ is a dynamical variable at time t, A_k^* the hermitian conjugate of $A_k(0)$, and the angular brackets denote an equilibrium average. Collective motions around equilibrium such as sound propagation and spin waves are also described by this type of time-correlation functions, if normal coordinates are taken as A_k [4, 5]. For simplicity we consider the classical-mechanical cases in the following.

The time evolution of the time-correlation functions can be determined very often from the linearized Langevin equation. In a simple case it takes the form

$$dA_k/dt \simeq i\omega_k A_k - \gamma_k A_k + R_k(t), \tag{2}$$

where ω_k is a frequency and γ_k a damping constant. $R_k(t)$ is a Langevin fluctuating force and is assumed to be completely independent of macroscopic modes; namely,

$$\langle R_k(t)\, G(A_k^*) \rangle = 0 \tag{3}$$

for an arbitrary function G of A_k^*. It is also assumed that the time correlation of $R_k(t)$ is extremely shortlived and is related to the damping constant γ_k by

$$\langle R_k(t)\, R_k^*(t') \rangle = 2\gamma_k \delta(t - t'), \tag{4}$$

where A_k is assumed to be normalized so that $\langle A_k A_k^* \rangle = 1$. Multiplying Eq. (2) by A_k^* and taking the average we obtain $\xi_k(t) = \exp[(i\omega_k - \gamma_k)t]$.

When we look at high-frequency regions in neutron scattering and ultrasonic attenuation, we often get frequency-dependent kinetic coefficients. In such a case a clear-cut separation of two time scales is not possible and the assumption (4) breaks down. Then the second term $\gamma_k A_k$ must be modified to have the memory effect. Near critical points, thermal fluctuations of macroscopic variables become anomalously large and a nonlinear coupling among macroscopic modes cannot be neglected. Then nonlinear terms must be added to Eq. (2), for instance, by replacing the first term $i\omega_k A_k$ by a nonlinear function $v_k(A_k)$. If the assumption (4) is still valid, then we obtain a nonlinear Markoffian equation. As shown by Kawasaki and others [6], such nonlinear terms bring about a frequency-dependent additional term to the damping constant.

In both of these two cases, we can derive an exact linear equation which has a simple structure similarly to Eq. [2]. That is [5],

$$dA_k/dt = i\omega_k A_k - \int_0^t \varphi_k(s)\, A_k(t-s)\, ds + f_k(t), \tag{5}$$

where $f_k(t)$ is a fluctuating force and satisfies

$$\langle f_k(t)\, A_k^* \rangle = 0, \tag{6}$$

$$\langle f_k(t)\, f_k^*(t') \rangle = \varphi_k(t - t'). \tag{7}$$

The kinetic coefficient is given by the Laplace transform of the memory kernel $\varphi_k(t)$,

$$\varphi_k(i\omega) = \int_0^\infty \exp(-i\omega t)\, \varphi_k(t)\, dt. \tag{8}$$

When the memory kernel $\varphi_k(t)$ is shortlived like (4), Eq. (5) is Markoffian and reduces to Eq. (2). In general, however, γ_k is not the kinetic coefficient which is observed in conventional experiments, but is a "bare" kinetic coefficient in the sense used by Zwanzig [7].

According to (3), $R_k(t)$ is uncorrelated with every power of A_k, whereas according to (6), $f_k(t)$ is uncorrelated only with the first power of A_k and includes terms nonlinear in macroscopic variables. As far as the time-correlation function (1) is concerned, however, this does not yield any inconvenience. Namely, if we multiply Eq. (5) by A_k^* and take the average, then the last term of (5) disappears. When a nonlinear coupling among macroscopic modes is important, $f_k(t)$ contains macroscopic processes through the nonlinear terms and the memory kernel has a macroscopic time scale in strong contrast to (4). This important difference between $R_k(t)$ and $f_k(t)$ comes up from the fact that in Eq. (2) the macroscopic motion of $A_k(t)$ is completely separated out from the microscopic rapid motion, whereas in Eq. (5) the projection of $A_k(t)$ onto the subspace spanned by A_k is extracted out.

In this paper we review first the relation between the non-linear Markoffian equation and the linear non-Markoffian equation by using the projection-operator method, and then the treatment of the memory kernel $\varphi_k(i\omega)$.

2. Projection onto subspace

2.1. General equations. By extending a fundamental idea in Langevin's theory of Brownian motion, we construct a prescription for extracting a relevant motion of dynamical variables from

their equations of motion. We use the projection-operator method which was introduced in the quantum theory of damping phenomena [8] and set forth by Zwanzig [9]. Let us consider a set of dynamical variables $\{A_\mu\}$ whose invariant parts are set to be zero, and split up their time evolution into two parts,

$$A_\mu(t) = \sum_\nu \Xi_{\mu\nu}(t)\, A_\nu(0) + A'_\mu(t). \tag{9}$$

The first part is a smooth function of time which linearly depends on the initial values of the set $\{A_\mu\}$, and corresponds to the systematic part of the Brownian motion [1]. The second part is a deviation from this smooth path, and corresponds to the fluctuating part of the Brownian motion. Let us introduce a Hilbert space of dynamical variables. Then the first part represents the projection of $A_\mu(t)$ onto the A subspace spanned by $\{A_\mu(0)\}$ if we set

$$\Xi_{\mu\nu}(t) = (A_\mu(t), A_\nu), \tag{10}$$

where the parenthesis denote the inner product, A_ν denotes $A_\nu(0)$, and $\{A_\mu\}$ is assumed to be an orthonormal set, satisfying $(A_\mu, A_\nu) = \delta_{\mu,\nu}$. In the classical case, the inner product is defined by [5, 10]

$$(F, G) = \langle F\, G^* \rangle = \int dx\, \rho(x)\, F(x)\, G^*(x), \tag{11}$$

where x denotes the representative point of the system in phase space and $\rho(x)$ is an equilibrium phase-space distribution function.

The time evolution of $A_\mu(t)$ is assumed to be given by

$$dA_\mu(t)/dt = i\mathscr{L} A_\mu(t), \tag{12}$$

where $\mathscr{L}$ is a time-independent linear operator. Then the second part of (9) is easily shown to take the form

$$A'_\mu(t) = \sum_\nu \int_0^t \Xi_{\mu\nu}(s)\, F_\nu(t-s)\, ds, \tag{13}$$

where $$F_\mu(t) = \exp[t(1-\mathscr{P})i\mathscr{L}]\,(1-\mathscr{P})i\mathscr{L} A_\mu(0). \tag{14}$$

$\mathscr{P}$ is the projection operator defined by

$$\mathscr{P} G(x) = \sum_\nu (G, A_\nu)\, A_\nu(x), \tag{15}$$

which satisfies $\mathscr{P}(1-\mathscr{P}) = 0$. $F_\mu(t)$ corresponds to the fluctuating force of the Brownian motion and is orthogonal to the A subspace;

$$(F_\mu(t), A_\nu) = 0. \tag{16}$$

It should be also noted that the time evolution of the fluctuating forces $F_\mu(t)$ is governed by the linear operator $(1-\mathscr{P})i\mathscr{L}$.

A linear equation of motion for $A_\mu(t)$ can be derived from (12). Namely, defining the frequency matrix

$$i\Omega_{\mu\nu} = (i\mathscr{L} A_\mu, A_\nu), \tag{17}$$

we obtain [11]

$$dA_\mu/dt = \sum_\nu i\Omega_{\mu\nu}A_\nu - \sum_\nu \int_0^t \Phi_{\mu\nu}(s)\, A_\nu(t-s)\, ds + F_\mu(t), \tag{18}$$

where $$\Phi_{\mu\nu}(t) = -(i\mathscr{L}F_\mu(t), A_\nu). \tag{19}$$

In particular, if $(\mathscr{L}F, G) = (F, \mathscr{L}G)$, then (19) leads to [5]

$$(F_\mu(t), F_\nu(t')) = \Phi_{\mu\nu}(t-t'). \tag{20}$$

The most important feature of (18) is the fact that its systematic part is linear. In the derivation of this equation, the variables $\{A_\mu\}$ are quite arbitrary except the condition that their time-evolution operator $i\mathscr{L}$ is a time-independent linear operator. They have to be chosen properly depending on the problem concerned. In the following we find several examples.

2.2. Linear non-Markoffian equation for the collective variables. Let us consider an orthonormal set of collective variables for describing the macroscopic state of the system and denote them by $\{A_k\}$, where A_{-k} is included to denote A_k^*. In isotropic Heisenberg ferromagnets they are the Fourier components of the spin density and the energy density, and in simple fluids they are the Fourier components of the number density, the energy density and the momentum density. But they have to be properly orthonormalized. The time evolution of these variables are governed by the Liouville operator L, which satisfies (LF, G) = (F, LG). Application of Eq. (18) to these variables thus leads to [5]

$$dA_k/dt = \sum_\ell i\omega_{k\ell}\, A_\ell - \sum_\ell \int_0^t \varphi_{k\ell}(s)\, A_\ell(t-s)\, ds + f_k(t), \tag{21}$$

where the coefficients and the fluctuating forces are obtained from (17), (20), and (14);

$$i\omega_{k\ell} = \langle \dot{A}_k\, A_\ell^* \rangle, \quad \varphi_{k\ell}(t) = \langle f_k(t)\, f_\ell^*(0) \rangle, \tag{22}$$

$$f_k(t) = \exp[t(1-P_A)iL]\,(1-P_A)iLA_k(0), \tag{23}$$

where P_A is the projection operator onto the subspace spanned by the collective variables $\{A_k\}$. Since $\langle f_k(t)A_\ell^* \rangle = 0$ from (16), the fluctuating forces $f_k(t)$ are orthogonal only to the first power of the collective variables and thus include terms nonlinear in the collective variables. When a nonlinear coupling among the collective variables cannot be neglected, therefore, the macroscopic motion is involved in the time evolution of $f_k(t)$ and $\varphi_{k\ell}(t)$ through the nonlinear terms.

2.3. Nonlinear Langevin equation for the collective variables. In order to separate the macroscopic motion completely from the microscopic motion, we take as $A_\mu(x)$ the distribution function $g_a(x)/\sqrt{w(a)}$, where

$$g_a(x) = \delta(A(x) - a) \equiv \prod_k \delta(A_k(x) - a_k), \tag{24}$$

$$w(a) = \langle g_a \rangle = \int dx\, \rho(x)\, \delta(A(x) - a), \tag{25}$$

where A and a denote the set $\{A_k\}$ and its values $\{a_k\}$, respectively. w(a) represents the equilibrium distribution of the $\{a_k\}$. Then the projection (15) takes the form

$$P_Z\, G(x) = \int dx'\, \rho(x')\, G(x')\, \delta(A(x') - A(x))/w(A(x)), \tag{26}$$

which is identical to the projection operator introduced by Zwanzig [12]. Then Eq. (18) leads to an equation for $g_a(t)$ which is obtained by adding to Zwanzig's generalized Fokker-Planck equation the fluctuating force

$$F_a(t) = -\exp[t(1-P_Z)iL] \sum_k R_k(0) \frac{\partial}{\partial a_k} \delta(A(0)-a), \tag{27}$$

where $$R_k(t) = \exp[t(1-P_Z)iL](1-P_Z)iLA_k(0). \tag{28}$$

From this equation for $g_a(t)$ we can obtain a nonlinear equation for $A_k(t)$ with the aid of

$$A_k(t) = \int a_k g_a(t)\, da. \tag{29}$$

When we can assume that

$$\langle R_k(t)\, R_\varrho^*(0)\, g_a \rangle / \langle g_a \rangle \simeq \langle R_k(t)\, R_\varrho^*(0) \rangle \simeq 2\gamma_{k\varrho}\delta(t), \tag{30}$$

$$w(a) = C \exp\left(-\frac{1}{2} \sum_k a_k a_k^*\right), \quad \text{(Gaussian)}, \tag{31}$$

we thus obtain a nonlinear Langevin equation,

$$dA_k/dt = v_k(A) - \sum_\varrho \gamma_{k\varrho} A_\varrho + R_k(t), \tag{32}$$

where $$v_k(a) = \langle \dot{A}_k g_a \rangle / \langle g_a \rangle, \quad \gamma_{k\varrho} = \int_0^\infty \langle R_k(t)\, R_\varrho^*(0) \rangle\, dt. \tag{33}$$

In similar approximations the equation for $g_a(t)$ takes the form

$$\partial g_a(t)/\partial t = D\, g_a(t) + F_a(t), \tag{34}$$

where D is the Fokker-Planck operator,

$$D = -\sum_k \frac{\partial}{\partial a_k} v_k(a) + \sum_k \sum_\varrho \gamma_{k\varrho} \left[\frac{\partial}{\partial a_k} a_\varrho + \frac{\partial}{\partial a_k} \frac{\partial}{\partial a_\varrho^*}\right]. \tag{35}$$

The fluctuating forces $R_k(t)$ and $F_a(t)$ are orthogonal to $g_a(0)$ and thus turn out to satisfy

$$\langle R_k(t)\, G(A^*) \rangle = \langle F_a(t)\, G(A^*) \rangle = 0 \tag{36}$$

for an arbitrary function G of A^*.

3. Relation between the nonlinear Langevin and the linear non-Markoffian equation

The time evolution of the time-correlation functions,

$$\xi_{km}(t) = \langle A_k(t)\, A_m^*(0) \rangle, \tag{37}$$

can be determined from either the linear non-Markoffian equation (21) or the nonlinear Markoffian equation (32). Next we discuss the relation between these two approaches by applying the linearization procedure of 2.1 to a nonlinear Markoffian equation. Using (29) the time-correlation functions can be written as

$$\xi_{km}(t) = \int a_k\, P(a,t)\, da, \quad P(a,t) \equiv \langle g_a(t)\, A_m^*(0) \rangle. \tag{38}$$

$P(a, t)$ is the probability distribution function of the $\{a_k\}$ with the initial condition $P(a, 0) = a_m^* w(a)$, and its time evolution is governed by the Fokker-Planck operator D. Using its adjoint operator Λ defined by

$$\int [\Lambda f(a)]\, g(a)\, da = \int f(a)[Dg(a)]\, da, \tag{39}$$

we transform (38) into

$$\xi_{km}(t) = \langle a_k(t)\, a_m^* \rangle \equiv \int a_k(t)\, a_m^*\, w(a)\, da, \tag{40}$$

where $$a_k(t) \equiv \exp(t\Lambda)\, a_k. \tag{41}$$

Λ is written as $\Lambda = \Lambda_0 + \Lambda_1$,

$$\Lambda_0 = \sum_k \sum_\varrho \left[(i\omega_{k\varrho} - \gamma_{k\varrho})\, a_\varrho \frac{\partial}{\partial a_k} + \gamma_{k\varrho} \frac{\partial}{\partial a_k} \frac{\partial}{\partial a_\varrho^*}\right], \tag{42}$$

$$\Lambda_1 = \sum_k v_k'(a) \frac{\partial}{\partial a_k}, \tag{43}$$

where $$v_k'(a) \equiv v_k(a) - \sum_\varrho i\omega_{k\varrho} a_\varrho. \tag{44}$$

We apply Eq. (18) to the time evolution of $a_k(t)$ by taking Λ as $i\mathscr{L}$, defining the inner product from (40), and introducing the projection operator

$$P_a G(a) = \sum_\varrho \langle G(a)\, a_\varrho^* \rangle\, a_\varrho. \tag{45}$$

Since $(\Lambda a_k, a_\varrho) = i\omega_{k\varrho} - \gamma_{k\varrho}$, we thus obtain

$$da_k/dt = \sum_\varrho (i\omega_{k\varrho} - \gamma_{k\varrho}) a_\varrho - \sum_\varrho \int_0^t \psi_{k\varrho}(s)\, a_\varrho(t - s)\, ds + q_k(t), \tag{46}$$

where $$\psi_{k\varrho}(t) = -(\Lambda q_k(t), a_\varrho) = \langle q_k(t)\, q_\varrho^*(0) \rangle, \tag{47}$$

$$q_k(t) = \exp[t(1 - P_a)\Lambda]\,(1 - P_a)\Lambda a_k = \exp[t(1 - P_a)\Lambda]\, v_k'(a). \tag{48}$$

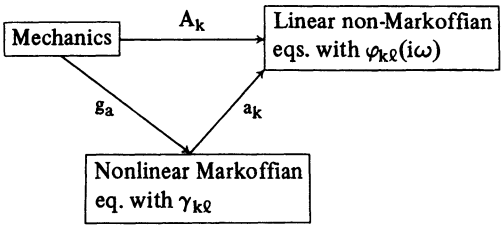
Since $v_k'(a) = (1 - P_a)v_k(a)$, it turns out that the fluctuating force $q_k(t)$ and the memory kernel $\psi_{k\varrho}(t)$ come out from the nonlinear coupling among the collective variables $\{a_k\}$. Eq. (46) has the same structure as Eq. (21). In fact, these two equations have an intimate relation. Since $\langle q_k(t) a_m^* \rangle = 0$ and $\langle f_k(t) A_m^* \rangle = 0$, we obtain two equations for $\xi_{km}(t)$ from (46) and (21). Comparing them we find that

$$\varphi_{k\varrho}(t) = 2\gamma_{k\varrho}\, \delta(t) + \psi_{k\varrho}(t). \tag{49}$$

The time evolution of $\varphi_{k\varrho}(t)$ and $\psi_{k\varrho}(t)$ are governed by the linear operators $(1 - P_A)iL$ and $(1 - P_a)\Lambda$, respectively. The projection operators P_A and P_a are related to each other by

$$P_A G(A(x)) = \int [P_a G(a)]\, \delta(A(x) - a)\, da, \tag{50}$$

and they are physically equivalent. The Langevin fluctuating force $R_k(t)$ is governed by $(1 - P_Z)iL$. Zwanzig's projection operator P_Z is quite different from both of P_A and P_a. This difference enabled us to have the two routes shown in the following graph:



The first route goes directly to the linear equation for $A_k(t)$ with the kinetic coefficient $\varphi_{k\ell}(i\omega)$. The second route goes first to the nonlinear Markoffian equation for $A_k(t)$ with $\gamma_{k\ell}$ through the linear Markoffian equation for $g_a(t)$, and then to the linear equation for $a_k(t)$ with $\bar{\varphi}_{k\ell}(i\omega) = \gamma_{k\ell} + \psi_{k\ell}(i\omega)$. These two routes are equivalent as far as the clear-cut separation of the two time scales is valid. The second route was developed recently by Zwanzig [7]. In this paper we reformulated this more exactly by using the projection-operator method.

4. Time-correlation functions

To study the memory kernels further, let us consider simple cases where $\xi_{km}(t) = \xi_k(t)\delta_{k,m}$. Then, since the matrices $\omega_{k\ell}$, $\gamma_{k\ell}(t)$, $\varphi_{k\ell}(t)$ are all diagonal, we have

$$d\xi_k/dt = i\omega_k\xi_k - \int_0^t \varphi_k(s)\xi_k(t-s)\,ds, \tag{51}$$

where $i\omega_k = \langle \dot{A}_k A_k^* \rangle$ and

$$\varphi_k(t) = \langle f_k^* \exp[t(1 - P_A)iL] f_k \rangle. \tag{52}$$

When we can use (49), we have

$$\varphi_k(t) = 2\gamma_k\delta(t) + \langle v_k'^* \exp[t(1 - P_a)\Lambda] v_k' \rangle. \tag{53}$$

If the nonlinear streaming velocity $v_k'(a)$ is small, then we may replace Λ in the second term of (53) by its unperturbed part Λ_0 to obtain

$$\psi_k(t) \simeq \langle v_k'^* \exp[t(1 - P_a)\Lambda_0] v_k' \rangle. \tag{54}$$

We can often assume that the leading nonlinear term is bilinear in the collective variables $\{a_k\}$ and take the form

$$v_k'(a) = -i \sum_\ell{}' V_{k\ell} a_\ell a_{k-\ell}, \tag{55}$$

where $V_{k,\ell} = V_{k,k-\ell}$ and $\sum_\ell'$ does not include the short-wavelength components of the collective variables. Then using (42), we can easily show that

$$\psi_k(t) \simeq 2 \sum_\ell{}' |V_{k\ell}|^2 \exp[-t(\gamma_\ell + \gamma_{k-\ell})]. \tag{56}$$

This is equivalent to the expression obtained by Zwanzig with the use of a second-order perturbation theory [7]. In simple cases, suffices k and ℓ denote the wave vectors of the Fourier components and we have $\gamma_k \sim k^2$, $|V_{k\ell}|^2 \sim k^2 \ell^x$ for small values of k, where x is a number. Then the integral over wave vector ℓ in the small wave number region leads to

$$\psi_k(t) \sim k^2 t^{-(d+x)/2}, \tag{57}$$

where d denotes the dimensionality of the system. This determines the long-time behavior of the memory kernel, and leads to 1/t in two dimensions if x = 0. Then the frequency-dependent kinetic coefficient diverges at low frequency as $\log(1/\omega)$ and is essentially non-Markoffian. This problem was investigated by a number of authors for various time-correlation functions [13, 7].

One way to avoid a perturbation calculation is to use a self-consistent approach. This can be obtained by replacing $\exp(-t\gamma_\ell)$ in (56) by $\xi_\ell(t)$ and then inserting $\varphi_k(t)$ thus obtained into (51). Then we have

$$d\xi_k/dt - (i\omega_k - \gamma_k)\xi_k \simeq -2 \sum_{\ell}{}' |V_{k\ell}|^2 \int_0^t ds \xi_\ell(s)\xi_{k-\ell}(s)\xi_k(t-s). \tag{58}$$

This is equivalent to the equations obtained by Kawasaki with the aid of the renormalization technique [6]. From (52) we can obtain an equation of similar structure. As an example let us consider the spin-correlation function in the paramagnetic phase of an isotropic Heisenberg ferromagnet, taking as A_k the Fourier components of the spin density. Then the fluctuating force f_k has the same form as (55), $-i\Sigma_\ell J_{k\ell} A_\ell A_{k-\ell}$. By approximating $t(1 - P_A)iL$ by tiL and replacing the four-spin correlations by products of the pair-spin correlations [14], we obtain

$$d\xi_k/dt \simeq -2 \sum_{\ell} |J_{k\ell}|^2 \int_0^t ds \xi_\ell(s)\xi_{k-\ell}(s)\xi_k(t-s), \tag{59}$$

where the sum includes the short-wavelength components. This equation is valid even if a separation of two time scales is not possible. This type of equation has been derived by a number of authors with the use of various approaches [15]. By solving (59) numerically for large values of k, Blume and Hubbard obtained the results which agree very well with Windsor's computer-simulation results at high temperature [16].

Finally let me add one word on the continued-fraction expansion of time-correlation functions [17, 18]. This type of expansion has been used with considerable success to study short-wavelength collective modes in liquids and magnetic systems [19].

References

[1] Chandrasekhar, S.: Rev. Mod. Phys. **15**, 1 (1943). Wang, M. C. and Uhlenbeck, G. E.: Rev. Mod. Phys. **17**, 323 (1945).

[2] van Hove, L.: Phys. Rev. **95**, 1374 (1954). Kubo, R. and Tomita, K.: J. Phys. Soc. Japan **9**, 888 (1954).

[3] Kubo, R.: J. Phys. Soc. Japan **12**, 570 (1957).

[4] Kadanoff, L. P. and Martin, P. C.: Ann. Phys. **24**, 419 (1963).

[5] Mori, H.: Prog. Theor. Phys. **33**, 423 (1965).

[6] Kawasaki, K.: Ann. Phys. **61**, 1 (1970).
Kawasaki, K.: paper presented at the International Conference on Statistical Mechanics, Univ. of Chicago, Chicago, 1971, and the references cited therein.

[7] Zwanzig, R.: paper presented at the International Conference on Statistical Mechanics, Univ. of Chicago, 1971.
[8] Heitler, W.: Quantum Theory of radiation. Oxford 1954.
Kohn, W. and Luttinger, J. M.: Phys. Rev. **108**, 590 (1957).
[9] Zwanzig, R.: J. Chem. Phys. **33**, 1338 (1960).
[10] In Ref. [5], the inner preduct is denoted by (F, G*).
[11] Mori, H. and Okamoto, H.: Prog. Theor. Phys. **40**, 1287 (1968).
[12] Zwanzig, R.: Phys. Rev. **124**, 983 (1961).
[13] Dorfman, J. R. and Cohen, E. G. D.: Phys. Rev. Letters **25**, 1257 (1970).
Ernst, M. H., Hauge, E. H., and van Leeuwen, J. M. J.: Phys. Rev. Letters **25**, 1254 (1970).
Pomeau, Y.: Phys. Rev. **A3**, 1174 (1971).
Kawasaki, K.: Phys. Letters **32A**, 379 (1970); **34A**, 12 (1971).
[14] Kawasaki, K.: J. Phys. Chem. Solids **28**, 1277 (1967).
[15] Résibois, P. and DeLeener, M.: Phys. Rev. **152**, 305 (1968); **152**, 318 (1968).
Wegner, F.: Z. Physik **216**, 433 (1968).
[16] Blume, M. and Hubbard, J.: Phys. Rev. **B1**, 3815 (1970).
Hubbard, J.: J. App. Phys. **42**, 1390 (1971).
[17] Mori, H.: Prog. Theor. Phys. **34**, 399 (1965).
[18] Dupuis, M.: Prog. Theor. Phys. **37**, 502 (1967).
[19] Sears, V. F.: Can. J. Phys. **47**, 199 (1969); **48**, 616 (1970).
Lovesey, S. W.: J. Phys. (C) **4**, 3057 (1971).
Lovesey, S. W. and Meserve, R. A.: a preprint.
Tomita, H. and Mashiyama, H.: a preprint.

Fluctuations near Phase Transitions in Restricted Geometries

S. Grossmann

1. Introduction

This talk shall be concerned with some special aspects of the physics of phase transitions. They are of general interest to synergetics and may provide a unifying language to a lot of cooperative phenomena. Although these are met in a large variety of manifestations the methods to treat them are always similar.

As this symposium shall be an opportunity of interdisciplinary exchange of ideas it might be polite to mention also some concepts and results, which are familiar to the experts. Beginning with Landau's static order parameter description of phase transitions the main concern will be the fluctuations of the order parameter field. I shall concentrate on systems of lower dimension than 3, although some statements are expected to hold for D = 3 too. Special examples will be superconducting films (D = 2) or filaments (D = 1) as well as lasers (D = 0), although – again – much broader applicability is given. Especially the behaviour of the specific heat is considered as well as the correlation times of the order parameter and its intensity near phase transitions.

2. The order parameter and its fluctuations

A phase transition in a system of many particles or quasi particles is the onset of a special macroscopic feature if the external conditions are changed. Examples are: (i) Decreasing the temperature T of a lattice of interacting permanent magnets leads to spontaneous magnetization. (ii) Decreasing T or, at sufficiently low temperature, decreasing a strong applied magnetic field B, the electrons in certain metals become superconducting. (iii) Increasing the mean velocity u of a viscous fluid flowing through a pipe leads to the onset of oscillatory behaviour and finally to fully developed turbulence. (iv) Increasing the pumping rate of laser active atoms in a resonance cavity causes the photons to condensate macroscopically in one or a few modes of the cavity: above a critical inversion $\nu = N_{excited} - N_{ground}$ the resonator becomes a laser. – I am sure that more examples will be given in the following lectures.

The examples I have mentioned are different ones from a thermodynamic point of view. The common feature is: Change an external parameter, a; at a certain "critical" value a_0 (chosen as $a_0 = 0$) an internal ordering appears, which is measured by a quantity Ψ, typical for the macroscopically observed feature in the ordered phase. Referring to the examples: $a \sim (T - T_c)$, $\sim (B - B_{c2})$, $\sim (u - u_{Reynolds})$, $\sim (\nu - \nu_{threshold})$ are the external parameters; the order parameter Ψ is given respectively by magnetization M, energy gap or number of Cooper pairs $\Delta \sim \langle \Psi_\uparrow \Psi_\downarrow \rangle$, amplitude of oscillation u_1, electric field strength E_λ of the lasing mode. Thus Ψ can be real or complex.

The actual value of the order parameter Ψ as a function of the external conditions, especially of a, is a dynamical balance of the ordering by interaction and the disordering by fluctuations in the microscopic degrees of freedom. It has therefore to be determined by statistical mechanical methods from the relevant microscopic Hamiltonian H_{mi}.

The fundamental physical quantity is the free energy. Without detailed calculations its qualitative features are understood with Landau's [1] famous idea: The microscopic motion manifests itself in some value of the order parameter Ψ. The macroscopic properties due to ordering should be functions of Ψ only. Especially that part of the free energy, ΔF, which is associated with ordering, can be represented by an expansion in $|\Psi|^2$, as the order Ψ is small near the transition.

$$\Delta F = a\,|\Psi|^2 + \frac{b}{2}\,|\Psi|^4,\ a \sim \text{external variable},\ b = \text{const.} \tag{1}$$

Just those microscopic configurations are realized that are favourable with respect to the energy it costs to have certain order Ψ in the system. Therefore the actual value of the order parameter is the one that minimizes ΔF.

$$\frac{\delta \Delta F}{\delta \Psi} = 0 \rightharpoonup |\Psi_0|^2 = \begin{cases} 0 & a > 0, \text{ disorder} \\ \dfrac{-a}{b} & a < 0, \text{ order} \end{cases} \tag{2}$$

But as ordering is a cooperative behaviour of many particles, it will not be constant in the system. In general only within the effective range of correlation a certain Ψ is realized. Particles far separated might prefer to establish another value of Ψ. Also at the same point but at a later time the microscopic movement might end up with another Ψ. Therefore Ψ will "fluctuate" in space and time. Eq. (1) might then be considered to be the free energy of all microscopic degrees of freedom in a correlated part $\Delta^3 r$ of the system at a given moment. The total energy is the sum of all parts.

$$H = \int d^3r \left\{ a\,|\Psi|^2 + \frac{b}{2}\,|\Psi|^4 + c\,|\nabla \Psi|^2 + d\,|\dot{\Psi}|^2 + \mathrm{Re}(\mu \cdot \Psi) \right\}. \tag{3}$$

The third term contributes the energy necessary to realize different Ψ at different points. Its order of magnitude can be estimated: Be Δx the distance in which Ψ changes appreciably. (3) tells us: There is no problem as long as $c(1/\Delta x^2) \lesssim |a|$. But disturbances with smaller wave length Δx cost too much energy and are only seldom realized. Thus $\sqrt{c/|a|} =: \xi$ is a typical correlation length of the order parameter.

The fourth term contributes the energy necessary to change Ψ in time. This energy arises, as the cooperative ordering has some effective inertia. The typical time for the molecules to reorganize is $\sim \sqrt{d/|a|}$. Changes on a smaller time scale are extremely seldom realized. This is expressed by the "kinetic" energy in (3).

The last term represents the effect of an external field $\mu(r)$, useful to calculate the susceptibility.

The main physical supposition underlying the energy density in Eq. (3) is its regularity (i) in Ψ as well as (ii) in the external parameters $a \sim (T - T_c)$ and μ together with b, c, d as constants. The justifying idea is that in finite parts of the system no nonanalyticity should be expected. But there are systems for which the assumption (ii) certainly is not true, e.g. the 2-dimensional Ising ferromagnet. Such will not be considered in the following.

3. The "energy principle"

The physical meaning of H is: it represents the free energy of the many particle system with the restriction that microscopic movement ends up in a certain cooperative field $\Psi(\mathbf{r}, t)$.

$$H(\Psi) = -\kappa T \ln \mathrm{Tr}_{(\Psi \text{ fixed})}\, e^{-\beta H_{mi}}. \tag{4}$$

H is a constrained free energy. The total free energy is found by summing over all possible fields of ordering.

$$e^{-\beta \Delta F} = \int \delta\Psi \cdot \mathrm{Tr}_{(\Psi \text{ fixed})} e^{-\beta H_{mi}} = \int \delta\Psi\, e^{-\beta H(\Psi)}. \tag{5}$$

Therefore H – forgetting its microscopic origin – represents the energy necessary to realize Ψ and the Boltzmann factor $e^{-\beta H(\Psi)}$ is the probability to find a certain field $\Psi(\mathbf{r}, t)$, $\dot{\Psi}(\mathbf{r}, t)$ [2].

This leads us to a twofold use of the cooperative energy H.

i). If the system is in a surrounding not changing in time, the fluctuations are stationary. This is the case in thermal equilibrium as well as in steady states far from equilibrium (eigen-oscillations being separated). Therefore one can ask for temporal mean values, mean square deviations, etc. of the order parameter, its modulus, etc. Eq. (5) tells us that – as is customary in statistical physics – we can calculate these time averages as ensemble averages, especially as the canonical average. The relevant energy as far as the cooperative effect is concerned is H.

Calling the sum[1]) over all possible fields the "trace", we have e.g.

$$\langle |\Psi(\mathbf{r}, t)|^2 \rangle = \mathrm{Tr}\left(|\Psi|^2 \frac{1}{Z} e^{-\beta H(\Psi)} \right), \quad Z \equiv \mathrm{Tr}\, e^{-\beta H(\Psi)} = e^{-\beta \Delta F}. \tag{6}$$

Here the kinetic energy cancels. The total free energy, including the order parameter fluctuations, is

$$F = -\kappa T \ln \mathrm{Tr}\, e^{-\beta H} + F_0. \tag{7}$$

The energy distribution parameter β often may be kept constant, $1/\kappa T_c$. Its role is played by the quality Q in a laser near threshold.

ii. Additional information is needed to calculate time correlation functions. The energy principle is used again to derive an equation of motion with H as Hamiltonian.

$$d\ddot{\Psi} + \gamma\dot{\Psi} + (a + b|\Psi|^2 - c\nabla^2)\Psi = 0. \tag{8}$$

A damping term is added ad hoc to take account of the coupling of the cooperative mode Ψ to the many microscopic degrees of freedom. Eq. (8) is to be read in (not time reversible) correlation functions, where damping occurs.

If a is the (dimensionless) relative distance to the transition, e.g. $a = (T - T_c)/T_c$ or $(\nu - \nu_{thresh})/\nu_{thresh}$, the characteristic time $\sqrt{d}$ of inertia as well as the damping time γ are physically expected to be given by the relevant microscopic mechanism. This is the thermal

---

[1]) More rigorous: the functional integral.

motion for superconductors or the atomic as well as the cavity damping ($\gamma_\perp^{-1}$ and κ^{-1} respectively) for lasers. In fact,

$$\left.\begin{aligned}\sqrt{d} &= \sqrt{\frac{7\zeta(3)}{16\pi^2}}\frac{\hbar}{\kappa T_c}\\ \gamma &= \frac{\pi}{8}\frac{\hbar}{\kappa T_c} = 1.7\sqrt{d}\end{aligned}\right\} \approx 10^{-12}\ \text{sec, superconductors [3],}$$

$$\left.\begin{aligned}\sqrt{d} &= [\kappa(\gamma_\perp + \kappa)]^{-1/2}\\ \gamma &= \kappa^{-1}\end{aligned}\right\} \approx 10^{-8}\ \text{sec, laser [4].}$$

The inertia term seems to be small as compared to the damping term. But this is correct only for the modulus $|\Psi|^2$, which is restored by the "force" $a + b|\Psi|^2$ being of order a. Then $d\,\partial_t^2\Psi/\gamma\partial_t\Psi \approx a \ll 1$ near the transition. The phase, $\arg\Psi$, instead, is not restored directly. Thus its movement is determined mainly by inertia; it is of diffusive type.

Time correlations may be determined from the equation of motion (8) by the following ideas. At least since the famous paper by Kadanoff and Martin [5] it is clear that an equation of motion fully determines the linear response as well as the fluctuation spectrum via fluctuation-dissipation-theorem. As we have a quasi canonical behaviour of the order parameter fluctuations, linear response theory applies well above and well below the transition. In the critical region itself ($a \approx 0$) the nonlinearity is essential and higher order time correlations are coupled to the order parameter correlation.

The following methods can then be used.

ii.1. The nonlinear (Langevin) equation. (8) is converted into a linear Fokker-Planck equation (assuming δ-correlated stochastic influence of the microscopic degrees of freedom). If the number of critical degrees of freedom is not small (e.g. a multimode laser, or a superconductor with infinitely many long wave length modes) the F.-P method is difficult. Then the following possibilities have proved to be good approximations for a laser [6].

ii.2. Solve the equation of motion with a given initial fluctuation, $\Psi(t; \Psi(t_0), \dot{\Psi}(t_0))$. The initial values are weighted according to the Boltzmann factor, i.e. once a fluctuation has occurred, it decays to equilibrium with the fully nonlinear motion, but not disturbed again by new fluctuations. This latter is even exact in the linear regime.

ii.3. Convert the equation of motion into a hierarchy of equations for correlation functions. A decoupling with respect to time can be achieved by

$$\frac{\langle A^n(t)A\rangle - \langle A^n\rangle\langle A\rangle}{\langle A^{n+1}\rangle - \langle A^n\rangle\langle A\rangle}, \text{ independent of n,} \tag{9}$$

as suggested by Stratonovich [7]. This decoupling expresses higher order time correlations through exact time correlations of the same (high) order. It has proven useful in laser line width calculations[1]).

---

[1]) A similar decoupling with respect to the r-dependence has been applied successfully by Marčelja to determine the diamagnetic susceptibility of Ni.

ii.4. The equation of motion can also be approximated by quasi linear response theory: one linearizes the fluctuations around the $t \to \infty$ solution, represented by the equilibrium value.

Some methods are very easy but nevertheless give rather good approximations, see below. They all express the idea, that the time development of a given fluctuation is essentially determined by a macroscopic equation of motion, Eq. (8), instead of being entirely random.

4. Restricted geometries

4.1. The energy principle taught us that, if local ordering occurs at all, the spatial coherence is at least of diameter $\sqrt{c/|a|}$ and lasts at least about $\sqrt{d/|a|}$. The correlation length at some distance $a \simeq 1$ from the transition point, $\xi_0 = \sqrt{c}$ is the microscopic coherence length, typical for each system, e.g. fluids, ferromagnets, or He^4 have $\xi_0 \approx 4$ Å, while Cooper pairs in superconductors are rather "large", $\xi_0 \approx \sqrt{\ell \cdot \xi_p} \approx 10^4$ Å in clean and $\approx 10^2$ Å in dirty metals (ℓ mean free path, ξ_p BCS coherence length).

If the extension R of a sample is larger than ξ, spatial variation in the ordering is possible. All wave numbers q with $1/R \lesssim q < 1/\xi$ occur, infinitely many for $\xi \ll R \approx \infty$. But if one or more linear dimension is smaller than ξ, there is coherence in this direction, i.e. no spatial fluctuation anymore. Therefore samples of thickness $R \lesssim \xi$ are 2-dimensional as far as ordering is concerned; filaments of cross area $R^2 \lesssim \xi^2$ are 1-dimensional; granulae of volume $\lesssim \xi^3$ (e.g. superconductors with diameter smaller than 1μ) are D = 0 dimensional systems. In this case the order parameter has a fixed r-dependence and can only fluctuate in time. The term $\sim |\nabla \Psi|^2$ does not contribute at all in (3) or as a const., depending on the boundary conditions.

While superconducting films, He^4-films, slightly larger magnetic systems (especially in sandwich form), and superconducting whiskers have been studied extensively [8, 9], experiments with granulae are not known to me (for theory see [10, 11]). But there exists another D = 0 dimensional system, the transition properties of which have been verified experimentally very recently: it is the laser.

The coherence length of a laser can be estimated by the "interaction" range of photons. Interaction of photons in lasers is possible via laser active atoms, i.e. during their decay time $\gamma_\perp^{-1}$. Thus $\xi_0 = c\,\gamma_\perp^{-1} \approx 1$ m $-$ 1 km and consequently $R \lesssim \xi_0$. The laser represents a single coherence cell of a many particle system. Its threshold properties are therefore rudiments of phase transition behaviour.

The laser's well studied static as well as time correlation properties (see below; [12, 13], [6], [4], and others) not only tell us how superconducting granulae would behave in the whole nonlinear regime. They may also provide a representative picture of a continuous but not singular transition in general, as it seems to occur for $D \leqslant 2$, at least in superconductors or He^4. They finally allow to compare approximative methods useful for $D > 0$ with the exact treatment.

4.2. Dimension D and width of transition region. Be $a = \alpha\epsilon$ with ϵ the relative distance from the transition. Reduce all lengths by the coherence length $\xi_0 = \sqrt{c/\alpha}$, the order parameter by $\sqrt{\alpha/b}$ (and times by $\tau_0 = \sqrt{d/\alpha}$).

$$\beta_c H = u_3 \int d^3x \left\{ \frac{1}{2}\epsilon |f|^2 + \frac{1}{4}|f|^4 + \frac{1}{2}|\nabla_x f|^2 + \frac{1}{2}|\partial_\tau f|^2 \right\}, \tag{10}$$

with

$$u_3 = 2\xi_0^3 \frac{\alpha^2}{b\kappa T_c} = 2\frac{\Delta c}{\kappa}\xi_0^3 = \frac{n^* \xi_0 \hbar^2}{m\kappa T_c}\chi. \tag{11}$$

$\Delta c = \alpha^2/bT_c$ is the specific heat jump, n^* the density of Cooper pairs, He-atoms etc, $\chi = 1$ or the Gorkov function. Restriction in geometry gives

$$d^3x = \left(\frac{R}{\xi_0}\right)^{3-D} d^D x, \qquad D = 3, 2, 1, 0.$$

One can write (10) in a form, in which a "law of corresponding states" becomes evident.

$$\beta_c H = \int d^D y \left\{ \frac{1}{2}\frac{\epsilon}{\epsilon_D}|\varphi|^2 + \frac{1}{4}|\varphi|^4 + \frac{1}{2}|\nabla_y \varphi|^2 + \frac{1}{2\epsilon_D}|\nabla_\tau \varphi|^2 \right\}. \tag{12}$$

As function of ϵ/ϵ_D all D-dimensional systems behave equal. The transition width is

$$\epsilon_D = \left[u_3 \left(\frac{R}{\xi_0}\right)^{3-D} \right]^{2/(D-4)}, \tag{13}$$

e.g. for He^4 we have $\xi_0 = 4$ Å, so $u_3 = 1/2$. Therefore the phase transition region is of order $\epsilon \approx 1$. The fluctuations are important nearly in the whole superfluid region. For electrons m is much smaller and $\xi_0 (\sim 1/m)$ is much larger, $\xi_0 \approx 10^3 - 10^4$ Å. Thus the transition region $\sim u_3^{-2}$ is very much smaller in superconductors. In general: the larger the coherence length the smaller the transition region. As impurities reduce the coherence length, they tend to broaden the transition region.

The restriction in geometry gives a factor $(R/\xi_0)^{3-D}$. If $\xi_0 \approx R$ as in He^4, the width ϵ_D of the transition region is not much affected for films or filaments. But for superconductors the restriction in geometry may enlarge ϵ_D considerably.

In detail it is

$$\epsilon_3 = \left(\frac{n^* \xi_0 \hbar^2}{m\kappa T_c}\chi\right)^{-2} = \left(\frac{2\Delta c_v}{\kappa}\xi_0^3\right)^{-2} \tag{14a}$$

$$\epsilon_2 = \left(\frac{n^* R \hbar^2}{m\kappa T_c}\chi\right)^{-1} = \left(\frac{2\Delta c_v}{\kappa}\xi_0^2 R\right)^{-1}, \tag{14b}$$

$$\epsilon_1 = \left(\frac{n^* R^2 \hbar^2}{\xi_0 m\kappa T_c}\chi\right)^{-2/3} = \left(\frac{2\Delta c_v}{\kappa}\xi_0 R^2\right)^{-2/3}, \tag{14c}$$

$$\epsilon_0 = \left(\frac{n^* R^3 \hbar^2}{\xi_0^2 m\kappa T_c}\chi\right)^{-1/2} = \left(\frac{2\Delta c_v}{\kappa}R^3\right)^{-1/2}. \tag{14d}$$

For pure tin one has $\xi_0 = 1{,}7 \cdot 10^3$ Å and $\Delta c_v/\kappa = 3{,}6 \cdot 10^{-5}$ Å^{-3}. If $R = 10^3$ Å one finds $\epsilon_3 \approx 10^{-11}$, $\epsilon_2 \approx 10^{-6}$, $\epsilon_1 \approx 5 \cdot 10^{-4} \approx \epsilon_0$. For amorphous (dirty) Bi-Sb one has [14] $\xi_0 \approx 80$ Å (also Δc_v seems to be somewhat smaller) and finds $\epsilon_3 \approx 10^{-4}$, $\epsilon_2 \approx 10^{-3}$ ($\approx \epsilon_1 \approx \epsilon_0$). For a laser we use $\hbar^2/2m\xi_0^2 = \alpha \Rightarrow \kappa$ (cavity damping), $\kappa T_c \Rightarrow Q$ (quality), and $nR^3 \Rightarrow \alpha/b$, the number of degrees of freedom. Then the relative threshold inversion is

$$\epsilon_0 = \left(\frac{\alpha^2}{bQ}\right)^{-1/2} = \frac{Q}{\kappa n_{thr}} \approx 10^{-4} \text{ (gas laser)}, \tag{15}$$

where $\sqrt{Q/b} = n_{thr}$ is the number of photons near threshold.

5. Examples: Static thermodynamic behaviour

So far the introductive, general part. I shall discuss now some special examples of recent interest, which fit into the description of phase transitions via Landau order parameter fluctuation.

5.1. Photon intensity of a two mode laser in the threshold region [15]. Let us consider a laser as a D = 0 dimensional system. If the atomic halfwidth $\gamma_\perp$ is broad enough, $\gamma_\perp \gtrsim c/L$, two modes may be amplified in the cavity. Then we have two order parameters, the field strengths $\Psi_\lambda(\lambda = 1, 2)$ of both modes. They have different threshold inversions (transition points) ν_λ. The fluctuation Hamiltonian is [15]

$$H = H_1 + H_2 + H_{int}, \text{ with } H_{int} = \frac{1}{2}\zeta\,|\Psi_1|^2\,|\Psi_2|^2. \tag{16}$$

The ratio ζ of the interaction between modes and the selfinteraction $\frac{1}{4}|\Psi_\lambda|^4$ is between 0 and 2/3 for a standing wave laser. The quality Q is assumed to be the same for both modes.

The physical meaning of $|\Psi_\lambda|^2$ is the number of photons in the mode λ. Therefore $\langle|\Psi_\lambda|^2\rangle$ gives the laser intensity as function of relative inversion $p = \epsilon/\epsilon_0$. At $p = 0$ the first mode becomes active; $p = \delta \sim (\nu_2 - \nu_1)$ is the bare threshold of the second mode. Fig. 1 shows the (exactly calculable) onset of the ordering intensity [15]. There is (i) an effective threshold shift $\delta_{eff} = \delta/(1 - \zeta)$, i.e. the second mode is suppressed by the first one, and (ii) a reduction in slope above both thresholds, $\langle|\Psi_\lambda|^2\rangle = p/(1 + \zeta)$.

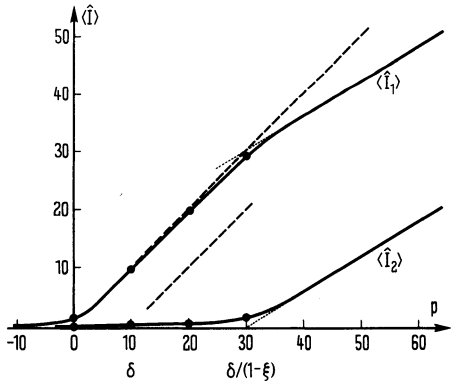


Fig. 1
Intensities of two mode laser vs. pumping rate. $\delta = 10$, $\zeta = 2/3$. For comparison, the dashed lines show the intensities without coupling, $\zeta = 0$.

The suppressing and sharing in pump intensity can also be seen in the transient properties. They have been calculated by solving the approximate Langevin equations of motion for $|\Psi_\lambda|^2$ with initial values $|\Psi_\lambda|^2 = 0$, see Fig. 2. The peak indicates the higher lasing of the first mode, as long as the second is not strong enough. Fig. 3 allows to compare with measurements of Meltzer and Mandel [16] on a single mode laser.

Figure 4 shows the photon correlation matrix $K_{\lambda\lambda'}(p) = \langle(\Psi_\lambda^2 - \langle|\Psi_\lambda|^2\rangle)(\Psi_{\lambda'}^2 - \langle|\Psi_{\lambda'}|^2\rangle)\rangle$. It tends to zero $\sim 1/\langle|\Psi_\lambda^2|\rangle^2$ well below threshold and is constant well above. There the relative correlations become smaller with increasing output, reflecting the non-Gaussian distribution. The effects of mode coupling are (i) again the threshold shift, (ii) a higher asymptotic value of the correlation.

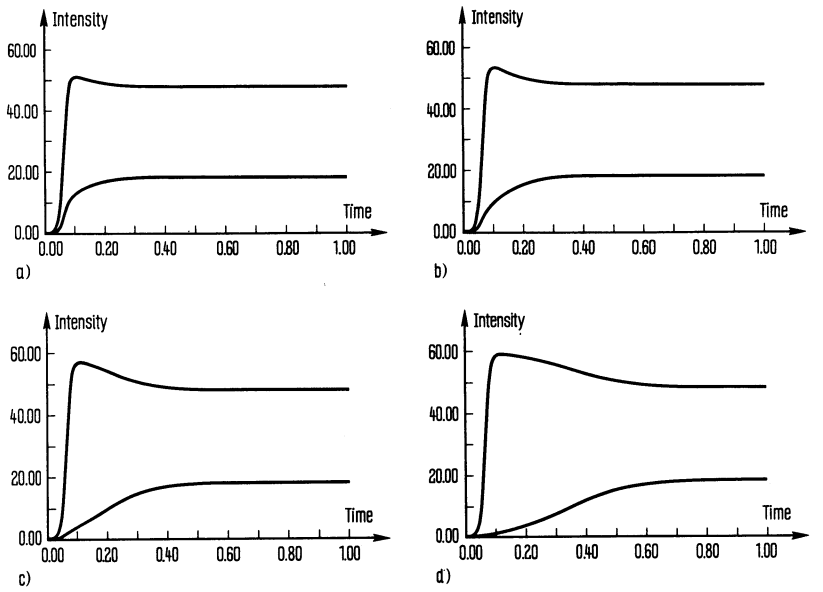


Fig. 2 Transient behaviour of intensities $|\Psi_\lambda|^2(t)$. $\delta = 10$; $p = 60$. The relative detuning is 0.33, 0.45, 0.71, and 1.0 resp.

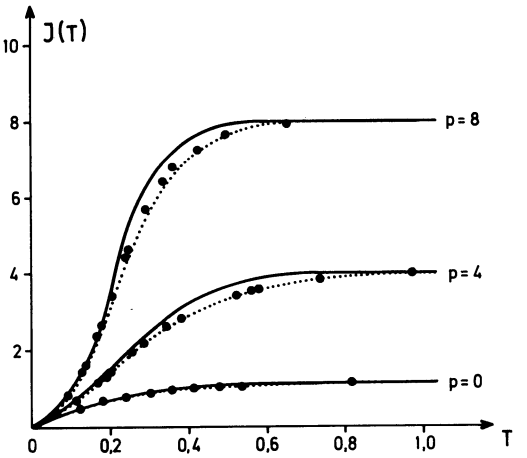


Fig. 3 Transient behaviour of single mode intensity, from Fokker-Planck-equation (dash-dotted [12]) and from Langevin approximation (full lines) for different pumping rates p.

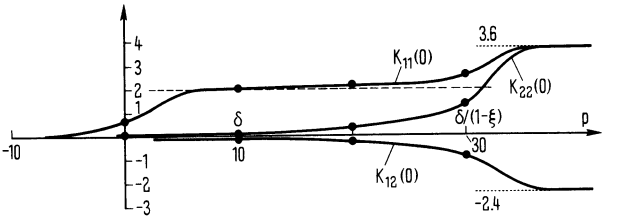


Fig. 4 Photon intensity correlations. $\delta = 10$, $\xi = 2/3$. The dashed line shows the asymptotic behaviour without interaction, the dotted lines for $\xi = 2/3$.

Besides the special meaning as a photon correlation, K represents an example of an autocorrelation function of the order parameter squared. It is $K \sim \partial^2 F/\partial a^2$. At thermal phase transitions $a \sim (T - T_c)$. Then the physical meaning of $-\partial F/\partial a \sim -\langle |\Psi|^2 \rangle$ is the entropy associated with ordering and $-\partial^2 F/\partial a^2 \sim -\partial^2 F/\partial T^2$ is the corresponding specific heat, called C instead of K. K reflects the onset of lasing by a smoothed jump, but there is no indication of a peak at the transition point. This is expected to be qualitatively quite different at thermal transitions in higher dimensional systems, e.g. the specific heat of bulk He^4 shows not only a peak but a (logarithmic) singularity at T_c. So, how is the change-over between D = 0 (entirely smooth) to D = 3 (highly singular)?

5.2. Specific heat of superconducting films. Because there are some infinite extensions left for D = 2, 1, one expects at least a peak in the specific heat at the phase transition.

Without fluctuations the free energy according to (1) is

$$\Delta F = -\frac{1}{2}\frac{a^2}{b} \cdot V, \text{ for } a = \alpha\epsilon < 0, \epsilon = \frac{T - T_c}{T_c}. \tag{17}$$

Therefore we get a jump in the specific heat:

$$\Delta C = T_c \partial_T(-\partial_T \Delta F) = V\frac{\alpha^2}{bT_c}. \tag{18}$$

The corrections according to the fluctuations have first been calculated for superconductors in a classical paper by Thouless [17] by a lengthy diagram summation. He found an increase of C above T_c of the form $\sim 1/\sqrt{\epsilon}$ with the very small prefactor, which we already know from Eq. (14a). Later on, Levanyuk [18], and Ferrell [19] got the same much easier starting from the Ginzburg-Landau-functional, Eqs. (3), (5) for arbitrary D. The idea was to neglect the order parameter interaction $\sim |\Psi|^4$.

One gets an onset of a singularity for $T > T_c$, increasing with decreasing D:

$$C \sim \epsilon^{-1/2}, \text{ bulk}; C \sim \epsilon^{-1}, \text{ film}; C \sim \epsilon^{-3/2}, \text{ filament}. \tag{19}$$

Evidently it should be $C \sim \epsilon^{-2}$, for D = 0. In fact, this is the asymptotic behaviour of K_{11} far below threshold.

In [20] we determined C also below T_c. The method was to linearize Ψ around the mean field value $\Psi_0 = \sqrt{-a/b}$, as was also done, e.g. by Rice [21] to study the (missing) long range order for D < 3 or by H. Schmidt [22] to calculate the conductivity transition. We found

$$\delta C_{\text{below}} = (\sqrt{2})^{D-2}\, \delta C_{\text{above}}. \tag{20}$$

This means the same onset of singularity if $T \nearrow T_c$ as (19). This is wrong for D = 0, where $K \sim C$ has shown no peak.

A Hartree treatment of the order parameter interaction $|\Psi|^4 = 2\langle |\Psi|^2 \rangle |\Psi|^2$ was introduced by Marčelja, Masker, Parks [23, 24] to treat the resistive transition of superconductors successfully. For $\langle |\Psi|^2 \rangle$ a familiar selfconsistency equation was postulated

$$\langle |\Psi|^2 \rangle = \int \frac{d^D q}{(2\pi)^D} \frac{\eta}{\exp[\eta\beta_c(a + b\langle |\Psi|^2 \rangle + cq^2)] - 1}$$

$$\approx \int \frac{d^D q}{(2\pi)^D} \frac{\kappa T_c}{a + b\langle |\Psi|^2 \rangle + cq^2}, \tag{22}$$

in quantum (with $\eta = 1$) or classical ($\eta \to 0$) version. Another postulate was given, what the free energy should be. Using this together with the solution of (22), they found for the specific heat Fig. 5, supporting the expectations.

At the beginning of 1972 the first specific heat measurements of films near T_c were reported by Zally and Mochel [14], see Fig. 6. There is no peak, although the authors have looked for one.

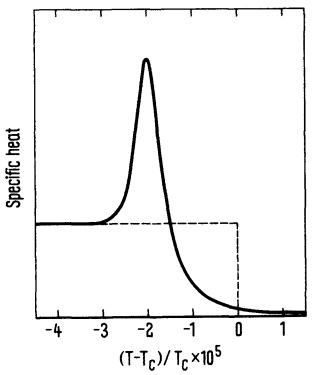
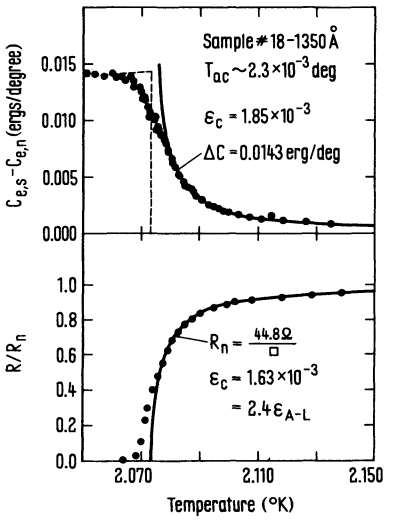


Fig. 5 Expected specific heat anomaly for superconducting films. (Parameters: R = 500 A, ℓ = 500 A.)

Fig. 6
Heat capacity and conductivity near T_c for a 1350 A film of amorphous $Bi_{0.37}Sb_{0.63}$ (from [14]).



Note the similarity of the observed specific heat with the laser correlation function K for D = 0. This encouraged us [25] to believe in a theory, which runs as follows[1]):

In the free energy (7) we approximate the interaction $|\Psi|^4 \equiv (|\Psi|^2 - n)^2 + 2n|\Psi|^2 - n^2$ by neglecting the square of the deviation from a parameter n. This latter is determined afterwards selfconsistently by minimizing the free energy F(n). The selfconsistency equation has the form (22), but with the difference that large q values are cut off at $\sim \xi_0^{-1}$ instead of the thermal wave length λ_{T_c} as in (22). This is achieved by choosing $\eta = 1/2\alpha\beta_c$. It considers the fact that the Ginzburg-Landau-functional (3) is the lowest order in an expansion $(\xi_0 \partial_r \ldots)^2$.

Differentiating this selfconsistent free energy one finds a specific heat, represented by the full line in Fig. 7. We have shifted the theoretical curve to find the bare T_c (which cannot be observed), and fitted the parameter ϵ_2 of the asymptotic behaviour

$$c = \frac{\epsilon_2}{2\pi}\frac{1}{\epsilon}, \quad \epsilon_2 = (0.125\, k_F^2\, \ell R)^{-1} \text{ (dirty limit).}$$

The shift comes out as 19 m°K. In general the effective T* is

$$\frac{T^* - T_c}{T_c} = -\frac{\epsilon_2}{2\pi}\left(\ln\frac{4\pi}{\epsilon_2} - 1\right). \tag{23}$$

[1]) After acceptance of our work for publication the same ideas were published by L. Gunther, L. W. Gruenberg, Solid State Comm., **10** (1972) 567.

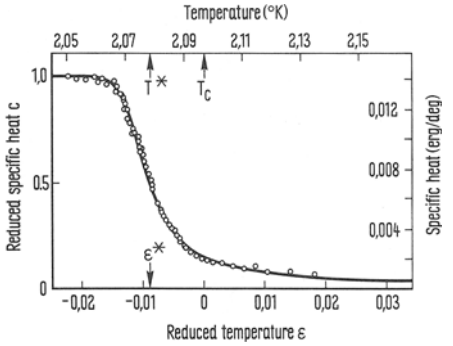


Fig. 7 Selfconsistent specific heat from [25] compared with experimental values from [14]. $\epsilon_2 = 1/110$.

5.3. Specific heat in He^4 films. The remarkable agreement between selfconsistent Hartree theory of the fluctuations of the Landau order parameter field and the observed smooth specific heat of superconducting films puts the question of the behaviour of He-films. This is shown in Fig. 8, together with the bulk specific heat. There is indeed no singularity.

To understand this we used Hartree approximation for F but with He^4 parameters. As then $\epsilon_2 \sim 1/5$ we also took care of the explicit T-dependence in Eq. (7). Fig. 9 shows the specific heat if $u_2 = \epsilon_2^{-1}$ changes from 110 for a superconducting film to 5 for a He^4 film.

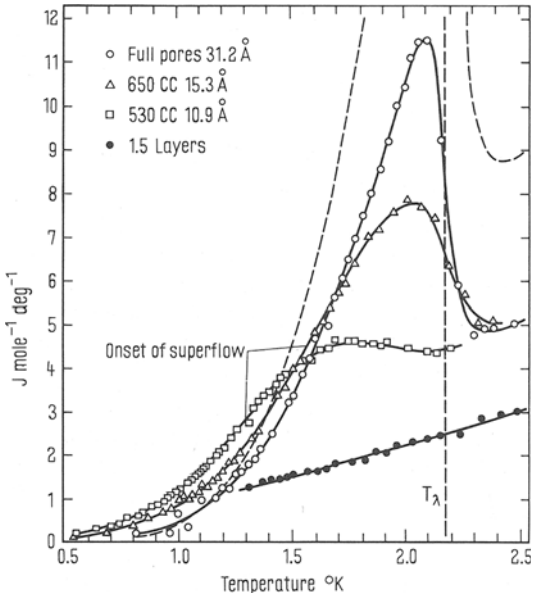


Fig. 8 Multilayer specific heat of He^4 absorbed on vycor from [26].

5.4. Is the selfconsistent Hartree approximation really sufficient? Although experiment and Hartree treatment of the free energy seem to agree, one might wonder what the higher order corrections contribute. They should be very important at least if D = 3. For comparison the specific heat in selfconsistent Hartree approximation in bulk superconductors is sketched in Fig. 10.

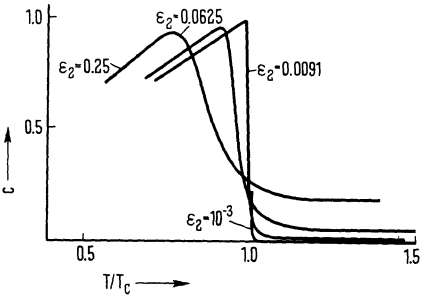


Fig. 9 Selfconsistent specific heat of films for increasing transition width.

A possibility to estimate the fluctuation contribution beyond the Hartree one is provided by superconducting filaments. The partition function Z has the shape of a Feynman path integral [27, 28]. As one can learn e.g. from [29] for an infinitely long one dimensional system the free energy F is described by the lowest eigenvalue of the anharmonic Schrödinger equation. Fig. 11 shows the recently [30] reported result for $\partial^2 F/\partial a^2$.

The bump of $\approx$ 10% might be characteristic for D = 1 and 2 and may have been overlooked in experiments so far?

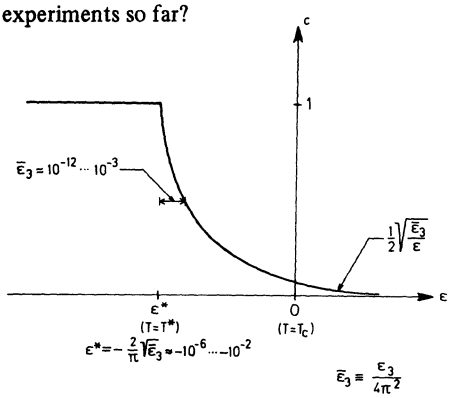


Fig. 10 Selfconsistent Hartree approximation for D = 3.

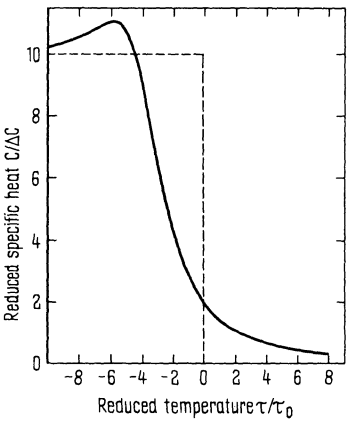


Fig. 11
Exact specific heat of a long filament versus reduced temperature from [30]. It is $\tau_0 = 2^{-2/3}\,\epsilon_1$.

6. Examples: Dynamical properties

The characteristical dynamical feature near phase transitions is the enlarged time scale of relaxation processes, called "critical slowing down". This can also be observed in various circumstances in restricted geometries.

6.1. Photon number correlation halfwidth. Let us consider D = 0 first, realized by a single mode laser. The photon number $|\Psi|^2$ fluctuates in time around the stationary value. The time correlation function $\langle \delta\,|\,\Psi(t)\,|^2\,\delta\,|\,\Psi(o)\,|^2\rangle$ can be calculated by the methods mentioned in Sect. 3.ii. While typical decay times of photon correlations are of order $10^{-7} - 10^{-8}$ sec, near the onset of lasing there is critical slowing down to $\approx 10^{-4}$ sec.

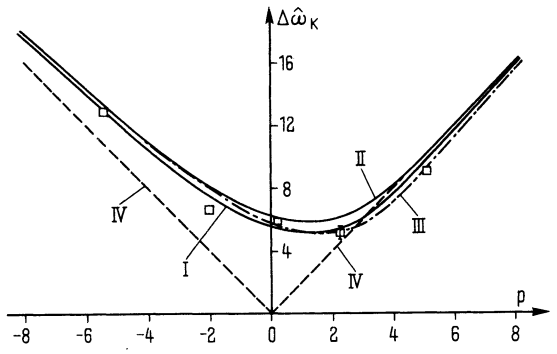


Fig. 12 Intensity correlation halfwidth vs. pump parameter. I. from equation of motion, II. by decoupling the hierarchy, III. solving the Fokker-Planck-equation [12, 13], IV. linear theory, i.e. without mode interaction. Experimental values are from Arecchi's gas laser [31], which had $\Delta\omega \approx 10^4$ Hz and $\approx 10^3$ photons at threshold.

Figure 12 shows the intensity correlation halfwidth first given by Risken (see [12]) compared with results from the different methods described above [6].

6.2. Slowing down in a coupled 2-mode laser. As an example of the influence of another mode coupled to a critical one let us consider the 2-mode laser with intensity coupling. Using the quasilinear method we found correlation halfwidths for $\langle \delta \mid \Psi_\lambda(t) \mid^2 \delta \mid \Psi_\lambda{}'(o) \mid^2 \rangle$ as represented in Fig. 13. There is a second critical slowing down, also in the non-critical mode, if the other one reaches its effective threshold $\delta/(1-\zeta)$. One furthermore observes an asymmetry in the gross behaviour of the correlation halfwidth: below both thresholds $\Delta\omega_\lambda = 2(-p_\lambda)$, both slow down independently; above both thresholds they speed up less, namely $\Delta\omega_2^1 = 0.231\, p \pm \frac{0.106\delta}{0.336\delta}$.

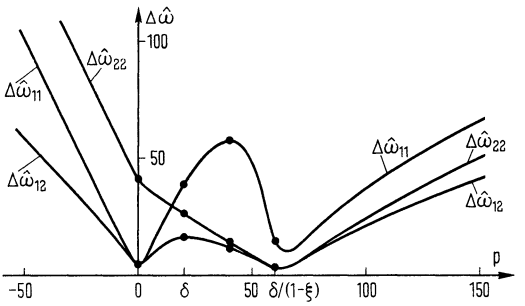


Fig. 13 Intensity linewidth [15] vs. pumping rate in the transition region; $\delta = 20$, $\zeta = 2/3$.

6.3. The order parameter linewidth. So far the order parameter "intensity" correlations have been looked upon. It is the photon number of a laser or the Cooper pair density $n_s \sim \langle \mid \Psi \mid^2 \rangle$ in superconductors etc. A different behaviour is shown by the fluctuation time of the order parameter itself, $\langle \Psi(t)\Psi^*(o) \rangle$, at least for a transition with complex Ψ.

Again there is critical slowing down if the transition is approached from the disordered phase. But within the ordered phase the correlation time does not speed up again, but gets longer and longer. This can be observed directly in the extremely sharp spectral linewidth of a laser, see Fig. 14. Below threshold Ψ fluctuates around $\Psi_0 = 0$; the restoring force $\sim$ a decreases on approaching the transition, therefore $\Delta\omega \sim a \sim 1/\langle \mid \Psi \mid^2 \rangle$. Above threshold Ψ fluctuates around $\Psi_0 = \sqrt{-a/b} \neq 0$. The motion of $\mid \Psi \mid$ speeds up, but decreases in magnitude if the output $\sim$ p gets larger. The phase instead, arg Ψ, which has no restoring force in the Hamiltonian (3) but a kinetic

energy $d \cdot \Psi_0^2 \mid \arg \Psi \mid^2$, travels around the valley of minimal "potential" energy. With increasing output $\langle \mid \Psi \mid^2 \rangle \sim \mid a \mid \sim \Psi_0^2$ the effective inertia of phase motion increases. Thus the diffusive phase motion decreases $\sim 1/\langle \mid \Psi \mid^2 \rangle$ and the linewidth again is $\sim 1/\langle \mid \Psi \mid^2 \rangle$, getting extremely sharp at large output.

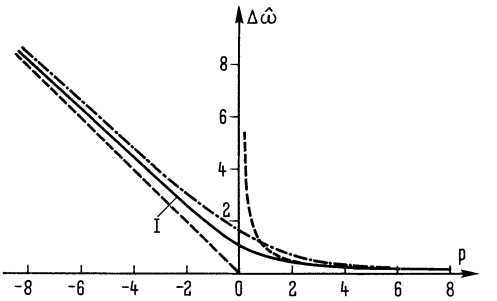


Fig. 14 Spectral linewidth vs. pump parameter from approximate solution of the Langevin-equation (8). Dashed lines: linear approximations; dot-dash line: Risken; Hempstead and Lax, solution of Fokker-Planck-equation.

The solution of the Langevin equation or the Fokker-Planck equation give for the (reduced) linewidth factor $\Delta\hat{\omega} \cdot \langle \mid \hat{\Psi} \mid^2 \rangle$ the constant values 1 above and 2 below threshold. This simply reflects the fact that phase diffusion only represents 1 degree of freedom of the order parameter. Near threshold there is a smooth transition, represented in Fig. 15. Only quite recently this has been confirmed by experiments. Gerhardt, Welling, and Güttner [32] [1]) were able for the first time to measure the natural laser linewidth in the threshold region.

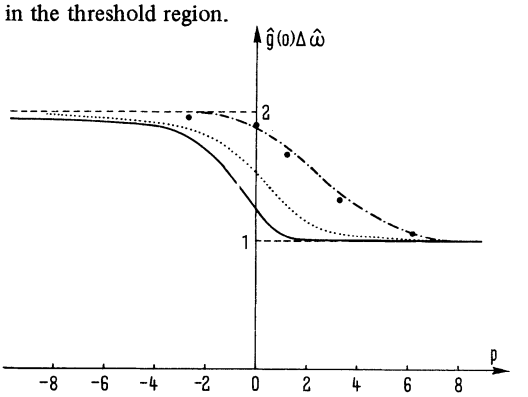


Fig. 15 Linewidth factor vs. pumping power. I. approximate solution of Langevin-equation, II. decoupling of hierarchy, III. solution of Fokker-Planck-equation by Risken; Hempstead and Lax. Experimental points [32].

6.4. Conductivity transition in superconductors. In superconductors the order parameter correlation time is not directly observable. Nevertheless it manifests itself in transport effects, the most prominent of which is the conductivity. The simple Drude picture of conductivity (applied to superconductors by A. Schmid [33]) gives

$$\sigma_s = \frac{e^2}{m} \sum_k n_k \tau_k. \tag{24}$$

It is $n_k = \langle \mid \Psi_k \mid^2 \rangle$ the number of Cooper pairs with wave number k and τ_k the life time of the ordering, more exactly, the correlation time.

---

[1]) I thank Prof. Welling for sending me a preprint of his work prior to publication.

Without the influence of the fluctuations one guesses $\tau_k = 0$ above T_c. But the spontaneous existence of order leads to a superconductivity σ_s even above T_c. This has first been observed by Glover [34] and theoretically interpreted as an effect of thermal order parameter fluctuations using G. L. theory by A. Schmid [33], together with current-current correlation formulae for σ_s by H. Schmidt [22] and microscopically by Aslamozov and Larkin [35]. All these authors treat the linear regime above T_c, getting from (24) the extremely simple result

$$\sigma_s = \frac{e^2}{16\hbar R}\frac{1}{\epsilon}, \qquad D = 2, \qquad \epsilon > 0. \tag{25}$$

An increasing number of papers has been concerned with σ_s meanwhile, giving experimental support as well as theoretical extensions[1]).

An approximate treatment of the whole transition region has been proposed by Marčelja, Masker, and Parks [24], taking

$$\tau_k = \text{const}\, n_k, \qquad \text{const} = \frac{1}{\gamma\kappa T_c}. \tag{26}$$

This is just the relation $\Delta\hat{\omega} \cdot \langle |\Psi^2| \rangle = \text{const}$, known from the laser, but without reduction (or other modification) in the nonlinear transition region. Nevertheless, qualitatively τ_k seems to be correct, as comparison of the Hartree σ_s with experiments indicate, see e.g. Fig. 16. Also the ω-dependence [22] in the linear and quasilinear regimes is correctly described by fluctuation theory [38].

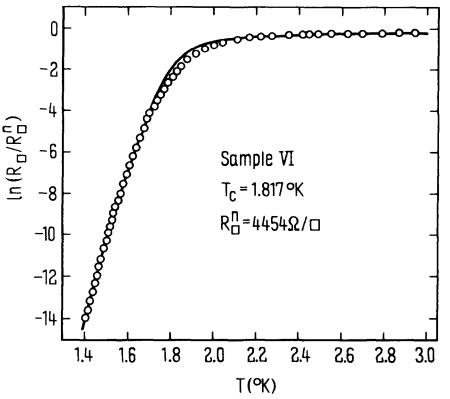


Fig. 16 Excess conductivity for dirty Al-films, R = 170 Å, T_c = 1.82°K, D = 2, from [24].

6.5. Coherence length. Finally I should mention the question of the order parameter coherence length, as it is connected with the statement, that there should not be any phase transition if $D < 3$. The spatial order parameter correlation function, $\langle \Psi^*(\mathbf{r})\, \Psi(\mathbf{r}') \rangle$, can be treated quite similarly as the time correlations. This has been done below T_c by Rice [21]. Analogously as for time correlations, the phase correlations are dominant. They decay in D = 1 (as $\exp\{-|\mathbf{r} - \mathbf{r}'|/\xi\}$) and D = 2 dimensions (as $\exp\{-[\ln(2Q|\mathbf{r} - \mathbf{r}'|)]/\pi\xi\}$), but not in bulk

[1]) For a survey up to 1970 see [8]. I should mention that diamagnetic effects are also treated by fluctuation theory, see e.g. [36, 37].

material. Similar statements hold for superfluid Helium [39]. This means that there is no long range order if $D < 3$ as well as no finite "order parameter" as defined by

$$\left(\lim_{|\mathbf{r}-\mathbf{r}'|\to\infty} \langle \Psi(\mathbf{r})\Psi^*(\mathbf{r}')\rangle \right)^{1/2} =: \bar{\Psi}_0.$$

The correlation length in Hartree approximation is found to be

$$\xi/\xi_0 = \epsilon^{-1/2}, \qquad T > T_c, \qquad \text{all D},$$

$$\xi/\xi_0 = \begin{cases} \infty & T \leqslant T^* < T_c \quad D = 3, \\ \dfrac{1}{\sqrt{2}} \exp\left(\dfrac{\pi}{\epsilon_2} \cdot |\epsilon|\right) & T \lesssim T_c \quad D = 2, \\ |\epsilon|/(\epsilon_1^{3/2}) & T \lesssim T_c \quad D = 1. \end{cases} \tag{27}$$

Films thus show a very strong increase in susceptibility $\sim \xi^2 \sim e^{2\pi|\epsilon|/\epsilon_2}$. This can be observed in magnetic films.

Despite the lack of a finite quasiaverage of the order parameter we have seen, that the correlation functions behave entirely different in the "ordered" phase. This point of view, namely, that the correlation functions instead of the order parameter should be used as an indicator of a phase transition, has been suggested and discussed extensively by Mikeska and Schmidt [40]. The nonexistence of long range order does not imply the absence of typical aspects of ordering, such as a raise in the specific heat or in the conductivity.

Acknowledgement: I thank Drs. P. H. Richter and C. Wissel for numerous discussions and especially for reading the manuscript (P.H.R.) and doing numerical computation for Fig. 9 (C.W.). I also thank H. Kümmel for numerical work for Fig. 2.

References

[1] Landau, L. D. and Lifschitz, E. M.: Statistical Physics. Oxford – New York – Toronto–Sydney 1959.
[2] Ginzburg, V. L.: Soviet Physics Solid State 2, 1824 (1960).
[3] Werthamer, N. R.: in Superconductivity, I, 1969, p. 321 (ed. R. D. Parks).
[4] Haken, H.: Handbuch der Physik, Bd. XXV/2c, Berlin-Heidelberg-New York 1970.
[5] Kadanoff, L. P. and Martin, P. C.: Ann. Phys. (N.Y.) **24**, 419 (1963).
[6] Grossmann, S. and Richter, P. H.: Z. Phys. **242**, 458 (1971).
[7] Stratonovich, R. L.: Topics in the Theory of Random Noise, Vol. II. New York 1967.
[8] Proc. 12th Int. Conference Low Temp. Phys., Kyoto, 1970) (ed. Eizo Kanda); see espec. P. C. Hohenberg, p. 211 and R. D. Parks, p. 217.
[9] Hohenberg, P. C.: EPS meeting 1972, Freudenstadt.
[10] Hurault, J. P., Maki, K., and Béal-Monod, M. T.: Phys. Rev. **B3**, 762 (1971).
[11] Mühlschlegel, B., Scalapino, D. J., and Denton, R.: Phys. Rev. Let. **26**, 707 (1971) and Preprint, 1972.
[12] Risken, H.: Fortschr. Physik **16**, 261 (1968).
[13] Hempstead, R. D. and Lax, M.: Phys. Rev. **161**, 350 (1967).

[14] Zally, G. D. and Mochel, J. M.: Phys. Rev. Let. **27**, 1710 (1971).
[15] Grossmann, S. and Richter, P. H.: Z. Phys. **249**, 43 (1971).
[16] Meltzer, D. and Mandel, L.: Phys. Rev. Let. **25**, 1151 (1970).
[17] Thouless, D. J.: Ann. Phys. (N.Y.) **10**, 553 (1960).
[18] Levanyuk, A. P.: Sov. Phys. Solid State **5**, 1294 (1964).
[19] Ferrell, R. A.: J. Low Temp. Phys. **1**, 241 (1969).
[20] Grossmann, S. and Richter, P. H.: Phys. Let. **33A**, 39 (1970).
[21] Rice, T. M.: Phys. Rev. **140**, A 1889 (1965).
[22] Schmidt, H.: Z. Phys. **216**, 336; (1968); **232**, 443 (1970).
[23] Marčelja, S.: Phys. Let. **28A** 180 (1968).
[24] Masker, W. E., Marčelja, S., and Parks, R. D.: Phys. Rev. **188**, 745 (1969).
[25] Grossmann, S., Richter, P. H., and Wissel, C.: Solid State Communications, 1972.
[26] Brewer, D. F.: J. Low Temp. Phys. **3**, 205 (1970).
[27] Langer, J. S.: Ann. Phys. (N.Y.) **41**, 108 (1967).
[28] Tucker, J. R., and Halperin, B. I.: Phys. Rev. **B3**, 3768 (1971).
[29] Feynman, R. P. and Hibbs, A. R.: Quantum Mechanics and Pathintegrals. New York 1965.
[30] Gruenberg, L. W. and Gunther, L.: Phys. Let. **38A**, 463 (1972) (March).
[31] Arecchi, F. T., Giglio, M., and Sona, A.: Phys. Let. **25A** 341 (1967).
[32] Gerhardt, H., Welling, H., and Güttner, A.: to appear in Z. Phys. (1972).
[33] Schmid, A.: Z. Phys. **215**, 210 (1968).
[34] Glover, R. E.: Phys. Let. **25A**, 542 (1967).
[35] Aslamozov, L. G. and Larkin, A. I.: Soviet Physics Solid State **10**, 875 (1968).
[36] Patton, B. R., Ambegaokar, V., and Wilkens, J. W.: Sol. State Commun. **7**, 1287 (1969).
[37] Lee, P. A. and Payne, M. G.: Phys. Rev. Let. **26**, 1537 (1971).
[38] Lehocky, S. L. and Briscoe, C. V.: Phys. Rev. **B4**, 3938 (1971).
[39] Kane, J. W. and Kadanoff, L. P.: Phys. Rev. **155**, 80 (1967).
[40] Mikeska, H. J. and Schmidt, H.: J. Low Temp. Phys. **2**, 371 (1970).

Instabilities and Phase-Transition-Like Phenomena in Physical Systems far from Thermal Equilibrium

Phase-Transition-Like Phenomena in Lasers and Nonlinear Optics

Robert Graham

Summary

We start with the simple physical idea that the laser threshold is characterized by the formation of a macroscopic electromagnetic field in the time domain. This idea is then used to set up a phenomenological theory of the threshold phenomena in lasers and nonlinear optics. For the single mode laser the results of the microscopic theory are reproduced, which are in agreement with all experimental data. In section 3 the microscopic theory is considered from the new point of view and some analogies and differences to mode-mode coupling theories of phase transitions are pointed out. In section 4 some recent theoretical results on multi-mode lasers are reviewed with the conclusion that mode-mode coupling effects near thresholds in multi-mode lasers should be measurable. Finally our phenomenological theory is applied to various light sources of nonlinear optics. The phenomenological theory predicts that the form of the photo-count distributions for all these sources is independent of the details of the microscopic interactions and will depend only on the phase-angle symmetries and the order (in a perturbation theoretical sense) of the basic nonlinear scattering process. In all cases, where a comparison with microscopic theories is possible, agreement is found. The results indicate the great usefulness of the language and the concepts of phase transition theory even for systems which are not in thermodynamic equilibrium, and which don't become critical in a strict sense.

1. Introduction

Optics is quite usually described in terms of a macroscopic theory, i. e. in terms of a set of equations for averaged, macroscopic electric and magnetic fields $\langle E(r,t)\rangle$, $\langle B(r,t)\rangle$.

$$\begin{aligned} &\frac{1}{c}\frac{\partial\langle E\rangle}{\partial t} = -\frac{1}{c} j(r,t) + \nabla \times \langle B\rangle \\ &\frac{1}{c}\frac{\partial\langle B\rangle}{\partial t} = -\nabla \times \langle E\rangle \\ &\nabla\langle B\rangle = 0; \qquad \nabla\langle E\rangle = \rho(r,t) \end{aligned} \tag{1.1}$$

$j(r,t)$ and $\varphi(r,t)$ are the macroscopic densities of current and charge, respectively.

In order to see more clearly, in which sense conventional optics is a macroscopic theory, let us look at a quasi-monochromatic, electromagnetic field, e. g., of the form

$$\langle E(r,t)\rangle = E_0(r,t) \quad \sin(\omega t - kr) \tag{1.2}$$

with an amplitude, which changes very slowly in space and time compared to the optical oscillations with frequency ω and wave vector k. Such a field could be generated, e. g., in a piece of ruby with a density $2 \cdot 10^{19}$ cm^{-3} of Cr^{3+} ions, which we can excite externally so that the Cr^{3+} ions emit light in their 6943 Å line. At room temperature this transition has a thermal width of $\sim 5 \cdot 10^{10}$ sec^{-1}. If the emitted light is not very intense it will have a coherence time of $\sim 2 \cdot 10^{-11}$ sec which defines a coherence length of 0.6 cm. Clearly, if the averages in Eqs. (1.1), (1.2) are understood as spatial averages, taken over a distance which is large compared to the interatomic distance of $\sim 4 \cdot 10^{-7}$cm and small compared to the coherence length of 0.6 cm, the averaging is justified. The fields are constant over long distances in space, i. e., spatially they look like quasi-conserved, macroscopic quantities. However, if we interpret the average as an average in time, over an interval of the order of the observation time, e. g., in a time resolved photograph of interference fringes (typically of the order of 10^{-6} sec) we would find that all averages in Eqs. (1.1), (1.2) are zero. The reason is that each time a new wave track is emitted by an atom, it has a new phase; the average over all phase factors vanishes. As a result, the fields behave in the time domain like rapidly fluctuating, non-conserved, microscopic quantities (cf. fig. 1a). S p a t i a l l y, conventional optics is a macroscopic theory; with respect to t i m e it has still to be considered to be a microscopic theory.

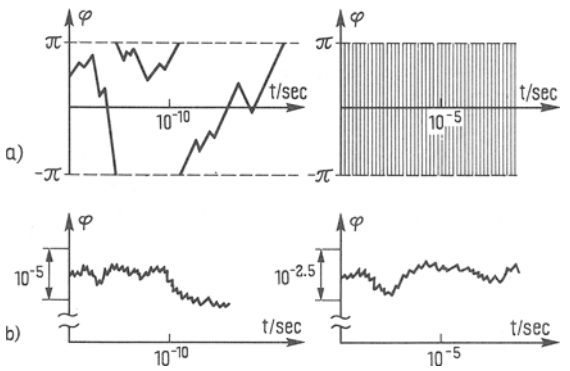


Fig. 1
Time scale dependence of the phase fluctuations of a laser field
a) below threshold
b) above threshold (note the change of scale φ)

Now the same piece of ruby is turned into a laser, by using much more effective excitation mechanism and providing mirrors to concentrate the light in only one or a few modes. Then the coherence time of the emitted light is no longer determined by the microscopic time of $2 \cdot 10^{-11}$ sec.; in fact, if all external noise sources could be eliminated, the coherence time could grow by more than 10 orders of magnitude to become as large as 1 sec.

The phase fluctuations of the field will then display a slow Brownian motion and will transform like the square root of the time scale in the vast range between the microscopic and the 1 sec time scale, because no intrinsic time constant of the system exists in that range

(cf. fig. 1b). On our time scale of observation ($\sim 10^{-6}$ sec) the field has turned into a quasi-conserved, macroscopic quantity.

Similar effects are known to occur in other systems. In liquid He undergoing the λ-transition suddenly a new macroscopic variable (or order parameter) appears, the macroscopic wave function, which is responsible for all the effects peculiar to superfluids. In fact, the analogy between macroscopic electromagnetic fields and the macroscopic wave function in superfluids can be carried through to a considerable degree [1a]. (cf. table 1). A similar analogy holds with respect to simple ferromagnetic systems [1b].

Table 1 Analogy between lasers and 1-dimensional superfluids

LASER	SUPERFLUID (1-dim.)
electromagnetic field	macroscopic wave function
coherence time	coherence length
t h r e s h o l d :	' q u a s i ' - p h a s e t r a n s i t i o n :
gain = loss	collective motion in balance with thermal random motion
increase in quantum efficiency	increase in conductivity
large intensity fluctuations, slowing down	large current fluctuations, slowing down
slowing down of phase fluctuations	slowing down of phase fluctuations
nearly broken phase symmetry	nearly broken phase symmetry
coherence over long but finite time intervals	coherence over long but finite distances
slow but free phase diffusion → → residual linewidth	slow but free phase diffusion → residual resistivity (viscosity)
interference fringes	interference effects
(mode structure) no analogon in the time domain because of lack of boundary conditions	flux quantization due to boundary conditions

However, two important differences should be noted:

i. In systems undergoing 'ordinary' phase transitions the order parameter becomes a macroscopic variable in the space and in the time domain simultaneously, since there the order para-

meter is always a static quantity (with frequency 0). An order parameter with finite frequency would necessarily dissipate energy, and can thus not appear in thermodynamic equilibrium. In the optical case, the transition from a microscopic to a macroscopic behaviour occurs only in the time domain. The space and time behaviour are here separated from each other, because the order parameter has a finite (optical) frequency. This is only possible since we are dealing with a stationary non-equilibrium state, in which a constant but finite energy dissipation is allowed.

ii. In the above discussion of a laser the time scale of observation played an important role in distinguishing between microscopic and macroscopic variables. This is different in systems with 'ordinary' phase transitions. For reasons discussed before, the time scale of observation would correspond there also to a length scale of observation. It is one of the striking properties of the critical point of phase transitions that the order parameter becomes macroscopic on a n y length scale[1]). The situation in optics is more analogous to systems with finite spatial extension or with restricted spatial dimensionality. E. g., in 1-dimensional superconductors (in which only the length in one dimension is large compared to the coherence length), it will again depend on the time and length scale of observation, whether one observes a non-zero macroscopic wave function or not. The reason for this analogy is, of course, that all coherent light sources built so far are at most 1-dimensional systems if compared with the coherence length.

Table 2 Common features of lasers and other dissipative structures (e. g. Bénard instability)

Stationary non-equilibrium states
Various branches of states of different symmetry
Thermodynamic branch has highest symmetry (least structure)
Thermodynamic branch is stable in linear vicinity of equilibrium
Instability (threshold) of symmetric branch occurs far from equilibrium; nonlinearity in deviation from equilibrium induces the instability
The state of the system changes continuously at threshold
The new branch of states has lower symmetry (more structure)
The transition is driven by fluctuations

So far, we compared a laser with equilibrium systems. A still closer analogy should hold if we look at stationary non-equilibrium systems. Glansdorff and Prigogine [3] recently gave a discussion of stationary non-equilibrium systems in hydrodynamics and chemistry, passing through instabilities and thereby changing the symmetry of their state. The spontaneous

[1]) This is the starting point for the derivation of scaling laws among critical indices [2].

structures appearing in the state with broken symmetry have been called dissipative structures. It is possible to show that the laser behaves in complete agreement with the Glansdorff-Prigogine theory [4]. Therefore, the coherence of laser light may, indeed, be called a dissipative structure (cf. table 2). Moreover, it can be expected that most of the general points made in the following can also be applied to other dissipative structures.

2. Phenomenological Theory[2])

From the discussion in section 1 two natural assumptions may be extracted:
i. the stationarity of the systems under consideration, ii. the existence of a few variables which move very slowly compared to all the remaining variables. In a reasonable approximation the slow variables can then be described by a Markoff process; by assuming that the slow variables change sufficiently smoothly (an excellent approximation for variables which are spatially macroscopic), we may even adopt a diffusion description and represent the Markoff process by a Fokker-Planck equation of the general form:

$$\frac{\partial W}{\partial t} = \sum_i \frac{\partial}{\partial \omega_i} \left(K_i W + \sum_k K_{ik} \frac{\partial W}{\partial \omega_k} \right) \tag{2.1}$$

The ω_i are the slow variables, $W(\omega, t)$ is their time dependent probability density, $K_i(\omega)$, $K_{ik}(\omega)$ are the (time independent) drift and diffusion coefficients, respectively. We introduce now three further general assumptions:
i. a stationary solution W^s of Eq. (2.1) exists; ii. no s t r i c t l y conserved quantities are among the variables ω; iii. the random process is such that all points in the ω-phase space can, in principle, be reached; (this contains assumption ii.).

As was shown by Lebowitz and Bergmann [5] it follows from these assumptions that the stationary distribution W^s is unique and stable, even for a general Markoff process. This result has very important consequences in our case. In fact, it is able to explain in a very simple way the striking feature of dissipative structures: that the passing through an instability can lead to a change in the symmetry of the system's state.

A first consequence of the uniqueness of the stationary distribution is the fact that W^s must have all the symmetries in phase space, which are present in the original equation (2.1) and therefore in the system. From this follows that a maximum or a minimum of the probability distribution is attached to all points in phase space which are left invariant by symmetry operations. Consider as an example a two-dimensional phase space with rotation invariance (cf. fig. 2). By the relation

$$W^s = \exp(-\phi^s) \tag{2.2}$$

we introduce a potential in this phase space with the property that the system will be found in a state with minimum ϕ^s more likely than in any other state. The origin in fig. 2 is left invariant by rotations; therefore, it has a minimum or a maximum of the potential attached to it.

---

[1]) Many results of this and the following sections have been derived and are explained in more detail in [4]. More complete references to earlier work in this area are given there.

Assume now that we start out with a system described by the potential in fig. 2a, where the most probable state has complete rotation invariance. Then we change the strength of some external forces acting on the system, without changing the rotation invariance in phase space.

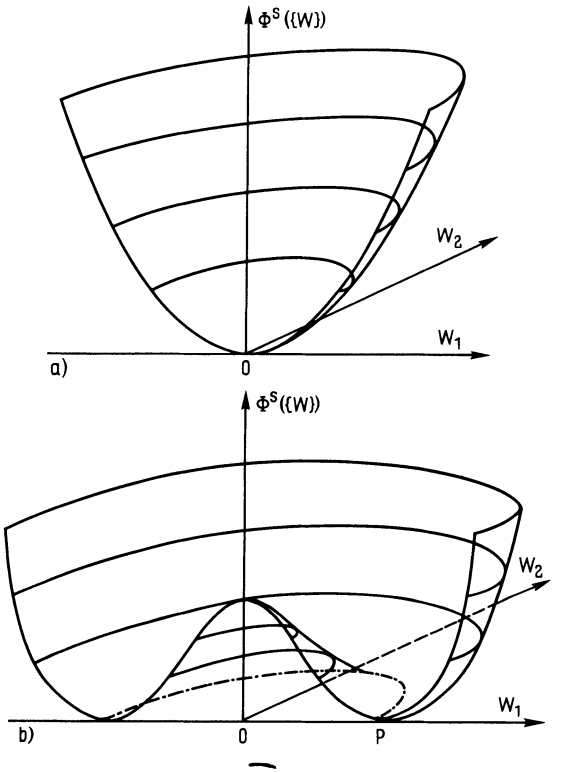


Fig. 2
Stationary potentials ϕ^s (cf. Eq. (2.2)) in a two-dimensional rotation-invariant phase space (cf. text)

Then, the origin in fig. 2 has always an extremum of ϕ^s attached to it. As long as it is a minimum, the symmetry of the most probable state cannot change. But eventually we may reach a point, where the minimum turns into a maximum. Then a new minimum must be formed somewhere else, since the potential has to increase at the far outside (because of the boundary conditions on W^s). This new minimum with necessity has a different symmetry, since the state(s) with the original symmetry are occupied by a maximum. We realize now why a symmetry change is so often connected with an instability. It is the same story, which is known already from the Landau theory of phase transitions [6]. In fact, the Landau theory may be used to construct a quantitative, phenomenological description of this transition [4]. We simply have to expand ϕ^s and the coefficients of Eq. (2.1) in a power series with respect to the deviations from the most symmetric state, making use of the general symmetry requirements. Instead of looking at the general case consider a single mode laser or any other optical oscillator like a parametric oscillator or a Raman Stokes oscillator (cf. Section 5). There, the complex mode amplitude β is the slowly varying, quasi-conserved quantity. Since no special phase is preferred, we have complete phase-angle invariance. (This is exactly the case described in fig. 2).

Expanding the potential ϕ^s in powers of β we obtain, observing phase angle invariance

$$\phi^s = a|\beta|^2 + b|\beta|^4 \tag{2.3}$$

a, b are phenomenological parameters, with

$$a \begin{cases} < 0 \text{ above} \\ = 0 \text{ at} \quad \text{threshold;} \\ > 0 \text{ below} \end{cases} \qquad b > 0 \tag{2.4}$$

We may put $a \sim (\lambda_c - \lambda)$ where λ is a measure of the efficiency with which the system is excited; λ_c is the threshold value. The potential (2.3) was first obtained by Haken [7] on the basis of microscopic considerations. The stationary distribution $W^s \sim \exp(-\phi^s)$ which we obtain from (2.3) was first predicted by Risken [8] and Lax [9], also as a result of a microscopic theory. It is in complete agreement with photo-counting measurements on lasers.

The dynamics is obtained by a similar argument. Expanding the drift and diffusion coefficients of Eq. (2.1) in powers of β and requiring that (2.3) gives the stationary distribution we obtain

$$\frac{\partial W}{\partial t} = -\left(\frac{\partial}{\partial \beta} q(a - 2b|\beta|^2)\beta W\right) - \left(\frac{\partial}{\partial \beta^*} q(a - 2b|\beta|^2)\beta^* W\right) + 2q\frac{\partial^2 W}{\partial \beta \partial \beta^*} \tag{2.5}$$

where, formally, β and β^* are considered as independent variables. In the drift term of Eq. (2.5) we dropped all terms describing a detuning of the laser. In the diffusion term it is sufficient to keep only the lowest order term of the expansion.

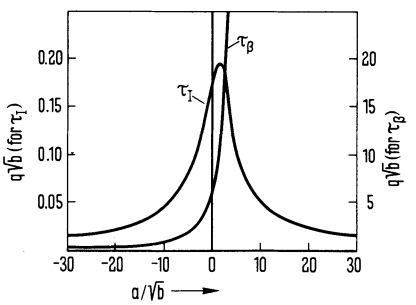


Fig. 3 Coherence time of phase (τ_β) and intensity (τ_I) fluctuations near laser threshold, according to Eq. (2.5) as a function of the pump parameter a (after [8])

Eq. (2.5) has also been obtained from a microscopic theory [8, 9]. Again all results derived from Eq. (2.5) are in agreement with experiments. In the present context it is of interest to look at the result for the coherence time of the complex mode amplitude, which is dominated by the phase coherence (see fig. 3.) As expected, this phase coherence time increases dramatically at

threshold, but always remains finite. This was first predicted by Haken [7]. The coherence time of the light intensity (fig. 3), which is independent of the phase is also increased considerably near threshold. This effect is reminiscent of the critical slowing down in the order parameter fluctuations near the critical point of a phase transition [10]. Indeed, Eq. (2.5) is completely equivalent to the corresponding equations of the Landau theory of equilibrium systems. Due to the simplicity of the present example it even displays the property of detailed balance, which in the present case, is a consequence of the symmetry rather than of the reversibility of the microscopic equations. In the phase transition case the critical slowing down (i. e. diverging coherence time) occurs because of the exact matching between collective motion and random thermal motion at the critical point. In the optical case we find a slowing down because of the matching between loss and stimulated emission gain at threshold. However, this matching never becomes critical for the following reason[1]): Since we have a finite frequency mode, we have to consider spontaneous emission. The loss at threshold equals the sum of the gain due to spontaneous and stimulated emission. The spontaneous part of the gain is smaller than the stimulated part by a factor of $\frac{1}{\bar{n}}$ ($\bar{n}$ = mean number of quanta in the mode). Only for infinite $\bar{n}$ an exact matching between loss and stimulated gain could be achieved. However, even in an infinitely long 1-dimensional laser rod with infinite t o t a l field energy, $\bar{n}$ would always remain finite, due to the fact that the phase coherence time is finite; the finite phase coherence time leads to a continuous spectrum of modes over which the total number of field quanta is distributed.

3. Microscopic theory.

The theory of section 2 has some obvious defects:

i. it covers only the threshold region;
ii. it is not even certain that the threshold region is described completely, for the following reason. Besides the mode amplitude considered before, there are other variables in the problem, the (complex) polarization v of the medium and the population inversion σ of the lasing transition. If we calculate these quantities in a simple theory without fluctuations we get the results shown in fig. 4. The important point is a discontinuity in the slopes at threshold. Of

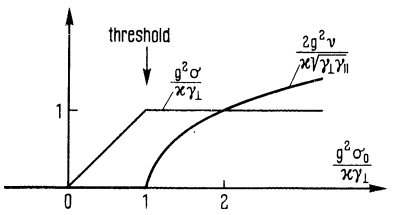


Fig. 4
Population inversion (σ) and polarization (v) as functions of the unsaturated inversion near threshold (fluctuations neglected)

course, this discontinuity is somewhat smoothed by fluctuations. However, it indicates that these variables also participate in the transition and should experience a slowing down. This

---

[1]) The following argument is due to Korenmann [11]. The author is indebted to Prof. P. C. Martin for drawing this reference to his attention and for a clarifying discussion.

expectation is, in fact, born out by a quasi-linear calculation (cf. fig. 5), If this slowing down of other variables is comparable to the slowing down of the amplitude β, the complete coupled process has to be considered. In fact, the situation is here quite similar to the mode-mode

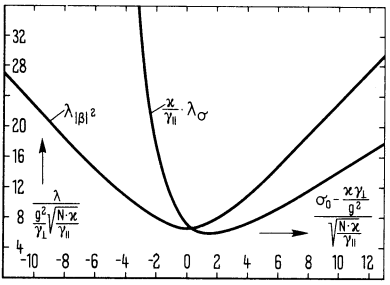


Fig. 5
Line widths of inversion (σ), and intensity fluctuations ($|\beta|^2$) as functions of the unsaturated inversion near threshold. (quasi-linear approximation, parameters of table 3, gas laser)

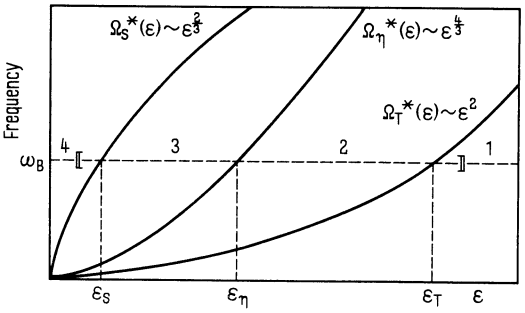


Fig. 6
Characteristic line widths of heat conduction mode (Ω_T^*), shear mode (Ω_η^*), and sound mode (Ω_s) in a simple fluid near the critical point as a function of the reduced temperature $\epsilon = (T - T_c'/T_c$ [12a]. ω_B is the Brillouin frequency (700 MHz) at which measurements were performed [12b] (from H. E. Stanley [12c]))

coupling effects in phase transitions [12] (see fig. 6). However, an important difference is the fact that in our case none of these times becomes critical, whereas at a real critical point all these times would go to infinity. Therefore, as long as these times remain short compared to the coherence time of the mode amplitude the theory of section 2 is applicable. For a real phase transition this theory would not be applicable sufficiently close to the critical point. Quantitative results about the size of these times can only be obtained by the microscopic theory. The microscopic theory for the laser [7] and related systems is quite similar to the kinetic theories of systems in or near thermal equilibrium. Its central goal is to derive a transport equation for a quasi-probability density of all relevant variables. This was done in a quantum mechanically exact way first by Haken, Risken, Weidlich [13]. From their transport equation one can extract in a very good approximation the following set of Langevin equations for the variables β, v, σ of a 2-level laser.

$$\begin{aligned}
&\left(\frac{d}{dt} + \kappa\right)\beta + igv = \Gamma_u \\
&\left(\frac{d}{dt} + \gamma_\perp\right) v - ig\beta\sigma = i\Gamma_v \\
&\left(\frac{d}{dt} + \gamma_{||}\right)(\sigma - \sigma_0) + 2ig\,(\beta v^* - v\beta^*) = \Gamma_\sigma
\end{aligned} \tag{3.1}$$

with

$$\langle \Gamma_i(t)\,\Gamma_j(t+\tau)\rangle = K_{ij}\,\delta(\tau) \qquad \text{and} \quad K_{ij}=0, \text{ except}$$
$$K_{\beta\beta^*}=\kappa; \quad K_{vv^*}=N\gamma_\perp; \quad K_{\sigma\sigma}=N\gamma_\parallel \tag{3.2}$$

The constants in (3.1), (3.2) are explained in table 3 and characteristic values for solid state (ruby) and gas (He-Ne) lasers are given.

Table 3 Laser parameters (cf. Eq. (3.1))

Parameter	definition	solid state (ruby)	gas laser (He-Ne)
κ	cavity width	$\sim 10^8\ \text{sec}^{-1}$	$\sim 10^7\ \text{sec}^{-1}$
$\gamma_\perp$	transverse atomic (homogeneous) linewidth	$\sim 10^{10}\ \text{sec}^{-1}$	$\sim 10^8\ \text{sec}^{-1}$
$\gamma_\parallel$	longitudinal atomic linewidth	$\sim 10^3\ \text{sec}^{-1}$	$\sim 10^8\ \text{sec}^{-1}$
g	dipole coupling constant	$\sim 10^3\ \text{sec}^{-1}$	$\sim 10^3\ \text{sec}^{-1}$
N	number of laser active atoms in lasing volume	$\sim 10^{19}$	$\sim 10^3\ \text{sec}^{-1}$
σ_0	unsaturated population inversion	at threshold $\sim 10^{12}$	at threshold $\sim 10^9$

The coherence times of the polarization and the inversion can be estimated by a quasi-linear analysis of these equations [14]. A result is shown in fig. 5. It turns out that the phenomenological approach is valid throughout the threshold region if the atomic decay times are sufficiently short compared to the photon life-time in the laser cavity[1]). But it is not difficult to construct cases where the slowing down of the atomic variables becomes important. This will modify the simple picture of section 2, at least for sufficiently short time scales, where non-Markoffian effects of the mode amplitudes may become important [15]. It would also be interesting to measure the slowing down of the atomic variables directly, possibly by its effect on the spontaneous emission of the laser atoms at a 90° angle, but it is not obvious how to do that.

---

1) The condition is $\frac{g^2}{\gamma_\perp}\sqrt{\frac{\kappa N}{\gamma_\parallel}} \ll \gamma_\perp, \gamma_\parallel$, i. e. a weak coupling between atoms and field is also necessary.

Not the least for this reason it is interesting that quite similar mode-mode coupling effects occur in multi-mode lasers, and should be much easier to measure there.

4. Multi-mode lasers

The phenomenological approach of section 2 can immediately be applied to multi-mode lasers, most simply to the case where the various modes have uncoupled, statistically independent phases [4]. The potential which one obtains for a two-mode laser is

$$\phi^s = -a_1 |\beta_1|^2 - a_2 |\beta_2|^2 + b_1 |\beta_1|^4 + 2b_{12} |\beta_1|^2 |\beta_2|^2 + b_2 |\beta_2|^4 \tag{4.1}$$

with relations for the coefficients a, b similar to (2.4). If the intensities of the modes are calculated for this case by neglecting fluctuations, i. e. by putting $\frac{\partial \phi^s}{\partial \beta_1} = \frac{\partial \phi^s}{\partial \beta_2} = 0$ the result of fig. 7 is obtained.

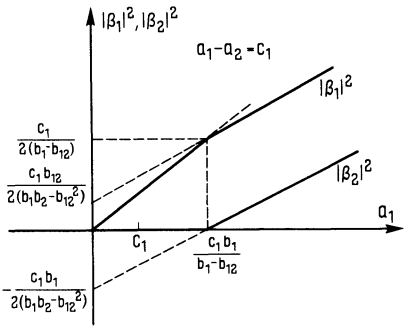


Fig. 7
Light intensities in a two-mode laser according to Eq. (4.1) (Fluctuations neglected)

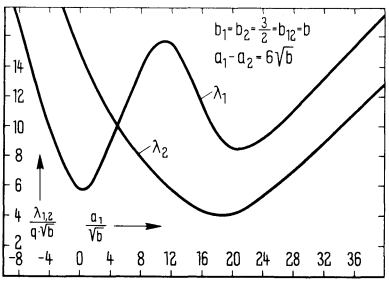


Fig. 8
Effective intensity line widths for a two-mode laser (after [16, 17]; $q_1 = q_2 = q$ defined as in Eq. (2.5))

The remarkable feature is again the discontinuous slope of the intensity in the first mode at the threshold of the second mode. Again it indicates that the first mode participates in the transition of the second mode, and slowing down should occur accordingly. The result of a numerical analysis of this slowing down is plotted in fig. 8[1]). The relaxation time of the first mode has a

[1]) A quasi-linear analysis was first worked out for this case in [16]. The present result was obtained from a numerical study in collaboration with W. A. Smith [17].

pronounced minimum at the threshold of the second mode. Only in the case where the thresholds of different modes are sufficiently far apart from each other, the phenomenological theory is applicable. Otherwise the coupled mode problem has to be solved. The slowing down predicted in fig. 8 should be measurable by presently existing photo-count techniques.

5. Stimulated scattering in nonlinear optics

Coherent light cannot only be generated by lasers, considered so far, but also by a whole bunch of various light sources based on nonlinear stimulated scattering of light in a variety of media [18].

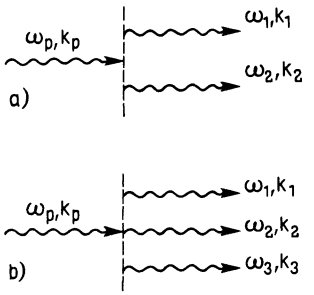


Fig. 9 Light scattering in nonlinear optics
a) parametric scattering in second order
b) parametric scattering in third order

An example is the optical parametric oscillator, which is based on the scattering process fig. 9a, involving three light quanta. Because of the uneven number of light quanta involved this process may only occur in media lacking inversion symmetry, like ferroelectrica. The gain in the parametrically excited modes can exceed the loss if laser pump sources are used and if mirrors reduce the losses in the generated modes. Raman and Brillouin scattering are described by the same diagram, if one of the generated photons is replaced by a phonon. If the two parametric modes are undistinguishable experimentally, subharmonic generation occurs with $\omega_1 = \omega_2$, $k_1 = k_2$. One can also imagine higher order parametric processes like fig. 9b, or third subharmonic scattering with $\omega_1 = \omega_2 = \omega_3$, $k_1 = k_2 = k_3$. In practice these higher order-effects have a very small cross section. Nevertheless, in view of the success in third harmonic generation [19], I think one may expect to be able to see these processes. All the coherent light sources based on the stimulated version of these scattering processes can now be treated by the phenomenological theory of section 2 in a very simple way. As an immediate implication we obtain that the form of the probability distributions for all these light sources should be independent of the details of the microscopic processes giving rise to induced emission.

The form of the distribution depends only on two general features of the microscopic interaction:

i. the phase-angle symmetry of the emitted light, because it affects the form of the power series expansion for ϕ^s;

ii. the order of the scattering process, (i. e. the number of generated quanta) since this determines in which order of the power series expansion the instability has to be introduced.

These features can most easily be understood by looking at second and third order scattering

processes (which, in practice, may be the only important ones). The process of fig. 9a can be described by a term

$$H_{int} = \gamma (b_p^+ b_1 b_2 + b_p b_1^+ b_2^+) \tag{5.1}$$

in the Hamiltonian (γ = coupling constant proportional to the second order susceptibility, b_p = annihilation operator of pump field, b_1^+, b_2^+ = creation operators of the parametric modes). Eq. (5.1) shows that only the sum of the phases of mode 1 and mode 2 is coupled to the phase of the pump field b_p. Each single phase is still arbitrary. Therefore, the distribution for the amplitude of mode 1 (or 2) alone has a phase angle invariance. Furthermore, since the interaction is of second order, it contributes to the second order terms in the expansion for ϕ^s. Therefore, we obtain the same power series as for the single mode laser and the form of the stationary distribution is the same. (Here and in the following the distribution has to be interpreted as the classical limit of a quasi-probability distribution). This result is fully substantiated by a microscopic theory of parametric oscillators [20]. Next, we don't distinguish mode 1 and mode 2 obtaining a Hamiltonian

$$H_{int} = \gamma (b_p^+ b_1^2 + b_p b_1^{+2}) \tag{5.2}$$

which describes subharmonic generation. In this case the phase of the subharmonic is coupled to the pump. Indeed, rotating the phase of the subharmonic by $\pi/2$ changes the sign of the Hamiltonian and we obtain damping, if we had gain before. But the Hamiltonian has still inversion symmetry $b_1 \to -b_1$ and the interaction contributes to the second order term. Therefore, the power series expansion for ϕ^s reads

$$\phi^s = a_1 |\beta_1|^2 + a_2^* \beta_1^2 + a_2 \beta_1^{*2} + b |\beta_1|^4 \tag{5.3}$$

with

$$a_2^* \sim \langle b_p^+ \rangle \tag{5.4}$$

A contour line plot of this potential is shown in fig. 10. A comparison of fig. 10 with fig. 2 demonstrates the influence of symmetry on the shape of the distribution.

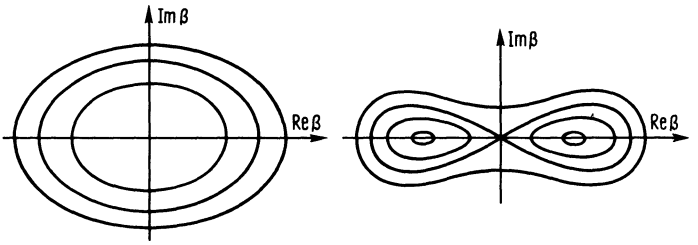


Fig. 10 The stationary potential ϕ^s for subharmonic scattering
a) below threshold
b) above threshold

Now, we consider a third order process (fig. 9b) with a Hamiltonian of the form

$$H_{int} = \gamma (b_p^+ b_1 b_2 b_3 + b_p^+ b_1^+ b_2^+ b_3^+) \tag{5.5}$$

The phase of each separate mode is again arbitrary, leading to a phase angle independent power series expansion for ϕ^s. However, this interaction process is now of third order in the mode amplitudes. For sufficiently small amplitudes this third order term is always small compared to other (second order) terms describing the damping of the modes. Therefore, the state with all amplitudes equal to zero will always be stable against small fluctuations. Only a large fluctuation can bring the third order term (5.5) into play and lead to a new state of the system.

The power series expansion for ϕ^s reads

$$\phi^s = a|\beta|^2 + b|\beta|^4 + c|\beta|^6 \tag{5.6}$$

The interaction (5.5) cannot contribute to the second order term, which therefore remains always positive $a > 0$. The lowest order in which (5.4) can contribute is the 4th order. An instability is introduced if b becomes sufficiently negative, therefore we put

$$b \sim (n_{pc} - \langle b_p^+ b_p \rangle) \tag{5.7}$$

c has always to be larger than zero in order to satisfy the boundary conditions.

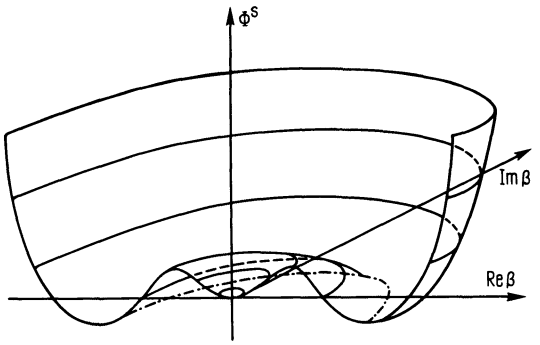


Fig. 11
The stationary potential ϕ^s for third order parametric scattering

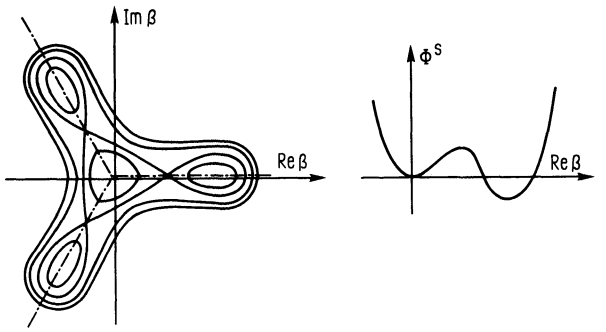


Fig. 12
The stationary potential ϕ^s for third order subharmonic scattering

Therefore we get the potential of fig. 11, which displays all the required features. This potential could be checked experimentally by photo-count methods. Finally, we consider the case of third subharmonic generation. The continuous phase angle invariance is then replaced by the discrete invariance against rotation of the phase by integral multiples of $2\pi/3$. Since the interaction is again of third order, the same considerations as above apply. Above threshold we obtain the potential shown in fig. 12, with a minimum at zero amplitude and three degenerate minima at finite amplitudes.

The methods of section 2 could now be used to analyze the dynamics of the system in this potential by numerical methods or by quasi-linear approximations.

Finally, it may be worth mentioning that for the special class of systems which have the property of detailed balance in the stationary state, it is possible to obtain directly the stationary solution of a microscopically determined Fokker-Planck equation [4] (see also [21]). As should be expected the phenomenological results presented here are in complete agreement with those microscopic results.

6. Conclusion

In the introduction we promoted the idea that the fundamental source of all the various threshold phenomena in lasers and other light sources of nonlinear optics is the fact that the electromagnetic field becomes a macroscopic wave function in the time domain.

In the sections 2 – 5 we have then shown that this idea is able to give a surprisingly simple and complete picture of the threshold phenomena, which can be checked either experimentally for the single mode laser, or theoretically for many special cases where microscopic theories could be developed. In sections 3 – 4 we have shown that even finer details of the threshold phenomena appear in a new light and can be better understood by the use of concepts from phase transition theory. In section 5 we made use of the phenomenological approach to predict very simple and universal properties of the amplitude distributions of various light sources based on induced nonlinear optical scattering. These results once more prove the striking property of the phase transition concept of being applicable, at least in a weak sense, to an extreme variety of problems.

References

[1a] Martin, P. C. in: Low Temperature Physics LT 9, part A (Ed. F. G. Daunt, et al.) New York 1965; Graham, R., Haken, H.: Z. Physik **237**, 31 (1970).

[1b] De Giorgio, V., Scully, M. O.: Phys. Rev. **A2**, 1170 (1970).

[2] Kadanoff, L. P., Physics **2**, 263 (1966).

[3] See e. g. Glansdorff, P., Prigogine, I.: Thermodynamic Theory of Structure, Stability, and Fluctuations. New York 1971.

[4] Graham, R.: Springer Tracts on Modern Physics, vol. 65. Berlin-Heidelberg-New York 1972.

[5] Lebowitz, J. L., Bergmann, P. G.: Annals of Phys. **1**, 1 (1957).

[6] Landau, L. D., Lifshitz, E. M.: Course of Theoretical Physics, vol. 5. London 1958.

[7] Haken, H.: Z. Physik **181**, 96 (1964). A detailed account of the theory initiated by that paper is given in: Haken, H.: Encyclopedia of Physics, vol. 25/2c, Berlin-Heidelberg-New York 1970.

[8] Risken, H.: Z. Physik **186**, 85 (1965); see also Risken, H.: Progress in Optics, vol. 8 (ed. E. Wolf) Amsterdam 1970.
[9] Hempstead, R. D., Lax, M.: Phys. Rev. **161**, 350 (1967). See also Lax, M. in: Statistical Physics, Phase Transitions and Superfluidity (ed. H. Chretian et. al.) New York 1968.
[10] Kawasaki, K.: Annals of Physics **61**, 1 (1970).
[11] Korenmann, V.: Annals of Phys. **39**, 72 (1966).
[12a] Kadanoff, L. P., Swift, J.: Phys. Rev. **165**, 310 (1968); **166**, 89 (1968).
[12b] Ford, N. C., Jr., Langley, K. H., Puglielli, V. G.: Phys. Rev. Lett. **21**, 9 (1968).
[12c] Stanley, H. E.: Introduction to Phase Transitions and Critical Phenomena. New York 1971
[13] Haken, H., Risken, H., Weidlich, W.: Z. Physik **206**, 355 (1967).
[14] See e. g. McCumber, D. E. Phys. Rev. **141**, 306 (1966).
[15] Haake, F.: Z. Physik **227**, 179 (1969).
[16] Grossmann, S., Richter, P. H.: Z. Physik. **242**, 458 (1971).
[17] Graham, R., Smith, W. A.: to be published.
[18] See, e. g., Bloembergen, N.: Nonlinear Optics, New York 1965.
[19] Maker, P. D., Terhune, R. W.: Phys. Rev. 137, A801 (1965).
[20] Graham, R.: Z. Physik **211**, 469 (1968). Louisell, W. H., White, D. R: Phys. Rev. **A1**, 1347 (1970).
[21] Graham, R.: Phys. Lett. **32A**, 373 (1970). Haken, H.: Opt. Electronics **2**, 161 (1970).

Dynamics of Current Instabilities

H. Thomas

The dynamic behavior associated with current instabilities in semiconductors is discussed. The important role of electrodynamic interactions is pointed out. In the mean-field approximation, these interactions can be represented by a linear feedback mechanism. Results obtained for two specific cases, the Gunn instability and the Erlbach instability, are reviewed. In these cases, there occurs a soft longitudinal dielectric relaxation mode and an undamping of transverse modes.

1. Introduction

Instabilities of a uniform current distribution may occur under two different conditions[1])[1]: If a principal value of the differential conductivity tensor σ goes to zero at a critical value E_c of the applied electric field, then the uniform-field state becomes unstable against spacecharge density fluctuations (voltage-controlled instability, Fig. 1). Originally, the consideration was restricted to

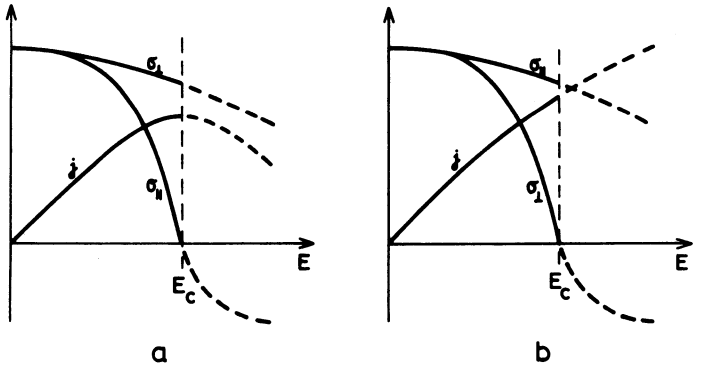


Fig. 1 Voltage-controlled instability
(a) in a direction parallel to the applied field,
(b) in a direction perpendicular to the applied field.

the case that the instability occurs for a direction parallel to the applied field (Fig. 1a). The conductivity tensor in high electric fields is, however, quite anisotropic, and one has also to consider the possibility that the instability occurs for a direction perpendicular to the applied

[1])In general, an instability can occur with respect to a finite wavelength. This would be reflected in the conductivity for corresponding wavevector **q**.

field (Fig. 1b). If, on the other hand, a principal value of σ becomes infinite, then the uniform-field state becomes unstable against current-density fluctuations (current-controlled instability, Fig. 2). The Gunn instability[2]) [2] in certain single-crystal semiconductors like GaAs and Ge is an example for the situation of Fig. 1a, the Erlbach instability [3] in Ge for the situation of Fig. 1b. Certain glassy semiconductors [4], on the other hand, seem to behave as in Fig. 2. There

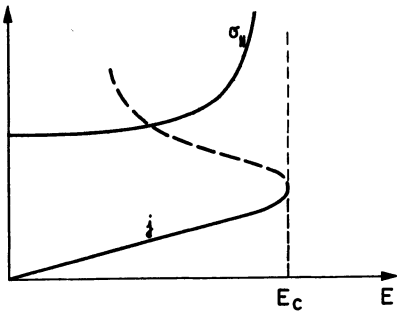


Fig. 2 Current-controlled instability.

exists, however, some evidence that these phenomena may be caused by structural changes due to local heating[3]) [5]. In this note, we shall concentrate on the voltage-controlled cases of the Gunn instability and the Erlbach instability.

These phenomena bear a certain resemblance to phase transition occurring in equilibrium systems. The dynamic behavior of an equilibrium system undergoing a phase transition is characterized by the existence of a soft mode, the frequency of which goes to zero at the stability limit, and of critical fluctuations of low frequency and large amplitude (for a review see Ref. [6]). It is of interest to determine whether analogous dynamic phenomena are associated with current instabilities, which occur in dissipative systems far from thermodynamic equilibrium.

For the voltage-controlled instabilities it is not obvious at first sight why $\sigma \to 0$ should cause an instability at all. Closer analysis shows that the instability is produced by the electromagnetic interactions between the carriers. For non-interacting carriers, the conductivity could become negative, without causing an instability. – The role of the interactions is studied in Sect. 2. In the mean-field approximation, they can be described in terms of a feedback mechanism, leading to a modification of the linear response and of the fluctuations. The effect is very anisotropic, corresponding to the different ways in which the interactions arise: For longitudinal perturbations, there occur electrostatic interactions between charge layers; for transverse perturbations, one has electrodynamic induction produced by the flux associated with the current sheets. – The treatment in this Section follows closely the presentation in Ref. [7].

In Section 3, we review briefly the dynamic behavior for the two cases of the Gunn instability and the Erlbach instability, which were studied in Refs. [8] and [9]. It is shown that the physical processes responsible for the instability affect not only the conductivity but also the real part of the dielectric constant significantly in the frequency range up to the relevant scattering frequencies. Under the feedback mechanism of the electromagnetic interactions, this leads to a critical undamping of normal modes and to a critical behavior of the fluctuations.

[2]) For a review see J. C. McGroddy, M. I. Nathan, J. E. Smith, Jr., IBM J. Res. Develop. 13, 543 (1969) and J. E. Smith, Jr., M. I. Nathan, J. C. McGroddy, IBM J. Res. Develop. 13, 554 (1969).
[3]) For a recent review see the Report of the Ad Hoc Committee on the Fundamentals of Amorphous Semiconductors, National Academy of Sciences, Washington, D.C. 1972.

2. Mean-field theory of electromagnetic interactions

We consider a system which has reached a steady state in the presence of a uniform static field **E**, and study its linear response to an additional external field $\delta \mathbf{E}^{\mathrm{ext}}(\mathbf{x},t)$. We assume homogeneous material[4]), and restrict the discussion to such values of E which give rise to uniform field and current distributions in the steady state. Then, the additional current induced by a plane wave field

$$\delta \mathbf{E}^{\mathrm{ext}}(\mathbf{x}, t) = \delta \mathbf{E}^{\mathrm{ext}}(\mathbf{q}, \omega)\, e^{i(\mathbf{q}\cdot\mathbf{x} - \omega t)} \tag{2.1}$$

is also a plane wave, the amplitude of which is given in terms of the current response tensor $\kappa(\mathbf{q}, \omega)$ by

$$\delta \mathbf{j}^{\mathrm{ind}}(\mathbf{q}, \omega) = \kappa(\mathbf{q}, \omega) \cdot \delta \mathbf{E}^{\mathrm{ext}}(\mathbf{q}, \omega). \tag{2.2}$$

It should be noted that for a system of interacting particles this current response tensor is different from the conventional conductivity tensor (see Eqs. (2.12) and (2.17)).

In addition to the induced current, there occur spontaneous current fluctuations $\delta \mathbf{j}^{\mathrm{fluct}}(\mathbf{x}, t)$ in the system. Because of the spatial uniformity and stationarity of the state considered, the current-current correlation tensor

$$S(\mathbf{x} - \mathbf{x}', t - t') = \langle \delta \mathbf{j}^{\mathrm{fluct}}(\mathbf{x}, t) \delta \mathbf{j}^{\mathrm{fluct}}(\mathbf{x}', t') \rangle \tag{2.3}$$

depends only on the differences $\mathbf{x} - \mathbf{x}'$ and $t - t'$. Its spatial and temporal Fourier transform is the fluctuation tensor

$$S(\mathbf{q}, \omega) = \frac{1}{(2\pi)^4} \int\int S(\mathbf{x}, t)\, e^{-i(\mathbf{q}\cdot\mathbf{x} - \omega t)}\, d^3x\, dt, \tag{2.4}$$

which determines the correlations between the Fourier amplitudes of the fluctuating current:

$$\langle \delta \mathbf{j}^{\mathrm{fluct}}(\mathbf{q}, \omega) \delta \mathbf{j}^{\mathrm{fluct}}(\mathbf{q}', \omega') \rangle = S(\mathbf{q}, \omega) \delta(\mathbf{q} + \mathbf{q}') \delta(\omega + \omega'). \tag{2.5}$$

The system can thus be represented as a linear transmission system with a built-in noise source (Fig. 3a). The characteristics of this system – the current response tensor $\kappa(\mathbf{q}, \omega)$ and the fluctuation tensor $S(\mathbf{q}, \omega)$ – will be functions of the static field **E**. It is the object of this paper to study their behavior when **E** approaches the stability limit of the uniform phase.

The linear response tensor $\kappa(\mathbf{q}, \omega)$ can be analytically continued into the complex ω-plane. The poles of the eigenvalues of $\kappa(\mathbf{q}, \omega)$ are the normal-mode frequencies $\Omega(\mathbf{q})$. In order for the steady state belonging to the applied field **E** to be stable, all normal modes must be damped, i.e. all $\Omega(\mathbf{q})$ must lie in the lower half of the ω-plane. As **E** approaches the stability limit of the uniform phase, we expect one or more normal modes to become undamped, i.e. one or more of the $\Omega(\mathbf{q})$ to move up to the real axis.

We start with a system of non-interacting electrons, the behavior of which is represented by the diagram of Fig. 3a with κ and S replaced by the independent-particle tensors κ^{free} and S^{free}. These tensors contain the effects of electron-lattice interactions, and can be obtained in principle from microscopic transport theory or by an independent-particle Green's function approach (see Refs. [10] and [11], and further references contained therein).

---

[4]) We disregard the atomic structure of the semiconductor and use a continuum description. This is a very good approximation for the long wavelengths considered.

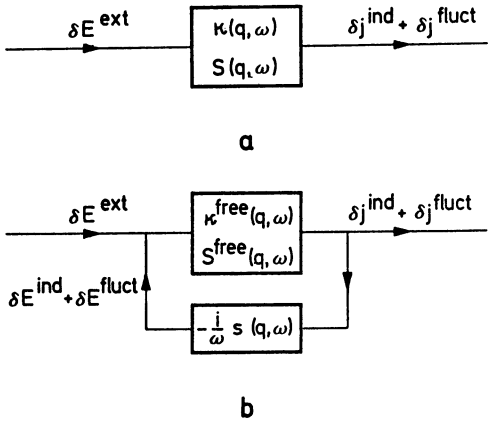


Fig. 3 (a) Representation of the system response as a linear transmission system, (b) Representation of the electromagnetic interactions by a feedback mechanism.

The electromagnetic interactions can be described by interaction fields acting on the electrons in addition to the external field $\delta \mathbf{E}^{\mathrm{ext}}$. The current contributions $\delta \mathbf{j}^{\mathrm{ind}}$ and $\delta \mathbf{j}^{\mathrm{fluct}}$ give rise to fields $\delta \mathbf{E}^{\mathrm{ind}}$ and $\delta \mathbf{E}^{\mathrm{fluct}}$ which can be written in terms of a dynamical screening tensor $s(\mathbf{q}, \omega)$ in the form

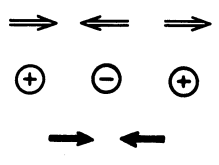
$$\delta \mathbf{E}^{\mathrm{ind}}(\mathbf{q}, \omega) = -\frac{\mathrm{i}}{\omega} s(\mathbf{q}, \omega) \cdot \delta \mathbf{j}^{\mathrm{ind}}(\mathbf{q}, \omega). \tag{2.6}$$

and

$$\delta \mathbf{E}^{\mathrm{fluct}}(\mathbf{q}, \omega) = -\frac{\mathrm{i}}{\omega} s(\mathbf{q}, \omega) \cdot \delta \mathbf{j}^{\mathrm{fluct}}(\mathbf{q}, \omega). \tag{2.7}$$

The electromagnetic interactions are thus represented by a feedback mechanism as shown in Fig. 3b.

a) Longitudinal waves:



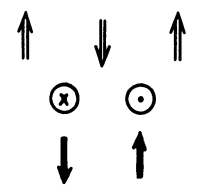
$$\delta j^{\mathrm{ind}}$$

$$i\omega \delta \varrho^{\mathrm{ind}} = iq\, \delta j^{\mathrm{ind}}$$

$$iq\, \delta E^{\mathrm{ind}} = 4\pi \delta \varrho^{\mathrm{ind}}$$

$$\delta E^{\mathrm{ind}} = -\frac{4\pi i}{\omega} \delta j^{\mathrm{ind}}$$

b) Transverse waves:



$$\delta j^{\mathrm{ind}}$$

$$-icq\, \delta B^{\mathrm{ind}} = 4\pi \delta j^{\mathrm{ind}} - i\omega \delta E^{\mathrm{ind}}$$

$$icq\, \delta E^{\mathrm{ind}} = i\omega \delta B^{\mathrm{ind}}$$

$$\delta E^{\mathrm{ind}} = -\frac{4\pi i}{\omega} \frac{\omega^2}{\omega^2 - c^2 q^2} \delta j^{\mathrm{ind}}$$

Fig. 4 Induced field produced by the electromagnetic interactions (a) for longitudinal waves, (b) for transverse waves.

The mean-field approximation is obtained by treating the response to the resulting field $\delta \mathbf{E}^{\mathrm{ext}} + \delta \mathbf{E}^{\mathrm{ind}} + \delta \mathbf{E}^{\mathrm{fluct}}$ in linear response theory, i.e. by neglecting the non-linear response of the system to its own fluctuations. Then, the fluctuations are linearly superimposed on the response to $\delta \mathbf{E}^{\mathrm{ext}}$, and we obtain independently for the two contributions,

$$\delta \mathbf{j}^{\mathrm{ind}}(\mathbf{q}, \omega) = \kappa^{\mathrm{free}}(\mathbf{q}, \omega) \cdot [\delta \mathbf{E}^{\mathrm{ext}}(\mathbf{q}, \omega) + \delta \mathbf{E}^{\mathrm{ind}}(\mathbf{q}, \omega)] \tag{2.8}$$

and

$$\delta \mathbf{j}^{\mathrm{fluct}}(\mathbf{q}, \omega) = \delta \mathbf{j}^{\mathrm{fluct,\,free}}(\mathbf{q}, \omega) + \kappa^{\mathrm{free}}(\mathbf{q}, \omega) \cdot \delta \mathbf{E}^{\mathrm{fluct}}(\mathbf{q}, \omega). \tag{2.9}$$

This method of representing the interactions as a feedback mechanism is quite analogous to the method discussed in Ref. [6] for equilibrium systems.

The conductivity tensor $\sigma(\mathbf{q}, \omega)$ is defined by the current response to the total non-fluctuating field $\delta \mathbf{E}^{\mathrm{ext}} + \delta \mathbf{E}^{\mathrm{ind}}$. Eq. (2.8) shows that in the mean-field approximation it is given by

$$\sigma(\mathbf{q}, \omega) = \kappa^{\mathrm{free}}(\mathbf{q}, \omega). \tag{2.10}$$

The dynamical screening tensor $s(\mathbf{q}, \omega)$ has to be obtained from Maxwell's equations [7]. We give here its form for the special cases of purely longitudinal and purely transverse waves. In the longitudinal case, the interaction field is produced by the space charge layers associated with the wave (see Fig. 4a), and one finds

$$s_{\ell}(\mathbf{q}, \omega) = 1. \tag{2.11}$$

We solve the feedback equations (2.8) and (2.9) for the resulting currents $\delta \mathbf{j}^{\mathrm{ind}}$ and $\delta \mathbf{j}^{\mathrm{fluct}}$ in terms of $\delta \mathbf{E}^{\mathrm{ext}}$ and $\delta \mathbf{j}^{\mathrm{fluct,free}}$, respectively. We thus obtain for the current response and, by making use of Eq. (2.5), for the fluctuation spectrum of the interacting system

$$\kappa_{\ell}(\mathbf{q}, \omega) = \frac{\sigma_{\ell}(\mathbf{q}, \omega)}{\epsilon_{\ell}(\mathbf{q}, \omega)} \tag{2.12}$$

and

$$S_{\ell}(\mathbf{q}, \omega) = \frac{1}{|\epsilon_{\ell}(\mathbf{q}, \omega)|^2} S_{\ell}^{\mathrm{free}}(\mathbf{q}, \omega). \tag{2.13}$$

Here

$$\epsilon_{\ell}(\mathbf{q}, \omega) = 1 + \frac{4\pi \mathrm{i}}{\omega} \sigma_{\ell}(\mathbf{q}, \omega) \tag{2.14}$$

is the longitudinal component of the dielectric tensor. The frequencies Ω_{ℓ} of the longitudinal normal modes (plasma modes) are thus obtained as solutions of

$$\epsilon_{\ell}(\mathbf{q}, \Omega_{\ell}) = 0 \tag{2.15}$$

In the transverse case, the interaction field is produced by electrodynamic induction of the magnetic flux sheets associated with the wave (see Fig. 4b), and one finds from Maxwell's equations

$$s_t(\mathbf{q}, \omega) = \frac{\omega^2}{\omega^2 - c^2 q^2}. \tag{2.16}$$

The current response and the fluctuation spectrum of the interacting system are in this case given by

$$\kappa_t(\mathbf{q}, \omega) = \frac{\omega^2 - c^2 q^2}{\epsilon_t(\mathbf{q}, \omega) \omega^2 - c^2 q^2} \sigma_t(\mathbf{q}, \omega) \tag{2.17}$$

and

$$S_t(\mathbf{q}, \omega) = \left| \frac{\omega^2 - c^2 q^2}{\epsilon_t(\mathbf{q}, \omega) \omega^2 - c^2 q^2} \right|^2 S_t^{\mathrm{free}}(\mathbf{q}, \omega). \tag{2.18}$$

The frequencies Ω_t of the transverse normal modes are obtained as the solution of

$$\epsilon_t(q, \Omega_t)\,\Omega_t^2 - c^2 q^2 = 0 \tag{2.19}$$

The mean-field approximation is justified as long as the fluctuations are small. Since we expect large fluctuations as E approaches the stability limit, the results will unfortunately become quantitatively unreliable just in the most interesting critical region.

3. Specific cases

The above considerations have shown that in the mean-field approximation the linear response and the fluctuation spectrum of a particular system can be found from the behavior of the corresponding system in the absence of electromagnetic interactions between the carriers according to Eqs. (2.12), (2.13) and (2.17), (2.18) in connection with Eq. (2.10). Thus, for the systems considered here, we have to investigate the transport properties of an electron gas in a semiconductor interacting only with the phonons and imperfections of the crystal. The treatment of this single-electron transport problem on a microscopic level, e.g. by means of the Boltzmann equation, is still quite complicated and requires powerful numerical methods. We have therefore studied the transport problem on a more macroscopic level by means of balance equations for the particle number and momentum densities, which describe the scattering processes in terms of phenomenologically introduced scattering rates. We briefly discuss here the two cases of the Gunn instability [8] and the Erlbach instability [9].

3.1. Gunn instability. This type of instability can occur in n-type semiconductors with a conduction band consisting of a high-mobility valley (1) which forms the bottom of the band, and a set of low-mobility valleys (2) which lie a few tenths of an electron volt higher in energy. At small values of the applied field, all electrons are in valley 1, and one finds ohmic conduction corresponding to its high mobility. As the field increases, the electrons are heated up, and when their energy becomes sufficiently high, a further increment of the field will give rise to competing contributions to the current: The drift velocity will continue to increase, but the carrier concentration in valley 1 will decrease due to electron transfer to the upper valleys. At the critical field, the two contributions cancel each other.

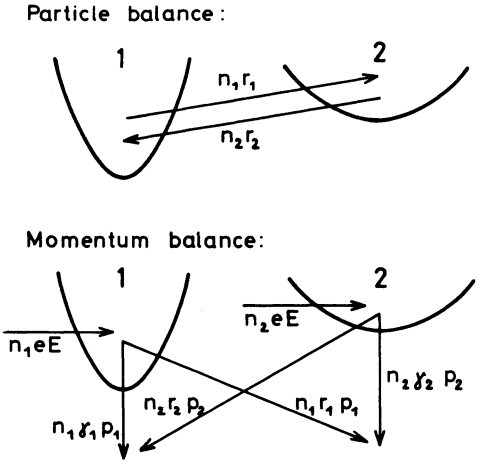


Fig. 5 (a) Particle density balance, (b) momentum density balance for a two valley model.

We treat the upper valleys formally as a single valley, and describe the state of the system by the particle densities n_1 and n_2 and the average drift velocities v_1 and v_2 in the lower and upper valleys. Particle conservation requires $n_1 + n_2 = n_c$, where n_c is the total carrier concentration in the conduction band. The change of the state with time is determined by the balance equations for the particle and crystal-momentum densities (Fig. 5). For uniform states, we write

$$\frac{\partial n_1}{\partial t} = -n_1 r_1 + n_2 r_2 = -\frac{\partial n_2}{\partial t} \tag{3.1}$$

and

$$\begin{aligned} \frac{\partial(n_1 m_1 v_1)}{\partial t} &= n_1 e\mathbf{E} - (\gamma_1 + r_1) n_1 m_1 v_1 \\ \frac{\partial(n_2 m_2 v_2)}{\partial t} &= n_2 e\mathbf{E} - (\gamma_2 + r_2) n_2 m_2 v_2 \end{aligned} \tag{3.2}$$

Here, we have introduced phenomenological scattering rates γ_1, γ_2 for intravalley scattering and r_1, r_2 for intervalley scattering. Their dependence on drift velocity has to be obtained from a microscopic theory; in Ref. [8] we have made use of results based on approximate solutions of the Boltzmann equation [11].

The conditions for a stationary state in the presence of a constant field $\mathbf{E}$ are obtained by setting the left hand sides of Eqs. (3.1) and (3.2) equal to zero. Solving these conditions for n_1, n_2 and v_1, v_2 in terms of $\mathbf{E}$ and substituting into the expression for the current

$$j = e(n_1 v_1 + n_2 v_2) \tag{3.3}$$

gives the static characteristic. It is found that the steep rise of r_1 with v_1 when transfer from valley 1 to valley 2 becomes energetically possible does indeed lead to a negative differential conductivity [8].

The linear response to a dynamic perturbation

$$\delta\mathbf{E}(t) = \delta\mathbf{E}\, e^{-i\omega t} \tag{3.4}$$

is obtained by solving Eqs. (3.1) and (3.2) for small deviations $\delta n_1 = -\delta n_2, \delta v_1, \delta v_2$ from the stationary state in terms of $\delta\mathbf{E}$. The frequency-dependent conductivity $\sigma_c(\omega; \mathrm{E})$ is then found from

$$\begin{aligned} \delta j &= e[(n_1 \delta v_1 + n_2 \delta v_2) + (v_1 - v_2)\delta n_1] \\ &= \sigma_c(\omega; \mathrm{E})\, \delta\mathbf{E}. \end{aligned} \tag{3.5}$$

One recognizes easily the two competing contributions to the current due to drift and transfer. The response of the bound electrons is described by the dielectric constant ϵ_L of the lattice, such that the total response of the semiconductor is given by the dielectric function

$$\epsilon(\omega; \mathrm{E}) = \epsilon_L + \frac{4\pi i}{\omega} \sigma_c(\omega; \mathrm{E}). \tag{3.6}$$

According to Eq. (2.15), the zeros of this function are the eigenfrequencies of the longitudinal modes. Their dependence on the applied field is studied in detail in Ref. [8].

For a discussion of the dynamical properties connected with the instability, we disregard for simplicity the contribution of the low mobility valleys to the current, and assume that the intervalley transfer rates are small compared to the momentum relaxation rates (Drude cutoff). The interesting phenomena occur for frequencies small compared to the Drude cutoff frequencies. In this frequency regime, we find for the real parts $\epsilon'(\omega)$ and $\sigma'(\omega)$ the approximate expressions

$$\sigma'(\omega) = \sigma_0\left[1 - \frac{\theta}{R}\frac{R^2}{\omega^2 + R^2}\right] \tag{3.7}$$

and

$$\epsilon'(\omega) = \epsilon_0 + 4\pi\sigma_0\theta\frac{1}{\omega^2 + R^2} \tag{3.8}$$

where

$$R = r_1 + r_2; \qquad \theta = v_1\frac{dr_1}{dv_1}. \tag{3.9}$$

As the field increases, θ rises steeply; at the critical field, $\theta = R$. The behavior of $\sigma'(\omega)$ and $\epsilon'(\omega)$ is shown in Fig. 6. Parallel with the decrease of the static conductivity $\sigma'(0) = \sigma_0(1 - \theta/R)$, the static dielectric constant $\epsilon'(0) = \epsilon_0 + 4\pi\sigma_0\theta/R^2$ rises from the value ϵ_0 which may be positive or negative, depending on the carrier concentration, to the large positive value $\epsilon_0 + 4\pi\sigma_0/R$. Significant changes occur in the whole frequency range $\omega \lesssim R$.

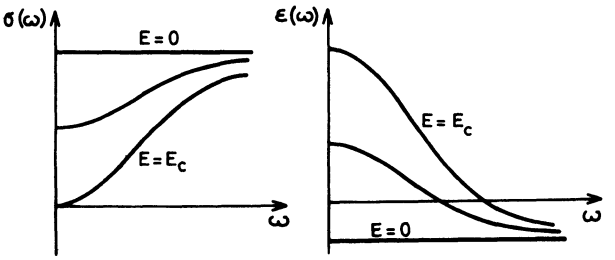


Fig. 6
Variation of the real parts of conductivity $\sigma'(\omega)$ and dielectric constant $\epsilon'(\omega)$ in the frequency range $\omega < R$ with applied field.

From Eq. (2.15), one finds for fields close to the critical field a longitudinal mode with a purely imaginary frequency

$$\Omega_l = -i\Gamma_d \tag{3.10}$$

where

$$\Gamma_d = 4\pi\frac{\sigma'(0)}{\epsilon'(0)} \tag{3.11}$$

is the dielectric relaxation rate. Since Γ_d goes to zero proportional to $\sigma'(0)$ as $E \to E_c$, we have thus found a soft dielectric relaxation mode. – The longitudinal fluctuation spectrum is given by

$$S_l(\omega) = \frac{1}{[\epsilon'(0)]^2}\frac{\omega^2}{\omega^2 + \Gamma_d^2}S_l^{free}(\omega). \tag{3.12}$$

For the transverse modes, one finds from Eq. (2.19) close to the critical field a frequency

$$\Omega_t = \pm v_t q - \frac{i}{2}\Gamma_d, \tag{3.13}$$

with a phase velocity

$$v_t = \frac{c}{\sqrt{\epsilon'(0)}} \tag{3.14}$$

and a damping proportional to Γ_d. Thus, also this transverse mode becomes undamped for $E \to E_c$ in the long-wave-length limit. – The transverse fluctuation spectrum is given by

$$S_t(q, \omega) = \frac{1}{[\epsilon'(0)]^2} \frac{(\omega^2 - c^2 q^2)^2}{(\omega^2 - v_t^2 q^2)^2 + (\omega \Gamma_d)^2} S_t^{\text{free}}(q, \omega). \tag{3.15}$$

3.2. Erlbach instability. This type of instability can occur in n-type semiconductors with a conduction band consisting of a set of equivalent anisotropic valleys. The simplest model consists of two equivalent valleys 1 and 2 with anisotropic axes pointing into different directions (Fig. 7).

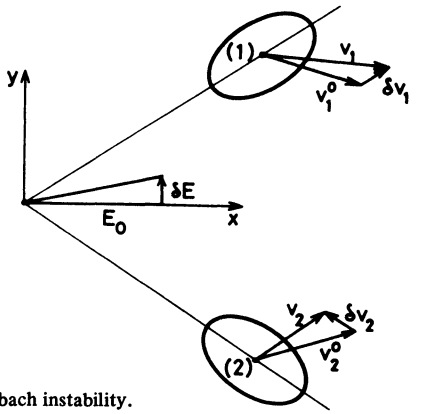


Fig. 7 Mechanism of the Erlbach instability.

The static field $\mathbf{E}_0$ is applied along the symmetry axis (x-axis). It induces drift velocity components

$$v_{1x}^0 = v_{2x}^0; \qquad v_{1y}^0 = -v_{2y}^0 \tag{3.16}$$

in the two valleys, and leaves the carrier concentrations unaltered:

$$n_1^0 = n_2^0 = n_0 \tag{3.17}$$

Thus, because of the anisotropy, the drift velocity is tilted in each valley with respect to the field direction, but the perpendicular contributions to the current cancel on account of symmetry. A small field δE_y perpendicular to $\mathbf{E}_0$ induces a linear change in the drift velocities

$$\delta v_{1x} = -\delta v_{2x}; \qquad \delta v_{1y} = \delta v_{2y} \tag{3.18}$$

leading to a positive contribution to the current change δj_y. However, due to this change the total drift velocity in valley 1 has become larger than the total drift velocity in valley 2, and the electrons are scattered more effectively from valley 1 to valley 2 than in the opposite direction. The resulting unbalance

$$\delta n_1 = -\delta n_2 \tag{3.19}$$

in the carrier concentration leads to a negative contribution to the current change:

$$\delta j_y = 2e[n_0 \delta v_{1y} + v_{1y}^0 \delta n_1]. \tag{3.20}$$

Thus, we find again a competition between contributions due to drift and transfer. At the critical field, the two contributions cancel.

The dynamic behavior of this system has been studied in Ref. [9] by means of balance equations for particle and crystal momentum densities similar to Eqs. (3.1), (3.2). In contrast to the previous section, the two valleys are equivalent in this case, and the anisotropy of the effective masses is of crucial importance and has to be taken into account. – For the intravalley and intervalley scattering rates, use has been made of the results of an approximate microscopic calculation [13]. It is found that the scattering rates increase sufficiently strongly with the applied field for an instability to occur. The behavior for fields close to the critical field is qualitatively very similar to the behavior discussed above for the case of the Gunn instability: One finds again a soft longitudinal dielectric relaxation mode, and undamping of transverse modes.

It should be noted that the considerations presented here apply only to the states of uniform field distribution present for fields below the critical field. In the case of the Gunn instability in the non-uniform state a high-field domain moves through the material with a velocity corresponding to the drift velocity of the carriers[5]). Such a domain can in principle be obtained as the solution of the (nonlinear) balance equations for non-uniform states. – In the case of the Erlbach instability, the structure of the non-uniform state is not known.

References

[1] Ridley, B. K.: Proc. Phys. Soc. **82**, 954 (1964).
[2] Gunn, J. B.: Solid State Comm. **1**, 88 (1963); IBM J. Res. Develop. **8**, 141 (1964).
[3] Erlbach, E.: Phys. Rev. **132**, 1976 (1963); M. Shyam, Kroemer, H.: Appl. Phys. Letters **12**, 283 (1968).
[4] Ovshinsky, S. R.: Phys. Rev. Letters **21**, 1450 (1968); for references to other work see H. Fritzsche: IBM J. Res. Develop. **13**, 515 (1969).
[5] Pearson, A. D.: IBM J. Res. Develop. **13**, 510 (1969); H. J. Stocker (private communication).
[6] Thomas, H.: IEEE Transactions on Magnetics **5**, 874 (1969).
[7] Thomas, H.: Lectures presented at the Conference on Fluctuation Phenomena, Chania, Crete, August 1969 (To be published).
[8] Pytte, E. and Thomas, H.: Phys. Rev. **179**, 431 (1969).
[9] Pfundtner, K.: Diplomarbeit Frankfurt 1972.
[10] Schlup, W. A.: Phys. kondens. Materie **8**, 167 (1968).
[11] Pytte, E.: Phys. kondens. Materie **9**, 211 (1969).
[12] Conwell, E. M. and Vassell, M. O.: Phys. Rev. **166**, 797 (1968).
[13] Reik, H. G. and Risken, H.: Phys. Rev. **124**, 777 (1961).

[5]) Actually, a Gunn domain is ordinarily nucleated in the non-uniform field region close to the cathode before the bulk stability limit E_c is reached.

Cooperative Phenomena in Data Processing[1])

Rolf Landauer and James W. F. Woo

Introduction

Classical physics was based on the presumption that there is a universe following exact laws of motion. This viewpoint allowed for the fact that we were able to measure that universe only imperfectly; this was presumed to be a human failure, and by the use of diligence, the failure could be arbitrarily diminished. The physicist has now learned, first through relativity, and more clearly through quantum mechanics, that the measurement process is not an idealizable entity separate from the laws of the real physical world. As far as the calculational process is concerned, the physicist has been less perceptive. The prevalent assumption has been that through the use of enough graduate students any calculation becomes feasible. Any calculational routine is presumed to be significant and meaningful, regardless of our physical ability to execute it. The advent of the modern computer has, perhaps, served to further remove the calculational process from the physicist's direct detailed awareness, and thereby aggravated the physicist's lack of attention to the subject. One clear exception is P. W. Bridgman, who in his writings recognized that the totality of physical operations relevant to science included "paper and pencil operations" [1–3].

In this paper, we want to point out that these operations on information are themselves susceptible to analysis. While the analysis is still in a very rudimentary state, it at least permits us to articulate some of the basic questions in the field. While the computer may have served to remove the step-by-step operations from the view of the research worker, it has, of course, sharpened the awareness of those of us who are actually in the computer technology business. Tremendous strides have been made within the last quarter century in computer technology producing a change in the time required by the elementary step from the original speeds of mechanical devices to today's devices which can switch in the 10^{-9} sec range. Additional factors of ten are visible in our current exploratory work. Computer components have simultaneously changed from the size of a vacuum tube to integrated circuits in which densities of 10^6 components per in^2 are possible. This very naturally poses the question: How far can these technological advances go, is there a limit that is imposed by the laws of physics?

---

[1]) We gratefully acknowledge permission for the quotation of selected short excerpts. Permission comes from the Bell System Technical Journal in the case of reference 8 (Copyright, 1948, The American Telephone and Telegraph Co.), and from the IBM Journal of Research and Development for references 13 and 19 (Copyright, 1960 and 1961, International Business Machines Corp.).

The question can be answered at two levels. One level, which we shall call a semipractical level, looks at present computer systems, and asks how can they be scaled? Discussions of this sort can be found in recent papers by R. W. Keyes [4], and by the author [5], as well as in a recent Committee report [6]. At a still more fundamental level, and this is the level of the present discussion, we try to idealize computer elements to their ultimate, and try to ignore all unsolved technical difficulties. We try to retain a level of generality which covers all the devices not yet invented. An earlier review by this author [7] has stressed the epistemological aspects discussed in this Introduction.

Before we go into the detailed analysis, we shall try to describe the epistemological questions in a general way. Our basic point is that all information processing activities, whether carried out in a computer, in the brain, or with paper and pencil, utilize physical degrees of freedom. Once we allow for the physical nature of the information, it immediately becomes subject to physical degradation, e. g. through thermal agitation. Not only is the information itself subject to the possibility of degradation, but the physical machinery dealing with it, the computer itself, is also subject to the physical degradation of its own structure, for example through diffusional decay of its components.

To illustrate the fallibility of the calculational process, consider the evaluation of π using pencil and paper. The mathematician has given us an idealized universe in which π, for example, is defined and clearly distinguished from all other numbers because we have rules for calculating π to an arbitrary number of places. Now we come back with such questions as "Do we really know what π is if the initial digits of π deteriorate away by atomic diffusion processes after a finite number of years and at a fairly well-defined stage of the calculation? " or "Can we really distinguish π from a very close numerical neighbor if we compare the numbers, digit by digit, by methods that are very reliable but can never be made completely foolproof? ".

The mere existence of information decay processes does not guarantee an insurmountable barrier. Human knowledge continues to increase despite the rather limited lifetime of any one individual; similarly, information transfer from computer to computer can be used to keep information "alive". The fact that a given computer has a limited reliability due to its internal thermal agitation does not necessarily prevent us from making another computer that, while not guaranteed to be error-free, is more reliable than the first one. The more reliable computer, however, is likely to be bigger or slower than the initial one. Thus we can invoke schemes whereby any calculation which has run long enough will be automatically transferred to a more reliable machine. After a certain time on the second machine, if the computation has still not run its course, we go to a third and still more reliable machine, etc.

The really basic question is whether data processing can be made a r b i t r a r i l y precise, i. e. can the deleterious effects of random processes be contained to any required degree? This question is not going to be conclusively answered by our subsequent discussions, but once we limit the computer to use a finite mass and to carry out computations at some minimal speed the answer is, at least in the author's opinion, likely to be a pessimistic one.

We are strongly suggesting that in the real (and probably finite) universe, the classical mathematical schemes which invoke arbitrarily many steps are not executable. The mathematician is likely to wave this aside as irrelevant. The mathematician believes that he declared his independence from the real universe by declaring that his formal systems were not necessarily intended to describe that universe. That, of course, is his privilege; we can all "do

our own thing". But that does not mean that the operations used in formal systems can be independent of the physical universe.

In our subsequent detailed analyses, we shall touch upon essentially three topics. We shall first discuss memory. A computer memory is much like a communications channel; it does not distort the information intentionally, and does not cause streams of information to interact with each other. The discussion of memory can therefore be based on information theory, as originally generated by Shannon [8]. In the second portion of the detailed discussions, we will discuss the energy dissipation of the nonlinear logic elements and show that dissipation is not just an unwanted by-product but an essential effect. In this second discussion, we will focus on computing devices, typified by cores, ferroelectrics, and superconductors. These can hold information without dissipation of energy.

In this second part, we will be touching upon subjects closely related to Szilard's theory [9] of measurement and its extensions as discussed by Brillouin [10]. These theories show that the measurement process is accompanied by a need to dissipate energy. The reason a new theory is needed is that these older works do not really make it clear which steps in the computing process are "measurements". In this second section, we will also recognize that if dissipation is required, then thermal noise sources must inevitably be present and will inquire about their effects on computer reliability.

In the final section, we will develop a theory of Brownian motion in dissipative bistable systems, since computers are in actual fact dependent upon transistor circuits, and these are dissipative devices.

1. Shannon's information theory

Before presenting new ideas, it will be advisable to remind the reader about the central result of Information Theory, as presented by Shannon [8] in 1948.

Shannon is concerned with defining the information handling capability of a source of signals. Instead of taking the number of possible messages to measure the diversity in the message space, he uses its logarithm to the base two. This is an obvious emulation of the entropy in statistical mechanics, and Shannon has no hesitation to call this measure the entropy of the source. It is an additive measure: if two independent message sources are either used simultaneously or one after another, the log of the total number of possible messages is the sum of the two individual logarithms. If a message consists of a series of independently transmitted symbols, we can then define an information content per symbol. In the simple case where we have an equally likely probability of "0" and "1" for the elementary message unit, it becomes comfortable to have this elementary unit assigned an information content of unity, this explains the base "two" for the logarithm. To take care of the case where the messages i (or symbols i) cannot be generated with equal probability, Shannon generalizes the definition of information to

$$H = -\sum_{i=1}^{n} p_i \log p_i, \tag{1.1}$$

again in accordance with the statistical mechanics notion of entropy.

Indeed Shannon shows that Eg. (1.1) is the only functional form which satisfies the following requirements, quoted directly from Shannon [8].

"1. H should be continuous in the p_i.

2. If all the p_i are equal, $p_i = 1/n$, then H should be a monotonic increasing function of n. With equally likely events there is more choice, or uncertainty, when there are more possible events.

3. If a choice be broken down into two successive choices, the original H should be the weighted sum of the individual values of H. The meaning of this is illustrated (Fig. 1). At the

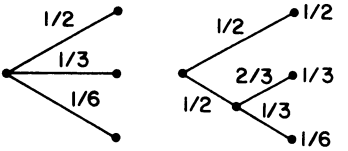


Fig. 1
Decomposition of a choice from three possibilities. (Ref. [8])

left we have three possibilities $p_1 = 1/2, p_2 = 1/3, p_3 = 1/6$. On the right we first choose between two possibilities each with probability 1/2, and if the second occurs make another choice with probabilities 2/3, 1/3. The final results have the same probabilities as before. We require, in this special case, that

$$H\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right) = H\left(\frac{1}{2}, \frac{1}{2}\right) + \frac{1}{2} H\left(\frac{2}{3}, \frac{1}{3}\right)$$

The coefficient 1/2 is because this second choice only occurs half the time."

Shannon then goes on to generalize this definition, for the case of a noisy information channel, where the symbols received are not necessarily identical to the symbols transmitted. In that case, the rate of information transmission is labelled as

$$R = H(x) - H_y(x) \tag{1.2}$$

where H(x) is the entropy (per unit time) of the source alphabet x, and $H_y(x)$ represents the loss during transmission. $H_y(x)$ is called a conditional entropy, or equivocation, and represents the entropy or diversity of the source x, for a given received message y, weighted by the probability of y. R is equal to the undiminished H(x) if the mapping of x into y is strictly one-to-one without error. If x and y are completely independent, then $H(x) = H_y(x)$ and R vanishes.

Again we can best explain by quoting from Shannon [8]:

"Suppose there are two possible symbols 0 and 1, and we are transmitting at a rate of 1000 symbols per second with probabilities $p_0 = p_1 = 1/2$. Thus our source is producing information at the rate of 1000 bits per second. During transmission the noise introduces errors so that, on the average, 1 in 100 is received incorrectly (a 0 as 1, or 1 as 0). What is the rate of transmission of information? Certainly less than 1000 bits per second since about 1% of the received symbols are incorrect. Our first impulse might be to say the rate is 990 bits per second, merely subtracting the expected number of errors. This is not satisfactory since it fails to take into account the recipient's lack of knowledge of where the errors occur. We may carry it to an extreme case and suppose the noise so great that the received symbols are entirely independent of the transmitted symbols. The probability of receiving 1 is 1/2 whatever was transmitted and similarly for 0. Then about half of the received symbols are correct due to chance alone, and we would be giving the system credit for transmitting 500 bits per second while actually no

information is being transmitted at all. Equally "good" transmission would be obtained by dispensing with the channel entirely and flipping a coin at the receiving point.

Evidently the proper correction to apply to the amount of information transmitted is the amount of this information which is missing in the received signal, or alternatively the uncertainty when we have received a signal of what was actually sent. From our previous discussion of entropy as a measure of uncertainty it seems reasonable to use the conditional entropy of the message, knowing the received signal, as a measure of this missing information.

In the example considered above, if a 0 is received the a posteriori probability that a 0 was transmitted is .99, and that a 1 was transmitted is .01. These figures are reversed if a 1 is received. Hence

$$H_y(x) = -[.99 \log .99 + 0.01 \log 0.01] = .081 \text{ bits/symbol}$$

or 81 bits per second. We may say that the system is transmitting at a rate $1000 - 81 = 919$ bits per second. In the extreme case where a 0 is equally likely to be received as a 0 or 1 and similarly for 1, the a posteriori probabilities are 1/2, 1/2 and

$$H_y(x) = -\left[\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2}\right] = 1 \text{ bit per symbol}$$

or 1000 bits per second. The rate of transmission is then 0 as it should be."

In general, then for a binary channel transmitting independent bits of information, with equal likelihood of "0" and "1", and with a probability q for an error in the transmission

$$R = 1 + q \, \text{ld} \, q + (1 - q) \, \text{ld} \, (1 - q), \tag{1.3}$$

gives the information per bit, with ld denoting $\log_2$.

So far we have exhibited plausible definitions, Shannon's main result concerns what we can do with "coding". Coding refers to the transformation of a message before transmission, into a different message. Coding is used, typically, to protect against error, and when used this way is always used to reduce the size of the independently usable message space. For example, we can use 9 bits to represent 2^9 different messages. Alternatively we can use eight of these nine bits to represent 2^8 messages and use the ninth bit to indicate whether the other eight bits have an even or odd sum. Then we can tell at the receiving end when one of the bits has been altered during transmission. Or to give another example, we can use nine bits to represent a message of four bits repeated twice, and use the ninth bit to indicate the parity of the four bits. In that scheme we can always reconstruct the original message, if only one bit has been altered. Powerful and economic coding schemes differ from our simple examples: they do not generally map the bits into themselves, but map the original message space into a portion of a larger space in a very unorderly way. At the receiving end, we will generally have to apply an unscrambling transformation to regenerate the original message.

Shannon's basic theorem can be stated as follows. As we lengthen the message, coding methods can be found for the error free transmission of information at a rate approaching that given by Eq. (1.2). Shannon's theorem has a non-constructive proof, it does not provide the coding methods but only asserts their existence.

The real point of Shannon's theory is that if we have noise problems, which give a small probability of error, then increasing signal power reduces the error probability, but does not

make it vanish. Using a modest extra power instead to transmit some additional bits can reduce the error to any desired level via the proper coding method.

Unfortunately in the computer we lack the powerful equivalent of the coding possibilities utilized by Shannon's theorem in the communication process. There are many schemes which show, by the use of extra computer components, how reliability can be improved. Generally, however, these schemes require multiplication in the component count by factors which are small integers, even though the original probability of component failure, or of intermittent error due to fluctuations, are relatively minute. Furthermore, while these schemes can greatly diminish the probability of an incorrect calculation, they cannot make the error probability approach zero, except by letting the component count become infinite. This deficiency may be a deficiency of invention, but probably it is more fundamental; computing is a more complex process than communication. Furthermore in Shannon's theory, the channel is the only source of errors; the coding and decoding equipment is presumed error free. As we approach the limits of reliability allowed by Shannon's theorem, these coding schemes become more complex, involving larger data blocks. This would not be a realistic view in a computer. Coding and decoding would not be very different from other computing steps, and just as prone to the influence of thermal fluctuations, for example.

Within the framework of information theory it can be shown [11] that in the presence of thermal noise an energy of $kT \log_e 2$ is needed per bit for information transmission. This is not surprising, one would expect an energy comparable to the thermal energy per degree of freedom. The association of kT with a degree of freedom in the computer is also most natural, and was probably first discussed explicitly by von Neumann [12].

Note, however, that in these early discussions, it was generally left unclear under what conditions the energy had to be dissipated, and when it could somehow be used over again. This point will be of major concern in Section 3.

2. The finiteness of memory

A computer has a place where it keeps the information it subsequently needs. This information consists not only of data, but also of the instructions which take the computer through its routine. This is the computer memory, and in the majority of existing computers is an array of doughnut shaped ferrite "cores", which are left magnetized in one direction or another. Memories, however, can also be made out of other magnetic materials, e. g. thin films, materials with cylindrical domains, ferroelectrics, or superconductors. In general, we need phenomena which are readily coupled to electrical circuitry, permit the existence of two (or more) preferred states, and in the transition between the two states involve energies small compared to chemical binding energies, so that the material is not disrupted. Thus, while in principle a single interstitial atom shiftable between two lattice sites can hold information very effectively, the information is not too accessible electrically. Furthermore, the impressive density of information involved in such atomic schemes is not too useful, we cannot be shifting too many such interstitials without blowing up our material. The typical cooperative phenomena we have mentioned give us one collective coordinate, readily electrically available, and distributable over many crystal cells, for energy dissipation and heat conduction purposes. Information can, of course, also be stored in active electrical circuits, e. g. via transistors. This latter kind of information storage has always been used in the working portion of computers aside from the

separate boxes called "memory". In recent years the rapid progress of integrated circuits has led to the use of transistors for the separate memory boxes as well.

We shall now enter into a discussion of John Swanson's work [13] on "Physical versus Logical Coupling in Memory Systems". Swanson's theory is concerned with the optimum utilization of matter for storage purposes. While the considerations are guided by the typical cooperative phenomena used in memories, the results apply, at least in their general qualitative features, equally to other forms: e. g. bistable transistor circuitry. We can best introduce the subject by quoting from Swanson's paper [13]:

"The existence of the optimum can be expected from the following crude line of thought. Assume that we are given a fixed amount, say a pound, of perhaps a ferromagnetic material, which can be used to manufacture bistable units capable of serving as memory elements. How finely shall we divide our pound? We can use the whole pound to make only one bit. The resultant bit, because of its massiveness, will be unlikely to lose its information, either through a tunneling process, or through thermal agitation, and will therefore be a very reliable bit. Nevertheless our whole memory is rather useless since it is only a single bit. At the opposite extreme, we can divide the pound into a great many units, each of which is so small that it loses its information almost immediately. Despite the large number of elements in such a memory, it is again almost useless. There is, therefore, presumably an intermediate degree of subdivision which provides for a maximum memory capacity, if the time over which we wish to preserve information has been specified."

To locate the optimum, we will invoke information theory, i. e. we will allow for the possibility of storing information in coded form. The reasonableness of this follows from the kind of considerations given in Section 1, with regard to the effectiveness of adding extra power to obtain redundancy, versus use of the same extra power to obtain additional signal amplitude. By making memory elements larger, we can make them more stable, but we can never make them completely immune to information loss that way. Coding, and the use of extra elements, i. e. redundancy, permits us, however, to make the memory (just like a communications system) arbitrarily reliable with the effective error probability diminishing to zero as the length of the coded block becomes infinite. Once we have our memory bits large enough to be relatively stable, it will then be more effective to add a little redundancy, rather than to make each bit more stable: The optimum solution will therefore invoke some coding.

Let V be the volume of information holding memory material available. We will want to maximize:

$$\frac{I}{V} = \frac{1}{v}\left[1 + q\,\mathrm{ld}\,q + (1-q)\,\mathrm{ld}\,(1-q)\right] \tag{2.1}$$

where v is the volume assigned to one bit, and q is the probability of information loss, over the period of time for which information preservation is required. I is the total information content of the memory. v is the independent variable, and we still have to discuss the functional dependence of q on v. In introducing this functional dependence, we will lose some generality. It is believed, however, that a choice can be made which is broadly applicable, and, furthermore, that the main features of the result do not depend critically on the nature of q(v). We shall neglect here the fact that there is, for all our cooperative phenomenon, some minimal volume, below which the cooperative effect does not really exist. Swanson's analysis recognizes the existence of this minimal volume, through the general form of his equations, but not really in his analytical manipulations.

We shall assume that there is a transition probability, γ per unit time, between the states "1" and "0", leading to a time-integrated error probability, which increases with time as:

$$q = \frac{1}{2}[1 - \exp(-2\gamma t)] \tag{2.2}$$

This still leaves us with the dependence of the transition rate γ on v. Consider a simple one-dimensional model, as shown in Fig. 1, in which the "0" and "1" state are represented by two wells. In the case of thermal agitation the rate of barrier crossing has the form $\nu e^{-U/kT}$ with U proportional to v. In the case of tunneling we have a transition rate

$$\nu \exp[-h^{-1} \int (2m[V-E])^{1/2} dx].$$

Since both m and $(V - E)$ scale with the volume, we again have a rate of the form $\gamma \sim \nu e^{-\mu v}$. As a final and third alternative, let us use our variable volume v to make up a number n, of subunits, each of volume v_o, and then constrain the subunits so that a transition for the set n is allowed only if all (or some other fraction, e. g. more than half) of the subunits make the transition independently. Let us assume that a characteristic time τ is involved in establishing the constraint. If γ_0 is the transition rate for one volume (and $\gamma_0 \tau \ll 1$), then within the time τ there is a probability $(\gamma_0 \tau)^n$, that all volumes will be impelled to make the transition, and a new overall transition rate, $\tau^{-1}(\gamma_0 \tau)^n$, will prevail. Again an exponential dependence on n or v results. Thus taking γ in the form $\nu e^{-\mu v}$ seems to be a plausible choice.

Now the actual attempt to maximize (2.1) has not yet yielded to an elegant analytical treatment. It can readily be shown that the derivative of (2.1) with respect to v is positive at small v and negative at large v, and that therefore a zero must exist in between. Swanson shows in fact that the solution cannot be too far from

$$\mu v \cong \log \nu T + \log\log \nu T, \tag{2.3}$$

where the interval for information preservation is written as T; the use of a capital letter emphasizes the fixed nature of T in the maximization process. Eq. (2.3) would correspond to

$$\gamma T = \nu T e^{-\mu v} \sim 1/\log \nu T. \tag{2.4}$$

Thus we would be dealing with a transition probability q, given by Eq. (2.2) which for small γT is equal to γT, hence equal to $1/\log \nu T$, according to (2.4). This quantity is small compared to unity. Thus we first of all recognize that the optimum occurs at volumes v giving relatively reliable elements; redundancy is not an effective way to remedy very unreliable elements.

From even a semipractical viewpoint, this optimum is probably not very inviting. Making the elements even a little larger will make them much more reliable, and can therefore eliminate much of the complexity of coding, which is after all not free. Coding is not free, equipment is needed for coding and decoding. Coding also costs us in the sense that the retrieval of even one bit from our memory requires the decoding of a large block, and spoils the independent accessibility of bits in a random access memory.

The discussion by Swanson emphasizes the finiteness of memory available from a finite amount of memory material. An even more fundamental viewpoint has been taken by Bremerman, in a series of closely related papers [14–16]; he relates the total energy, $m c^2$, available in a closed system (possibly the whole universe) to the maximum rate of information transmission it can

supply in a channel limited by quantum noise. In commenting on Bremerman's results, it should be pointed out that the maximum r a t e of information transmission does not necessarily correspond to the maximum total amount of information. In fact it is easily shown that the maximum rate is obtained in a burst of energy release which is so short that the total amount of information transmitted is only about o n e bit.

Ligomenides [17] and Marko [11] have also discussed quantum noise. We shall summarize the discussion by Marko. The channel capacity of a transmission line is an increasing function of the bandwidth and signal power and a decreasing function of the noise power [8]. The bandwidth is limited by the total energy available, $m c^2$, while the noise power is limited by quantum mechanics. In this way, Marko finds the maximum time integrated capacity of a channel is

$$T = \sqrt{W/W_1} \tag{2.5}$$

where W is the action (energy x time) provided fo the channel and $W_1 = (\log 2)^2 \, h/2 = 1.59 \times 10^{-34}$ Joule seconds. If we use 10^{56} grams as the mass of the universe and take the age of the universe as $4 \cdot 10^9$ years, Marko finds $7 \cdot 10^{60}$ bits.

Note that this is the bit number available if the whole world's energy had been transmitted through one channel. Since $T \sim W^{1/2}$ we gain by dividing the available action among several channels. This arises because the lower powered sub-channels will occupy a smaller frequency range, and therefore will be facing a smaller quantum noise $h\nu$. There is a limit to this process, once $h\nu < kT$, thermal noise becomes the limit. In that case, the channel capacity is directly proportional to the energy, so that it is immaterial whether one channel or many are used. We have, from Marko [11], for the time integrated channel capacity, using an energy U, and limited by thermal noise:

$$T = U \, (kT_0 \log 2)^{-1}. \tag{2.6}$$

Now it is not clear what is a reasonable "effective" temperature for the universe. But we are interested in large exponents, and their order of magnitude; errors of 10^5 either way are not too significant. Let's take $T_0 = 1\ ^{\circ}K$. This yields $T \sim 10^{93}$ bits.

We do not want to overemphasize the significance of these sorts of limits. They are, after all, written down without a careful understanding of the mechanisms that generate the information stream, and the mechanisms that react to it. In particular, these estimates picture the energy of the universe organized along one-dimensional channels, they are in no sense measures of the possible number of states of the universe; that would take us to enormously larger numbers.

These estimates do, however, demonstrate one point. All such considerations are likely to emerge with a limit which is some combination of the physical constants of nature, and will result in a number with an exponent more likely to be 100 than 10^6.

These numbers seem huge compared to the storage capacities of existing machines, which are in the 10^6 to 10^{12} range. They are, however, rather minute compared to the numbers of ways of rearranging things generated easily by modest sounding combinatorial problems. Ashby [18], for example, cites a picture made up of 20 x 20 spots, 400 altogether. This is a much more primitive grid than we have in television, it's only about enough resolution to present a number or letter without artificially distorting its appearance. This very modest grid already permits 2^{400} different pictures, or 10^{120}.

If we talk about the number of possible sets selected from this picture selection, it becomes two raised to 10^{120}. Bremermann [15] also points to the $10^{2.4 \times 109}$ different possible variations of the human genetic structure.

3. Irreversibility and heat generation in the computing process

As mentioned earlier, and as amplified through Eq. (2.6), it is natural to a s s o c i a t e one bit with kT, or kT $\log_e 2$. But an energy involvement is not necessarily an energy dissipation. It is the purpose of the present section to sharpen up these notions using the ideas developed in one of the author's own earlier papers [19], from which we quote:

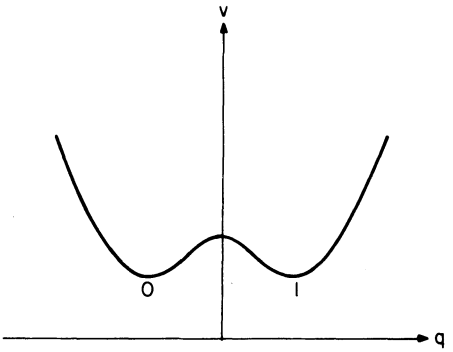


Fig. 2
In this bistable potential well q is a generalized coordinate representing the quantity that is switched.

"Our main argument will be a refinement of the following line of thought. A simple binary device consists of a particle in a bistable potential well shown in Fig. 2. Let us arbitrarily label the particle in the left-hand well as the ZERO state. When the particle is in the right-hand well, the device is in the ONE state. Now consider the operation RESTORE TO ONE, which leaves the particle in the ONE state, regardless of its initial location. If we are told that the particle is in the ONE state, then it is easy to leave it in the ONE state, without spending energy. If on the other hand we are told that the particle is in the ZERO state, we can apply a force to it, which will push it over the barrier, and then when it has passed the maximum we can apply a retarding force, so that when the particle arrives at ONE, il will have no excess kinetic energy, and we will not have expended any energy in the whole process, since we extracted energy from the particle in its downhill motion. Thus at first sight it seems possible to RESTORE TO ONE without any expenditure of energy. Note, however, that in order to avoid energy expenditure we have used two different routines, depending on the initial state of the device. This is not how a computer operates. In most instances a computer pushes information around in a manner that is independent of the exact data which are being handled, and is only a function of the physical circuit connections.

If, however, we permit the potential well to be lossy, this becomes easy. A very strong positive initial force applied slowly enough so that the damping prevents oscillations will push the particle to the right, past ONE, regardless of the particle's initial state. Then if the force is taken away slowly enough, so that the damping has a chance to prevent appreciable oscillations, the particle is bound to arrive at ONE."

The operation "RESTORE TO ONE" considered above is an example of more general class of logical operations which we will call logically irreversible, because they throw away information. Any device with two binary inputs and only one output is logically irreversible. Our argument, which will be developed in more detail below, is that computers require logical irreversibility, and this in turn is accompanied by physical irreversibility and by dissipative effects. Our presentation will be more in the nature of a plausibility argument than a proof.

The first point to be developed concerns the need for logical irreversibility. The simplest argument, but also the least satisfactory, comes from the fact that all known actual computers do use logically irreversible functions in abundance.

The second, and slightly more convincing argument considers a particular class of computers, namely those using logical functions of at most two input variables at a time. There are only sixteen different functions of this type, and they are well understood [20]. Consider a matrix as shown:

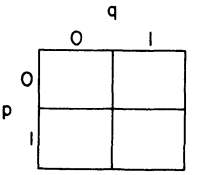


Fig. 3

in which the inputs are represented along the two sides and the four empty boxes contain the outputs. Each empty box must hav a "0" or "1" in it. This gives us the $2^4 = 16$ different ways of filling in the matrix. Now consider our computer as a machine which holds N binary pieces of information, and periodically goes through a cycle which maps each of its 2^N possible internal states onto new states. If we do not permit ourselves to use logically irreversible functions, then the space of 2^N initial states must be mapped, one-to-one, onto the same space, rather than just a subspace. But in the 2^N possible states each bit position has a "1" and a "0" appearing with equal frequency. Hence the reversible computer can utilize only those truth functions whose matrix exhibits the outputs "1" and "0" with equal frequency. Only six out of the sixteen possible matrices meet this criterion, and it is well known [20] that the six involved are not a complete set that permit synthesis of all other truth functions.

Of course, we can use more general devices in a machine than were presumed in the above paragraph. Let us still, however, assume our machine is of the cyclic form envisioned above. Consider some logical function which has to be performed, and which taken by itself is irreversible. We can easily supplement the needed function by additional outputs such that the totality of outputs yields a reversible function. Thus our logic function generates the outputs really needed, as well as the extra bit of information which would allow us to reverse the calculation. The extra bits, even though not needed, must be saved through all subsequent steps; and furthermore, all of its compatriots from earlier and later cycles must also be saved. This requires an exponentially growing number of stages. A general purpose computer, however, is a device which in principle can run for any required length of time without the designer having to allow for a maximum number of steps which it can carry out. Indeed even if we did have enough information capacity to save the inessential intermediate results, we would only be delaying the irreversible operation, we would have to wipe out the saved information eventually anyway to prepare the computer for its next calculation.

Now we must still connect logical irreversibility with physical irreversibility. Consider "RESTORE TO ONE" as a typical example. It is completely analogous to magnetizing an ensemble of initially random spins. In the alignment process, the entropy associated with the spins is reduced, this must turn up somewhere else as heat, it is most naturally taken up by the other degrees of freedom of the lattice in which the spins are embedded. Fig. 4 illustrates

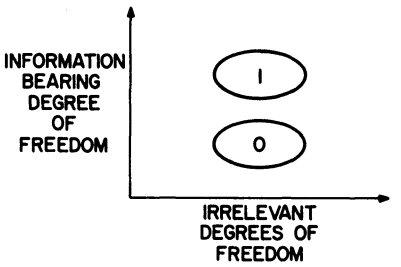


Fig. 4
Symbolic representation of phase space for "zero" and "one".

the situation. The vertical coordinate represents the information bearing degree of freedom, the horizontal coordinate symbolizes all the other irrelevant degrees of freedom. In "RESTORE TO ONE" we map 0 and 1 into closely adjacent volumes of phase space which both correspond to the value 1 for the information bearing coordinate. The total volume assigned to a closed system in phase space is incompressible. Hence, a diminution in the spread of the information bearing coordinate results in an expansion in the other coordinates. In order to handle information in a standardized fashion we let the two adjacent volumes representing 0 and 1 diffuse into each other. Thus, the entropy is increased and we have heating. We may ask ourselves why we cannot permit the information bearing coordinate itself to spread out. One might do this (assuming it isn't a quantized degree of freedom, such as spin), during an initial transient stage, but certainly we cannot, in a long computation, keep making the information bearing coordinate more uncertain at every successive step; computers depend on the existence of well standardized, recognizable "0" and "1" signals. In principle, the possibility exists of letting the spread in the information bearing coordinate increase, without however letting the two spaces encroach upon each other; we let the "0" and "1" spaces grow on their far apart sides. In that case, we can consider the information bearing degree of freedom made up of two parts. One is a binary digit (or spin component) telling us which half space of the information coordinate is involved; the other coordinate is positive and continuous and defines the exact position within the half space. We have now defined a new continuous coordinate which actually takes up the entropy increase. In a steady computation, however, this heating will eventually reach a new equilibrium, and the continuous half space coordinate will pass on entropy to other degrees of freedom as fast as it is generated.

To return to the quantitative consideration of "RESTORE TO ONE", consider a statistical ensemble of bits in thermal equilibrium. If these are all reset to ONE, the number of states covered in the ensemble has been cut in half. The entropy therefore has been reduced by $k \log_e 2 = 0.6931\ k$ per bit. The entropy of a closed system, e. g. a computer with its own power sources, cannot decrease; hence this entropy reduction must appear elsewhere as a heating effect, supplying 0.6931 kT per restored bit to the surroundings.

Other logically irreversible operations will yield different values for the minimal heating, but they are typically of order kT. If we are given an information stream to "RESTORE TO

ONE" which does not contain "0" and "1" with equal probability, then the possibility exists of tailoring the operation to the input, and achieving a lesser minimal heating.

The minimal dissipation, kT, is of course not really fixed; T is somewhat under our control. Lowering the temperature of the computer itself by means of a refrigerator won't help the minimal energy requirements; more energy will be dissipated in the refrigerator than will be saved in the computer. (From a practical standpoint, however, there are advantages to a system in which the heat generation doesn't occur in the logic apparatus itself.) To really save dissipation, we must plant the computer in a lower temperature environment. We doubt if it is realistic to assume that the universe has arbitrarily cool places available, more likely there is some minimal realizable ambient temperature.

This whole section has been based on a picture of a computer which is restricted in two ways: (1) We have assumed that a computer is essentially a clocked or driven mechanism, going through a sequence of steps at a predictable rate. The rate may be intentionally slow, to reduce losses through viscous damping, but it is not unregulated. (2) Furthermore, we have assumed that the computer is a quasi-steady state machine, i. e. the same forces and interaction potentials manifest themselves at each step, and without looking at the detailed information content of the machine one cannot measure how far along its computational course it is. Recent work by Charles Bennett [21] demonstrates that a computer need not satisfy these restrictions. In particular it can proceed through its calculation in a diffusive way alternating forward steps and backward steps. It can in fact go through a computation as an activated process where in the initial stages the computer is driven backwards, and is carried forward against these driving forces only through chance fluctuations. Bennett points out that a computer of this sort can, in fact, require minimal dissipations far lower even than the ones we have discussed, namely a few kT for the whole computation.

4. Thermal fluctuations

If dissipation is an inevitable part of the computing process, then the information bearing degree of freedom must be coupled to the thermal reservoir consisting of many other degrees of freedom. This has two implications. One is that these extra degrees of freedom must be present, there is at least some difficulty with a "molecular computer" in which a very high percentage of the degrees of freedom are utilized. Furthermore we must face the consequence of fluctuation-dissipation theory, noise must be present in our computer.

Let us look at the consequences of this noise. Consider the bistable potential well of Fig. 2. The existence of noise tells us that a particle in state "zero" has a finite probability of jumping to the "one" state. The transition rate [22–24] is proportional to $e^{-\Delta/kT}$ where Δ is the height of the potential barrier as seen by the particle at "zero". Thus, for reliable storage, we require $\Delta \gg kT$. We also know that in equilibrium, the ratio of the number of particles in state "zero" to the number in state "one" is $\exp[-(u_0 - u_1)/kT]$, where u_0 and u_1 are the respective well potentials. For reliable switching we require this ratio to be large (if we switch to "one"). Thus $u_0 - u_1 \gg kT$. This suggests that in switching we must bring particles from a state with high energy to one with energy lower by many times kT and that this energy difference must be dissipated.

Elaborations of this idea are contained in papers by Ligomenides [25], and by Neyman [26].

Neyman, for instance, starts from an entropy increase per switching event,

$$\Delta S > k \log_e r \, \Delta I, \tag{4.1}$$

where k is Boltzmann's constant, I the information, and $r = 1/p_1$ where p_1 is the probability of an error due to thermal fluctuations in an "individual measurement". This equation has an uncertain origin, though Brillouin [27] and Ligomenides [25] both reach a similar looking equation by considering error probabilities in measuring the excitation of a harmonic oscillator, a process which is hard to connect with our typical binary information processing schemes. Eq. (4.1) leads Neyman to an energy loss

$$\Delta E \geqslant kT \log_e r \, \Delta I, \tag{4.2}$$

which at first sight has a high plausibility.

Actually, the above suggestions on energy requirement for switching are incorrect. This was pointed out by Keyes and Landauer [28]. They noted that the full well difference ($|u_1 - u_0|$) must be dissipated only if one well is suddenly lowered with respect to the other. However, if the well is lowered slowly, the particles can relax and make the transition at an intermediate stage when the energy difference is small. Only relatively few particles will then dissipate large energies by jumping later in the switching process. A detailed analysis shows that in this way only $kT \log_e 2$ must be dissipated in restoring an ensemble with equal numbers of 0 and 1 members to 1.

Keyes and Landauer [28] discuss a hypothetical computer, which is a modification of an invention by von Neumann [29]. von Neumann's invention is described in detail by Wigington [30]. The computer described by Keyes and Landauer uses wells periodically and continuously brought from a monostable state to a bistable one and back again to monostability (Fig. 5).

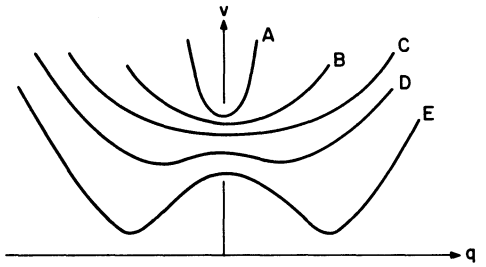


Fig. 5
Potential is varied cyclically in time from A to E and returned to A.

A particle in a flat bottomed well, near the transition point from monostability to bistability is particularly susceptible to external forces. In analogy to the soft mode in ferroelectrics, we label such wells as being in the "soft" state. Wells to be influenced are coupled to other wells which act as inputs. This coupling causes the well to be influenced to be tipped while in the soft state. Meanwhile the particles which act as sources of the tipping force are in very deep wells, and thus protect the information against fluctuations. A more careful analysis shows that all this can be done with a dissipation of $kT \log_e 2$ per input variable per logical step with arbitrary reliability, but at the expense of very slow processing. That is, we can dissipate very little energy if we can take arbitrarily long per computation step. Thus in contrast to Neyman's conclusion, increasing reliability in itself does not require unbounded energy consumption.

The tradeoff between speed, energy dissipation and reliability has been analyzed for the Keyes-Landauer model. In this analysis [31], the particle in a well is assumed to move in a viscous medium. Motion of the information bearing particle through this medium is the only source of dissipation. The time required to adjust to changes in potential becomes long if the viscosity becomes too large or too small. It is found that for an optimum choice of viscosity the energy dissipated is

$$E_d \cong \frac{v}{\sqrt{kT/m}} W + kT \log_e 2 \tag{4.3}$$

and the error probability is

$$p \cong \exp\left(-\frac{W}{kT}\left[1 - \sqrt{3}\,\frac{v}{\sqrt{kT/m}}\right]\right). \tag{4.4}$$

Here v is the velocity of the particle motion given by the distance between "zero" and "one" positions divided by the switching time and W is the depth of one potential well relative to the other. The factor $\frac{v}{\sqrt{kT/m}}$ appearing in the exponent in p is due to the distribution not having attained equilibrium at the completion of switching. $\frac{v}{\sqrt{kT/m}}$ is assumed to be much less than one.

We see that for slow switching (v small) the Keyes-Landauer results are obtained. For fast switching, the energy dissipation approaches that predicted by Neyman but the error probability can be considerably bigger. Unfortunately, it is not known whether these results have an applicability transcending the particular model which led to them. It is quite possible that there are hypothetical devices which will give better performance.

5. Active circuits

Up to now we have confined ourselves to systems which can hold information statically, i. e. without continuous energy dissipation. Actual computer circuits use transistors and bistable circuits which continually dissipate energy. In this section, we shall make an attempt to treat such a system without, however, detailed conclusions of great generality. While our detailed calculation has very limited generality, the authors do believe that many of the physical features can be preserved in a more general theory. The main idea we wish to express is that just as the Boltzman factor is a key to the understanding of the equilibrium problem, a similar factor can play the same role for the non-equilibrium situation. Our calculation is a simplified version of one given in an earlier paper by one of the authors [32].

As an example of a bistable active circuit, consider Fig. 6, showing an Esaki diode, and its I–V characteristic. That figure also contains the characterization of the battery in series with a resistance, as given by the straight line called a "load-line" in electrical engineering jargon. This line gives the voltage available at the device, as a function of the current through the resistor. Where the two curves intersect, at A, B, and C, we have states of the system satisfying all the circuit equations. A and C are stable states, B is unstable. If we draw force versus distance, for a bistable potential well, we find a very similar situation, as shown in Fig. 7.

For the two stable states, A and C, of the tunnel diode circuit, we can now ask questions resembling those for the bistable well remembering, however, that we are now dealing with an

unsymmetric well system. Thus we can ask whether A or C is the really preferred state, and we can ask what is the time required for relaxation to a steady state ensemble distribution. That will be the limit for the time during which the system can be used to preserve information.

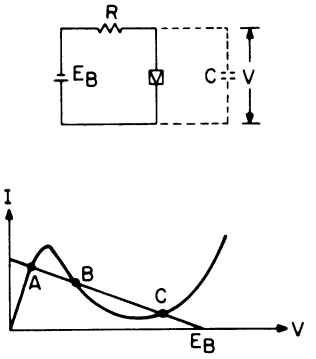


Fig. 6
An Esaki diode and its capacitance in series with a resistance and voltage source. The I–V plot gives the diode characteristic and load line. (Ref. [32])

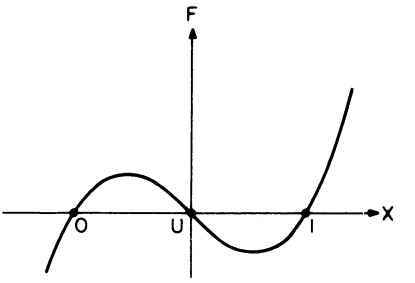


Fig. 7 Force against distance for the bistable well shown in Fig. 2. (Ref. [19])

The system of the type shown in Fig. 6 is one in which the charge on the Esaki diode capacitance is the stochastic variable. A detailed analysis is easier for the circuits of Fig. 8; in

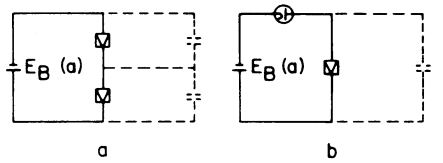


Fig. 8
Esaki diode in a is in series with another Esaki diode. In b it is in series with a thermionic vacuum diode, drawing its full emission current. (Ref. [32])

these, the resistance has been replaced by elements which let only an integral number of electrons pass. The circuit of Fig. 8a is the one discussed by Goto et. al. [33]. If we have a probability distribution ρ (N) for N electronic charges on the diode capacitance, then the time variation of ρ, as well as the steady state distribution, can be derived from the master equation:

$$\frac{\partial \rho\,(\mathrm{N})}{\partial t} = -\alpha\rho\,(\mathrm{N}) + \beta\rho\,(\mathrm{N}+1) + \gamma\rho\,(\mathrm{N}-1), \tag{5.1}$$

where the coefficients α, β, and γ depend upon the probability of the electronic charge on the diode capacitance changing by $\pm$ 1. Now in the system under consideration it turns out that it is n o t legitimate to replace Eq. (5.1) by its continuum differential equation form. Nevertheless, if we permit ourselves to make this approximation, we preserve all the general features of the discussion without a distressing analytical complexity, and can discuss the system much more easily. The reader is referred to the original paper for the complexities of the more accurate treatment which, however, leads to the same end result, as given in Eq. (5.19) of the present paper.

We then find, in accordance with the usual treatment of stochastic phenomena [34] that there is a current of ensemble members in q-space (q is the fluctuating capacitive charge = − eN) given by by

$$j_q = \rho(q) \lim_{\Delta t \to 0} \frac{\langle \Delta q \rangle_{Av}}{\Delta t} - \frac{1}{2} \frac{\partial}{\partial q} \{ \rho(q) \lim_{\Delta t \to 0} \frac{\langle \Delta q^2 \rangle_{Av}}{\Delta t} \}.$$

Here $\langle \Delta q \rangle_{Av}/\Delta t$ gives the behavior of the ensemble average. The second right-hand term includes the effects which arise from the nonuniform behavior of ensemble members; it is clearly related to diffusion. A more detailed interpretation will be given after some rearrangement of the terms involved. We can in fact write this equation in the form:

$$j_q = \rho(q) \langle \dot{q} \rangle_{Av} - \frac{\partial}{\partial q}(D\rho), \qquad (5.2)$$

$$\langle \dot{q} \rangle_{Av} = \lim_{\Delta t \to 0} \langle \Delta q \rangle_{Av}/\Delta t \quad \text{and} \quad D = \frac{1}{2} \lim_{\Delta t \to 0} \frac{\langle \Delta q^2 \rangle_{Av}}{\Delta t}.$$

At this point, it is interesting to note that there does not seem to be universal agreement on the form of the diffusion terms in systems where the diffusion coefficient is a function of the coordinate of motion. In some papers diffusion is taken to be of the form $D\,\partial\rho/\partial x$, in others $\partial(D\rho)/\partial x$. This situation is discussed in detail in Appendix A. Suffice it here to say that it is only the total transport current j_q which has to be correct, we can rewrite the Equation (5.2) equally well as

$$j_q = \rho \left(\langle \dot{q} \rangle_{Av} - \frac{\partial D}{\partial q} \right) - D \frac{\partial \rho}{\partial q}. \qquad (5.3)$$

It is this form, as is demonstrated in the appendix, which gives in the term $\left(\langle \dot{q} \rangle_{Av} - \frac{\partial D}{\partial q} \right)$ the motion of ensemble members due to the forces which favor a nonuniform ensemble distribution. $\langle \dot{q} \rangle_{Av}$, without the correction term $-\partial D/\partial q$, also includes terms due to mobility gradients, which arise from the fact that the ensemble member can jump farther from a given point in one direction than the other; but these terms do not correspond toward tendencies for nonuniform equilibrium distributions. The authors are indebted to H. Thomas for calling attention to the fact that Eq. (5.3) leads to simpler equations than Eq. (5.2) used in the originally published analysis [32]. D, for our Esaki diode circuit, can be derived from the known noise theories for this device [35]. Let us denote $(\langle \dot{q} \rangle_{Av} - \partial D/\partial q)$ by v_q. The Equation (5.3) for the steady state:

$$j_q = \rho v_q - D\,\partial\rho/\partial q = 0, \qquad (5.4)$$

can easily be integrated to

$$\rho = C \exp \int \frac{v_q}{D} dq. \qquad (5.5)$$

To bring out the strong analogy with the Boltzmann distribution, we rewrite the latter, with the use of the Einstein relation:

$$\rho(x) = \exp(-V/kT) = \exp\left(- \int \frac{\nabla V}{kT} dx\right)$$

$$= \exp\left(-\int \frac{\mu \nabla V}{\mu kT}\, dx\right) = \exp\left(\int \frac{\dot{x}\, dx}{D}\right). \tag{5.6}$$

Thus terms of the form $\int (v_q/D)\, dq$ take the place of potential barriers. This is very plausible: the numerator, v_q, in the integrand represents restoring forces, the denominator, D, represents the noise behavior which permits deviation from states A and C.

Let us now discuss some additional consequences of Eq. (5.4) needed subsequently for the discussion of approach to the steady state. At A, B, and C, the macroscopic motion, v_q, vanishes; the population is a maximum (or minimum) and locally the force towards a variation of the equilibrium population with q disappears. As we move away from A and C, the macroscopic motion provides for a restoring force. Near B, instead, we develop a velocity taking us away as we depart from the exact point of instability. This yields

$$v_q = -(q - q_A)/\tau_A \quad \text{near A,}$$

$$v_q = (q - q_B)/\tau_B \quad \text{near B,}$$

and

$$v_q = -(q - q_C)/\tau_C \quad \text{near C.} \tag{5.7}$$

In the tunnel diode case, the time constants are all well known resistance capacitance products. The exponents near A, B, and C, that enter into Eq. (5.5) are then easily evaluated, e. g.

$$\int_{q_A}^{q} \frac{v_q\, dq}{D} = -\frac{(q - q_A)^2}{2D_A \tau_A}, \text{ with } D(A) = D_A; \tag{5.8}$$

similarly for C and B, except that the sign is reversed in the case of B. Thus, near A we have an ensemble density:

$$\rho(q) = \rho(A) \exp(-q^2/2D_A\tau_A), \tag{5.9}$$

and a total population obtained by integrating (5.9) to yield

$$N(A) = \rho(A)(2\pi D_A \tau_A)^{1/2}. \tag{5.10}$$

Relaxation to the steady state. Now let us consider how deviations from Eq. (5.6) relax with time. We shall assume, in accordance with other treatments [22–24] of Eq. (5.5), as well as with common sense, that the equilibration process within each well is fast compared to the processes which require motion over the unlikely state B. Thus

$$\rho = \beta \rho_{ss} \tag{5.11}$$

where ρ_{ss} is the steady-state distribution of Eq. (5.6), and β is essentially constant within each well, and changes only in the vicinity of B. For a current j, in the vicinity of B we have

$$j = \rho v_q - D_B\, \partial\rho/\partial q, \tag{5.12}$$

with $v_q = (q - q_B)/\tau_B$. Substituting $\rho = \beta\rho_{ss}$ into the above we find

$$j = -D_B \rho_{ss} \frac{\partial \beta}{\partial q}, \tag{5.13}$$

or

$$\beta = \int -\frac{j}{D_B} \frac{1}{\rho_{ss}} dq. \tag{5.14}$$

In view of the rapid variation of ρ_{ss} with q, through $\rho_{ss} = \rho\,(B) \exp(-[q - q_B]^2/2D_B\tau_B)$, with integral in (5.14) has contributions only from the immediate vicinity of B, yielding

$$\beta\,(C) - \beta\,(A) = -\frac{j}{D_B} \frac{(2\pi D_B \tau_B)^{1/2}}{\rho\,(B)}. \tag{5.15}$$

(The reason that the Fokker-Planck approximation works is actually due to the fact that the variation of β is confined to a region in which ρ_{ss} is a minimum, and therefore varying more slowly than elsewhere. The departures from Fokker-Planck behavior are therefore limited to the exact form of ρ_{ss}, and if we are careful to cast our results in a form which does not explicitly represent this variation, we obtain agreement between the Fokker-Planck results and the exact theory.) The integrated ensemble populations near A and C are given bs (5.10) and (5.11) as

$$N\,(A) = \rho_{ss}\,(A)\,\beta\,(A)\,(2\pi\, D_A \tau_A)^{1/2},$$

$$N\,(C) = \rho_{ss}\,(C)\,\beta\,(C)\,(2\pi\, D_C \tau_C)^{1/2}. \tag{5.16}$$

Define "free energies of activation" ΔA and ΔC such that

$$e^{\Delta A} = N_{ss}\,(A)/N_{ss}\,(B),$$

and

$$e^{\Delta C} = N_{ss}\,(C)/N_{ss}\,(B), \tag{5.17}$$

in which $N_{ss}\,(B)$ is a fictitious population, defined in formal analogy with (5.10) as $\rho_{ss}\,(B)\,(2\pi D_B \tau_B)^{1/2}$. $N_{ss}\,(B)$ is a measure of the exponential inaccessibility of B and the phase space (or breadth) associated with the range over which β varies. Invoking these definitions takes Eqs. (5.15) and (5.16) into the form

$$\beta\,(C) - \beta\,(A) = -j\,(2\pi\tau_B)/N_{ss}\,(B),$$

$$N\,(A) = \beta\,(A)\, e^{\Delta A}\, N_{ss}\,(B),$$

and

$$N\,(C) = \beta\,(C)\, e^{\Delta C}\, N_{ss}\,(B). \tag{5.18}$$

The equilibration rate, $1/\tau$, is found by subtracting out the equilibrium portion of the distribution, thus leaving a distribution which integrates to zero, and asking at what rate this decays through the current j. This equilibration rate is thus found to be:

$$(1/\tau) = (2\pi\tau_B)^{-1} [\exp(-\Delta A) + \exp(-\Delta C)]. \tag{5.19}$$

Note that τ_B^{-1} is a measure of the ease with which ensemble members move past the maximum at B.

To make Eq. (5.19) physically plausible, consider the case of a set of symmetrical states A and C. Any unbalance must be adjusted through flow over the barrier, hence exponential factors of the form $\exp[-\int (v_q/D)\, dq]$ appear and account for the lesser density of ensemble members at the barrier. The remaining non-exponential factor (including the non-exponential parts of $e^{-\Delta A}$ and $e^{-\Delta C}$) is

$$\frac{1}{2\pi} \frac{(D_B/\tau_B)^{1/2}}{(D_A\tau_A)^{1/2}} \tag{5.20}$$

$(D_A\tau_A)^{1/2}$ is a measure of the phase space, near A, which is occupied. The relaxation time must be inversely proportional to this population at A, which has to be adjusted, in the relaxation process. The diffusion gradient is maintained over a distance $(D_B\tau_B)^{1/2}$, near the barrier peak, and the diffusion current is therefore $D_B/(D_B\tau_B)^{1/2}$ or $(D_B/\tau_B)^{1/2}$, as shown in the numerator of (5.20).

In the original paper [32], it was pointed out that Esaki diodes with a cross-section greater than 10^{-11} cm^2 cross section would be adequate to be practically useful information holding devices. The main point, however, is that just as in a static device, protection is available through increase of an exponent; modest scaling can take us a long way. Consider the exponent, $\int (v_q/D)\, dq$, as we simultaneously scale up the cross-section of all the electrical devices involved. We can convert the integral to one over a voltage range. The physical behavior of the device as a function of voltage, and therefore the voltage range under consideration, will remain independent of device area. Thus:

$$\int \frac{v_q}{D} dq = \int \frac{1}{D} \left(\frac{dq}{dt}\right) dq = \int \frac{1}{D} \left(\frac{dq}{dV}\right)^2 \frac{dV}{dt}\, dV. \tag{5.21}$$

The effective diffusion constant is proportional to the mean square fluctuation of the number of charges so that D increases linearly with cross-sectional area. dq/dV, a capacitance, scales in the same manner. Therefore, the integrand as a whole scales up linearly with the cross-sectional area.

Note that one of the original schemes that suggested the time dependent potentials of Fig. 5, was in fact based on the circuits [33] of Fig. 8a, which are clearly characterized by the analysis given in this section. In that scheme the battery shown in Fig. 8a is replaced by an oscillating "pump" potential centered about zero. For large enough pump voltages the two series diodes represent a bistable arrangement, with one diode in a high voltage state, and the other in a low voltage state. Resistors then couple the midpoint between the two diodes in question to elements which have reached their bistable state at an earlier time, and which in turn influence the element in question toward one direction or the other.

The barrier between the two favored states increases as we scale up the cross-section of the circuit elements, and thus the probability of undesired spontaneous information decay can be made as small as desired. Furthermore the probability that a circuit settles into an erroneous state which is opposed to the influence exerted upon the circuit in question by the circuits which acted as information sources can also be shown to diminish with increasing cross-section. Thus the computation can be made arbitrarily reliable by choosing a large enough cross-section.

In the tunnel diode, we have an example of a system whose steady state nonequilibrium behavior is remarkably similar to that of equilibrium systems. In both cases the distribution function has an exponential form. We also note that the scaling of the relaxation barrier discussed above is similar to the behavior of the barrier between the various ground states of a ferromagnet in the absence of an external field. In the ferromagnet, as we go to the "thermodynamic limit", i. e. make the magnet large, and if we are in the presence of zero applied field, then the jump rate between the degenerate lowest energy states goes to zero. However, in the case of the laser, another system displaying the above-mentioned distribution function analogy, the relaxation rate does not vanish as the system becomes large [36], and this has been brought up as an indication that these analogies between dissipative systems and equilibrium states are incomplete. In our opinion it only indicates that the laser system has a somewhat different character.

We now turn our attention to another bistable active circuit, the subharmonic oscillator. We will see that under favorable circumstances, its distribution function can again be given the appearance of the Boltzmann factor.

The subharmonic oscillator [37], which can be a computer component, is a system of two coupled harmonic oscillators, one with frequency ω and the other with frequency 2ω. The complex amplitude at ω will be denoted by c. A small linear dissipation is assumed to be present. If the device is driven by an external field at frequency 2ω, the system displays monostability for low and bistability for high input power. In the latter case, the two stable states are 180° apart in the phase of their oscillations.

In contrast to some of the tunnel diode circuits where the Fokker-Planck approximation is valid only near the points of stability and instability, it applies here throughout the range of interest. If the relaxation time associated with the high frequency mode is short compared to that of the low frequency mode, the resulting equations become particularly simple and the steady state distribution can be obtained. It turns out to have the form

$$\rho\,(c, c^*) \sim e^{-V(c, c^*)}. \tag{5.22}$$

The function V is analogous to the free energy divided by the temperature. Below a threshold amplitude for the "pump" signal at 2ω, V has one minimum and above it has two. The height of the barrier between the minima in the bistable region increases with increasing input power. Thermally activated transitions between these two states can be calculated using the same method as in the equilibrium theory. Again the transition rate depends exponentially on the height of the barrier between the two states. Near threshold, V behaves like the Ginzburg-Landau free energy of a system near a second order phase transition. In this region, fluctuations in c are large and it becomes easy to influence the signal.

Thus, arbitrarily reliable computing can be obtained with this device as well as with the tunnel diode. We merely proceed slowly through the threshold and then "freeze" the situation once the desired distribution between two possible states has been reached. In the present case, the act of freezing can be accomplished by a large enough input power 2ω.

We have discussed two cases for which a Boltzman type distribution obtains for the steady state distribution. Is this a general phenomena? This has been partially answered by Graham and Haken [38]. They showed that if a Fokker-Plank description is valid, then the steady state distribution is of the Boltzman type if and only if detailed balance is satisfied. The subharmonic oscillator in which the high frequency mode relaxes quickly compared to the low frequency

mode satisfies detailed balance. In the case of the particular tunnel diode circuit shown in Fig. 6, the detailed balance condition is actually not satisfied. This is one reason for emphasizing the circuits in Fig. 8.

We have seen that arbitrarily reliable computing is available from active circuits as well as from dissipationless elements. For very slow computing, where we have to contend with a very small effective bandwidth of noise, this is not too surprising a conclusion.

We have, however, not come to any conclusions about the minimum dissipation of dissipative devices. Further progress in the analysis of the nonequilibrium problem is severely hampered by the lack of knowledge of two things. The first is the relation between the change in entropy and heat dissipation. In the steady state, the entropy of the system as defined by $-\mathrm{kTr}\,(\rho \log \rho)$ is time independent. If we now change an external parameter, there is an accompanying change in entropy. Is this change in entropy related to the heat dissipated above and beyond the steady state dissipation? Secondly, we need to know whether something like the second law of thermodynamics applies to this change in entropy. That is, if entropy is removed from one degree of freedom, must this entropy appear somewhere else, or is it only necessary to require that the total production of entropy (including the steady state production) be positive?

6. Unsolved problems

Since, in a sense, our analyses have just scratched the surface, is isn't hard to list questions that remain unanswered. Nevertheless, we will list some queries that do not seem hopeless, and at the same time seem to be essential steps in the further progress of the subject.

Within the framework of our physical theory, we have ignored the quantization of the electromagnetic signal. A well developed quantum theory of the linear channel exists today, but nothing is understood about the nonlinear interactions typical of information streams in computers. In the linear communications process one photon, for example, can be used in one of many time slots, and therefore can carry more than one bit of information. Do computing systems permit similar economies? What about the typical voltage transition step propagating along computer transmission lines? How many photons does it need? Unfortunately we have become used to "one photon" as representing a wave packet, rather than the photon decomposition of base band pulses.

The analysis of minimum dissipation clearly leaves much unsaid, in particular it tells us nothing of wide applicability about computation at finite rates and whether it can be made "foolproof" without requiring unbounded energy consumption per logical step. This is probably the key question in terms of the epistemological implications of this subject.

Appendix: Diffusion in the presence of spatial inhomogeneity

Diffusion currents are sometimes specified as

$$-D\frac{\partial n}{\partial q}, \tag{A.1}$$

and at other times as

$$-\frac{\partial}{\partial q} Dn. \tag{A.2}$$

If D is independent of q, the two are equivalent. In actual fact, however, D need not be a constant, and a real disagreement exists. Form (A.1) is used, for example, by LeClaire [39], and generally by people concerned with atomic diffusion. Form (A.2) is used in the hot electron literature, for example by W. P. Dumke [40], and by Butcher, Fawcett, and Ogg [41]. Stratton [42] acknowledges the existence of the two forms for the diffusion current. A. D. LeClaire [43], without explicitly addressing this problem, touches upon closely related points. Manning [44] tries to resolve this issue, and makes points which are very similar to ours but in a somewhat different context.

A simple-minded glance at the derivations [34] of the Fokker-Planck equation seems to support Eq. (A.2). The Fokker-Planck treatments consider particles moving away from a point q, in the time interval Δt. Let the mean displacement of these particles be $\langle \Delta q \rangle_{Av}$ and let $\langle \Delta q^2 \rangle_{Av}$ be the mean squared displacement in the same time interval Δt. Then these treatments yield a particle current

$$j = n(q) \lim_{\Delta t \to 0} \frac{\langle \Delta q \rangle_{Av}}{\Delta t} - \frac{1}{2} \frac{\partial}{\partial q} \left\{ n(q) \lim_{\Delta t \to 0} \frac{\langle \Delta q^2 \rangle_{Av}}{\Delta t} \right\} . \tag{A.3}$$

A casual inspection of the second term in (A.3) indicates a diffusion current of the form:

$$-\frac{\partial}{\partial q} Dn,$$

with $$D = \frac{1}{2} \lim_{\Delta t \to 0} \frac{\langle \Delta q^2 \rangle_{Av}}{\Delta t}$$

and seems to agree with (A.2). Note that hereafter we will omit $\lim_{\Delta t \to o}$ for brevity.

On the other hand, form (A.2) suggests that a mere spatial variation of mobility due to scattering center density, in the presence of a uniform carrier concentration, would generate diffusion currents, and this seems most implausible.

The basic answer to these apparent contradictions is that the "diffusion current", unlike the total particle current is not a well-defined unambiguous quantity. Whether (A.1) or (A.2) is correct depends on how the remaining term or terms in the particle flux are written.

Consider now, in detail, the second right-hand term in Eq. (A.3). It can be rewritten as:

$$-D(q) \frac{\partial n}{\partial q} - n(q) \frac{\partial D}{\partial q} \tag{A.4}$$

The first term $-D(q) \frac{\partial n}{\partial q}$ is just the expression (A.1). The second term is one which we shall show cancels against a similar contribution coming from the term $n(q) \langle \Delta q \rangle_{Av} / \Delta t$.

Let $P(q, q')$ give the probability that within the interval Δt a jump from q to q' takes place. Then we have

$$\langle \Delta q \rangle_{Av} = \int_{-\infty}^{\infty} P(q', q)(q' - q)\, dq',$$

$$\langle \Delta q^2 \rangle_{Av} = \int_{-\infty}^{\infty} P(q', q)(q' - q)^2\, dq'. \tag{A.5}$$

Let us further decompose $P(q', q)$ into symmetric and anti-symmetric parts

$$P(q', q) = P_g + P_u$$

with
$$P_g = \frac{1}{2}[P(q', q) + P(q, q')]$$

$$P_u = \frac{1}{2}[P(q', q) - P(q, q')]. \tag{A.6}$$

In equilibrium we would expect, from detailed balancing,

$$n(q)\,P(q', q) = n(q')\,P(q, q') \tag{A.7}$$

and if furthermore the concentrations are independent of q, as we would expect in equilibrium, and in the absence of forces causing some kind of transport, then

$$P(q', q) = P(q, q'),$$

or
$$P_u = 0.$$

We will label all sources of nonuniform distribution as transport terms. In our terminology P_u causes transport, P_g does not.

1. The symmetric part: P_g. The contribution of P_g to the second term of (A.3) via

$$-\left(\frac{\partial}{\partial q} n(q)\right) \cdot \frac{\langle \Delta q \rangle^2}{\Delta t}$$

represents real diffusion. The remaining term, having the appearance

$$-n(q)\frac{\partial}{\partial q} D(q)$$

as already suggested cancels the P_g contribution to the first right-hand term of (A.3).

Consider the drift term:

$$n(q) \int_{-\infty}^{\infty} P_g(q', q)\,(q' - q)\,dq'. \tag{A.8}$$

The terms from $q' > q$ and their reflection about q, can be combined. They tend to cancel since the factor $(q' - q)$ changes sign at $q' = q$. If $P_g(q', q)$ were dependent only on the jump distance $(q' - q)$, then the terms would cancel exactly. Utilizing the fact that

$$P_g(q - \delta q, q) = P_g(q, q - \delta q) = P_g(q + \delta q, q) - \delta q \frac{\partial P_g}{\partial q}(q + \delta q, q), \tag{A.9}$$

where the last term represents differentiation at fixed δq, we find for (A.8)

$$n(q) \int_0^{\infty} [\delta q \frac{\partial}{\partial q} P_g(\delta q, q)]\,\delta q\,d\delta q. \tag{A.10}$$

We have, in the above, written $P(\delta q, q)$ for $P(q + \delta q, q)$. The factor δq in [ ] multiplying the derivative arises from the incomplete cancellation process, the next δq is the one kept from the

original equations. The above can be rewritten as

$$\frac{1}{2} n(q) \int_{-\infty}^{\infty} (\delta q)^2 \frac{\partial}{\partial q} P_g(\delta q, q)\, d\delta q, \tag{A.11}$$

keeping in mind that

$$\frac{\partial}{\partial q} P_g(\delta q, q)$$

represents a derivative at fixed δq, it doesn't vary appreciably over the interval of integration. If we do permit $\partial P/\partial q$ to vary, it results in terms proportional to $(\delta q)^3$ and higher powers. If the Fokker-Planck treatment is to be valid, these must vanish faster than Δt. The derivative can be taken outside the integral sign, and thus the above integral cancels the

$$n(q) \frac{\partial}{\partial q} D(q)$$

term insofar as it arises from P_g.

2. The anti-symmetric part: P_u. This term clearly contributes to the first right-hand term of Eq. (A.3). In fact, after the cancellation shown above, it is the only remaining source of that term. We want to show that this drift term is the only place where P_u enters. Consider the extreme right-hand term of (A.3)

$$-\frac{1}{2}\frac{\partial}{\partial q} \{ n(q) \int_{-\infty}^{\infty} P_u(\delta q, q) (\delta q)^2\, d\delta q \} \tag{A.12}$$

As before, we can combine terms with the same absolute value of δq. These would once again cancel if $P_u(\delta q, q)$ were strictly odd as a function of δq. Since it is not, we can, however, use

$$\begin{aligned} P_u(-\delta q, q) &= -P_u(\delta q, q - \delta q) \\ &= -P_u(\delta q, q) - \frac{\partial P_u}{\partial q} \delta q, \end{aligned} \tag{A.13}$$

to yield

$$\begin{aligned} &\int_0^{\infty} [\delta q \frac{\partial}{\partial q} P_u(\delta q, q)] (\delta q)^2\, d\delta q \\ &= \frac{1}{2}\frac{\partial}{\partial q} \int_{-\infty}^{\infty} P_u(\delta q, q) (\delta q)^3\, d\delta q. \end{aligned} \tag{A.14}$$

Now, however, we invoke the fact that if the Fokker-Planck treatment is at all applicable $\langle \Delta q^3 \rangle_{Av}$ goes to zero faster than Δt. Otherwise, the expansion leading to Eq. (A.3) wouldn't be valid. Thus, the P_u contribution to the $\langle \Delta q^2 \rangle$ terms is negligible.

We have thus shown that if the transport terms are associated with the tendency toward non-uniform concentrations, then they are related only to P_u. In that case the diffusion term contains only P_g contributions and has the form $D\, \partial n/\partial q$.

References

[1] Bridgman, P. W.: The Logic of Modern Physics, New York 1927.

[2] Bridgman, P. W.: The Nature of Thermodynamics, Cambridge, Mass. 1943.

[3] Bridgman, P. W.: A Physicist's Second Reaction to Mengenlehre, Scripta Math., **2**, 224 (1934).

[4] Keyes, R. W.: Power Dissipation in Information Processing, Science **168**, 796 (1970).

[5] Landauer, R.: The Future Evolution of the Computer, Physics Today **23**, 22 (1970).

[6] National Materials Advisory Board: Fundamentals of Amorphous Semiconductors, National Research Council, National Academy of Sciences, Washington D. C. 1972.

[7] Landauer, R.: Wanted: a physically possible theory of physics, IEEE Spectrum **4**, 105 (1967).

[8] Shannon, C. E.: A Mathematical Theory of Communication, Bell System Technical Journal, **27**, pp. 379–432, pp. 623–656 (1948). Reprinted also in: Shannon, C. E. and Weaver, W.: The Mathematical Theory of Communication, Urbana, Ill. 1949.

[9] Szilard, L.: Über die Entropieverminderung in einem thermodynamischen System bei Eingriffen intelligenter Wesen, Z. Physik **53**, 840 (1929).

[10] Brillouin, L.: Science and Information Theory, New York 1956.

[11] Marko, H.: Physikalische und biologische Grenzen der Informationsübermittlung, Kybernetik **2**, 274 (1965).

[12] von Neumann, J.: Theory of Self-Reproducing Automata, Urbana, Ill. 1966.

[13] Swanson, J. A.: Physical versus Logical Coupling in Memory Systems, IBM J. Res. Develop. **4**, 305 (1960).

[14] Bremermann, H. J.: Quantum Noise and Information, in: Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability **4**, 15–20. Berkeley, Calif. 1967.

[15] Bremermann, H. J.: Quantitative Aspects of Goal-Seeking Self-Organizing Systems, in: Progress in Theoretical Biology (F. M. Snell ed.), New York, pp. 59–77 (1967).

[16] Bremermann, H. J.: Optimization through Evolution and Recombination, in: Self-Organizing Systems (M. C. Yovits, G. T. Jacobi, and G. D. Goldstein, eds.), Washington D. C. pp. 93–99 (1962).

[17] Ligomenides, P. A.: Wave-mechanical uncertainty and speed limitations, IEEE Spectrum **4**, 65 (1967).

[18] Ashby, W. R.: Some Consequences of Bremermann's Limit for Information-processing Systems, in: Cybernetic Problems in Bionics (H. L. Oestreicher and D. R. Moore, eds.), New York, pp. 69–76 (1968).

[19] Landauer, R.: Irreversibility and Heat Generation in the Computing Process, IBM J. Res. Develop. **5**, 183 (1961).

[20] Hilbert, D. and Ackermann, W.: Principles of Mathematical Logic, New York, p. 10 (1950).

[21] Bennett, C. H.: (private communication).

[22] Kramers, H. A.: Brownian Motion in a Field of Force and the Diffusion Model of Chemical Reactions, Physica **7**, 284 (1940).

[23] Landauer, R., and Swanson, J. A.: Frequency Factors in the Thermally Activated Process, Phys. Rev. **121**, 1668 (1961).

[24] Langer, J. S.: Theory of Nucleation Rates, Phys. Rev. Letters **21**, 973 (1968).

[25] Ligomenides, P. A.: Wave Mechanical Limitations on Information Retrieval Rate, Proc. I.R.E.E. Australia **29**, 65 (1968).

[26] Neyman, M. S.: The Negentropy Principle in Information-Processing Systems, Telecommunications and Radio Engineering (English translation of "Radiotekhnika"), **21**, 68 (1966).

[27] Reference 10, p. 195.

[28] Keyes, R. W., Landauer, R.: Minimum Energy Dissipation in Logic, IBM J. Res. Develop. **14**, 152 (1970).

[29] Von Neumann, J.: Non-Linear Capacitance or Inductance Switching, Amplifying and Memory Organs, U. S. Patent #2, 815, 488, issued December 3, 1957.

[30] Wigington, R. L.: A New Concept in Computing, Proc. IRE **47**, 516 (1959).

[31] Landauer, R. and Woo, J. W. F.: Minimal Energy Dissipation and Maximal Error for the Computational Process, J. Applied Phys. **42**, 2301 (1971).

[32] Landauer, R.: Fluctuations in Bistable Tunnel Diode Circuits, J. Appl. Phys. **33**, 2209 (1962).

[33] Goto, E., Murata, K., Nakazawa, K., Nakagawa, K., Moto-oka, T., Matsuoka, Y., Ishibashi, Y., Ishida, H., Soma, T., and Wada, E.: Esaki Diode High Speed Logical Circuits, IRE Trans. on Electronic Computers **EC-9**, 25 (1960).

[34] Chandrasekhar, S.: Stochastic Problems in Physics and Astronomy, Revs. Modern Phys. **15**, 1 (1943).

[35] Tiemann, J. J.: Shot Noise in Tunnel Diode Amplifiers, Proc. IRE **48**, 1418 (1960).

[36] Graham, R.: in a Discussion Session, Meeting of the German Physical Society, Essen, Sept. 30, 1971; Thomas, H. (private communication).

[37] Woo, J. W. F. and Landauer, R.: Fluctuations in a Parametrically Excited Subharmonic Oscillator, IEEE J. of Quantum Electronics **7**, 435 (1971).
This paper contains references to earlier work on the same subject.

[38] Graham, R. and Haken, H.: Generalized Thermodynamic Potential for Markoff Systems in Detailed Balance and far from Thermal Equilibrium, Z. Physik **243**, 289 (1971).

[39] LeClaire, A. D.: Progress in Metal Physics 1, New York, p. 306 (1949).

[40] Dumke, W. P.: On a Current Instability Resulting from the Diffusion of Hot Electrons, Appl. Phys. Letters **11**, 314 (1967).

[41] Butcher, P. N., Fawcett, W., and Ogg, N. R.: Effect of field-dependent diffusion on stable domain propagation in the Gunn effect, Brit. J. Appl. Phys. **18**, 755 (1967).

[42] Stratton, R.: Diffusion of Hot and Cold Electrons in Semiconductor Barriers, Phys. Rev. **126**, 2002 (1962).

[43] LeClaire, A. D.: Random Walks and Drift in Chemical Diffusion, Phil. Mag. **3**, 921 (1958).

[44] Manning, J. R.: Correlated Walk and Diffusion Equations in a Driving Force, Phys. Rev. **139**, A126 (1965).

Theory of Dissipative Structures

I. Prigogine and R. Lefever

1. Introduction

One of the main reasons to develop thermodynamics beyond the equilibrium range lies in the need for a physico-chemical basis of biological order. Not only would one like to explain the maintenance of low entropy situations and subtle metabolic regulation processes but also one wants to provide a physico-chemical basis of evolution towards structures of increased complexity.

On the contrary of biological evolution, in physics, one is familiar with the idea that a macroscopic system evolves towards an equilibrium state of maximum disorder. This is a consequence of the second law which amounts to say that in an isolated system the entropy increases monotonously until it becomes a maximum.

If we now consider instead of an isolated system, a system in contact with an energy reservoir at temperature T, we still can describe its behavior in classical terms. Using the free energy F defined by $F = E - TS$, where E is the internal energy, we have that at equilibrium F has to be a minimum. On the other hand the probability for the system to be found with an energy F is given by the celebrated Boltzmann formula. It implies that when the temperature T is low, only the low energy states will be observed. On the contrary, when T increases, the contributions of the entropic and energetic terms to the free energy become comparable and the population of the different energy levels tends then to become equal. A possibility therefore exists for the formation of low entropy structures, provided the temperature is maintained sufficiently low. This ordering principle is responsible for the appearance of equilibrium ordered structures such as crystals as well as for phase transition phenomena.

It cannot however explain biological structures. Their probability of formation at ordinary temperatures under equilibrium conditions is vanishingly small. In biology, we necessarily are confronted with open systems in which the exchanges with the external world play the capital role.

This fact is for example illustrated by the observation that the early embryogenesis correspond to the period of highest energy dissipation in development. The rate of heat production, which roughly speaking should furnish a measure of the entropy production is in chicken's eggs about 6 kcal/day gr at the fourth day of development. At the 16th day of development it drops to about 1 kcal/day gr. This is a very global result but nevertheless it examplifies clearly that the most crucial steps in the establishment of biological structures and functions which occur during the early stages of differentiation are non equilibrium phenomena lying beyond the methods of equilibrium thermodynamics and statistical mechanics.

The thermodynamic theory of open systems has now been developed since many years (for a historical account, see Ref. [2]). An extended version of the second law has to be considered [2], so that the entropy variation during a time interval dt takes the form

$$dS = d_eS + d_iS \qquad d_iS \geqslant 0 \tag{1}$$

where d_eS is the entropy flow due to exchanges with the surroundings. The characteristic feature expressed by the second law is that the entropy increase d_iS due to changes inside the system, is never negative. It is zero when the system undergoes reversible (or equilibrium) transformations only, but it is positive if the system is subject to irreversible processes. On the contrary the flux term d_eS has no definite sign. As a result, during evolution smaller entropy situations that initially may appear and be maintained indefinitly provided a steady state such that $dS = 0$ or $d_eS = -d_iS \leqslant 0$ can be reached. Thus, in principle at least, it suffices to supply to a system a sufficient amount of negative entropy flow to maintain it at an ordered state.

These arguments cannot, of course, suffice for solving the problem of biological order. One would like not only to show that the second law is satisfied as a whole ($d_iS \geqslant 0$) but to precise in addition the mechanism of appearance, maintenance and further evolution of states of low entropy and high coherence.

This is the problem we tried to develop during the last few years and which one would like to discuss in some detail here below. (For a general presentation see Ref. [3].)

2. Dissipative structures

One of the main conclusions of the theory we want to develop here, is that there exist systems showing two types of behaviors: a tendency to a state of maximum disorder for one class of situations and a coherent behavior in a second class. The destruction of structures prevails always in the neighborhood of thermodynamic equilibrium. In contrast, creation of structures may occur far from equilibrium, beyond the states showing the usual thermodynamic behavior.

This duality is well known in fluid dynamics. Typical examples are the onset of turbulence or of convection in Bénard problem. However in contrast to these two phenomena where external forces and mechanical motions have to be taken into account, we have investigated systematically purely dissipative systems like chemical networks of biological interest. The surprising result was that they share most of the properties of hydrodynamic instabilities with the additional important feature that the variety of regimes beyond instability is much greater in chemical kinetics. This is clearly a consequence of the practically infinite number of ways through which nonlinearity may appear in chemical systems (autocatalysis, cross-catalysis, activation, inhibition, etc.) Beyond instability these systems may then become spontaneously inhomogeneous and present an ordered distribution of chemical constituents in space. Under different conditions the concentrations may show sustained time oscillations or in some systems exhibit a multiplicity of steady state values. Examples will be discussed below.

In all these phenomena, an ordering mechanism not reducible to the equilibrium principle appears. For reasons to be explained later, we shall refer to this principle as order through fluctuations. One has structures which are created by the continuous flow of energy and matter from the outside world. Their maintenance requires a critical distance from equilibrium, i.e. a minimum level of dissipation. For all these reasons we have called them "dissipative structures"[4].

3. Model Systems

3.1. Chemical systems. The first models for chemical instability were presented by Turing [5]. His work was dealing with the specific subject of morphogenesis and the schemes proposed were rather complex. Since then the relation of this type of behavior to thermodynamics and its wide applications in biology have been studied in much greater detail. We shall briefly summarize here the results obtained from the following simple scheme (cf. Refs. [6], [7], [8], [9]).

$$\begin{aligned} &A \rightleftarrows X \rightleftarrows E \\ &B + X \rightleftarrows Y + D \\ &2X + Y \rightleftarrows E \end{aligned} \tag{2}$$

Notice the presence of the autocatalytic step (c) which is largely responsible for the unusual behavior of the system. A, B, D, E are the initial and final products, the concentrations of which are maintained constant in time.

(a) At or near thermodynamic equilibrium, system (2) admits a unique steady state solution which is always stable with respect to arbitrary perturbations.

(b) Let us now drive the system far from equilibrium by making the concentrations of D and E vanishingly small in the system. Still for time independent boundary conditions, one can define a steady state corresponding to the continuation of the equilibrium like behavior. The branch of states defined in this fashion will be referred to as thermodynamic branch. Depending on the type of perturbations to which the system is subject and on the values of the parameters, such as A, B, the diffusion coefficients D_A, D_B, D_X and D_Y different situations will be realized. This is illustrated by the non-equilibrium phase diagram of Fig. 1.

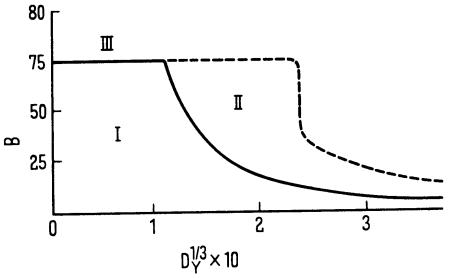


Fig. 1

Non equilibrium phase diagram for scheme (2). A, D_A, D_X have been held constant and D_B was taken so large that the distribution of B was maintained uniform in the system. Numerical values: $D_A = 0.197$, $D_X = 1.05 \times 10^{-3}$, $D_B \to \infty$. Only one dimensional diffusion has been considered. The values of A at both boundaries have been chosen equal to $\bar{A} = 14.0$.

(c) When the values of B and D_Y correspond to domain II of the diagram, the thermodynamic branch is unstable with respect to fluctuations in chemical composition. A new steady state corresponding to a regular spatial distribution of X and Y appears (cf. Fig. 2).

The properties of this low entropy spatial dissipative structure are clearly different from the usual behavior near thermodynamic equilibrium. In addition to the coherent character exhibited by such states, one can show that the final configuration depends to some extent on the initial perturbation. This primitive memory effect makes these structures capable of storing information accumulated in a remote past.

(d) For $D_B \gg D_A \gg D_Y \simeq D_X$ we have an unstable region III in which the system is driven to a new state depending both on space and on time. In the course of one period there appear wave-fronts of composition which propagate in the reaction volume, first outward and then, after

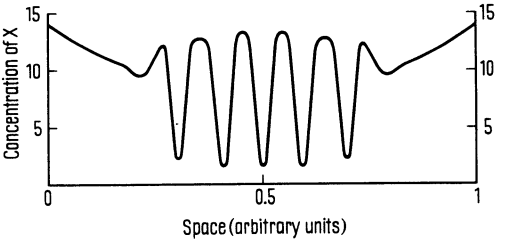


Fig. 2

Localized dissipative structure for model 2. Numerical values: $D_X = 1.05 \times 10^{-3}$, $D_Y = 5.25 \times 10^{-3}$, $D_A = 197 \times 10^{-3}$, $D_B \to \infty$, $B = 26$ $\bar{A} = \bar{X} = 14, \bar{Y} = 1.86$.

reflecting on the boundaries, to the interior. Moreover, at each point in space the system performs quasi-discontinuous oscillations in time. This space-time dissipative structure provides therefore a primitive mechanism for propagating information over macroscopic distance, in the form of chemical signals.

3.2. Dissipative structure in an artificial membrane. To illustrate the discussion of the preceding section, we would like to report briefly the behavior of an artificial membrane in which a dissipative structure may easily be observed experimentally [10].

The system consists in an artificial membrane on which two enzymes have been reticulated homogeneously: glucose oxydase and urease. This membrane constitutes really a macroscopic system of 100 mμ thickness. The activity of urease v_u and glucose oxidase v_g as a function of pH is reported in Fig. 3 and Fig. 4. We observe that the maximum of v_g is shifted to a lower pH than

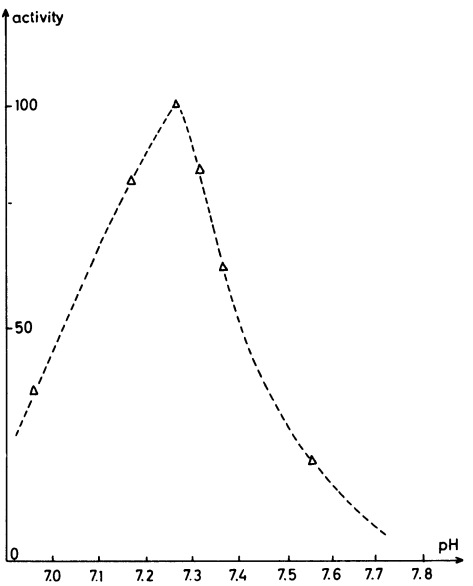


Fig. 3 Activity of urease as a function of pH in presence of an excess of substrate.

for v_u. Moreover we have the additional property that the urease reactions tends to increase pH; while the glucose oxidase reaction on the contrary tends to lower pH; $v_g \sim -\, d(pH)/dt$. Therefore the rate of variation of pH versus pH has the form given by Fig. 5. Accordingly in the

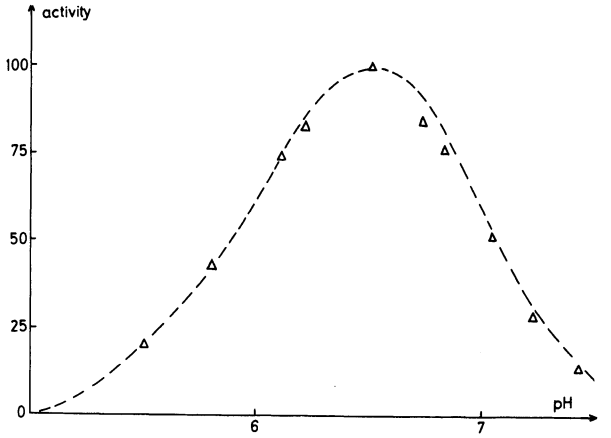


Fig. 4 Activity of glucose oxidase as a function of pH in presence of an excess of substrate.

neighborhood of pH_0, the system is in good approximation described by the equation

$$\frac{\partial pH}{\partial t} = D\frac{\partial^2 pH}{\partial r^2} + KpH - a$$

which admits as steady state solution (thickness of the membrane equal $2\pi/\sqrt{k}$)

$$pH(r) = pH_0 + A \sin \sqrt{k} r$$

A is an arbitrary constant depending on the diffusion coefficient between the membrane and the bulk solutions. Although the system is structurally homogeneous and placed in a symmetrical environment, it spontaneously breaks this symmetry and divides itself in two parts: one part in which the urease reaction is mainly proceeding, and another part in which the glucose oxidation is taking place. A differentiation in space and in function has been achieved.

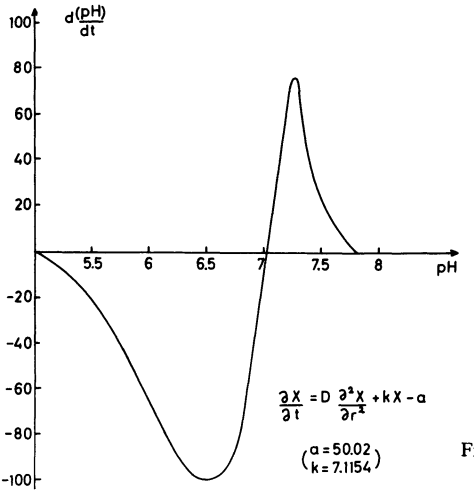


Fig. 5 Rate of variation of pH, as a function of pH.

4. The mechanism of instabilities: order through fluctuations

The models of the preceding section have illustrated the great diversity of situations that may arise in non-linear systems far from equilibrium. We shall see here that in spite of this, there is a general thermodynamic theory underlying the mechanism by which a system is driven to the new regime beyond the instability of the thermodynamic branch.

The central quantity of thermodynamics of irreversible processes is the entropy production. Its evaluation for systems for which a macroscopic description in terms of a limited number of local variables is possible, takes the remarkable form:

$$P = \frac{d_i S}{dt} = \sum_\rho J_\rho X_\rho \geqslant 0 \tag{3}$$

The J_ρ are the flows of the irreversible processes and the X_ρ are the corresponding forces. In the neighborhood of equilibrium P becomes a quadratic form in X'_ρs and provided the system is subject to time-independent boundary conditions, one has:

$$\frac{dP}{dt} \leqslant 0 \tag{4}$$

This implies a minimum entropy production at the steady state and the stability of this state. P acts then as a type of universal thermodynamic potential.

One shows easily that inequality (4) breaks down for states far from thermodynamic equilibrium. However it is still possible to obtain a general inequality in this domain if one decomposes P into two terms:

$$\frac{dP}{dt} = \sum_\rho X_\rho \frac{dJ_\rho}{dt} + \sum_\rho J_\rho \frac{dX_\rho}{dt} = \frac{d_J P}{dt} + \frac{d_x P}{dt} \tag{5}$$

One can now show that in the whole domain where (3) is valid and provided the system is subject to time-independent boundary conditions,

$$\frac{d_x P}{dt} \leqslant 0 \tag{6}$$

In contrast to (4), inequality (6) does n o t imply the stability of the steady state, because $d_x P$ is in general not the differential of a state function. Instead, (6) can be used as a stability condition for such states. The result is that stability is ensured, whenever

$$\sum_\rho \delta J_\rho \delta X_\rho \geqslant 0 \tag{7}$$

δJ_ρ, δD_ρ are the excess flows and forces due to the deviation of the state of the system from the reference regime. On the other hand in the domain of validity of (3) one shows that $\delta^2 s$ is a quadratic form obeying the inequality $\frac{1}{2}\delta^2 s \leqslant 0$ and, is related to (7) by the equality:

$$\frac{d}{dt}\left(\frac{1}{2}\delta^2 s\right) = \sum_\rho \delta J_\rho \delta X_\rho \tag{8}$$

Thus according to Lyapounov's theorem, the reference state will be stable provided (7) is satisfied, i.e., provided the excess entropy increases in time.

On the other hand, recent investigations [11] have shown that Einstein's fluctuation theory may be extended to non-equilibrium states, at least for wide classes of model systems. The probability of small fluctuations around a reference state is then given by:

$$P \propto \exp[(\delta^2 s)_0/2k] \tag{9}$$

where $(\delta^2 s)_0$ is evaluated around the reference state. Accordingly, for an unstable state, we see that starting with $(\delta^2 s)_0 < 0$ and $\partial/\partial t(\delta^2 s)_0 < 0$, P will first decrease. The system will then evolve to a new regime corresponding to the most probable state.

These considerations permit to stress the importance of the statistical element in the appearance of the new regime. In the region of instability, the average description following deterministic laws breaks down. The evolution of the system becomes essentially a stochastic process in the sense that the final state will depend on the probability of creating initially a fluctuation of a given type. Of course, once this probability is appreciable, the system will reach eventually a state which will be unique, once the boundary conditions are specified, and stable. This state may then be the starting point for further evolution.

5. Biological examples

In this last section we would like to comment on some examples of biological problems in which dissipative effects and non-equilibrium play a crucial role. The choice for such systems is very big at any level in biology, be it the molecular level, the cellular level or the supracellular level. Roughly speaking however the ideas outlined in the previous sections may be useful in the understanding of two classes of problems: first we have problems related to the discussion of the functional order observed in actual living systems. This refers to the physico-chemical basis of maintenance of life; second we have the problem of emergence of the actual structures (nucleic acids, proteins, cells, etc.,) from an inert prebiotic mixture of very simple molecules. This is the problem of prebiotic evolution. Much attention has been given to both questions recently and the detailed study of specific systems can be found in the references as well as in review articles [12, 13]. Here we shall limit ourselves to some general considerations about the behavior of cooperative membranes and the evolution of prebiotic molecules.

5.1. Stability of cooperative membranes. These last years it has been suggested that cooperative effects might exist in biological membranes and be associated to the phenomenon of nerve excitation [14]. The general idea is that the two states of permeability of the nerve membrane (low permeability and high permeability) correspond to two distinct conformations of membrane units such that the transition from one state to the other is a cooperative phenomenon. Accordingly the equilibrium constant for the transition from the non-permeable to the permeable state has the form (molecular field approximation) $K = \exp(\epsilon - \eta\langle r\rangle)$ or $K = \ell\Lambda^{\langle s\rangle}$ $(\Lambda = \exp(-\eta))$ where ϵ, η are positive constants and $\langle r\rangle$ is the fraction of membrane unit in the permeable conformation. We admit that the membrane is an isothermal lattice placed between two baths of different electrochemical potential and that each unit carries at least two specific ionic binding sites, one on the inner face of the membrane and one on the outer face. The transport ions across this system can then be decomposed into two processes: first the transport J_m across the membrane itself occurring when one ion jumps from one site to the other; second the diffusion processes J_{do} and J_{di} of the ions from the immediate neighborhood of the membrane in the bulk

solutions on both sides. It can be shown that these processes are conveniently described by the equations:

$$J_{d_i} = \gamma(\alpha_i - \alpha)$$
$$J_{d_o} = \gamma(d_o - \beta)$$
$$J_m = \langle r \rangle(\beta\, e^{x/2} - \alpha\, e^{-x/2})/(1 + \alpha\, e^{-x/4})(1 + \beta\, e^{x/4})$$

α_i, α_o are the concentrations in the bulk solutions; α, β are the concentrations near both sides of the membrane X = FV/RT with V the potential difference across the membrane. The fraction $\langle r \rangle$ as a function of α, β, ℓ and Λ is given by

$$\langle r \rangle = (1 + \alpha\, e^{-x/4})(1 + \beta\, e^{x/4})/[(1 + \alpha\, e^{-x/4})(1 + \beta\, e^{x/4}) + \ell\Lambda^{\langle r \rangle}] \qquad (11)$$

A steady state is attained in the system when $J_{di} = -J_{do} = -J_m$.

From the point of view of our discussion here it is sufficient to consider the case where V = 0. The departure of this system from equilibrium is then simply expressed by the concentration difference between bulk solutions $(\Delta\alpha)_t = \alpha_0 - \alpha_i$. The interesting result we would like to stress is the following: once the membrane lattice is placed in a non-equilibrium environment, i.e. once the chemical gradient $(\Delta\alpha)_t$ is increased, instabilities may appear which have no equilibrium analogue and are the result of an amplification of cooperativity by energy dissipation. Indeed, if we go back to the behavior of the system under equilibrium conditions, Eq. (11) reduces immediately to

$$\langle r \rangle = (1 + \alpha)^2/[(1 + \alpha)^2 + \ell\Lambda^{\langle r \rangle}]$$

One verifies easily that no equilibrium phase transition exists as long as $\ell n\Lambda \geqslant -4$, so that $\langle r \rangle$ is a one valuated function of α. As shown in Fig. 6, a multivaluated $\langle r \rangle$ function appears here as the result of the functioning of the system inserted in a concentration gradient. When the deviations

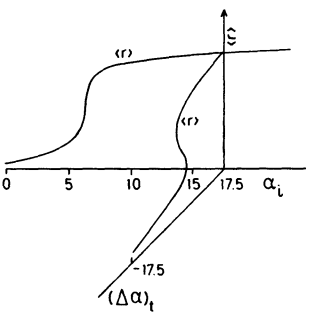


Fig. 6

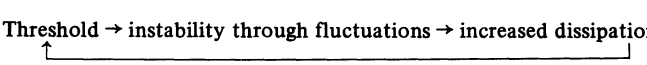
In the plane corresponding to thermodynamic equilibrium (i.e., for $\alpha_i = \alpha_o = \alpha = \beta$) we have plotted $\langle r \rangle$ as a function of α_i. Only one value of $\langle r \rangle$ is then accessible to the membrane for a single value of α_i. In a plane perpendicular to the plane of the figure, we have plotted $\langle r \rangle$ as a function of the overall gradient $(\Delta\alpha)_t$ for a given value of $(\Delta\alpha)_t$. The figure shows how the behavior of the system is modified when the membrane is brought into environmental asymmetry.

of the overall gradient from equilibrium are small, the conformational curve $\langle r \rangle$ corresponds to an extrapolation of the equilibrium situation. One may therefore say that the membrane state lies on the "thermodynamic branch". On the contrary for bigger values of the gradient, the membrane, in our model, exhibits a low conductance corresponding to a new type of steady state solution separated from the former one by a discontinuity.

5.2. The evolutionary feedback. In most of our preceding discussions we presented mechanisms by which self-organisation could be maintained in biological systems. We now would like to suggest how these ideas could apply to the problem of biological evolution. A new evolutionary

principle which would replace Darwin's idea in the context of prebiotic evolution has been proposed recently by Eigen [16]. It amounts to optimizing a quantity measuring the faithfulness or quality of macromolecules in reproducing themselves via template action. An alternative description of prebiological evolution has been proposed recently [13].

The main idea is to look for the possibility that a prebiological system may evolve through a whole succession of transitions leading to a hierarchy of more and more complex and organized states. We have shown that such transitions can only arise in non-linear systems which are maintained far-from-equilibrium. As a result, if the system is to keep the ability to evolve through successive instabilities, a mechanism has to be developed whereby each new transition favors further evolution by increasing the non-linearity and the distance from equilibrium. An obvious mechanism for this is that each transition enables the system to increase the entropy production. We may visualize this evolutionary feedback as follows:



It is important to insist on the complementarity between this increasing dissipation behavior, which we believe is essential for prebiological evolution, and the commonly observed tendency of physico-chemical systems near equilibrium to evolve towards a state of minimum dissipation or entropy production.

Let us outline briefly such a formulation of the evolution problem in terms of stability theory [17]. We shall be especially interested in the formulation of thermodynamic criteria for stability and evolution. We shall remain as general as possible, referring for the discussion of particular models until to the original articles [17, 18].

We consider a set of n interacting chemical substances X_i (i = 1, . . . n) and assume that all X's may exist in relatively abundant quantities. The time evolution of $\{X_i\}$ is described by the chemical rate equations:

$$\frac{dX_i}{dt} = F_i^e(X_1, \ldots, X_n) + \bar{F}_i(x_1, \ldots, x_n) \tag{12}$$

The term F_i^e describes the flow of matter (e.g. in the form of monomers) from the surroundings and $\bar{F}_i$ the reactions inside the system. It will be assumed that F_i^e is constant throughout the system which will be maintained uniform in space.

We assume that (12) admits at least one asymptotically stable steady solution. As is well-known, this implies that all n roots of the characteristic equation of (12) have negative real parts.

Suppose now that we take into consideration the formation, by random fluctuations, of "error copies" of X_i which we call Y_j (j = 1, . . ., m). The order of the differential system describing the joint evolution of X and Y will increase by m. Let us take for simplicity m = 1. Then the characteristic equation of the enlarged system will contain a correction term of $O(\epsilon)$, $\epsilon > 0$ multiplying either the $(n + 1)^{st}$ power of the characteristic root, or the term of the characteristic equation which is independent of the characteristic root. Thus, the characteristic equation of (12) will be recovered in the limit $\epsilon \to 0$ and the effect of the new substance in the evolution of X will disappear. For $\epsilon \neq 0$ but small, the equation will have (n + 1) roots of which n must have values close to those of the original characteristic equation and in particular have the same sign for the real parts. The stability of the steady state of the enlarged system can only be

effected by the new root, ω_{n+1}. For ϵ sufficiently small this root will have the following ϵ-dependence:

$$\omega_{n+1} \propto \frac{1}{\epsilon} \tag{13a}$$

if the correction term multiplies the $(n+1)^{st}$ power of the characteristic root, or

$$\omega_{n+1} \propto \epsilon \tag{13b}$$

if the presence of the new substance adds a constant term in the characteristic equation. Depending on the values of the parameters and on the kinetics, the coefficient in front and the r.h.s. in Eq. (13a) or (13b) may have a positive real part. Thus we reach the important conclusion that by taking into account a small, and at first sight "spurious" parameter corresponding to the addition of a new variable, the stability properties may change even if the original systems were stable with respect to random perturbations of the $\{X\}$ variables. In different terms, one has here a problem of structural stability of the reduced system (12) with respect to perturbations that introduce a side reaction which increases the order of the differential system. It is noteworthy that in case (13a) the departure from the unstable regime will be very fast, whereas for (13b) the unstable mode will evolve very slowly.

The next point is to find the equations of evolution of the $\{X\}$ and $\{Y\}$ variables that correspond to cases (13a) or (13b). Clearly, (13a) implies an evolution of Y according to

$$\epsilon \frac{dY}{dt} = G(\{X\}, Y, \epsilon) \tag{14a}$$

This will affect the evolution of $\{X\}$ which now will be given by the system

$$\frac{dX_i}{dt} = F_i^e + F_i(\{X\}, Y, \epsilon) \tag{15a}$$

Of course F_i will have to be such that for $\epsilon = 0$, i.e. for

$$G(\{X\}, Y, 0) = 0 \tag{16}$$

we get $$F_i(\{X\}, \bar{Y}(\{X\}), o) \equiv \bar{F}_i(\{X\}) \tag{17}$$

Notice that in Eq. (14a) there is no flow of mass term included. This is natural, since Y is produced solely by $\{X\}$ and is therefore essentially (except for diffusion terms which can be taken as small as we please) a closed system.

On the other hand, the form (13b) implies

$$\frac{dY}{dt} = G_1(\{X\}) + \epsilon G_2(\{X\}, Y, \epsilon) \tag{14b}$$

and correspondingly

$$\frac{dX_i}{dt} = F_i^e + \bar{F}_{i1}(\{X\}) + \epsilon F_{i2}(\{X\}, Y, \epsilon) \tag{15b}$$

Equations (14) and (15) have been constructed in such a way that the appearance of the new substance Y may give rise to an instability of the original system. Mathematically, this would imply that the solution of (14) and (15) does not remain for all times in a neighborhood of $0(\epsilon)$ of the solution of (12). Thus we will have evolution through unstable transitions of the

original system upon addition of new substances. This evolution may lead, e.g. to a new state of high Y concentration which will be dominated by the new substances. Let us now comment on the class of problems covered by this formulation. Consider first the situation described by (14a) and (15a).

The most natural realization of these equations in the context of prebiotic evolution is, probably, the case where a new function appears in the system. For instance, one of the error copies of X may serve as template for the synthesis of a substance which in turn catalyzes further the production of this error copy. One may expect this catalytic step to be more efficient than the previously existing mechanism of synthesis. This will be translated by the appearance in the r.h.s. of the equation for Y, of terms containing some large kinetic constant k. Dividing through by k and setting $\epsilon = k^{-1} \ll 1$, one would obtain an equation of the form (15a).

An alternative interpretation is suggested by rewriting Eqs. (14a), (15a) in the equivalent form

$$\frac{dY}{dt} = G(\{X\}, Y, \epsilon)$$

$$\frac{dX_i}{dt} = F_i^e + F_i(\{X\}, Y, \epsilon) \tag{18}$$

We have introduced a new time variable $\tau = t/\epsilon \gg t$. Thus Eqs. (14a) and (15a) describe systems which evolve according to two different time scales: A rapid motion describing the mechanism of production of the new substances. During this motion one could set, approximately, $\{X\} = \{\bar{X}\} = \text{const}$. And a slower evolution in the space $\{X\}$ variables, after Y has already attained a value given by Eq. (16). Notice that this picture of widely separated time scales is in agreement with the microscopic, all-or-none character of mutation. It should be built into any stochastic theory of evolution through mutation and selection. It is only because one is limited here to a macroscopic description, that one has to introduce reaction mechanisms with artificially imposed time scale differences giving rise to kinetic equations of the type (18). One expects that these equations will predict a very sharp selection during the evolution. This is in agreement with the numerical results of some models analyzed by Eigen.

Finally, let us turn to the class of problems described by (14b) and (15b). In the case of instability, these equations predict a slow evolution of the mutant substance to the new regime. Comparing with Eigen's equations we see that ϵ may be interpreted here directly as an average mutation rate. Thus the instability is attributed here directly to the copying errors. In contrast, in the previous case of Eqs. (14a) and (15a) the role of mutations was primarily to couple the old and the new variables. The evolution through instabilities was attributed to another element such as, e.g., a new enzyme appearing in the system. In principle, both types of situation are conceivable in a prebiotic medium.

It should be emphasized that the picture of evolution as a succession of instabilities is meaningful only if the original system (12) becomes unstable for arbitrarily small ϵ (formally, in the limit as $\epsilon \to 0$). In the context of prebiotic evolution, this can only be an idealization of the real situation. If ϵ is interpreted as a mutation rate, one could imagine easily values of the order of 0.1 or more (see numerical examples discussed by Eigen, reference [16]). And if ϵ is interpreted as the inverse of a rate constant in an enzyme-catalyzed reaction one would expect again a finite value owing to the rather poor catalytic ability of the primitive enzymes. Thus, the results derived from Eqs. (14) and (15) can only indicate the general trend rather than the actual kinetics of prebiotic evolution.

References

[1] Zotin, A. I. and Zotina, R. S.: J. Theor. Biol. **17**, 57 (1967).
[2] Prigogine, I.: Etude thermodynamique des Phénomènes Irréversibles. Liège 1947.
[3] Glansdorff, P. and Prigogine, I.: Thermodynamic Theory of Structure, Stability and Fluctuations. New York 1971.
[4] Prigogine, I.: Structure, Dissipation and Life, Communication at the First International Conference: Physique Théorique et Biologie (ed. Marois) Amsterdam 1967.
[5] Turing, A. M.: Phil. Trans. Roy. Soc. (London) **B237**, 37 (1952).
[6] Prigogine, I. and Lefever, R.: J. Chem. Phys. **48**, 1695 (1968).
[7] Lefever, R. and Nicolis, G.: J. Theor. Biol. **30**, 267 (1971).
[8] Herschkowitz-Kaufman, M. and Platten, J. K.: Bull. Acad. R. Belg. (Cl. Sci.) **57**, 26 (1971).
[9] Prigogine, I. and Nicolis, G.: Quart. Rev. Biophys. **4**, 107 (1971).
[10] Thomas, D.: Private Communication.
[11] Nicolis, G. and Prigogine, I.: Proc. Nat. Acad. Sci. (U.S.A.) **68**, 2102 (1971).
[12] Nicolis, G.: Adv. Chem. Phys. **19**, 209 (1971).
[13] Prigogine, I., Nicolis, G., and Babloyantz, A.: (to appear in Physics Today).
[14] Changeux, J. P., Thiéry, N., Tung, I., and Kittel, C.: Proc. Nat. Acad. Sci. (U.S.A.) (1967).
[15] Blumenthal, R., Changeux, J. P., and Lefever, R.: J. Membrane Biol. **2**, 351 (1970).
[16] Eigen, M.: Naturwiss. **58**, 465 (1971).
[17] Babloyantz, A. and Nicolis, G.: (to be published).
[18] Babloyantz, A.: (to be published).

A Gibbs Ensemble Approach to Biochemical Kinetics

Edward H. Kerner

Introduction

In the present note a crude first sketch will be attempted of a Gibbs-ensemble view of large, open biochemical networks. The network finally in issue is of course the one in nature that is the interior of the living organism, a great cauldron of 10^5 or 10^6 or more chemical species in a continual state of intra-action and of interchange with the exterior. It is not the regime of some 10^{16} interacting atoms per cell over which ensembles are to be built, but just this lower-order regime of 10^6 chemical species whose concentrations are the dynamical variables in coupled motion under those strong differential laws called chemical kinetics. The reason is that the 10^6 "chemical coordinates" in the "chemical dynamics" here stand within effective observational reach while embracing at once the central biochemical phenomena, like enzyme catalysis. The aim is to move on down still further from the 10^6 reduced coordinates to some few ensemble moduli toward global oversight of biological process. The ensemble, as merely methodological tool for scanning differential systems, seems worth trying at the still certainly complex reduced level of chemical dynamics.

Suppose we look with purely chemical eye into the cellular interior, assaying steadily over a very long time the concentration (or total number of molecules) of some selected few of the numerous biochemical types. The assay will be a kind of noise that must be expected to show on the one hand short-term and probably somewhat erratic fluctuations, and on the other hand longer-term near-periodicities corresponding to cell splitting. In this myopic chemical view, however, the stunning morphology of the splitting does not enter explicitly; only the single-cell chemical interior on through one (any one) long linear chain of the morphological descendants comes under observation; it is just the single cell-in-continuo, as one entity pulsating in time over very many morphological cell-lifetimes, that is under analysis.

Now granting the uniformity of the exterior, it is clear that the long-time interior biochemical assay of the single cell-in-continuo also will have a certain uniformity, viz, the ups and downs of concentration present something like a stationary record in the curve of concentration-vs-time: two long strips of the curve are substantially the same in their mean characteristics, and it cannot be told which strip preceded which – as for any stationary process, you cannot tell what time it is, the process being time-translationally invariant in the mean. The stationarity is of course the

feature that hints to a possible ensemble approach. By contrast with the cell-in-continuo, the closed chemical complex may show some swings in concentrations, but as the system inevitably glides to equilibrium the non-stationarity becomes grossly evident, there is no question at all in distinguishing the remote future behavior from the remote past.

There are of course in this picture serious counter-indications to an ensemble attempt. At any level of observation, including the above micro-chemical one, the near-periodicity associated to cell division is going to have to show through significantly. This appears to be very anergodic, very contrary, seemingly, to the whole idea of getting at time averages through ensemble averages. The cell-in-continuo does make biochemical noise, but also, on a grosser time-scale, does run with reasonable regularity, a noisy clock that rings the hours moderately well though ticking the seconds poorly. Plainly at least two quite different time scales have to enter the theoretical description (or, ensemble-wise, at least two ensemble moduli will be needed), with the long scale appropriately sharp while still allowing an estimate for its experimentally significant unsharpness. Out of respect for the strength of known chemical laws, it will not do to fabricate ad hoc "biochemical oscillators" here; the issue is, rather, to test whether open chemical kinetics as a cogent theory of time-evolution of chemical populations in non-equilibrium is itself equal to the task of portraying the cell-in-continuo.

Open kinetics

Let us write down the laws of mass-action controlling chemical concentrations $z_1, z_2, \ldots, z_n$ under rules for reactions of the most primitive but universal types, and augment them with terms allowing material influxes and outfluxes in the simplest reasonable way:

$$\dot{z}_i = \epsilon_i - \sigma_i z_i + k_{i\alpha} z_\alpha + K^i_{\alpha\beta} z_\alpha z_\beta \equiv Z_i(z). \tag{1}$$

[summation on repeated Greek indices understood]

The first term ϵ_i permits growth of species i due to influx from the exterior; owing to an assumed constant exterior, the input rate ϵ_i is itself properly taken as constant. The second term describes outward transport of i to the exterior at a rate naturally proportional to the interior concentration z_i. The third and fourth terms are the conventional mass-action laws for the fundamental reaction types: $k_{ii} z_i$ ($k_{ii} \leqslant 0$) tells of i-decrease owing to all possible types of chemical decay, for instance all the decompositions $i \to a + b$ and isomerizations $i \to i'$; while $k_{ij} z_j$ ($k_{ij} \geqslant 0$) describes i-increase due to the like of $j \to i + a$ (including isomerization $j \to i$). Similarly, terms like $K^i_{ab} z_a z_b$ ($K^i_{ab} \geqslant 0$) describe i-increase from $a + b \to i$, and $K^i_{ia} z_i z_a$ ($K^i_{ia} \leqslant 0$) i-decrease from, typically, $i + a \to j$. None other than these elementary conventional steps in k, K need be considered, as anything more complicated can be built up from them through introduction of suitable chemical complexes, intermediates, excited states and the like, albeit at the expense of greater chemical speciation: the quadratic non-linearity in the differential laws seems as high as it is ever necessary to go, and it expresses anyway that it is binary collisions that can be taken as fundamental. The universal format seems easily worth any extra speciation.

The principal "bio" component of the kinetics is deposited in the transport terms $\epsilon_i - \sigma_i z_i$. If we strike these out, the remainder in k, K is ordinary closed-system kinetics where any starting configuration z(0) in z-space ends up at some static equilibrium endpoint $z(\infty)$. Are the simple transport terms equal to preventing such stasis? The answer seems to be Yes: it has been shown by Chance and by Higgins [1] by computer that the simple transport terms, in modest networks,

can give genuinely oscillatory regimes of motion; moreover the oscillation has fairly direct experimental meaning. More recently it has been shown [2] on the theoretical side that to some degree equations (1) can be handled like (generalized) Riccati equations, allowing sometimes, Riccati-wise, a transformation to exact linearity (in second-order equations) where possibilities for sustained oscillation comes partially into view.

Besides assuming (a) that the chemical dynamics (1) admits quasi-oscillatory motion, we are going to assume at this stage (b) that the volume is fixed (so that concentration and total number of molecules are effectively the same), and (c) that in the interior there is spatial uniformity; (b) and (c) clearly need amendment at a later stage, now in progress. Finally (d) it will be assumed that (1) holds even for low concentrations. This raises a serious question, since the mass-action laws are really of the nature of statistical laws, correct in the mean and at high enough concentrations to ignore their small fluctuations; on principle a stochastic formulation of kinetics [3] should be employed, but this is out of hand mathematically even in very simple cases. It is good to remember here that Newtonian mechanics also holds only in the (quantal) mean; yet classical ensembles over Newtonian theory speak cogently, if not always correctly and completely, to the global view of complex physical systems – the inner conceptual core, hinging on the invention of notions like temperature and entropy, comes very well through the quantal averaging. Notice that, unlike physical ensemble theory where the aim is to account for bulk properties of matter, the ensemble view of (1) looks at the z_i as "microscopic" but observable coordinates whose noisiness is the aim of ensemble-statistical assay.

Ensembles

The base for Gibbs ensemble theory is Liouville's theorem and knowledge of a conservation law (with ergodicity usually a separate unproved assumption). In chemical dynamics (1) we have neither of these in any evident way. A way around is as follows. The divergence of the velocity-field **Z** in z-space is

$$\operatorname{div} \mathbf{Z} = \frac{\partial Z_\gamma}{\partial z_\gamma} = k_{\gamma\gamma} - \sigma_\gamma + 2K^\gamma_{\gamma\beta} z_\beta \equiv k + K_\beta z_\beta,$$

and clearly does not vanish (i.e. no Liouville theorem). However the geometrical meaning of div **Z** is that for a small sac δV of phase points in z-space making up a fluid flowing under $\dot{\mathbf{z}} = \mathbf{Z}(z)$,

$$\operatorname{div} \mathbf{Z} = \frac{1}{\delta V}\frac{d}{dt}(\delta V) = k + K_\beta z_\beta.$$

Integrating formally,

$$\delta V = (\delta V)_0 \exp \int_0^t (k + K_\beta z_\beta)\, dt$$

$$= (\delta V)_0 \exp t\{k + K_\beta \bar{z}^t_\beta\},$$

where $\bar{z}^t_\beta$ is the time average of z_β over (0, t). If we assume, under our picture above of the cell-in-continuo, that the dynamics is faithful to this picture and therefore giving stationary behavior as regards time-averaging of $z_\beta(t)$, then after sufficient time $\bar{z}^t_\beta$ is fixed and independent of the time-origin. Hence δV presently is uniform over the phase space, merely changing in scale equally everywhere by the factor exp t {const}. When now we consider a distribution

P(z)dV of phase fluid, representing in typical ensemble fashion an appropriate Gibbsian probability density, with P a function of a conserved quantity (not containing t explicitly), the exp t {const} simply falls away upon normalization. It is to say that δV is effectively invariant. This has only to do with the motion having stationary character.

Next we consider conservation laws. If $\epsilon_i - \sigma_i z_i$ is dropped (closed kinetics) we have in fact a family of simple conservation laws, $\nu_{p\alpha} z_\alpha$ = number of atoms of type p (e.g., hydrogen, oxygen, etc.) = conserved; here $\nu_{p\alpha}$ is the number of atoms of type p in molecular species α. These lack what can be called "global" character: none in general involves all z_α and none stands to be easily measured directly, as one entity. An obviously important and "global" conservation law is, however, the linear combination

$$m_\gamma \nu_{\gamma\alpha} z_\alpha = M_\alpha z_\alpha = M = \text{conserved mass},$$

(m_γ = atomic weight, M_α = molecular weight). It is clear that for the cell-in-continuo M stays in some sensible approximation a conserved quantity, a fundamental gross coordinate whose slow variations are yet to be taken into account.

Now over a time interval long enough for numerous fine-scale changes in z_α to occur, but short enough for M to be counted a good conservation law, we may try roughly a canonical type distribution function

$$P \sim \exp - \frac{M_\alpha z_\alpha}{\theta} \tag{2}$$

directly in z-space (basing on the argument before that dV can be treated as effectively invariant) in order to get rough averages of z_α (or functions thereof). The averages can be understood as crude time-averages over the fore-mentioned time interval if the walk of a representative phase point over the plane $M_\alpha z_\alpha$ = const is reasonably comprehensive (i.e. ergodic). (The assumption of ergodicity here may be appreciably less defensible than the treatment of dV as invariant and M as conserved). If we were to treat the family $\nu_{p\alpha} z_\alpha$ on the same basis we would have to introduce a family of moduli,

$$P \sim \prod_p \exp - \frac{\nu_{p\alpha} z_\alpha}{\theta_p},$$

i.e. a separate "hydrogen temperature", "oxygen temperature", etc. Because only M is visibly global, this family will be set aside.

There is an important restriction on the global modulus θ. From (1) we have

$$M_\alpha \dot{z}_\alpha = M_\alpha \epsilon_\alpha - M_\alpha \sigma_\alpha z_\alpha > M_\alpha \epsilon_\alpha - \sigma_0 M_\alpha z_\alpha \tag{3}$$

where σ_0 is the largest of all the σ_α (the terms in k, K go away because of $M_\alpha z_\alpha$ being a conservation law of the closed kinetics). Thence

$$M(t) > M(0)\, e^{-\sigma_0 t} + \frac{M_\alpha \epsilon_\alpha}{\sigma_0},$$

that is, M has the lower bound $M_\alpha \epsilon_\alpha / \sigma_0 \equiv M_0$: only the phase-space beyond the plane $M \geqslant M_0$ is accessible. To avoid this clumsy partition of phase space, we note, in a familiar argument, that $\exp(-M/\theta)\, dV \sim \exp(-M/\theta)\, M^n\, dM$ for the volume element dV taken as that between infinitesimally separated mass planes, viz, the canonically distributed fluid peaks with enormous

sharpness at the plane $M = n\theta$. Consequently we can take care of the restriction $M \geqslant M_0$ simply by placing a lower bound on θ,

$$\theta \geqslant \theta_0 \equiv \frac{M_\alpha \epsilon_\alpha / n}{\sigma_0} = \frac{\text{mean mass input rate}}{\text{maximal mass output rate}}.$$

With this bound on θ it does no harm in any phase integral to integrate over all of phase-space freely (subject of course to $z_\alpha \geqslant 0$, the inherent domain of variation of z_α under (1)). We conclude that, because of its openness, the chemical-kinetic system necessarily cannot be "cooled" below the intrinsic "kinetic-temperature" limit θ_0. The canonical mean chemical level according to (2) is $\bar{z}_i = \theta/M_i$ which is also the r.m.s. value $\{\overline{(z_i - \bar{z}_i)^2}\}^{1/2}$: the crude ensemble theory is saying that (i) the large molecular-weight molecules are generally much less populous than the small molecular-weights, (ii) the fluctuations in chemical level are very roughly comparable with the mean chemical level and neither can be made arbitrarily small, because of θ_0.

The gross coordinate M calls into play in (3) its natural mate $J \equiv M_\alpha \sigma_\alpha z_\alpha$ = gross mass output current,

$$\dot{M} = \epsilon_0 - J \qquad (\epsilon_0 \equiv M_\alpha \epsilon_\alpha). \tag{4}$$

Using (1), J moves according to

$$J = M_\lambda \sigma_\lambda \epsilon_\lambda + M_\lambda \sigma_\lambda k'_{\lambda\beta} z_\beta + M_\lambda \sigma_\lambda K^\lambda_{\alpha\beta} z_\alpha z_\beta \tag{5}$$
$$(k'_{ij} = k_{ij} - \sigma_i \delta_{ij})$$

We can try for the behavior of M, J now on the coarse time scale where their variations come into play by using (2) to average over the fine-scale ranging of z_α in (5),

$$J(\text{coarse}) = M_\lambda \sigma_\lambda \epsilon_\lambda + M_\lambda \sigma_\lambda k'_{\lambda\beta} \frac{M}{nM_\beta} + M_\lambda \sigma_\lambda K^\lambda_{\alpha\beta} \frac{M^2}{n^2 M_\alpha M_\beta} \tag{6}$$
$$\equiv \alpha_0 + \beta_0 M + \gamma_0 M^2,$$

where the equipartition rule $\bar{z}_i = \theta/M_i = \bar{M}/nM_i \simeq M/nM_i$ from (2) has been exploited, as well as $\overline{z_\alpha z_\beta} = \overline{z_\alpha} \cdot \overline{z_\beta}$ from P's separability. In this modo grosso we find, decoupling (4) and (6),

$$\ddot{M} = -\frac{\partial}{\partial M}\left(\alpha_0 M + \frac{1}{2}\beta_0 M^2 + \frac{1}{3}\gamma_0 M^3\right) \equiv -\frac{\partial}{\partial M} U(M),$$

for the coarse-scale M motion, in terms of the composite parameters $\alpha_0, \beta_0, \gamma_0$ (note their explicit dependence on system size n). The cubic in M is a kind of "potential energy" for M's conservative motion. The latter clearly has the possibility to be stable, slow, anharmonic oscillation for suitable weakly-restricted $\alpha_0, \beta_0, \gamma_0$. In the present order of approximation, the fine-and coarse-scale motions have been separated. An expression of their coupling can be brought in now by admitting that the coarse-scale conservation law

$$\frac{1}{2}(\epsilon_0 - J)^2 + U(M) \equiv \Omega = \text{conserved}$$

has itself to be scanned over different initial data, namely, the oscillator is in a type of heat-bath. We can guess to distribute it in the genuinely Hamiltonian phase space J,M according to

$$P^*(J, M)\, dJ\, dM \sim \exp - \frac{\{\frac{1}{2}(J - \epsilon_0)^2 + U(M)\}}{\varphi} dJ\, dM,$$

with a "slow-motion" modulus φ. Thinking of the cubic U(M) with a maximum, say, at $M_1 > 0$ and a minimum at $M_2 > M_1$, then, in order to keep M from running off to the left of M_1 in unbounded motion, Ω has to have a lid on it, $\Omega \leqslant U(M_1)$. This means, just conversely to the θ- or fast-motion modulus, that φ must be bounded from above, $\varphi \leqslant \varphi_0$.

Let us remark that the present transparently rough sketchwork cannot yet produce clear computational answers but only broad hints at best as to what may be the main character of fine and coarse motions abstracted from the main chemical dynamics. We have introduced θ_0, φ_0, and ϵ_0, $\alpha_0, \beta_0, \gamma_0$, – gross ensemble and composite parameters that, however, are put forth not ad hoc but, attemptedly, as visible, estimatable condensates from an a priori chemical dynamics. The assumptions (b) and (c) above need amendment before a more detailed ensemble look is taken at the cell-in-continuo.

References

[1] Chance, B. et al.: Ann. N. Y. Acad. Sci. **115**, 1010 (1964).
Higgins, J.: Proc. Nat. Acad. Sci. **51**, 989 (1964).
Higgins, J.: Ind. and Eng. Chem. **59**, 19 (1967).

[2] Kerner, E. H.: Bull. Math. Biophy., in press, June 1972.

[3] McQuarrie, D. A.: Stochastic Approach to Chemical Kinetics. London 1967.

Models of Population Growth, Diffusion, Competition and Rearrangement[1])

Elliott W. Montroll and Bruce J. West

1. Survey of mathematical models of population growth and competition without diffusion

Interest in population growth models started at the end of the 18th century by which time it was observed that the population of Europe seemed to be in an exponential growth mode. Records over the period 1200-1700, while scanty, indicated some fluctuation up and down, but no significant trend. The 18th century author most quoted on these matters is Malthus [1]. However, Thomas Jefferson and others made similar observations. The exponential Malthusian law of population growth is summarized in the elementary differential equation

$$dN/dt = kN \tag{1}$$

where N(t) is the population at time t and k is a constant. Verhulst [2] (1844) noted that the growth in population was not unbounded as modeled by Eq. (1). He, therefore, modified Eq. (1) to provide a mechanism for the retardation of population growth. The equation he proposed was

$$dN/dt = kN(\theta - N)/\theta \tag{2}$$

which limits the population growth to the level θ. The solution of (1) is

$$N(t) = N(0)\, e^{kt} \tag{3}$$

and that of Eq. (2) is

$$N(t) = \theta N(0)/\{N(0) + [\theta - N(0)]\, \exp(-kt)\} \tag{4}$$

where N(0) is the population at time t = 0. It should be noted that Eq. (4) reverts to Eq. (3) as $\theta \to \infty$; that is, as the upper bound to the population becomes infinite.

The general structure of the rate of increase for a saturating population may be written as

$$dN/dt = kNG(N/\theta). \tag{5}$$

Here G(x) is a saturation-inducing function such that

$$G(x) \to 0 \quad \text{as} \quad x \to 1 \tag{6}$$

The general form [4]

$$G(x) = (x^{\alpha} - 1)/\alpha$$

---

[1]) This study was partially supported by the U.S. National Science Foundation and the New York State Education Department.

leads to the Verhulst equation when $\alpha = 1$ and when $\alpha = 0$ to a second traditional form of the saturation function due to Gompertz [3] (1825) (although his equation was developed to model the mortality rates in elderly people rather than population growth)

$$dN/dt = -kN \log[N/\theta], \tag{7a}$$

linear in $v \equiv \log(N/\theta)$,

$$\dot{v} = -kv \tag{7b}$$

The above models, as well as those with the more general forms of G(x) were constructed to predict the growth of a single species in an environment with limited resources. A population which is growing can circumvent the saturation inherent in Eq. (5) by diffusing into nearby unoccupied territory, as the resources are depleted at their existing locations. The diffusion does not stop the saturation, but it does retard it. R. A. Fisher [5] considered a variation of this mechanism in his work on genetics. He was interested in the propagation of a virile mutant in a population which lived in a linear habitat (e.g., along a seacoast). We consider population diffusion and rearrangement models in the remainder of this report.

Equation (5) can be extended to include the effect of other species, the economy, the weather, etc., on the population growth rate of a single species. Probably the most primitive way of doing this is to add a random forcing term (which we choose to be proportional to N in order to always scale the growth rate by the instantaneous population). Then one finds [4, 6]

$$dN/dt = kNG(N/\theta) + NF(t) \tag{8}$$

or, if we let $v = \log(N/\theta)$

$$dv/dt = kG(e^v) + F(t). \tag{9}$$

In the Gompertz case with $G(x) = -\log x$,

$$\dot{v} = -kv + F(t). \tag{10}$$

When F(t) is postulated to be generated by a Gaussian random process, one can find a Fokker-Planck equation for the probability that v achieves a level v(t) after time t. This equation can be transformed to look like the Schrödinger equation [6]. The Gompertz saturation term corresponds to a harmonic oscillator potential and the Verhulst form of G(x), $G(x) = (1 - x)$, corresponds to the Morse potential. The equilibrium distribution for the Gompertz case is Gaussian [6] and for the Verhulst case [7] is Poisson.

A more refined way of exhibiting the influence of other species is by adding interaction terms explicitly to (5) as has been done by Lotka [8] and Volterra [9], as well as other authors. The statistical mechanics of such interacting systems has been investigated by Kerner [10] as well as the authors of Ref. [6]. The n species Volterra equation is

$$dN_i/dt = k_iN_i + \beta_i^{-1}N_i \sum_{j=1}^{n} a_{ij}N_j \qquad i = 1, 2, \ldots, n \tag{11}$$

The quadratic term represents binary collisions between i and j with a_{ij} being positive if i eats j and negative when j eats i. The β_i's represent exchange rates of the various species so that $\beta_i^{-1}/\beta_j^{-1}$ is the ratio of i's lost (or gained) to j's gained (or lost). Volterra postulated the a_{ij} to be

antisymmetric $a_{ij} = a_{ji}$. He showed the existence of constant of the motion for his dynamical system. One of the authors [4] has generalized (11) to a new form

$$dN_i/dt = k_i' N_i + \beta_i^{-1} N_i \sum_{j=1}^{n} a_{ij}(N_j^{\alpha} - 1)/\alpha \tag{12}$$

which becomes the Volterra system if $\alpha = 1$ and

$$k_i' = k_i + \beta_i^{-1} \sum_j a_{ij}.$$

On the other hand, if $\alpha \to 0$, (12) becomes

$$\beta_i d \log(N_i/q_i)dt = \sum_{j=1}^{n} a_{ij} \log(N_j/q_j) \tag{13}$$

where $\{q_j\}$ represents the set of steady state populations obtained from (12) by setting all $dN_i/dt = 0$. Then, in terms of the new variables,

$$v_i = \log(N_i/q_i) \quad \text{or} \quad N_i = q_i \exp v_i \tag{14}$$

$$\beta_i \dot{v}_i = \sum_{j=1}^{n} a_{ij} v_j \tag{15}$$

which can be solved by usual methods of solving linear rate equations. As in the Volterra case, this system has a constant of the motion and solutions of (13) resemble those obtained by machine calculation on the Volterra set.

With the above background material in mind, we are now in a position to explore some mechanisms of aggregation and diffusion of populations. In the limited space available here we will only be able to introduce some models and give a few characteristics of each. Their more systematic analysis will be published elsewhere.

2. Some models for spacial population redistribution

Our interest in this section is confined to the redistribution of populations rather than to population growth. Correlated growth and redistribution will be discussed in the next section. We start our discussion here with

2.1. Clannish random walkers [11]. Let us suppose that we have two species of random walkers identified by A and B and that they execute a random walk on a line that is divided into adjacent segments, each of length a. Furthermore, we suppose that ρ is the uniform number density of walkers per unit length (members of either species being counted indiscriminately) on the line. The total number of walkers on each segment is $N \equiv \rho a$. As a consequence of the random walk, species concentrations will be redistributed, but N will remain unchanged. If τ is the average time required per step of length a, we define $N(ja; \nu\tau)$ as the number of walkers of species A on the jth segment at a time $\nu\tau$ after the walk begins, and $f(ja; \nu\tau)$ as the fraction of walkers on the jth segment which are A's.

We postulate the "home" of the A's to be segments of large positive j, and that of the B's to correspond to large negative j's. The clannishness is characterized by a parameter ψ, which on our segmented line we write as

$$\psi = \alpha a/\tau. \tag{1}$$

We choose α so that the probability of an A walker at segment k stepping to segment k ± 1 in time τ

$$\frac{1}{2}\{1 \pm \alpha[1 - N^{-1}N(ka;\nu\tau)]\}.$$

A walker who finds himself completely surrounded by members of his own species [such that $N(ka;\nu\tau) = N$] steps with equal probability in either direction. One in the midst of strangers steps toward his home region biased with a probability $\frac{1}{2}(1 + \alpha)$ and in the opposite direction with probability $\frac{1}{2}(1 - \alpha)$. The number of A's on the kth segment after $(\nu + 1)$ steps is

$$N(ka;[\nu+1]\tau) = \frac{1}{2}N([k-1]a;\nu\tau)\{1 + \alpha[1 - N^{-1}N([k-1]a;\nu\tau)]\} +$$

$$+\frac{1}{2}N([k+1a;\nu\tau)\{1 - \alpha[1 - N^{-1}N[k+1]a;\nu\tau)]\},$$

or

$$\{N(ka;[\nu+1]\tau) - N(ka;\nu\tau)\}/\tau$$

$$= (a^2/2\tau)\{N([k-1]a;\nu\tau) - 2N(ka;\nu\tau) + N([k+1]a;\nu\tau)\}/a^2 -$$

$$- (\alpha a/\tau)\{N([k+1]a;\nu\tau)(1 - N^{-1}N([k+1]a;\nu\tau)) -$$

$$- N([k-1]a;\nu\tau)(1 - N^{-1}N([k-1]a;\nu\tau))\}/2aN. \tag{2}$$

This equation becomes a partial differential equation in the limit $\tau \equiv dt \to 0$, $a \equiv dx \to 0$, $\alpha \to 0$ if the limit is taken such that D and ψ defined by

$$D = a^2/2\tau \quad \text{and} \quad \psi = \alpha a/\tau \tag{3}$$

are finite. It is convenient to introduce the density of A's at x at time t

$$f(x, t) = N^{-1}N(x, t) \tag{4}$$

to obtain the nonlinear partial differential equation

$$\partial f/\partial t = (\partial/\partial x)\{D(\partial f/\partial x) - \psi f(1 - f)\}. \tag{5}$$

This equation is not a new one. It is the basic equation of cascade separation processes such as thermal diffusion and molecular effusion separation of isotopes [12-14]. One of its interesting features is that in the steady state with $\partial f/\partial t = 0$, the equation

$$df/dx = (\psi/D)f(1 - f) \tag{6}$$

has exactly the same mathematical form as the Verhulst equation for population growth except that here the growth is given as a function of position along the line rather than of time. The solution of Eq. (1.2) given by (1.4) can be written in a slightly different form, that is

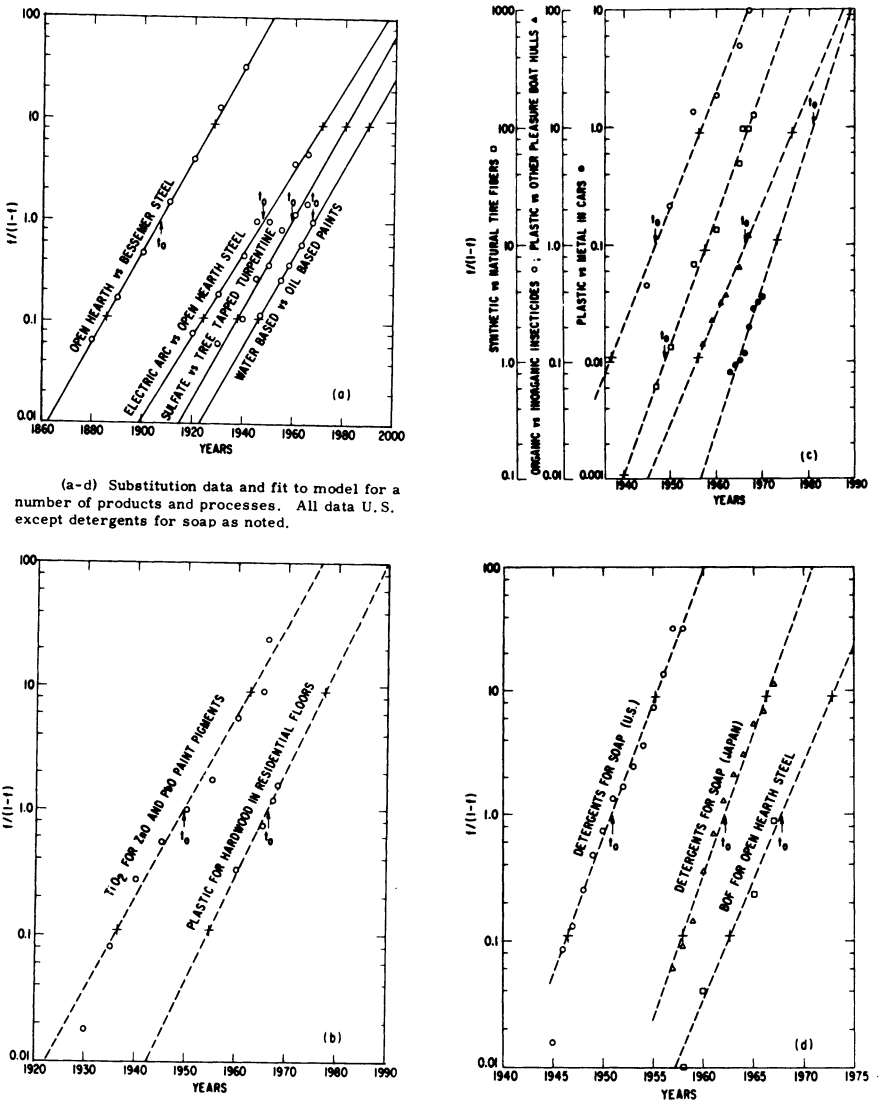
$$\log f/(1 - f) = \log f_0/(1 - f_0) + kt; f = N/\theta \tag{7}$$

Hence on logarithmic graph paper a plot of $f/(1 - f)$ as a function of t becomes a straight line. Similarly the solution of (6) is

$$\log f/(1 - f) = \log f_0/(1 - f_0) + (\psi x/D). \tag{8}$$

Fisher and Pry [15] have made the remarkable observation that when a new industrial process or product is introduced and takes hold to the extent of representing 10% of the output, it eventually takes over the rest of the market following the logistic law (4) or (7) as exhibited in Fig. 1 for several replacements.

A possible social application of Eqs. (6) and (8) is for the investigation of the development of ethnic attitudes in American cities. On certain streets in large cities one has at various times had



(a-d) Substitution data and fit to model for a number of products and processes. All data U.S. except detergents for soap as noted.

Fig. 1 Rate of Replacement of one technology by another. Data taken by J. C. Fisher and R. C. Pry, Ref. [15].

situations in which one ethnic group has lived at one end of the street and another at the other end. The "thickness" of the interface in which the change from one group to the other occurs would give a measure of the ratio (ψ/D) in (8). One could compare these values from year to year for various pairs of ethnic groups. (D/ψ), which has the units of a length might be called the penetration distance of one group into the other. A small (D/ψ) value would suggest that the groups try to avoid each other, while a large value would indicate that they are quite compatible. Common names associated with various ethnic groups could be used as tracer elements in examining old residential patterns through old city directories. Current appraisals could be made through a detailed census of residents on various streets.

Another nice feature of the differential equation (5) is that it can also be solved exactly in the non-steady state by applying a nonlinear transformation in the dependent variable [13, 14, 11].

Suppose that $u(x, t)$ is a solution of the diffusion equation $u_t = Du_{xx}$. Then relate u to f through the nonlinear transformation

$$u = \exp \int^x g(f)\, dx \tag{9}$$

where $g(f)$ is an arbitrary, as yet undefined, function of f. Then one finds with $g'(f) \equiv \partial g/\partial f)$

$$\int^x g'(f) f_t\, dx = D[g'(f) f_x + g^2(f)] \tag{10}$$

where, after differentiation with respect to x,

$$f_t = D\{f_{xx} + (g''/g')f_x^2\} + 2Dgf_x \tag{11}$$

Since the special case

$$g = (\psi/D)\left(f - \frac{1}{2}\right) \tag{12}$$

yields (5), we find that the solution of (5) is obtained from (9) after noting that

$$u_x/u = (\psi/D)\left(f - \frac{1}{2}\right) \tag{13}$$

or

$$f = \frac{1}{2} + D(u_x/u)/\psi \tag{14}$$

At time $t = 0$,

$$u(x, 0) = \exp \int^x (\psi/D)\left[f(x, 0) - \frac{1}{2}\right] dx \tag{15}$$

Hence the solution of (5) is given by (15) where $u(x, t)$ is the solution of the diffusion equation which corresponds to the initial distribution (15). A detailed discussion of special cases is given in Ref. [13].

These ideas can be extended to other condensation and separation models. We now consider:

2.2. Random walkers with a tendency for aggregation. In a mixed system of two components, diffusion tends to randomize the system. On the other hand, intermolecular forces are often such that aggregation is the preferred state. Here we shall be concerned with a nonlinear random walk model which can be used to describe rate processes in nonlinear and macroscopic population

systems characterized by this competition. Our model has the basic characteristics given in the first paragraph of our discussion of clannish walkers, except that a new alternative bias is introduced. We assume that at each step the walker goes to one of his nearest neighbor segments, following a bias to walk in the direction in which the concentration of A's is higher.

The bias is chosen so that the probability of step from segment j to j + 1 is

$$\frac{1}{2}\{1 + \mu N^{-1}[N(j+1,\nu) - N(j-1,\nu)]\, g(N_j/N)\} \tag{16a}$$

and from segment j to j − 1 is

$$\frac{1}{2}\{1 + \mu N^{-1}[N(j-1,\nu) - N(j+1,\nu)]\, g(N_j/N)\} \tag{16b}$$

The interpretation of these terms is as follows:

(a) If $N(j+1,\nu) > N(j-1,\nu)$ then the transition $j \to j+1$ is favored over $j \to j-1$, and vice versa when the inequality is reversed.

(b) As $\mu \to 0$, or as $N(j+1,\nu) \to N(j-1,\nu)$, the bias vanishes.

(c) We have left g(f) to be an arbitrary function. If it is set to be a constant, say 1, then the bias depends only on the difference in the fraction of A's in the (j + 1)st and (j − 1)st segments. This would be true if $f_j \equiv N_j/N$ be 0.1 or 0.9. One might expect in some cases that the bias might be less when a segment is occupied by 90% A's than by 10% A's so that one might choose

$$g(N_j/N) = [1 - (N_j/N)^{\alpha}]/\alpha. \tag{17}$$

It will turn out that this decision can be postponed until later since the solution of the resulting nonlinear equation for our random walk can be reduced to quadratures in any case.

We find using (16a) and (16b) that the number of walkers in the jth segment at time $(\nu + 1)$ is (with $N_j \equiv N(j,\nu)$)

$$N(j,\nu+1) = \frac{1}{2}N(j-1,\nu)\{1 + \mu N^{-1}[N(j,\nu) - N(j-2,\nu)]\, g(N_{j-1}/N)\} +$$

$$+\frac{1}{2}N(j+1,\nu)\{1 + \mu N^{-1}[N(j,\nu) - N(j+2,\nu)]\, g(N_{j+1}/N)\}$$

Now let $f(j,\nu) \equiv N(j,\nu)/N$ and $\Delta F(j) \equiv F(j+1) - F(j)$

Then
$$\tau^{-1}\{f(j,\nu+1) - f(j,\nu)\} = \frac{1}{2}(a^2/\nu)\{f(j-1,\tau) - 2f(j,\nu) + f(j+1,\nu)\}/a^2 +$$

$$+\frac{1}{2}(a\mu/\tau)f(j-1,\nu)\{\Delta f(j-1,\nu) + \Delta f(j-2,\nu)\}g(f(j-1,\nu))/a -$$

$$-\frac{1}{2}(a\mu/\tau)f(j+1,\nu)\{\Delta f(j,\nu) + \Delta f(j+1,\nu)\}g(f(j+1,\nu))/a.$$

If we let $\tau \to 0$, then the left hand side becomes $\partial f/\partial t$. Then, if we also let

$$D = a^2/2\tau \quad \text{and} \quad \psi = a\mu/\tau$$

where D and ψ remain finite as $a \equiv dx \to 0$ and $\tau \equiv dt \to 0$ we find that

$$\partial f/\partial t = D\partial^2 f/\partial x^2 - 2\psi(\partial f/\partial x)\partial[fg(f)]/\partial x. \tag{18}$$

This partial differential equation has already been investigated by one of the authors [11] for arbitrary preassigned functions g(x, t). Let us first choose

$$g(f) \equiv 1/2; \tag{19}$$

then we have

$$f_t = Df_{xx} - \psi f_x^2 \tag{20}$$

whose solution can be obtained from that of the diffusion equation

$$c_t = Dc_{xx} \tag{21}$$

by setting $f = -(D/\psi)\log c$ (22)

or $c = \exp(-\psi f/D)$, (23)

since the solution of the diffusion equation is in an unbounded region.

The intitial value solution of the diffusion equation is

$$c(x, t) = \frac{1}{2(\pi tD)^{1/2}} \int_{-\infty}^{\infty} c(y, 0)\exp\{-(x-y)^2/4Dt\}\,dy \tag{24}$$

Hence
$$f(x, t) = -(D/\psi)\log\left\{\frac{1}{2(\pi tD)^{1/2}} \int_{-\infty}^{\infty} \exp[(-\psi/D)f(y, 0) - (x-y)^2/4Dt]\,dy\right\}. \tag{25}$$

The second interesting case we consider corresponds to

$$g(f) \equiv 1 - f \tag{26}$$

The bias associated with a concentration gradient disappears when f = 1 and is a maximum when f vanishes. Our basic differential equation then has the form

$$f_t = Df_{xx} - 2\psi(1-2f)f_x^2 \tag{27}$$

which is of the form

$$f_t = Df_{xx} + f_x^2 F(f) \tag{28}$$

whose solution can be related to that of the diffusion equation $u_t = Du_{xx}$ through the transformation [11]

$$u = \int^{f} \exp\left\{\int^{f_1} D^{-1}F(f_2)\,df_2\right\} df_1 \tag{29}$$

In our case

$$F(f) \equiv -2\psi(1-2f) \tag{30}$$

so that $u = \int^{f} \exp -\{(2\psi/D)f(1-f)\}\,df$ (31)

or $u\exp(\psi/2D) = \int^{f} \exp\left\{(2\psi/D)\left(f - \frac{1}{2}\right)^2\right\} df$ (32)

This expression can be inverted to yield f as a function of u. The quantity

$$W(x) = e^{-x^2} \int_0^x e^{t^2}\, dt \tag{33a}$$

is called Dawson's integral. It has been well tabulated and has simple series expansions for large and small x. It is also related to the error function through

$$-\frac{2i}{\sqrt{\pi}} \int_0^z e^{t^2}\, dt = 1 - \operatorname{erfc}(-iz) = \operatorname{erf}(-iz). \tag{33b}$$

Another form of g(f) which we will encounter later is

$$g(f) = -\alpha f^{-1} \log(f/a) \tag{34}$$

which for small f gives a large bias to escape from one's associates. When f = a, the bias disappears and for f > a there is a tendency to cluster. This set of policies would be associated with the idea that the lower the concentration of one's fellows, the more one is being watched and oppressed but, as the population density builds up, the anonymity of a crowd makes life more bearable. Then Eq. (18) takes the form (with $\alpha\eta' = \eta$)

$$f_t = Df_{xx} - 2\eta f_x^2/f. \tag{35}$$

Various other policies in which g(f) changes sign can be selected.

2.3. Random walkers who abhor congestion. The next model of population distribution to be considered is that of an ensemble of walkers belonging to a single species and which abhor congestion. When executing their random walk, they would exhibit a slight bias in the direction of lower concentration. For the analysis of this problem, we would set up an expression similiar to (16) and proceed in the same manner that was used therein. However, in this case, μ is taken to be negative. We let $\mu = -\nu$, ν being positive. Then

$$f_t = Df_{xx} + 2\eta' f_x\, d[fg(f)]/dx \tag{36}$$

where $\eta' = \lim a^2\tau/\tau$ as $a \to 0$ and $\tau \to 0$.

It remains to chose an expression for g(f).

An interesting class of g(f)'s would be such that, starting with a fixed bias at f = 0, have it increased until $g(0) \to \infty$. An example of such a function is

$$g(f) = -f^{-1} \log[1 - (f/\theta)]. \tag{37}$$

Then
$$f_t = Df_{xx} + 2\eta(f_x^2)/[1 - (f/\theta)]. \tag{38}$$

Another such example is

$$g(f) = (1 - f/\theta)^{-1}. \tag{39}$$

Then
$$f_t = Df_{xx} + 2\eta(f_x)^2/[1 - (f/\theta)]^2. \tag{40}$$

Since this equation is of the form (28), it can also be reduced to quadratures.

3. Population growth and diffusion

We now consider a one-component population which both grows and diffuses. We start our discussion by adding a Verhulst population growth term to the diffusion model (2.40) for species which abhor congestion. Then

$$f_t = Df_{xx} + 2\eta(f_x)^2/[1 - (f/\theta)] + kf[1 - (f/\theta)]. \tag{1}$$

Two special cases of this equation have already been discussed in the literature. The first, with the special value of η, $\eta = D/\theta$ by one of the authors [16], and the second, with $\eta = 0$ by R. A. Fisher [5]. The case with $\eta = D/\theta$ can be reduced to quadratures while the Fisher case must be solved numerically.

When $\eta = D/\theta$, (1) becomes

$$f_t = D\{f_{xx} + 2f_x^2/(\theta - f)\} + kf(\theta - f)/\theta \tag{2}$$

The introduction of the new variable h(x, t) defined by [16]

$$e^{kt}\, h(x, t) = f(x, t)/[\theta - f(x, t)] \tag{3}$$

can easily be shown to satisfy the diffusion equation

$$h_t = Dh_{xx} \tag{4}$$

whose solution in an infinite domain is

$$h(x, t) = \frac{1}{2(\pi tD)^{1/2}} \int_{-\infty}^{\infty} h(x', 0) \exp\{-(x - x')^2/4Dt\}\, dx' \tag{5}$$

where h(x, 0) is related to the initial concentration distribution f(x, 0) by

$$h(x, 0) = f(x, 0)/[\theta - f(x, 0)]. \tag{6}$$

Hence
$$h(x, t) = \frac{1}{2(\pi tD)^{1/2}} \int_{-\infty}^{\infty} \frac{f(x', 0) \exp[-(x - x')^2/4Dt]\, dx'}{[\theta - u(x', 0)]} \tag{7}$$

and
$$f(x, t) = \theta\, e^{kt} h(x, t)/[1 + e^{kt}\, h(x, t)]. \tag{8}$$

There are several interesting initial conditions which give some insight into the character of the solutions of (2).

The first initial distribution which we discuss is one in which saturation exists on the negative half line to within a distance of about $1/\beta$ from the origin and no mutant population exists at points beyond a distance of about $1/\beta$ to the right of the origin. A representation of such an initial distribution is

$$u(x, 0) = \theta/(1 + e^{\beta x}). \tag{9}$$

Then
$$h(x, 0) = e^{-\beta x}. \tag{10}$$

If this is substituted into (5), one finds that

$$h(x, t) = e^{-\beta(x - \beta Dt)} \tag{11}$$

and
$$u(x, t) = \theta/\{1 + \exp \beta[x - (\beta D + k/\beta)t]\}. \tag{12}$$

The rate at which the mutant front propagates is

$$dx/dt = (\beta D + k/\beta). \tag{13}$$

In this case in which saturation exists behind the mutant front, the shape of the diffusion front remains invariant as the wave propagates. It is interesting to note that the propagation rate is determined by the initial slope at $x = 0$. If the initial front is sharp, (β large), the propagation is dominated by diffusion while, if it is broad, (small β), it is dominated by saturation development. When both processes make the same contribution, the propagation velocity has a minimum value which is

$$(dx/dt)_{min} = (2kD)^{1/2} \tag{14}$$

and which corresponds to $\beta = (k/D)^{1/2}$. A similar result was noted by Fisher in connection with (1) with $\eta = 0$. He found that no solution of the form $p = f(x + vt)$ could be found for which $|v| < (2kD)^{1/2}$.

As a second example we discuss an initial distribution of mutant which is peaked at the origin at a fraction η of the saturation level θ, and which extends over a length a.

Then $$f(x, 0) = \eta\theta/[\eta + (1 - \eta)\exp(x^2/2a^2)], \tag{15}$$

so that $$h(x, 0) = [\eta/(1 - \eta)]\exp(-x^2/2a^2), \tag{16}$$

and from (5)

$$h(x, t) = \frac{\eta a \exp[-x^2/2(a^2 + 2Dt)]}{(1 - \eta)(a^2 + 2Dt)^{1/2}}. \tag{17}$$

Therefore, from (8)

$$f(x, t) = \theta\eta a[\eta a + (1 - \eta)(a^2 + 2Dt)^{1/2}\exp\{-kt + x^2/2(a^2 + 2Dt)\}]^{-1}. \tag{18}$$

When $|x| \ll \{2kt(a^2 + 2Dt)\}^{1/2}$ the exponential term in the denominator can be neglected and one has saturation. The propagating population front at any time can be identified with the value of x which makes the argument of the exponential vanish, i.e.,

$$x = \pm\{2kt(a^2 + 2Dt)\}^{1/2}, \tag{19}$$

so that with increasing t the velocity of propagation approaches

$$dx/dt = \pm 2(kD)^{1/2} \tag{20}$$

independently of a; i.e., of the dispersion of initial distribution.

This case differs from (10) in which a saturation region exists initially, for in that case the propagation velocity depends on the initial shape of the wave while in the everywhere unsaturated case it becomes independent of the initial distribution. It is interesting to note that (20) is just the minimum propagation velocity which can be achieved by an initial distribution which has a saturated region (see Eq. (13)).

The manner in which the initial distribution (15) propagates seems to be typical of that of any initial distribution whose Fourier transform exists.

An appropriate choice of time and length scales, as well as units of f, allows us to reduce the four parameter equation (2) to one which involves a single parameter. Let

$$u = f/\theta, \tau = k\theta t, z = x(k\theta/D)^{1/2} \text{ and } \gamma = \eta\theta^2/D$$

Then (2) becomes

$$u_t = u_{xx} + 2\gamma u_x^2/(1-u) + u(1-u).$$

The analysis given above in the derivation of (8) corresponds to the choice $\gamma = 1$.

The next model of population growth with diffusion which we consider is that obtained by adding a Compertz growth term to (2.35). Then we find that

$$f_t = Df_{xx} - 2\eta f_x^2/f - kf\log(f/\theta) \tag{21}$$

When $\eta = 0$, this becomes an analogue of the Fisher equation, but with the Gompertz term replacing the Verhulst. As in the case of Eq. (1), the special case $\eta = \frac{1}{2}D$ is exactly solvable and will be considered here. This new model is given some emphasis because it can be extended to yield a solvable model of prey and predator interactions.

Note that when $\eta = \frac{1}{2}D$, (21) can be rewritten as

$$\frac{\partial\log(f/\theta)}{\partial t} = D\left\{f^{-1}\frac{\partial}{\partial x}\left[f\frac{\partial\log(f/\theta)}{\partial x}\right] - \left(\frac{\partial\log(f/\theta)}{\partial x}\right)^2\right\} - k\log(f/\theta). \tag{22}$$

Hence, if we let $v = \log(f/\theta)$, then this equation becomes

$$v_t = Dv_{xx} - kv. \tag{23}$$

This transforms into the diffusion equation in h(x, t) with

$$h = v\exp kt. \tag{24}$$

Hence, if the diffusion equation in h is solved with appropriate boundary and initial conditions,

$$f(x, t) = \theta\exp\{h(x, t)\,e^{-kt}\}. \tag{25}$$

We consider two initial conditions here. First suppose that

$$f(x, 0) = \theta\exp(-e^{\beta x}) \to \begin{cases} 0 \text{ as } x \to \infty \\ \theta \text{ as } x \to -\infty \end{cases} \tag{26}$$

with $f(0, 0) = \theta/e$. It is easy to verify that at time t

$$f(x, t) = \theta\exp\{-\exp\beta[x - (k\beta^{-1} - D\beta)t]\} \tag{27}$$

so that the velocity of the moving population front is $dx/dt = (k\beta^{-1} - D\beta)$.

Another simple initial conditions is

$$f(x, 0) = \theta\exp\{-(x^2 + \gamma^2)/2\sigma^2\}. \tag{28}$$

Then $$h(x, 0) = -(x^2 + \gamma^2)/2\sigma^2. \tag{29}$$

The solution of the diffusion equation with this initial condition is

$$h(x, t) = -(x^2 + \gamma^2 + 2Dt)/2\sigma^2. \tag{30}$$

Hence, $$f(x, t) = \theta \exp\{-(x^2 + \gamma^2 + 2Dt)\, e^{-kt}/2\sigma^2\} \tag{31}$$

which, at a given location (x constant), saturates as $t \to \infty$. The dispersion $\sigma(t) = \sigma \exp(\frac{1}{2}kt)$ grows exponentially with time.

Notice that the travelling population waves (11) and (27) are of the form $f(x, t) = f(x - vt)$ which corresponds to saturation wave propagating with velocity v without changing its character. Such waves are called solitons. Another example of such a wave is the solution

$$\begin{aligned} f(x - vt) &= \frac{1}{2}\{1 - \tanh \alpha(x - vt)\} \\ &= \{1 + e^{2\alpha(x - vt)}\}^{-1} \end{aligned} \tag{32}$$

of the equation

$$f_t = Df_{xx} + kf(1 - f)[1 + \epsilon(1 - 2f)] \tag{33a}$$

when $$\alpha = (k\epsilon/4D)^{1/2} \quad \text{and} \quad v = (kD/\epsilon)^{1/2}. \tag{33b}$$

Compact solitons which move through the medium as a localized undistorted packet can be obtained from equations for the derivations f_x of (1), (21), and (33a). For example, if we differentiate the Fisher equation obtained from (1), by setting $\eta = 0$ we find

$$(f_x)_t = D(f_x)_{xx} + kf_x(1 - 2f)$$

so that the variable $p = f_x$ which satisfies

$$p_t = Dp_{xx} + kp\left(1 - 2 \int^x p\, dx\right)$$

has a localized undistorted packet solution which propagates in either the positive or negative directions.

4. Predator and prey interaction with diffusion

We can conclude this report with the presentation of a solvable model of a prey-predator process in which diffusion occurs. We combine our diffusion process characterized by (3.21) with our predator-prey process characterized by (1.12) with $\alpha = 0$. Our new basic equation is

$$\frac{\partial N_j}{\partial t} = D_j\left\{\frac{\partial^2 N_j}{\partial x^2} - \frac{2\eta_j}{N_j}\left(\frac{\partial N_j}{\partial x}\right)^2\right\} + k_j N_j + N_j \sum_{\varrho} a_{j\varrho} \log N_\varrho \tag{1}$$

which, in the special case $2\eta_j = D_j$ can be transformed into a set of linear equations. In this case (1) becomes

$$\frac{\partial \log N_j}{\partial t} = D_j\left\{\frac{1}{N_j}\frac{\partial}{\partial x}\left\{N_j \frac{\partial \log N_j}{\partial x}\right\} - \left(\frac{\partial \log N_j}{\partial x}\right)^2\right\} + k_j + \sum_{\varrho} a_{j\varrho} \log N_\varrho \tag{2}$$

If we let $\{q_K\}$ be a set of solutions of (2) which are constant in time and space so that

$$k_j = \sum_K a_{j\varrho} \log q_\varrho \tag{3}$$

then, after defining V, by

$$V_j \equiv \log(N_j/q_j) \tag{4}$$

we find $$\partial V_j/\partial t = D_j \partial^2 V_j/\partial x^2 + \sum_\varrho a_{j\varrho} V_\varrho. \tag{5}$$

Upon introduction of the Fourier transform variables $v_j(k, t)$ defined through

$$V_j(x, t) = \int_{-\infty}^{\infty} v_j(k, t) \exp ikx \, dk \tag{6}$$

we find that

$$\dot{v}_j(k, t) = -D_j k^2 v_j(k, t) + \sum_\varrho a_{j\varrho} v_\varrho(k, t). \tag{7}$$

This is a set of linear rate equations with constant coefficients whose solutions are easily obtained. They can then be inverted to yield the $V_j(x, t)$ which, by exponentiation (see Eq. (4)), give us expressions for N_j. While the analysis is standard, it can become lengthy so we will not describe it nor derive its consequences here. That will be done in another publication.

Most of the equations discussed here have immediate generalization to two- and three-dimensional diffusion processes.

References

[1] Malthus, T. R.: An Essay on the Principle of Population as it Affects the Future Improvement of Society, 1798.
[2] Verhulst, P. F.: Mem. Acad. Roy. Bruxelles, **28**, 1 (1844).
[3] Gompertz, B.: Phil. Trans. of Roy. Soc., **115**, 513 (1825).
[4] Montroll, E. W.: Some Math. Problems in Biol. (Am. Math. Soc.), **4**, 101 (1972).
[5] Fisher, R. A.: Ann. Eugen., 7, 355 (1937).
[6] Goel, N. S., Maitra, S. C., and Montroll, E. W.: Rev. Mod. Phys., **43**, 231 (1971).
[7] Leigh, E. G., Jr.: Some Math. Problems in Biol. (Am. Math. Soc.), **1**, (1969).
[8] Lotka, A. J.: Elements of Mathematical Biology, 1924. Repr. New York 1956.
[9] Volterra, V.: Lécon sur la théorie Mathematiques de la lutte pour la vie. Paris 1931.
[10] Kerner, E. H.: Bull. Math. Biol., **217**, (1959).
[11] Montroll, E. W.: Lectures in Theoretical Physics (Theor. Phys. Inst., Univ. of Colorado), X-A, 531 (1968).
[12] Cohen, K.: Science and Engineering of Nuclear Power. London 1949, vol. 2, p. 19.
[13] Montroll, E. W. and Newell, G. F.: J. App. Phys., **23**, 184 (1952).
[14] Majumdar, C. D.: Phys. Rev. **81**, 844 (1951).
[15] Fisher, J. C. and Pry, R. C.: Practical Applications of Technological Forecasting in Industry (ed. M. J. Cetron) Chichester-New York 1971.
[16] Montroll, E. W.: J. App. Prob., **4**, 281 (1967).

Selforganization of Nucleic Acids and the Evolution of the Genetic Apparatus[1] [2]

Hans Kuhn

It is unclear how biological systems could evolve during a time of 10^9 years given by earth history [1–7]. Even the simplest systems that can be imagined to evolve to more complicated ones must have the property of self-reproduction, and this is only possible for systems which already have an appreciable complexity. They must have a device similar to the genetic apparatus of the known organisms, a machinery of highest skill and ingenuity. How was it possible, that such systems evolved? Can this evolution be explained on the basis of physical chemistry, and in this case is it a common process under appropriate environment conditions of a process of extremely low a priori probability?

Possibilities of formation of selforganizing systems were considered by the mathematicians v. Neumann [8] and Turing [9]. Eigen [10] has put forward a general theory of selforganization of matter and the evolution of macromolecules, based on the concepts of irreversible thermodynamics.

In the following the problem of selforganization and evolution of the genetic apparatus is considered from a somewhat different point of view by proposing a model pathway consisting of many little steps (Fig. 1a). Any one of these steps is obtained from the previous step by asking: what seems to be the most obvious way the system could behave in an appropriate environment. One invents possible ways, estimates the time required for this, and then chooses the fastest possibility as the next step.

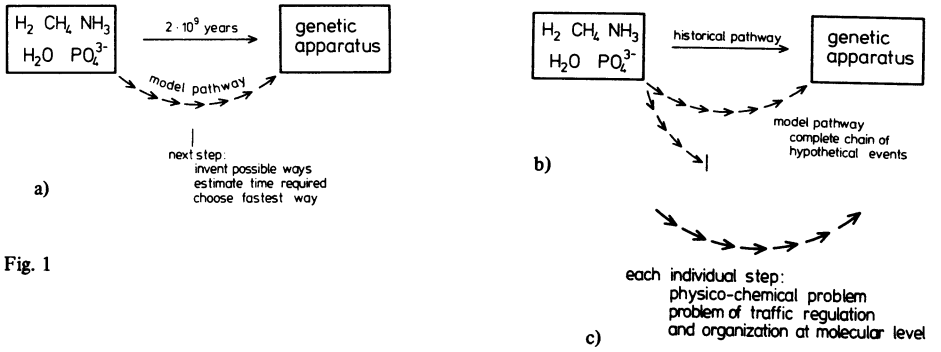


Fig. 1

[1]) The legends of the figures are at the end of this article on page 174.

[2]) Copyright: Verlag Chemie, Weinheim. Reprint from: Angewandte Chemie 84 (1972), with kind permission of the publisher.

In this proposed model each step is simply considered as a usual physico-chemical process – it is not intended to find any generalizing concept for selforganization and evolution. In contrary, it is hoped to see the essential points particularly well in the case of a simple model and to get automatically to selforganizing systems.

The proposed procedure is analogous to what is done in quantum chemistry: there one tries to approach an understanding of the chemical bond by inventing appropriate test functions as approximations of the wave functions, here one invents test model pathways. In quantum chemistry, very speculative test functions have proved to be extremely fruitful, and this should give courage to speculate on test pathways.

It seems essential to have a model pathway that gives a complete chain of hypothetical events. Many assumptions on steps of early evolution one can make, seem off hand quite reasonable, but lead to a dead end when proceeding further in the described manner (Fig. 1b).

It cannot be expected to find the historical pathway by this procedure, but by finding any one such model pathway, which is short enough to fulfill the requirements given by earth history, it is demonstrated that the early biological evolution can in principle by explained on the basis of physical chemistry. This philosophical implication stimulates to present a model in a stage, where many steps must be speculative. An experimental justification of the important steps seems essential and should be attempted.

The model should lead to the known genetic apparatus. This machinery is responsible for the replication of the DNA, the transcription of the message given in any part of the DNA by the sequence of the nucleotide bases to the messenger RNA and the translation from the four letter alphabet of the nucleotide bases to the 20 letter alphabet of the amino acids in the protein, using the transfer RNA's, charged with the amino acids (Fig. 2).

In the proposed model pathway each individual step is a physico-chemical process, but, more important, these steps are steps towards the solution of problems of traffic regulation and organization on the molecular scene (Fig. 1c). In managing daily life traffic situations, one

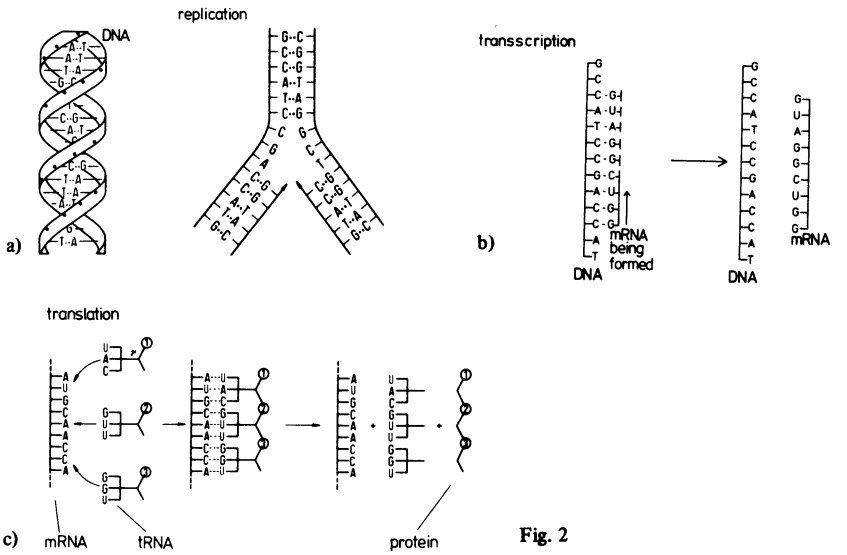


Fig. 2

does not need to know all constructional details about the participating objects, and will start with reasonable model assumptions for simplifying the job. Similarly, the model to be described in the following is based on several assumptions which seem reasonable.

An enzyme free formation of a complementary RNA strand at the matrix of a given strand is assumed to occur in the model environment. The reading error per nucleotide base is taken to be 1/100, i. e. on the average one out of a hundred bases is not the complementary one. An enzyme free replication of RNA can be assumed from experimental evidence, even if at the present there are some difficulties in defining these conditions exactly. W = 1/100 is what follows from the careful analysis given by Eigen and coworkers (Fig. 3a).

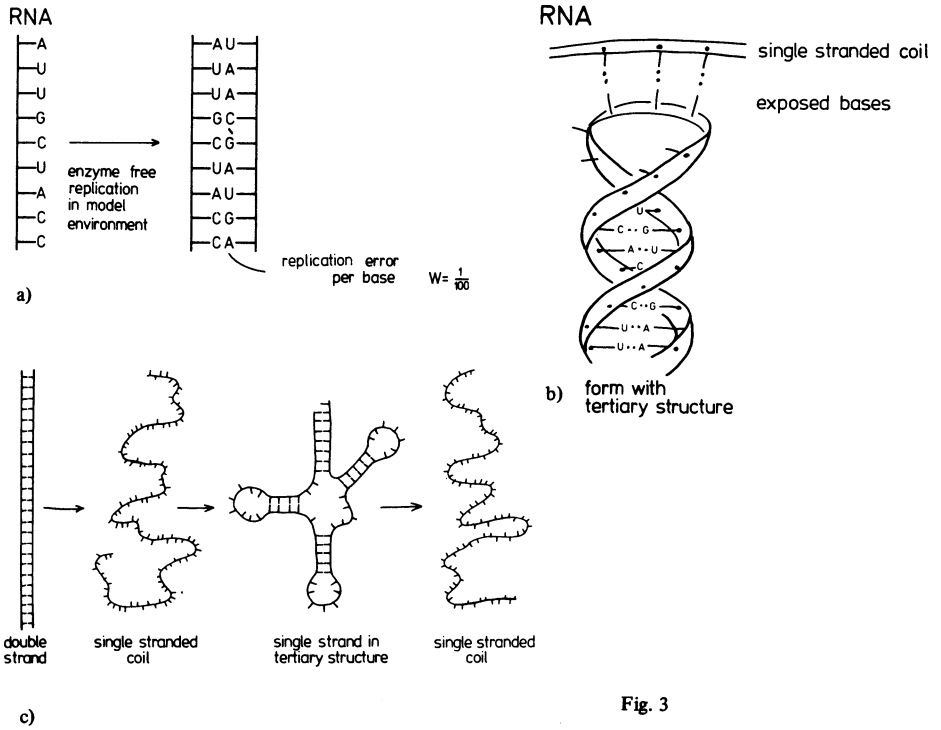


Fig. 3

Furthermore, we shall assume strong fluctuations in the model environment, such that transitions from double strands to single stranded coils and forms with tertiary structure occur. Some transitions of this kind are known experimentally (Fig. 3b).

Finally, we shall assume weak bonding between single stranded coils and exposed areas of forms with tertiary structure to occur in the model environment. Again, this assumption corresponds to what has been found experimentally (Fig. 3c).

In the beginning, of course, we cannot assume to have RNA helices, but energy-rich nucleotides, capable to polymerize (Fig. 4).

This follows from the well-known model experiments by Miller and many other authors simulating primitive earth conditions [1–7].

We shall assume a strong periodic fluctuation of the environment, given by the day-and-night period or for instance by the tides or the period due to the seasons of the year. A solution of appropriate compounds may be diluted and dried out or heated up and cooled down periodically and energy-rich compounds may appear and disappear periodically.

We assume a change between phases where polymers of nucleotides are formed and phases where they gradually decompose by hydrolysis. We have alternations between phases of formation and phases of decay.

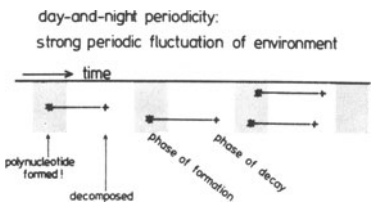


Fig. 4

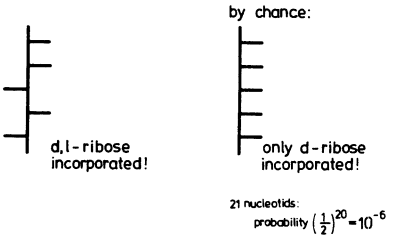


Fig. 5

Ribose, which contributes to the back-bone of the polymer, exists in two mirror image forms and we must expect that both forms, d and 1 are incorporated into the polymers. A polymer molecule may be formed, which by chance contains purely d ribose molecules (Fig. 5).

Such a molecule has particular new properties. We can assume that a double helix formation at the matrix of the molecule is possible in the model environment, which was not the case with the other polymer molecules due to the missing cooperativity there.

The double helix is particularly resistent to hydrolysis and will survive the decay phase, in contrary to the other polymers (Fig. 6).

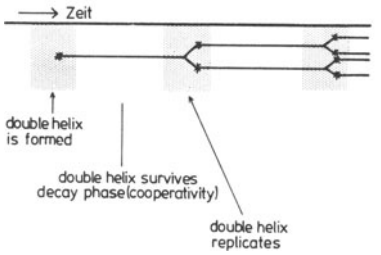


Fig. 6

When entering the next formation phase a replication of the double helix will occur, if the environment is assumed to be changing in an appropriate way. More and more such double helices will be produced in the following periods. They use up the energy-rich nucleotides and, thus, suppress the formation of the spontaneously occuring polymers.

By the initial formation of this double helix we have entered a first phase of selection. The divergent phase (incorporation of more and more different polymers with d and 1 ribose) has turned into a convergent one, where double helices are formed selectively.

We can expect that identical double helices now formed will condense from time to time in the way shown in Fig. 7a. Forms which have intramolecular complementary base pairing are obtained automatically. By further condensation (Fig. 7b) forms similar to transfer RNA molecules should occur. Transfer RNA molecules are known to form very specific tertiary structures.

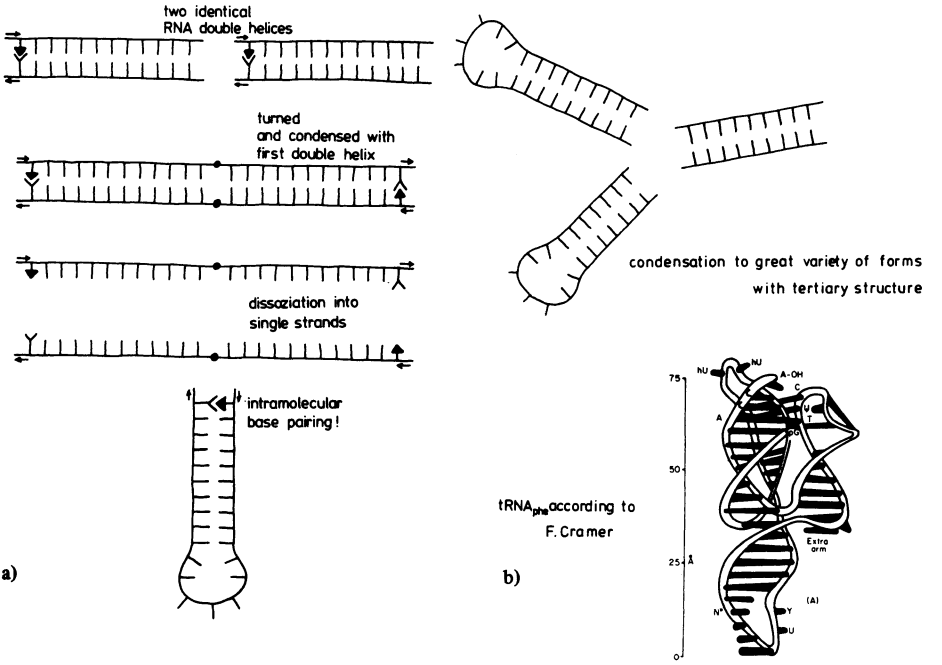


Fig. 7

A great number of different forms with very different tertiary structures will be built. Thus, again we are entering a divergent phase of evolution. A great variety of transfer RNA – like molecules will evolve.

We can ask: do we get to a selection of those molecular forms which are particularly resistant against hydrolysis due to their particular tertiary structures.

For answering that question let us first assume for a moment that we have just one form with a certain tertiary structure and that this form replicates without reproduction error. We have the replication during the formation phase, then the building of a tertiary structure, and afterwards a hydrolysis of part of the molecules during the decay phase (Fig. 8). Part of the molecules will be left over and will replicate again when entering the next formation phase. An increasing number of molecules will be obtained, but then saturation will be reached. There will be the same number Z of molecules at the end of each decay phase (Fig. 9).

Now let us assume that a molecule with differing tertiary structure is obtained by a replication error (Fig. 10). This mutant shall be somewhat more resistant against hydrolysis. Then the multiplication factor r of the mutant is somewhat larger than 1. r is the average number of

molecules of the mutant at the end of the next decay period. The multiplication factor of the normal form is 1 since we have reached saturation.

The number of molecules of the mutant, after the next period, is r^2. The mutant replaces the normal form after n periodes, that is, after n periodes the number of individuals of the mutant is about Z: $r^n = Z$.

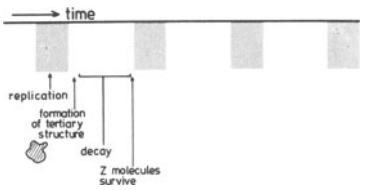


Fig. 8

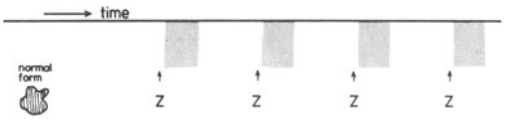


Fig. 9

For instance: if $Z = 10^6$ (this means that the preferential mutant appears among 10^6 individuals of the normal form) and $r = 2$ the mutant replaces the normal form in 20 periods, if $r = 1.01$ this number is 1400 (Fig. 10).

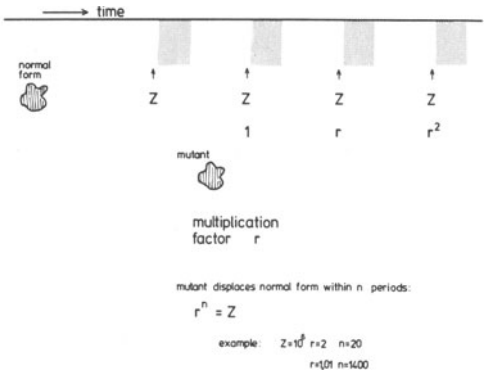
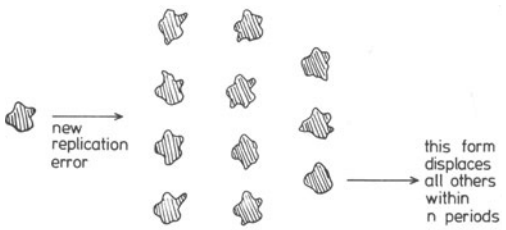


Fig. 10

Now after a while we will again have reproduction errors (Fig. 11). Mutants of very different shapes will occur and by far the most are less resistant than the form already present – it is extremely improbable to get an improvement, due to an error by chance, of a partially adapted



condition for selection: new replication error usually no earlier than after n periods

Fig. 11

structure. The isometric form in Fig. 11 will be more resistant than the parent form and has a chance to survive and to be selected in n periods.

However, such a selection is only possible if the reproduction error does not come too early, otherwise the probability of getting only detrimental new mutants is too large. The condition for selection is that the new error usually does not come earlier than after n periodes, that is after the preferential mutant has selected and is present in many individuals.

Now let us call P the probability of error for the reproduction of an individual. Then the condition for selection is that the number of reproduction errors in n successive replications is still not larger than one, that is $P \cdot n \leqslant 1$ (Fig. 12).

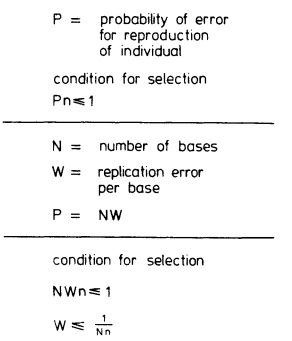


Fig. 12

Now N is the number of nucleotide bases in our RNA molecule and W the reproduction error per base. Then $P = N \cdot W$ (provided P is small compared to 1). The condition for selection can then be written, (introducing the second into the first equation) $NWn \leqslant 1$; $W \leqslant 1/NW$. Now in our case the number of bases was of the order of 50 and n about 20. Therefore W must be smaller than 1/1000.

However, we saw that actually $W = 1/100$. It follows that RNA molecules of particular tertiary structure do not select.

Each preferential form will be spoiled by new reproduction errors, long before the form has had time to select.

Therefore we must expect a divergent evolution of these transfer RNA – like molecules with very different tertiary structures.

It must be expected, that among this great variety of molecules sometimes we would find molecules fitting together by chance (Fig. 13a). Such aggregates, when entering the replication phase, will fall apart, and each partner will replicate. In the following phase the molecules will assume their typical tertiary structure and then assemble to form the organized aggregate.

During the replication process of the molecules a reproduction error may happen. (Fig. 13b). Then the error copy does not fit into the aggregate. It will diffuse away and will be substituted by an error free copy.

By this trick we get an error free reproduction of the aggregate in spite of the appreciable probability of errors in replicating the component molecules. With such aggregates the condition for selection can thus be fulfilled.

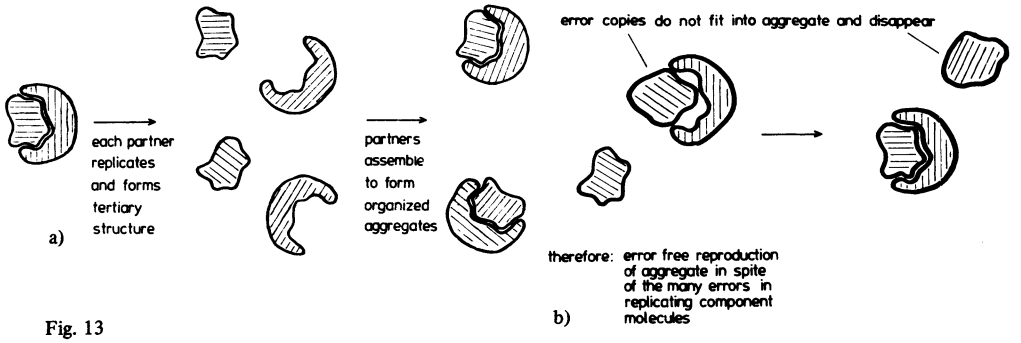


Fig. 13

However, the component molecules must find each other to form the aggregate within a time small compared to the period. This is possible only if we assume that the molecules are enclosed in a compartment (Fig. 14). The compartment facilitates the component molecules finding together. The compartment may enclose the associates, while single molecules may escape through pores in the compartmental envelope.

In case the period is a day, the time available for the formation of the aggregate is of the order of minutes. It can be easily estimated that the size of the compartment then must be of the order of 10^{-13} cm^3; this compares with the size of a bacterium. Compartments of such a size may be given for instance in the space between clay particles.

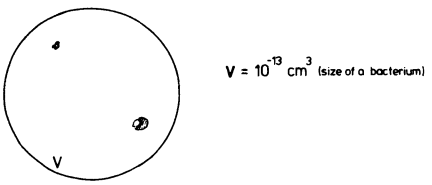


Fig. 14

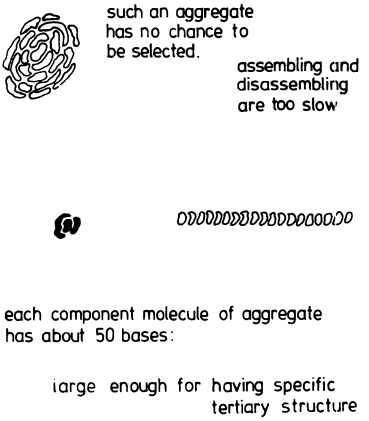


Fig. 15

We must expect to have again a highly selective, convergent phase of evolution and the question is: what kind of aggregates will be selected.

The aggregate on the left of Fig. 15 has no chance. The disentangling of all molecules when entering the replication phase takes too much time, and the fitting together as well. Assembling and disassembling are too slow.

Only small isometric aggregates have a chance to survive, and larger aggregates where the components are arranged one-dimensionally.

We can make a simple statement concerning the size of the component molecules: each molecule must have about 50 bases (Fig. 15): such a molecule is small enough for frequent correct copying: A replication error occurs in one out of 100 nucleotides; thus, a molecule should have not more than about 50 nucleotides for being copied correctly reasonably often.

On the other hand the molecules must be large enough for having specific tertiary structures, necessary for forming highly specific contact regions. This is the case for molecules with 50 nucleotides, and such molecules should select. This predicted size compares just about to the actual size of transfer RNA molecules.

Among the one-dimensionally arranged aggregates those are highly preferential that can be formed, similar to crystal needles, from a nucleation molecule and many identical molecules, matching together, the line-up molecules (Fig. 16).

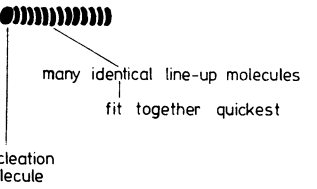


Fig. 16

Identical molecules fit together quickest and such aggregates build up much quicker than aggregates with different molecules, where one has to wait much longer until the aggregate has assembled.

There still is a difficulty in building up such an aggregate from a dilute solution of line-up molecules: In a partially assembled aggregate the contact region for matching the next line-up molecule is not found easily by this molecule. The diffusion to a given point is a slow process.

Now let us consider a nucleation molecule which has an open end with statistical sequence, forming a single strand without tertiary structure, in addition to the part of the strand organized into a tertiary structure forming the nucleation center of the aggregate (Fig. 17a). This strand, the collector strand, favours loose attachment of the line-up molecules. It is easily seen that this additional element highly accelerates an assembling of the line-up molecules and facilitates the organization into the ordered aggregate. The result can be easily approached quantitatively by considering the corresponding diffusion problem. The collector strand is an

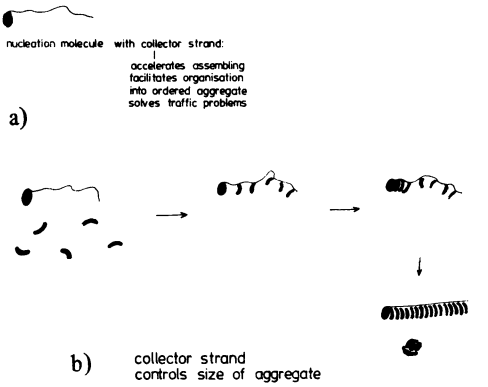


Fig. 17

important element for solving traffic problems by picking up these molecules and leading them to the assembling site.

The molecules are collected from the solution and then orderly arranged (Fig. 17b). The collector strand controls the size of the aggregate. The two types of aggregates that should be selected in this way, the one-dimensional and the small isometric aggregates, are shown in Fig. 17b below.

In these aggregates the regions necessary for contact between the components are kept constant by the trick of exact fit, but the region at the surface of the aggregates may change without any disadvantage for the system and a population is obtained where this region shows a great variety.

We are again in a divergent phase of evolution where many different systems have about the same chance to survive.

Hollows for small molecules will appear by chance, and many different catalytic reactions will occur (Fig. 18).

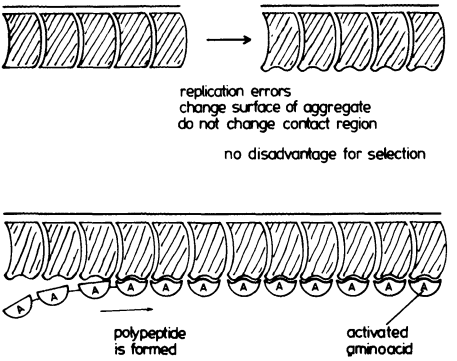


Fig. 18

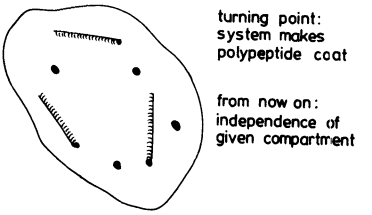


Fig. 19

One of these reactions may be the formation of a polypeptide at the surface of an aggregate of line-up molecules. We can assume that activated amino acids are in the solution. The assumption that such a polymer should be formed in this way, seems reasonable. Katchalsky and coworkers have found the cytolytic formation of polypeptides at the surface of a very specific clay, montmorillonite. In the case considered here, the surface of the organized aggregate is assumed to take the place of montmorillonite.

The reaction would not be of importance at this stage, if it would not lead to a very particular situation:

Such a polypeptide will make a coat. It is well-known from the work of Fox on microspheres that unspecific polypeptides form coats.

This again brings a turning to a convergent phase of evolution. From now on the system is independent of a given compartment. It is no more bound to the space between, say small particles (Fig. 19). The systems can spread all over within a region of appropriate fluctuation in the environment.

A high selection pressure towards systems with increasing precision in the polypeptide synthezising apparatus must occur, since such systems make their coat more quickly, and each

time those systems will survive, that make an appropriate coat quickest. Thus, an important turning point to a convergent phase is reached.

An isometric aggregate may enter into the scene as a cooperating element and accelerate the peptide bond formation.

The nucleation molecule with the collector strand is now the primitive messenger RNA, the line-up molecules are the primitive transfer RNA's, and the isometric aggregate is the primitive ribosome (Fig. 20).

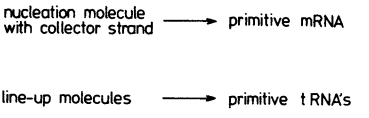


Fig. 20

This selection pressure in the direction of a precise fitting of groups at the peptide bond formation site, leads to a system which controls the supply of the transfer RNA's to the place of peptide synthesis. We may assume, that a mechanism is developed for realizing this precise fitting, according to which each three bases of the primitive mRNA are assigned to a base triplet on each primitive tRNA (Fig. 21).

Further selection pressure in the same direction – existing mechanism improved – leads to a better contact of molecular parts at the reaction site through complementary base pairing. We shall assume in the model, that the complementarity, at this stage of precision, is restricted to the bases between the middle position of the triplet and the respective place on the primitive mRNA (Fig. 22).

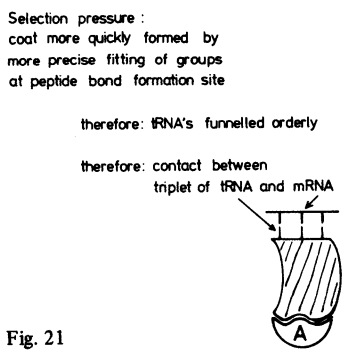


Fig. 21

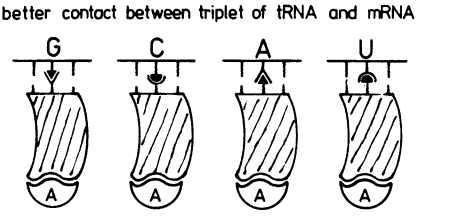


Fig. 22

For this kind of improvement of contact, four different sorts of tRNA's, which differ in their anticodon, are necessary.

The four sorts will gradually change their tertiary structure due to replication errors. A certain specifity in binding amino acids will result.

Therefore, the system produces polypeptides with a certain sequence. But: the code cannot be fixed yet: the sequence on the primitive messenger RNA is spoiled by transcription errors. The

condition for selection is not fulfilled with the existing error rate of 1/100. An error rate of 1/1000 is needed to fulfill this condition (Fig. 23).

At this point we do not get out of the difficulty that the code, even being present, cannot be fixed.

Since reading errors in the collector strand have no disadvantages at this stage of evolution, a population of systems is obtained, which produce polypeptides of all different sequences in the four amino acids or four groups of amino acids. Again, a divergent stage of evolution is reached. We can assume that one of our systems in the population, by chance, produces an enzyme for the replication process, a polymerase, which reduces the error rate by a factor of ten, as compared to the enzyme-free replication. The requirement for selection, an error rate of 1/1000 or less, then is fulfilled. The code for 4 amino acids or 4 groups of amino acids can gradually be fixed (Fig. 24).

primitive tRNA's :

errors by chance
change contact region
for aminoacid

a certain specificity
in binding aminoacids results :

thus : system produces polypeptides
with certain sequence

but : code cannot be fixed.
Sequence on primitive mRNA
spoiled by replication errors :
condition for selection
not fulfilled
$W=\frac{1}{1000}$ needed !
$W=\frac{1}{100}$ given !

Fig. 23

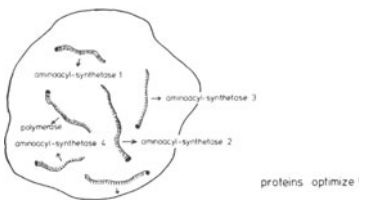
polypeptide assists in RNA replication: $(W=\frac{1}{100} \rightarrow W=\frac{1}{1000})$

code for 4 aminoacids or 4 groups of aminoacids
gradually fixed

other polypeptides may act as aminoacyl-synthetases

Fig. 24

Once having taken this barrier a strongly convergent evolution starts. The exact succession of steps in the gradual fixation of the code is unessential. We can assume the gradual formation of amino acyl transferases which help increasing the specifity in binding amino acids to the four transfer RNA's, but the steps of this interplay is unimportant – as soon as this turning point is reached the further development of the genetic code follows as a necessity. The steam engine of evolution is turned on. These primitive proteins, thus formed, will optimize more and more (Fig. 25).



note: refinement of existing mechanism.
abruptly mechanism serves an
entirely new purpose

Fig. 25

selection pressure towards larger proteins
of given sequence

need for better contact at peptide bond formation site

complementary base pairing at first position
in triplet

turning to build up code for additional amino acids !

Fig. 26

The way the genetic code came on should be noted: an existing mechanism was refined due to the strong selection pressure in the direction of forming the coat quicker by increasing the precision at the peptide bond formation site. Abruptly the mechanism served an entirely new purpose, the production of polypeptides with fixed sequence.

The systems that form the best enzymes will survive, and good enzymes will have larger molecular weights. Therefore, there will soon be a strong selection pressure towards systems forming larger proteins of fixed sequence (Fig. 26). Systems survive, which have the smallest probability to produce nonsense proteins by misreading.

For this reason systems will select, which show better contact between anticodon and primitive messenger RNA and this should be reached by complementary base pairing also at the first position in the triplet. This necessity of evolution towards a more precise bonding of amino acids by the protein synthesizing apparatus should again lead to a turning point: the building up of the code for additional amino acids.

We are now at the point where more and more cooperating enzyme systems of high specificity and skill will evolve. By a difficulty this evolution must soon come to an end. In the replication phase besides the original strands the complementary RNA strands are formed. They are useless in the protein synthesizing apparatus und produce a waste removal problem. The formation of these complementary strands is avoided by incorporation of DNA into the genetic process. We can assume that a pair of enzymes E_1 and E_2 is evolved (Fig. 27). E_1 produces the complementary DNA at the template of RNA and E_2 produces RNA at the template of this DNA. In later evolution steps a RNA linking protein may evolve and in this way the total information for the individual is merged in one DNA strand. This change has a great selective value. Finally we can assume a DNA polymerase to appear. In this way DNA becomes the carrier of genetic information. E_1 degenerates and E_2 evolves to the transscriptase.

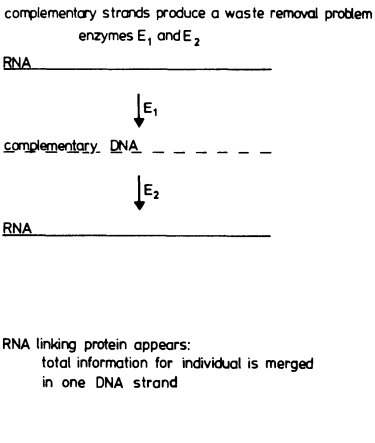


Fig. 27

We are now in the situation where we have all elements of the genetic apparatus. The further development of the genetic apparatur should now be a necessary consequence: the improvement of the ribosome, the introduction of start and stop signals, the further evolution of the code, and the evolution of the replication and transscription apparatus.

Now let us consider such a system with a fully developed genetic apparatus. It produces many proteins with fixed functions (forming a highly organized team), and we can assume that it produces one protein with sequence by chance, not yet belonging to the team (Fig. 28a). It can be easily estimated using the simple approximation mentioned in the beginning that the fittest individual in the population is selected in about $n = 10^3$ generations. The replication error per base then is $W = 1/nN$ where N (the number of nucleotide bases used for instructing the protein) is about 10^3. Therefore, $W = 10^{-6}$ (Fig. 28b). The number of generations for instructing the protein is easily calculated and a value of 10^8 is obtained, this corresponds to about 10^5 years for a generation per day. For the instruction of the 10^3 proteins in a bacterium a time of about 10^8 years is required, and this is not greater than the time allowed for earth history.

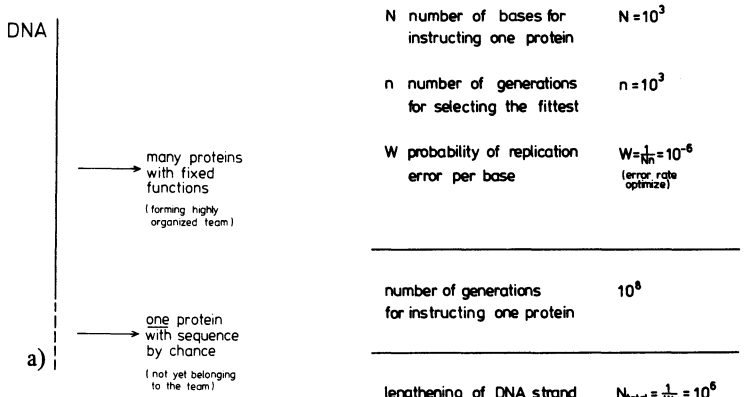


Fig. 28

With this theoretical value for the replication error rate per base, $W = 10^{-6}$, we get an upper limit for the number of bases in the DNA strand of $N_{total} = 1/W = 10^6$. At larger values of N_{total} no error free reproduction is to be expected, and the forms must disappear by deadly errors. This value of N_{total} corresponds to just about the content of informational DNA in bacteria.

At this point we again reached a limit, something must happen in order to increase the content of information. Let us discuss (as a speculative way to overcome this difficulty) the possibility that a system is formed, which has a triple storing of information and develops a sort of repair mechanism by which the three strands are constantly repaired so that they always have the same sequence (Fig. 29).

Usually the errors are repaired by this mechanism, and only those errors are fixed, which occur in two strands at the same position. The probability for that is $(10^{-6})^2 = 10^{-12}$. The empirical value of the error rate in the replication of a higher organism is about 10^{-11}.

The idea of this approach was: don't worry too much about general formulations, go ahead and build a model. We considered a model pathway (Fig. 30) where each step was obtained by asking: What seems the most obvious way the physico-chemical system could behave in an appropriate environment. Appropriate was: with strong periodic fluctuations. This was an essential point in the model. One then invents possible ways, estimates the time required and chooses the fastest way. The intention is: planning hypotheses with unbounded phantasy, followed by strictest possible control.

With this procedure we were lead to the essentials of the genetic apparatus. Each step was considered to be a usual physico-chemical process. No step of low probability had to be postulated. The formation of selforganizing systems and the evolution of a mechanism with the essential features of the genetic apparatus appears to be a quite normal process under appropriate external and internal conditions when allowing enough time.

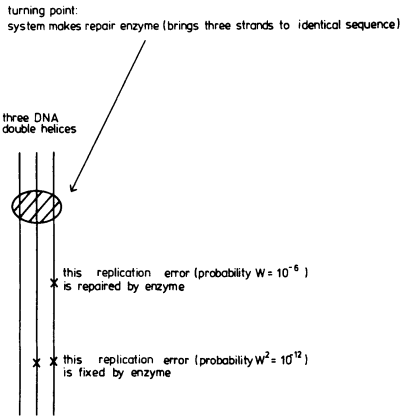


Fig. 29

We may ask: why did this procedure lead us to this end and not to quite different solutions of this phantasy and free-invention-game. This is partially due to the subjectivity of the procedure. What one invents is influenced by what one already knows.

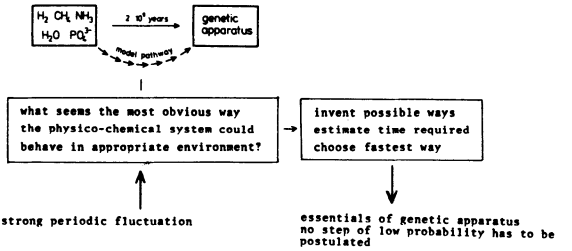


Fig. 30

On the other hand: the essential steps were solutions of traffic regulation problems, and in many such steps one is probably lead to similar solutions on very different model pathways, which are just best solutions of a particular problem. It is not pure chance that the same principle is used in the tape recorder and in the genetic apparatus.

Repeatedly a change between divergent and convergent phases of evolution was found (Table 1). In a divergent phase many different forms have similar chance to survive, and we find a great variety in the population. A gradual refinement of existing mechanisms occurs. At a turning point a mechanism abruptly serves an entirely new purpose. A convergent phase follows which

is strongly selective, and the selection is clearly directed, since those chance mutations survive, which serve the new purpose best. The sequence of changes between these phases given by the model are summarized in Table 2.

Table 1

divergent phase
many different forms have similar chance to survive
great diversity in population
gradual refinement of existing mechanisms
turning point
abruptly mechanism serves an entirely new purpose
convergent phase
strongly selective clearly directed selection pressure

Table 2 Turning points from divergent to convergent phases of evolution

cause	effect
polynucleotide with only d-ribose incorporated	molecules can be replicated
tertiary structured RNA-molecules form aggregates faulty molecules cannot be fitted aggregates are reproduced with low rate of error	systems with specific functions can be selected
polypeptide synthesis on surface of aggregate	selfcompartimentation of systems now possible
complementary pairing of bases at triplet of primitive tRNA's and primitive mRNA	proteins with specific functions can evolve; genetic code
formation of complementary strands of RNA avoided	DNA carrier of genetic information; information for organism merged in one strand
triple storage of information reduces error of base transcription to 10^{-12}	unification to system of large informational capacity

The single event leading to a turning point is very improbable, but its occurrence in a large population appears as a necessity. For instance: It is very improbable by a chance mutation of a system according to Fig. 17b, to obtain an aggregate that makes a coat, but it is very probable that such a system appears among a reasonably large population within a certain time interval. Since a selfmade coat seems to be a necessity for a further evolution of these systems, the turning point should necessarily appear in an appropriate environment.

The important steps being those at the turning points, it must be estimated very carefully in which ways the condition for selection can be fulfilled at each of these points. On the other hand, we have made no effort to give a detailed model describing the evolution during the convergent phases, for instance, a detailed picture of the evolving ribosome. There, the system must necessary evolve in the direction of the selection pressure and it is unimportant for the general view discussed in this paper exactly what solution of the respective traffic regulation problem the system finds. The features at the turning points, that lead to a breakthrough are relatively easy to grasp by quantitative estimates except for the last turning point in Table 2. These features depend on few model assumptions which are feasible from experimental grounds indicating lines of an experimental approach. Thus, a direct experimental attack should be possible and would be essential.

There is a discrepancy between common sense and result of the detailed model calculation. Intuitively one tends to say: such a complicated thing as the genetic apparatus that operates with an extremely clever strategy can never be the result of events happening merely by chance. This discrepancy is due to the large number of steps and the large number of individuals: the course of events cannot be grasped as a whole. It seems necessary to go through all the tedious little steps to find the result familiar.

This more philosophical aspect is a main issue of this paper. Its intention is, by using chains of theoretical models in quantitative terms, to find principle ways of selforganization. It is not attempted to find the historical pathway, but to encourage a search for as many different pathways as possible.

References

[1] Calvin, M.: Chemical Evolution. Oxford 1969.

[2] Rutten, M. G.: The Origin of Life. Amsterdam – London – New York 1971.

[3] Ponnamperuma, C.: Exobiology. Amsterdam – London 1972.

[4] Buvet, R. and Ponnamperuma, C.: Molecular Evolution 1, Chemical Evolution and the Origin of Life. Amsterdam – London 1971.

[5] Crick, F. H. C.: J. Mol. Biol. **38**, 367 (1968).

[6] Orgel, L.: J. Mol. Biol. **38**, 381 (1968).

[7] Monod, J.: Le hasard et la nécessité. Paris 1970; German transl.: Zufall und Notwendigkeit. München 1971.

[8] v. Neumann, J.: Theory of self-reproducing automata. Urbana, Ill. 1966.

[9] Turing, A. M.: Mind **59**, 433 (1950).

[10] Eigen, M.: Die Naturwissenschaften **58**, 465 (1971).

[11] Watson, J. D.: Molecular Biology of the Gene, second edition, New York 1970; Harbers, E.: Nucleinsäuren. Stuttgart 1969; Schreiber, G.: Angew. Chem. **83**, 645 (1971).

[12] Naylor, R.; Gilham, P. H.: Biochemistry **8**, 2722 (1966).

[13] Baldwin, R. L.: Accounts of Chemical Research **4**, 265 (1971); Scheffler, I. E.; Elson, E. L.; Baldwin, R. L.: J. Mol. Biol. **36**, 291 (1968).

[14] Uhlenbeck, O. C.; Baller, J.; Doty, P.: Nature **225**, 508 (1970). Lewis, J. B.; Doty, P.: Nature **225**, 510 (1970).

[15] Calvin, M.: Proc. Roy. Soc. (London) Ser. A **288**, 441 (1965); Schneider-Bernlohr, H.; Lohrmann, R.; Sulston, J.; Orgel, L. E.; Miles, H. T.: J. Mol. Biol. **37**, 151 (1968).

[16] Cramer, F.: Progress in Nucleic acid research **11**, 391 (1971).

[17] This selection process can easily be simulated in a computer program using chance numbers, see: Försterling, H. D.; Kuhn, H.: Physikalische Chemie in Experimenten. Weinheim 1971;
Försterling, H. D.; Kuhn, H.; Tews, K. H.: Angew. Chem. (in print).

[18] The selfassembly of organized aggregates as postulated in this paper is well-known. Viruses and the ribosome have been obtained by mixing the component molecules in solution (see: Butler, P. J. G.; Klug, A.: Nature New Biol. **229**, 47 (1971);
Nomura, M.: Scientific American, December 1969, and in: The Neurosciences. 2nd Study Program, (ed. F. O. Schmitt), New York 1970, pg. 913; Garret, R. A.; Wittmann, H. G.; in: Aspects of Protein Biosynthesis (ed. T. B. Anfinsen), New York 1971.

[19] Paecht-Horowitz, M.; Berger, J.; Katchalsky, A.: Nature **228**, 636 (1970); Paecht-Horowitz, M.: Citat 6, p. 245.

[20] Fox, S. W.: Science **132**, 200 (1960); Naturwissenschaften **56**, 1 (1969).

[21] Laird, C. D.: Chromosoma (Berl.) **32**, 378 (1971).

[22] Ohta, T., Kimura, M.: Nature **233**, 118 (1971); King, J. L.; Jukes, T. H.: Science **164**, 788 (1969).

Figure captions

Fig. 1 Historical pathway of evolution and model pathways, consisting of many individual physico-chemical steps. A model pathway must describe the hypothetical course of events as an uninterrupted chain of causative events. Otherwise it may not be judged whether the pathway represents a meaningful approach

Fig. 2 Genetic apparatus

a) DNA-double strand (C = Cytosine, G = Guanine, T = Thymine, A = Adenine). Replication of a DNA-strand by an enzyme system (replicase). Complementary bases are T and A as well as G and C

b) Formation of messenger-RNA (m-RNA) on the matrix of a DNA-strand by an enzyme system (transcriptase). The base thymine of the messenger-RNA is replaced by Uracil (U)

c) Protein biosynthesis on the ribosome. Transfer-RNAs (t-RNA) for amino acids 1, 2, 3 with anticodon triplet, which is characteristic for the amino acid. Orientation of the triplets on the messenger-RNA, synthesis of the protein, release of t-RNAs without amino acids and messenger-RNA

Fig. 3 Model assumptions concerning the behaviour of RNA

a) RNA is supposed to perform enzyme free replication under model conditions: energy-rich mono- or oligonucleotides are supposed to associate with the open chain conformation in such a way, that complementary bases form pairs and then proceed to condensate

b) RNA-transitions by appropriate changes of environment. Double strands are assumed to change to single stranded open chain conformation, from the latter to a conformation with tertiary structure given a suitable base sequence, and then back to the open chain conformation

c) During intervals, in which forms with tertiary structure appear, weak bonds between exposed bases of these species and the open chain single strands are assumed to be possible

Fig. 4 Starting conditions: periodical fluctuation of conditions in surrounding media: alternation between formation phases (formation of polymers from energy-rich mono- or oligonucleotides) and phases of decay (hydrolysis of polymers)

Fig. 5 Origin of RNA-molecules of uniform chiralty. Random formation of a molecule being sufficiently large and containing just d-ribose. The molecule serves as template for the formation of a double helix. Those molecules containing both d- and l-ribose do not have this property because of lacking cooperativity. The double helix is stabilized by cooperative effects, thus is able to multiply and has a longer life span

Fig. 6 Double helix molecules multiply and displace all the others. The originally divergent phase (formation of polymers with many different sequences of d- and l-ribose) has thus turned into a convergent phase during which d- or l-ribose is incorporated selectively

Fig. 7 Formation of RNA-molecules by random condensation of double helices (drawn with omission of helical screwing of the strands)

a) Formation of hair pin molecules (from top to bottom): identical double strands, double strand on the right turned by 180^{o} and condensated with the other one, strands separated from each other, bases form intramolecular pairs

b) Condensation of hair pin molecules, and possibly double helices (on the right), to give transfer-RNA-like molecules. Example of a transfer-RNA with a tertiary structure as proposed by Cramer

Fig. 8 Behaviour of molecules as shown in Fig. 7b in the model environment. Phases of formation: replication, transition to a tertiary structure; phases of decay: partial hydrolysis of the molecules

Fig. 9 At the end of each decay phase there are, after arriving at a state of saturation, about Z molecules present

Fig. 10 Appearance of a mutant having the multiplication factor r. Number n of generations during which the mutant displaces the normal form

Fig. 11 Appearance of new mutants, after the mutant of Fig. 10 has become prevalent: most of them have no advantage, since it is highly improbable that a species, which has proved to be fit, should be further improved by a random replication error. The most favourable mutant (most resistant to hydrolysis) displaces all other species within n periods

Fig. 12 Conditions for selection

Fig. 13 a) Species of molecules, which happen to match, form aggregates. During formation phases the individual molecules separate and replicate. Afterwards they reassemble to form aggregates: this particular type of aggregate multiplies

b) Error copy does not fit and is replaced by one that has been reproduced correctly. Thus, error free reproduction of aggregate. Condition for selection (Fig. 12) can be fulfilled for aggregates

Fig. 14 Compartimentation: aggregates must be formed fast enough in order to attain the condition for selection: compartimentation avoids diffusion of matching molecules

Fig. 15 Considerations about the nature of selecting aggregate types

Fig. 16 Aggregates of two types of RNA with definite tertiary structure (nucleation molecule and many identical line-up-molecules). This type of aggregate has selection advantages

Fig. 17 Nucleation molecule (with tertiary structure) equipped with an open chain strand without tertiary structure (collector strand). It is a selection advantage for the aggregate: the collector strand facilitates construction of an organized aggregate. The collector strand reduces disturbances due to incorrect entanglement of line-up-molecules, and, thus helps to solve traffic problems arising during formation of aggregates

Fig. 18 Gradual alteration of the aggregate according to Fig. 17. Catalyzing properties, e. g. polymerization of activated amino acids, on the surface of the aggregate

Fig. 19 System with polypeptide coat is independent of given compartment

Fig. 20 –

Fig. 21 Improvement of the device for polypeptide synthesis for the purpose of faster production of the envelopes: linkage between triplets of the t-RNA and the m-RNA

Fig. 22 Complementary base pairing between the middle position of the triplet and the respective place on the m-RNA

Fig. 23 Aminoacid specificity as a side effect of a random alteration of the surface of the t-RNAs

Fig. 24 Code for four aminoacids: replicase is formed by chance, and the fixation of the code results

Fig. 25 System of several protein synthesizing units. The proteins form an increasingly organized team

Fig. 26 Code for additional aminoacids

Fig. 27 DNA becoming important for the genetic apparatus

Fig. 28 System with fully developed genetic apparatus. Time requirement for the instruction of an additional protein, which is not incorporated into the functional network as of yet.
Upper limit of genetic informational capacity of model systems. If this limit is exceeded, too many replication errors per generation would occur

Fig. 29 Rate of replication errors is reduced by triple coding and reparation mechanism, which provides for the same base sequence of the three double strands

Fig. 30 –

Computer Simulation of Helix-Coil Transition

N. Ogita, A. Ueda, T. Matsubara, F. Yonezawa and H. Matsuda

1. Introduction

The investigation of the dynamical properties governing the change of the secondary structure of protein molecules such as the helix-coil transition would play an important rôle in understanding the molecular mechanism of enzyme reactions. It is well known that the secondary structure of a polypeptide undergoes a phase change from a helix-structure to a coil structure depending upon the environmental conditions such as the temperatur and the pH of the solvent. Theoretical and experimental studies of the equilibrium properties of this phase change shows that the essential characteristics of the phase change are well understood on the basis of the Zimm-Bragg model [1] or the Lifson-Roig model [2]. It is therefore natural to assume that the kinetics of the helix-coil transition may also be well described by the same models. Several attempts to construct the kinetic model of helix-coil transition have been made. The aim of the present papar is to give a brief review of such previous attempts and to compare their results with those of a recent computer simulation experiment carried out by our group.

2. Statistical models

Before entering into the details of the kinetics, we give in this section a brief sketch of the statistical models of helix-coil transition so far proposed. A typical polypeptide is shown schematically in Fig. 1. From each carbon atom connected to a radical R, two bonds C–C and C–N branch out. Around these bonds high-polymer molecular segments can execute rotations, so that the molecule as a whole assumes various configurations. Let us number the radicals in a certain order $j = 1, \ldots, N$ and let (ψ_j, ϕ_j) be a set of rotation angle about the axes, C–C and C–N, connected to the j-th radical. When the polypeptide molecule is in the α-helix state, the values of (ψ_j, ϕ_j) are limited to a certain narrow region in phase space. We shall allude to this by saying "t h e j - t h r a d i c a l i s i n t h e h - s t a t e".

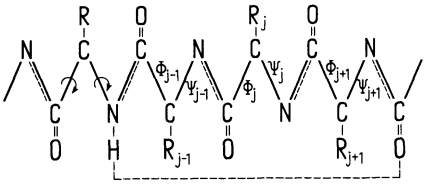


Fig. 1
The structure of polypeptide.
The dotted line represents a hydrogen bond which is formed when the molecule is in the α-helix state.
(ψ, ϕ) represent the internal rotation angle around two rotation axes.

If (ψ_j, ϕ_j) lie outside the above region in phase space, we shall say, "it is in the c-state". When three successive radicals, R_{j-1}, R_j, R_{j+1} are all in the h-state, we assume that a hydrogen bond is formed between N and 0 as shown in Fig. 1. In this case we shall say, "the j-th hydrogen bond is in the 1-state". If there exists at least one c-state among the three radicals (R_{j-1}, R_j, R_{j+1}), we assume that the hydrogen bond is broken and shall say, "the j-th hydrogen bond is in the 0-state" Zimm and Bragg represented each configuration of the molecule by a sequence of numbers, 0 or 1, focussing their attention mainly on the state of the hydrogen bonds (Z–B model) [1]. On the other hand Lifson and Roig chose to represent the configurations of the molecule by a sequence of the letter, h or c, focussing on the states of the radicals (L–R model) [2].

The statistical weight of each configuration is assigned according to the following rules. In the L–R model, each radical in the c-state is given a statistical weight factor u. Each radical in the h-state is assigned a factor w when it is surrounded by two h-state radicals, and a factor v otherwise. A configuration which is expressed by a sequence of h and c has a thermodynamic probability proportional to the product of the factors assigned to each radical. The first and second rows of Table 1 contain a sample of configuration and its corresponding statistical weight according to the L–R model.

Table 1

(i)	A configuration of the L–R model	c c h h h h c h h h c c h c h h c c
(ii)	Statistical weight of (i)	u u v w w v u v w v u u v u v v u u
(iii)	The configuration of the Z–B model corresponding to (i)	0 0 0 1 1 0 0 0 1 0 0 0 0 0 0 0 0 0
(iv)	Statistical weight of (iii)	1 $1\sqrt{\sigma}$ s s $\sqrt{\sigma}$ $1\sqrt{\sigma}$ s $\sqrt{\sigma}$ 1 1 1 1 1 1 1 1
(v)	A configuration of the L–R model corresponding to (iii)	c c h h h h c h h h c c c c c c c c

It is readily verified that a given configuration in the L–R model uniquely determines a configuration in the Z–B model. An example is given in the third row (iii) of the Table 1 which is uniquely derived from the configuration given in the first row (i). The converse of this correspondence is however not true: the configuration in the fifth row (v) will also yield the configuration given in (iii). Thus a given molecular configuration in the Z–B model corresponds to more than one state in the L–R model. In practice, however, since $v/u \ll 1$ under ordinary condition, configuration (v) has much larger statistical weight than configuration (i) and hence is more important. Therefore in order that both the L–R and Z–B models give the same statistical weight to a configuration, it would be sufficient to choose the statistical weight factor in the Z–B model in such a way that both configurations, (iii) and (v), have the same thermodynamic probability.

Thus the rules for assigning the statistical weights in the Z–B model are: (1) a factor of unity is assigned to a 0-state which is not a neighbour to any 1-state, (2) a factor, $s = w/u$. to every 1-state and (3) a factor, $\sqrt{\sigma} = v/u$, to a 0-state next to a 1-state.

There is no state in the L–R model which corresponds to a state in the Z–B model in which three 0-states appear in succession. Such a configuration cannot be realized from the structural point of view. However, as is easily supposed, such a state is not important so long as $\sigma \ll 1$. As a matter of fact, it turns out that the equilibrium value of any thermodynamical quantity such as the helix content, θ, is a function of N, s and σ, and becomes independent of the model when $\sigma \ll 1$. In this limit σ is determined solely from the molecular structure, whereas the change of environment is reflected in the value of s. It was demonstrated that for poly (γ-benzyl-L-glutamate), the value $\sqrt{\sigma} = 0.0141$ reproduces all the experimental results reasonably well. Furthermore when $N \gg 1/\sqrt{\sigma}$, the N-dependence of thermodynamical quantities disappears, and it is known that a sharp transition from the helix to the coil state takes place when s passes the critical region which is about unity with the width σ.

3. Time development of helix-coil transition [3]

We consider an ensemble of many polypeptides, each of which consists of N segments. The possible configurations of a molecule are enumerated in a definite order, $i = 1,2,\ldots,2^N$. Let $f_1 = f_i(t)$ be the fraction of molecules in the i-th state and assume that time variation is governed by a set of equations

$$\frac{df_i}{dt} = \sum_k \alpha_{ik} f_k \qquad (i, k = 1, \ldots, 2^N) \tag{3.1}$$

where α_{ik} is the transition probability per unit time from the k-th state to the i-th state. α_{ik}'s are functions of the environmental parameter, s. Due to the normalization, $\sum_k f_i = 1$, there exist in principle, $2^N - 1$ different relaxation time τ_s ($s = 1,2,\ldots,2^N - 1$) derived from the equations (3.1). We define, as one of thermodynamical quantities, the helix content such that

$$\theta = \frac{1}{N} \sum_i n_i f_i \tag{3.2}$$

where n_i is the number of hydrogen bonds included in the i-th state. Then the time variation of θ is generally given by

$$\theta(t) - \theta(\infty) = \sum_s \beta_s e^{-t/\tau_s}. \tag{3.3}$$

When $N \gg 1$, it is not easy to evaluate all the relaxation times (τ_s), but the mean value of the inverse of τ_s and the mean square of the inverse of τ_s may be determined from relations

$$\frac{1}{\tau^*} = \frac{\overline{1}}{\tau} = \frac{\sum \beta_s/\tau_s}{\sum \beta_s} = \frac{1}{\Delta\theta}\left(\frac{d\theta}{dt}\right)_{t=0} \tag{3.4}$$

and

$$\frac{\overline{1}}{\tau^2} = \frac{\sum \beta_s/\tau_s^2}{\sum \beta_s} = -\frac{1}{\Delta\theta}\left(\frac{d^2\theta}{dt^2}\right)_{t=0} \qquad (\Delta\theta = \theta(\infty) - \theta(0)). \tag{3.5}$$

Now if we assume that the ensemble, which at $t = 0$ is in equilibrium state under a given environment, is suddenly put into another environment, it moves to a new equilibrium state. In

this circumstance, the right hand sides of (3.4) and (3.5) are easily evaluated from the knowledge of the initial distribution (f_i (0)). Thus we can make a rough estimate of the order of magnitude of the relaxation time. In order to do this, we need a model to give explicit forms for α_{ik}.

On the basis of the Z–B model, Schwarz proposed [3] a dynamic model in which the configuration of the polypeptide at each instant is represented by a sequence composed of "0" or "1", and the elementary process for the time variation of the configuration is chosen to be a single jump that makes one of hydrogen bonds change from "0" to "1" or vice versa. The rate constants of the elementary processes depend on the local configuration at the point the single jump takes place. In Table 2, the second column contains the important elementary processes considered by Schwarz together with their rate constants. Given a rate constant for a forward reaction, that of the reverse process is determined uniquely by the principle of detailed balance.

Table 2

		(i) Z–B model	(ii) L–R model
(I)	Development process of helix or coil part	$001 \underset{k_B}{\overset{k_F}{\rightleftharpoons}} 011$	c c h h h $\xrightarrow{k_f}$ c h h h h
(II)	Nucleation process of helix part	$000 \underset{\gamma_H k_B}{\overset{\sigma\gamma_H k_F}{\rightleftharpoons}} 010$	c c c c c $\longrightarrow$ c h h h c $\sim k_f^3/k_b^2\,(0) \equiv \sigma k_f$ (c c c c c $\longrightarrow$ c c h c c $\longrightarrow$ c h h c c $\longrightarrow$ c h h h c)
(III)	Nucleation process of coil part	$111 \underset{\gamma_C k_F}{\overset{\sigma\gamma_C k_B}{\rightleftharpoons}} 101$	h h h h h — h h c h h $k_b\,(3) \equiv k_b\,(1)/s^2$ (1 1 1 1 1 $\longrightarrow$ 1 1 0 1 1 $\longrightarrow$ 1 0 0 1 1 $\longrightarrow$ 1 0 0 0 1) $\sim \sigma\gamma_C k_B/s^2$

It is also possible to construct a dynamic model on the basis of the L–R model. In this model the elementary process is a single jump which makes one of radicals change its state from "h" to "c" or vice versa. In the third column of the Table 2, the processes corresponding to those given in the second column on the basis of Z–B model are indicated. If we denote by k_f the rate constant of a single jump c → h in which no energy is required to break the hydrogen bond, then the reverse process h → c has different rate constant k_b depending on the number of hydrogen bonds broken by this process. By making use of the principle of detailed balance, the rate constant of a process in which n hydrogen bonds are broken is determined as

$$k_b\,(n) = k_f\, s^{-n\frac{n-1}{\sigma^2}} \qquad (n = 0, 1, 2, 3). \tag{3.6}$$

In the Table 2, the process indicated in (II ii) is not an elementary process but a composite one as explicitly shown in the parenthesis. However, when $\sigma \ll 1$, it turns out that $k_f \ll k_b$ (0) and hence the intermediate states which appear in the course of the process may be in equilibrium

with each other so that the rate constant of the total composite process is approximately given by k_f^3/k_b^2 (0). Similarly the process (III ii) is a composite process composed of the elementary processes given in the parenthesis, and has an approximate rate constant $(\sigma\gamma_C\, k_B)\,(k_B/k_F)^2 = \sigma\gamma_C\, k_B/s^2$. It is observed from the Table 2 that upon taking $k_F = k_f$, $k_B = k_b$ (1) and $\gamma_B = \gamma_C = 1$, the two dynamic models become approximately equivalent.

Poland and Scheraga have proposed a realistic model based on the L–R model by introducing more elementary processes. It can be shown, however, that as long as $\sigma \ll 1$, all the proposed models become nearly equivalent and give the same results. A simple manipulation reveals that near the critical region, $s \cong 1 + 0(\sqrt{\sigma})$, the averaged relaxation time τ^* defined in (3.3) is given by

$$\frac{1}{\tau^*} = k_F\,[(s-1)^2 + 4\sigma] \tag{3.7}$$

independent of the model. The maximum value of τ^* is therefore

$$\tau^*_{max} = \frac{1}{4\sigma\, k_F}\,. \tag{3.8}$$

If the rate constant for the development process of the helix part is governed by the rate of the hydrogen bond generation which in turn is controlled by a diffusion process, the value of k_F is estimated as $10^{10} \sim 10^{11}$ sec, which gives $\tau^*_{max} \cong 10^{-7}$ sec.

4. Computer simulation

4. 1. Object of simulation. The relaxation time for the structure change of a protein molecule has been estimated experimentally by various observations. It is inferred directly from the experiments of T-jump method [5], dielectric relaxation [6] and ultrasonic absorption [7], and indirectly from NMR [8] and tritium-hydrogen exchange in aqueous solution [9]. From the macroscopic observation of the helix-coil transition such as the direct experiments mentioned above, it is always concluded that the relevant relaxation time is shorter than 10^{-5} sec and in rather good agreement with the theoretical value of the initial relaxation time τ^* cited in the last part of the previous section.

There is however a question as to how the initial relaxation time τ^* can be related to the observed relaxation time.

There seems to be no obvious relation between them. In order to have a definite answer we need more elaborate theoretical calculation of the dynamical behaviour. Computer simulation would be a useful tool here, as the problem seems eminently suited for computer calculation. It is well known that the biological functions of proteins, such as enzyme activities, require suitable flexibility, and this molecular flexibility or motility of proteins is believed to be due to thermal fluctuation around the averaged structure. Computer simulation would give us in a direct way some knowledge about the thermal fluctuations of a model of a protein molecule. Consequently we conducted a computer simulation of the helix-coil transition of a protein molecule based on the L–R model. A similar computer simulation based on the Schwarz model was carried out by Craig and Crothers [10].

4. 2. Method of simulation and results. Our computer simulation was performed through the following steps (see the flow chart given in Fig. 2):

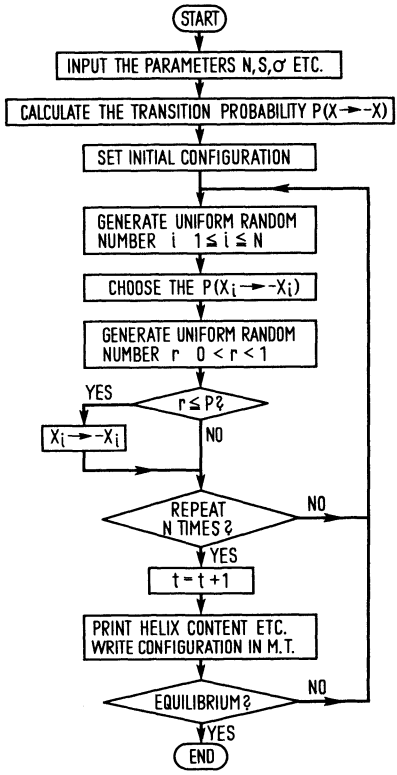


Fig. 2
Flow chart for the simulation.
The two states of each segment is described by a variable x = ± 1.

(1) Give the necessary parameters such as N, s, σ as input data.
(2) Calculate the transition probabilities in advance for all the possible combinations of the local configurations, and store them.
(3) Set an initial configuration in the form of bits information.
(4) Select a segment by a random number.
(5) Observe the state of the segment selected and of its two neighbours, and find the transition probability corresponding to the observed configuration.
(6) Compare this probability with another random number between 0 and 1. If the probability is larger than the random number, change the state of the segment.
(7) Repeat the processes (4) ~ (6) N times, and shift a unit time.
(8) Out put the helix content, θ, and the molecular configuration.
(9) Repeat the whole processes until the system attains an equilibrium state.

The initial condition adopted is that all the segments are in the h-state at t = 0.

Fig. 3 shows the time variation of a quantity

$$\log | [\theta (t) - \theta_f]/[1 - \theta_f] |$$

for three different environments s = 0.95, 1.0 and 1.05, where θ_f is the value of θ for the final equilibrium state. The values of the other parameters used in this simulation are: $\sigma = 0.01$, $k_f = 0.03$, N = 200.

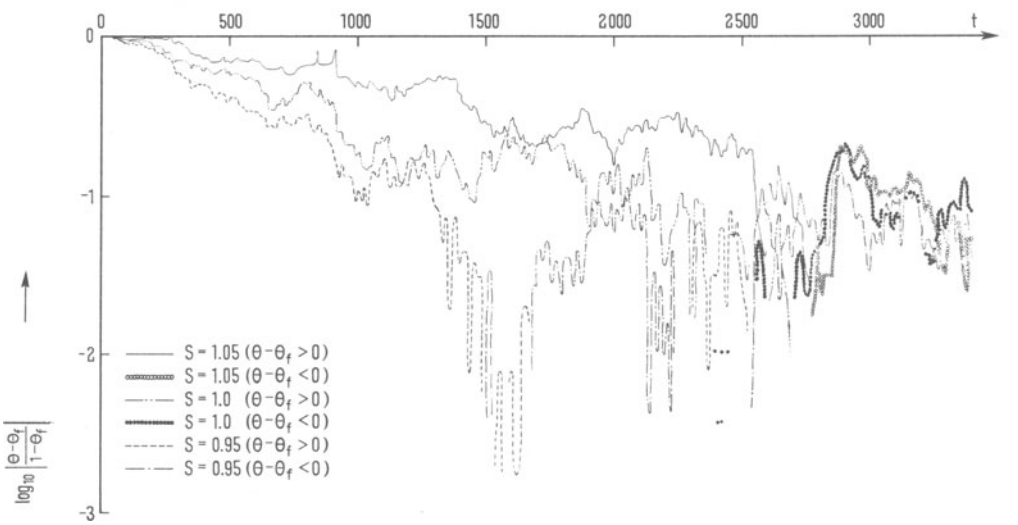


Fig. 3 Computer simulation based on the L–R model. Helix content versus time for three different environments: s = 0.95, 1.0 and 1.05. The values of the other parameters are N = 100, $k_f = 0.03$, $\sigma = 0.01$.

Although there exist large fluctuations due to the finiteness of the number of segments, it is seen from these curves that the time variation of θ (t) $- \theta_f$ may, on the average, be described by a single exponential function of time. We have estimated the relaxation times from these curves and summarized them in the second column of the Table 3. Theoretical values of τ^* calculated from (3.7) are also indicated in the third column. It is noted that τ and τ^* agree within a factor of two.

Table 3

s	τ	τ^*
0.95	~500	785
1.00	~650	835
1.05	~1250	785

Fig. 4 is an example of the time variation of molecular configuration where only the helix parts, that is the parts of segment in h-state with hydrogen bond formation, are shown by black lines. We can clearly observe the presence of thermal fluctuation in the structure.

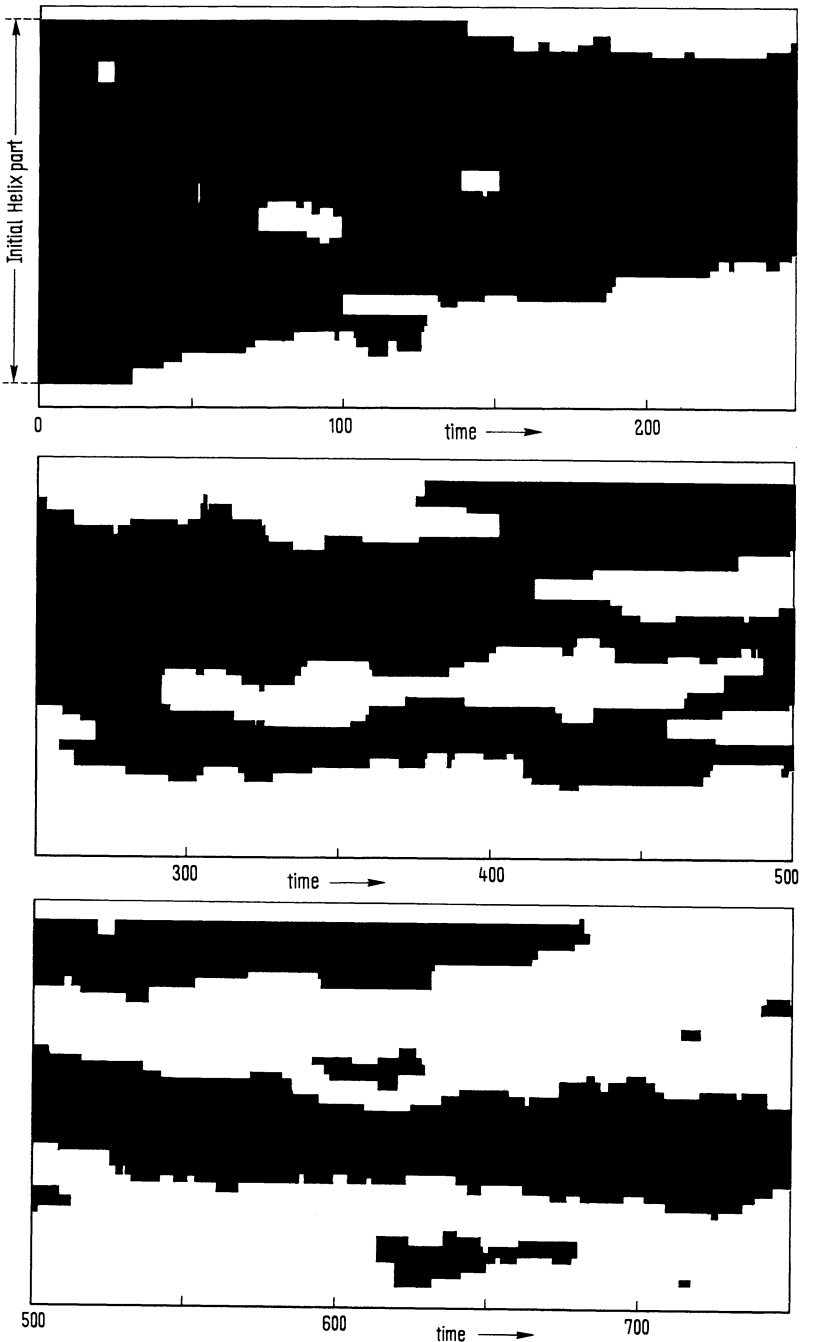


Fig. 4 Time variation of the configuration of a protein molecule. Only the helix part is shown by black lines.

5. Discussion

The results reported in this paper are as yet preliminary and many things should be done in future. But computer simulation seems to be a powerful tool for studying dynamical properties of biopolymers. There is no difficulty in principle to extend the present methods to more elaborate models with more realistic interactions. So far we have discussed only the case of homopolymer in which all the radicals are the same kind. From the practical point of view, however, it is more important to study the dynamical properties of copolymers which contain many kinds of radical. According to the tritium-hydrogen exchange experiment [11], there is some evidence that copolymers reveal complicated behaviour in their molecular flexibility differing from that of homopolymers. Since the theoretical study of copolymers is at present extremely difficult, it will be interesting to apply the computer simulation method to the case of copolymers.

References

[1] Zimm, B. H. and Bragg, J. K.: J. Chem. Phys. **28**, 1246 (1958); **31**, 526 (1959).
[2] Lifson, S. and Roig, A.: J. Chem. Phys. **34**, 1963 (1961).
[3] Schwarz, G.: J. Mol. Biol. **11**, 64 (1965).
[4] Poland, D. and Scheraga, H. A.: J. Chem. Phys. **45**, 2071 (1966).
[5] Lumry, R., Legare, R. and Miller, W. G.: Biopolymers **2**, 489 (1964).
[6] Schwarz, G. and Seelig, J.: Biopolymers **6**, 1263 (1968).
[7] Zana, Z., Candau, S. and Cerf, R.: J. Chim. Phys. **60**, 869 (1963). Burk, J. J., Hammes, G. G. and Lewis, T. B.: J. Chem. Phys. **42**, 3520 (1965). Nakajima, H. and Wada, Y.: Rep. Progr. Polym. Phys. Japan, **11**, 263 (1968).
[8] Bradbury, E. M., Robinson, C. C. and Rattle, H. W. E.: Nature **216**, 862 (1967). Bradbury, E. M., Robinson, C. C., Goldman, H. and Rattle, H. W. E.: Nature **217**, 812 (1968). Ullman, R.: Biopolymers, **9**, 471 (1970).
[9] Ikegami, A., Yamamoto, S. and Osawa, F.: Biopolymers **3**, 555 (1965).
[10] Craig, M. E. and Crothers, D. M.: Biopolymers **6**, 385 (1968).
[11] Ikegami, A. and Kono, N.: J. Mol. Biol. **29**, 251 (1967).

Mechanisms of Pattern Recognition by the Visual System of Insects[1])

Werner Reichardt

During the past years investigations have been undertaken in our laboratory with the aim to determine and to analyze spontaneous preferences of flying houseflies (Musca domestica) for elementary patterns in their optical environment; Reichardt and Wenking (1969), Reichardt (1970), Reichardt (1971).

1. Experimental Conditions

During the experiments a test-fly is suspended by a torque compensating device and mounted on the axis of a cylindrical panorama. The output voltage of the compensator, which is proportional to the instant flight torque signal, generated by the test-fly during fixed flight, is coupled to a fast servomotor whose shaft carries a ringpotentiometer and a panorama. Concerning the coupling process it was taken into account that the moment of inertia Θ of the fly around its vertical axis and the friction k between air and the fly's wings is strongly reduced by the experimental fixed flight conditions in still air. The dynamics under free flight conditions, including the parameters Θ and k, are approximately simulated by an electronic device, positioned between the compensator output and the servomotor input. The ringpotentiometer, mechanically connected to the servomotor shaft, serves as an indicator of the angular position ψ of the cylindrical panorama during the tests. The panorama is homogeneously illuminated and carries on its inner surface the individual test patterns. Under these operational conditions a torque signal $\pm$ D, generated by a test-fly, results in an angular velocity $\mp\, d\psi/dt$ of the panorama. A strongly simplified version of the experimental set up is presented in Fig. 1.

Due to the cylinder geometry of the fly's panorama, we are simulating only those retinal displacements of pattern images which result from a rotation of the animal in free flight. This limiting condition is the better fulfilled in free flight the more distant the optical environment is positioned from the fly, since the amount of every retinal displacements, due to translatory relative motions, is inversely proportional to that distance.

---

[1]) The account presented here is a very short version of two articles, one by Reichardt (1972) and one by Poggio and Reichardt (1972) to be published in „Die Naturwissenschaften" and in „Kybernetik".

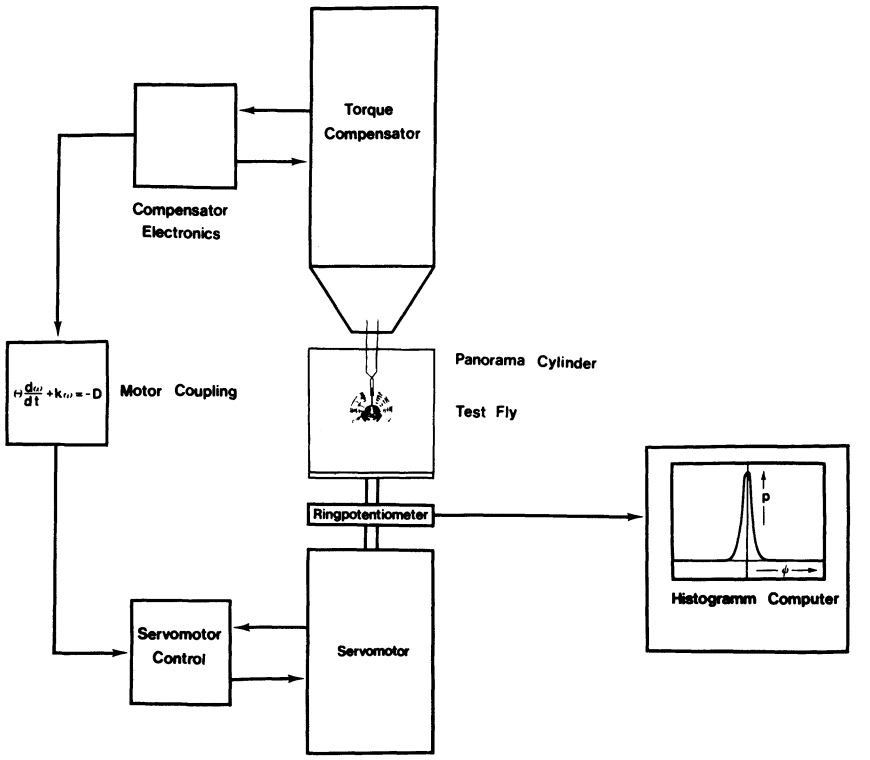


Fig. 1
Principle components of experimental set up, used in the pattern recognition tests. A test-fly is suspended by a torque compensator and mounted on the axis of a cylindrical panorama, which carries on its inner surface the individual test patterns. The panorama is homogeneously illuminated by stationary light. Average brightness of the panorama amounted throughout the tests to about 3000 Apostilbs (1 Apostilb = 10^{-4} Lambert). The output voltage D of the compensator is proportional to the instant flight torque signal, generated by the test-fly around its vertical axis. A fast servo-motor drive system, whose shaft carries a ringpotentiometer and the panorama, receives its input voltage ω from the motor coupling unit, which consists of an operational amplifier, performing an input-output relation in accordance with the first order differential equation $\Theta \frac{d}{dt} \omega + k\omega = -D$. The coupling between the test-fly and the panorama is determined by the parameter Θ/k, where $\Theta \approx 1.5 \cdot 10^{-3}$ gr $\cdot$ cm^2 is the fly's moment of inertia around its vertical axis and k a coefficient simulating friction. The ringpotentiometer voltage is fed into a 400 channel histogram on-line computer. The computer generates probability of positioning distributions for different testpatterns during the experiments.

2. Results

When the inner side of the panorama carries a vertically oriented black stripe and the fly's torque signal is coupled to the servomotor drive system, the panorama is rotated until the stripe has reached the average position $\psi = 0$ which is defined by the fly's direction of flight. A typical histogram, recorded during the stationary phase of fixation, is given in Fig. 2. This

specific fixation behaviour is observed also with individual grey stripes in a large contrast range if the average brightness of the panorama is kept at a high level. Conversely this also holds for the case of contrast inversion, namely when the stripe is brighter than its surroundings. There is experimental evidence that for equal average pattern brightnesses, positive and negative stripe contrast belongs to a class of fixation invariance.

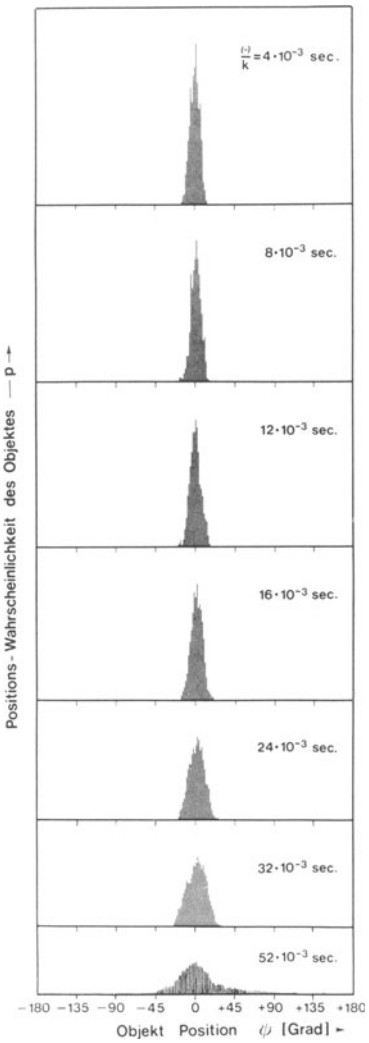


Fig. 2
Fixation of a 5 degree, vertically oriented, black stripe (object). The probability of object position p is plotted as a function of the position ψ of the object. $\psi = 0$ designates the direction of flight of the fixed flying test-fly. Parameter of the test is Θ/k, which determines the coupling-in conditions. For $\Theta/k = 4 \cdot 10^{-3}$ sec, a torque signal of 1 dyne · cm, generated by a test-fly, results in an angular rotation of 152.7 degrees per second of the panorama. The number of statistically independent data per histogram amounts to at least 240. With increasing coupling, the width of the distributions increase. This observation is based on the fact, that the fly's ability to perceive motion, declines for very high relative speeds between the object and the fly's compound eyes. Throughout the test, the fly's head was mechanically fixed onto its thorax.

If the pattern consists of two identical, vertically oriented, stripes separated by an angle $\Delta\psi$, the test-flies fixate an imaginary center line between the two stripes if $\Delta\psi$ amounts to less than 40 degrees. Also under these conditions there is o n e equilibrium position for the two stripe configuration. The test-flies behave as if the two stripes form a single pattern. The behaviour of the flies is quite different for separation angles $\Delta\psi > 40$ degrees. Then they are

able to fixate t w o imaginary lines, positioned in the neighborhood of one or the other stripe. These lines are fixated with equal probability. If $\Delta\psi^*$ describes the angular separation between the two fixation positions, then $\Delta\psi^*$ is always smaller than $\Delta\psi$, except for the case that $\Delta\psi$ amounts to 180 degrees where $\Delta\psi^*$ equals $\Delta\psi$. Under the condition $\Delta\psi > 40$ degrees, there are t w o equilibrium states for the configuration; that is to say the test-flies behave as if the two stripe pattern has broken into t w o parts. A representative set of two stripe experiments is given in Fig. 3.

An interesting finding is that only the lower parts of the compound eyes – below the equatorial line – contribute to the fixation of objects or pattern, whereas the upper parts – above the equatorial line – do not; Reichardt (1972).

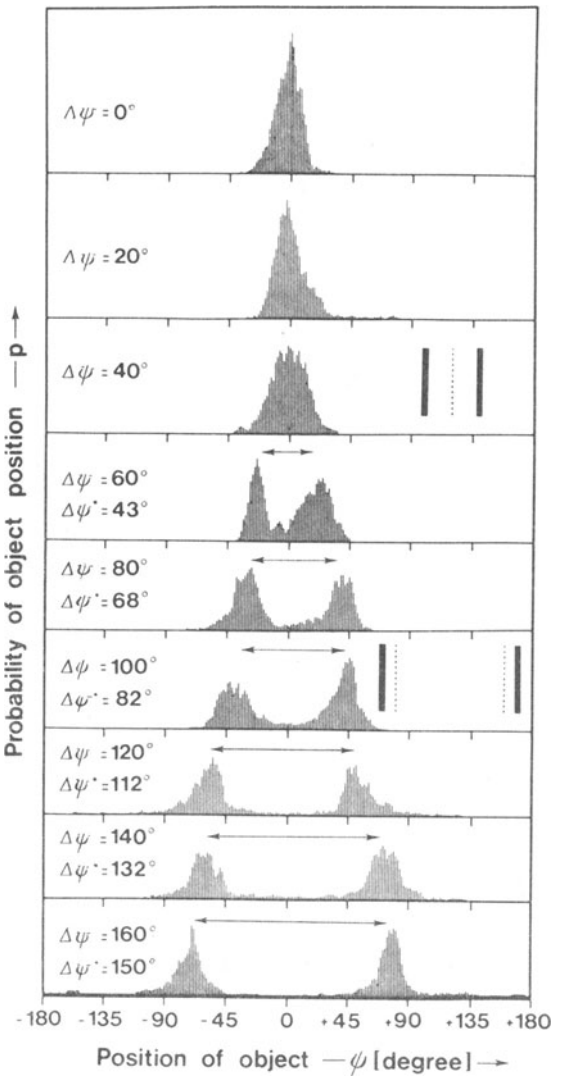


Fig. 3
Fixation of a two stripe pattern. The two identical black, vertically oriented, stripes have an angular width of 5 degrees. Stripe separation angle (from center-line to center-line) is designated by $\Delta\psi$. The probability of pattern (object) position p is plotted versus the position ψ (symmetry line between the two stripes) of the pattern (object). Coupling throughout the experiment amounted to $\Theta/k = 12 \cdot 10^{-3}$ sec. For stripe separation angles $\Delta\psi \leqslant 40$ degrees, the test-flies generate one maximum distributions, centered at about $\psi = 0$, whereas for separation angles $\Delta\psi > 40$ degrees, two maxima distributions are observed, whose centers of gravity separations are given by $\Delta\psi^*$. $\Delta\psi^*$ is always smaller than $\Delta\psi$, except for the case $\Delta\psi = 180$ degree, where $\Delta\psi^* = \Delta\psi$. This case is not included in the figure. The number of independent data per distribution amounts to at least 240. Further explanations are given in the text.

In this connection another striking observation has been made: If a small stripe segment is positioned below the equatorial line, its fixation is improved when the upper parts of the compound eyes are confronted with a 360 degree "pattern" of visual noise. The quality of fixation however is strongly reduced when the visual noise is placed below the equatorial line.

If, on the other hand, the visual noise fills the upper and the lower parts of the compound eyes, fixation is about what it was when only the stripe segment is presented. A typical set of experiments, describing fixation of a stripe segment without and in the presence of visual noise is given in Fig. 4.

Except for the latter findings the results described so far are based on, and can be described in terms of the following elementary observations: a) The test-flies generate torque signals,

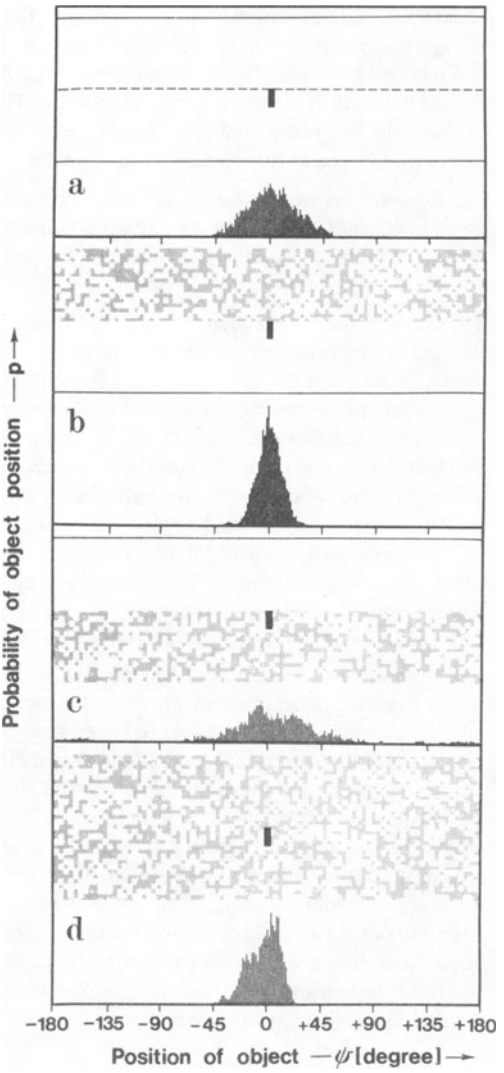


Fig. 4
Fixation of a small object (vertically oriented, black stripe segment) without and in the presence of visual contrast noise. Noise contrast amounted to about 50 %. Coupling throughout the experiment was $\Theta/k = 8 \cdot 10^{-3}$ sec.

a) Upper part: panorama; an object is positioned below the equatorial line (dotted) of the fly's compound eyes. Lower part: histogram of object fixation.

b) Upper part: panorama; the upper region of the panorama contains visual noise, whereas in the lower region only the object is present. Lower part: histogram of object fixation. Under these conditions the quality of the fixation, expressed by the half-width of the distribution, has improved, compared to a).

c) Upper part: panorama; the visual noise has been moved from the upper region into the lower region.
Lower part: histogram of object fixation. The quality of object fixation has strongly decreased.

d) Upper part: panorama; the visual noise covers both regions.
Lower part: histogram of object fixation. The quality of object fixation has improved under these conditions and is again comparable to case a).

The number of independent data per histogram mounts to at least 120.

irrespective of their optical environment, if the environment does not move with respect to the fly's compound eyes. The histogram of this torque signal is nearly gaussian. b) If the object or pattern exposed to a test-fly is stationary but illuminated with flickering light, no significant changes, either in the center of gravity or in the shape of the torque histogram is observed. From a) and b) it follows that the fixation of objects or patterns must be due to relative movements of these configurations with regard to the fly's compound eyes. c) This can be shown when a single stripe panorama is moved clock- and counter-clockwise around a position ψ with a small swing amplitude of $\pm \epsilon$ degrees. Under these experimental conditions the test-flies try to turn towards the direction of the position ψ, indicating that the progressing stripe motion (from front to back, with regard to one of the compound eyes) elicits a stronger induced torque signal than the regressing stripe motion (from back to front).

Due to the findings summarized in a) and c), it follows that the cause of the stripe transport is the symmetric torque fluctuation and the asymmetric induced torque response to progressing and regressing motion. The stripe reaches an equilibrium state at the fixation position $\psi = 0$, around which it fluctuates in a random motion. This position is defined by the symmetry line between the two compound eyes. This is the point where, when the stripe passes across it, regressive movement turns into a progressive one.

The resultant average torque signal, responsible for the stripe transport into the position of fixation depends strongly on the angular coordinate ψ. Its amount increases about linearly with increasing $|\psi|$, reaching extrema for $\psi = \pm 20$ degrees and falling off to zero at $\psi = \pm 180$ degrees.

The shape of the ψ-dependence of the fly's resultant average torque signal is in accordance with the findings reported in connection with the two-stripe experiments if one takes into consideration, that the two eyes of the fly with their associated parts of the visual system operate independently with respect to motion perception. For stripe separations $\Delta\psi < 40$ degrees, the ψ-dependence of the resultant average torque leads towards a stabile fixation of the center line between the two stripes, since the two stripes are positioned between the two extrema at $\psi = \pm 20$ degrees. Whereas for separation angles $\Delta\psi > 40$ degrees, this position is a labile one, as the two stripes are positioned beyond the extrema. Under the latter condition two stable positions exist, when one of the two stripes is positioned between $\psi = \pm 20$ degrees, and the other beyond the extrema of the resultant average torque characteristics. The breaking up of a pattern into two parts is therefore a consequence of a stability criterium, caused by the ψ-dependence of the fly's resulting average torque signal.

So far we have not discussed yet the observations, described in connection with Fig. 4. The fact that a stripe segment is fixated when it is positioned below the equatorial line of the two compound eyes, mainly rests – as has been stated before – on the observation that in this region the induced torque response is an asymmetric one. Fixation is not observed when the segment is positioned above the equatorial line, in accordance with the finding that in this region the induced torque is a symmetric one. This is crucial for the explanation of the observation that the presence of visual noise in the upper region improves fixation of a stripe segment, positioned in the lower region. Since the number of light receptors of the compound eyes receiving contrast signals from the moving stripe segment is quite small, the average amplitude of the symmetric torque fluctuation, generated by the fly under open loop conditions, does not decrease strongly under closed loop conditions, when the segment is fixated. Consequently the standard deviation of the fixation distribution, plotted in Fig. 4a, is relatively large. If however visual noise is positioned in the upper region of the panorama, its

movement generates contrast signals which are received by all those receptors located above the equatorial line of the compound eyes. The overall amount of these signals, contributing to symmetric induced torque responses, or equivalently the increase of the loop amplification by the fly's visual system, strongly reduces the average torque fluctuation amplitude, resulting in an improvement of the fixation of the stripe segment. The effect is demonstrated in Fig. 4b. Quite different is the situation when the visual noise "pattern" is positioned below the equatorial line. The principle causes for an improvement in fixation remain the same, however due to the asymmetry of the induced torque response triggered in the lower eye regions, each elementary contrast value of the visual noise contributes to its own fixation. As these values are scattered around 360 degrees of the panorama, they impose a strong statistical perturbation upon the ψ-dependent average torque characteristic which is responsible for the fixation of the stripe segment and its quality. The perturbation effect, due to the presence of visual noise in the lower region, is overriding the amplification effect, which results in a reduction of the quality of fixation, as can be seen in Fig. 4c. In Fig. 4d, finally, we have presented the case when visual noise covers both regions, above and below the equatorial line of the compound eyes. Here it seems that the improvement in the segment's fixation is slightly overriding the opposite effect, which is due to the fact that contrast elements, present in the upper region, contribute only to an improvement of the fixation, as has been stated before.

The experimental results of our investigations, carried out so far, suggest that there exists a many to one correspondence between patterns and ψ-dependent resultant average torque characteristics which determine the fixation behavior of the flies and their abilities to recognize and to discriminate patterns.

References

Poggio, T. and Reichardt, W.: Spontaneous Fixation of Elementary Patterns without and in the Presence of Contrast Noise by the Visual System of the Fly Musca domestica. Kybernetik (1972, in preparation).

Reichardt, W.: The Insect Eye as a Model for Analysis of Uptake, Transduction, and Processing of Optical Data in the Nervous System. The Neuroses, Second Study Program (F. O. Schmitt Editor). New York 1970.

Reichardt, W.: Visual Detection and Fixation of Objects by Fixed Flying Flies. Pattern Recognition in Biological and Technical Systems. Berlin–Heidelberg–New York 1971.

Reichardt, W.: Verhaltens-Analyse der Fixierung elementarer Muster durch das visuelle System der Fliege Musca domestica. Die Naturwissenschaften (1972, in preparation).

Reichardt, W. and Wenking, H.: Optical Detection and Fixation of Objects by Fixed Flying Flies. Die Naturwissenschaften **56**, 424–425 (1969).

A Cooperative Model of Stereoscopic Depth Perception

Bela Julesz

1. Introduction

It is usually assumed that Ising-models of physics represent some of the simplest examples of cooperative phenomena. The interesting properties of cooperative phenomena, such as order-disorder transitions, multiple states, hysteresis, etc., can be deduced from Ising models, but only through complex computations. The mathematical solution of the simplest Ising-models required the concentrated efforts of some of the best physicists for decades, and it seems that this complexity prevents an intuitive grasp of these phenomena.

In this paper I will describe a model that is conceptually much simpler than any Ising-model, yet exhibits all the properties of cooperative phenomena from state transitions to hysteresis. Because of its simplicity it is possible to gain a deep insight into the structure of this model and understand its workings on an intuitive level. This model is the result of twelve years of research in stereoscopic depth perception (Julesz, 1971) and thus describes some structural properties of the visual cortex. In spite of rapid progress in the neurophysiology of stereoscopic depth perception (also called stereopsis) most of the results obtained by single microelectrode recordings are local, while the psychological findings are rather global. Therefore my model is not a neurophysiological model, but a classical physical model, using magnetic dipoles and mechanical springs as building blocks. It is remarkable how such a simple model can describe a gamut of observed perceptual phenomena. It might be amusing if some phenomena were found in physics that could also be described by this model.

2. Some basic phenomena of stereopsis

In this section I try to review (and often demonstrate) the main phenomena of stereopsis that a model must emulate. Stereopsis is based on binocular disparity, the geometrical fact that three-dimensional objects cast slightly shifted corresponding projections (in the horizontal direction) on the left and right retinae. That this parallax shift is adequate to perceive stereoscopic depth was shown by Wheatstone (1838) through his invention of the stereoscope. That the horizontal registration of corresponding points in the left and right retinal projections is not the result of convergence movement of the eyes alone was shown by Dove (1841). While large disparities are eliminated by convergence-divergence movements, a few minutes of arc binocular disparity between the left and right images is cortically registered. Dove presented stereoscopic image pairs in a brief flash, using as a light source a brief electric spark. The

resulting afterimages persisted on the retinae for some time, though the duration of the spark was too brief for any convergence movement to take place. These "retinally stabilized" stereoscopic afterimages yielded a vivid depth percept, thus demonstrating that exact registration is not the result of eye movements, but of the central nervous system (CNS).

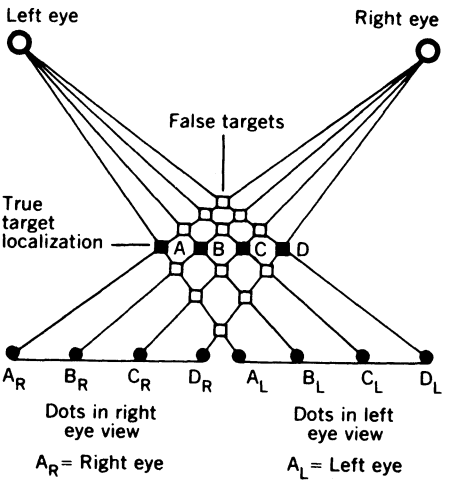


Fig. 1
Ambiguities of stereoscopic depth localization (From Julesz, B. 1968, Experiments in perception Psychol. Today, 2, 16–32)

While these experiments showed that binocular disparity is the basis of stereopsis, the other fundamental problem of how this binocular disparity information is extracted was not raised. If the corresponding points in the left and right retinal projections are individually labeled, (e. g. corresponding point pairs have the same brightness, size or color) there is no problem. In this case binocular disparity is simply the horizontal coordinate difference between the corresponding left and right dots. But what happens if there are many similar dots that lie on the same horizontal line in the left and right views? As Fig. 1 shows, even 4 points lead to 16 different binocular disparity values (i. e. 16 possible localizations) out of which only 4 are right, while the remaining 12 give rise to phantom targets. When the number of identical dots increases the ratio of correctly localized dots (correct disparities) to phantom dots (incorrect disparities) rapidly diminishes to zero. Psychologists never explicitly asked this question, and somehow assumed that the ambiguities are resolved by form recognition. For instance, in a crowd a certain face out of thousands is recognized in the left eye's view and, in turn, the same face is recognized in the right eye's view, while the rest is high-school trigonometry. In order to show the incorrectness of this view, in 1959 I introduced the random-dot stereogram technique in psychology (Julesz, 1960 a, b). These random-dot stereograms were based on animal camouflage, but because of being generated by digital computers and high resolution optical transducers, all monocular cues of gaps, shadows, and other imperfections were eliminated in the two views. As Fig. 2 shows, the left and right image of such a random-dot stereogram appears as a uniformly random texture, yet after stereoscopic fusion (by using a prism or crossing the eyes) a center diamond is seen in vivid depth hovering over the background. This demonstration shows that no form recognition is necessary for the CNS to eliminate the phantom dots and localize the true dots. In these displays the left and right areas are identical except for the center diamond that is also identical (consisting of black and white dots of equal probabilities) but is shifted horizontally in one eye's view relative to the

other eye's view as if it were a solid sheet. It is surprising that before this demonstration no one turned to monocularly camouflaged stereograms (that were available much before my computer-generated ones) in order to show that stereopsis does not require monocular form recognition and is a much simpler process than previously assumed.

Of course, computer-generated random-dot stereograms are much more flexible stimuli than those that occur in real life. For instance, one could still argue that the labelling of corresponding points is accomplished by form recognition. Perhaps this labeling does not require Gestalten (macro-patterns) as disproved by the experiment of Fig. 2, but micro-patterns that are similar in the two images of Fig. 2. However, the experiment of Fig. 3 disproves even this hypothesis (Julesz, 1960b). Here the left and right images are correlated, except for 16% of the dots that

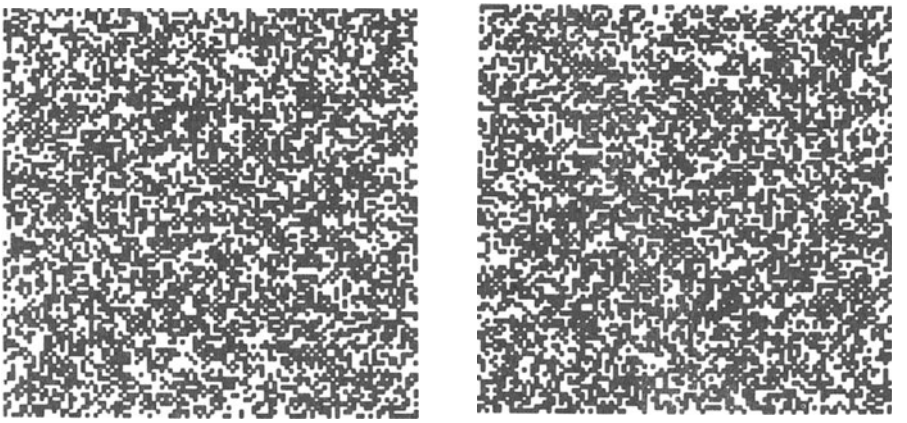


Fig. 2 Random-dot stereogram which when monocularly viewed appears as an aggregate of random dots. However, when stereoscopically fused (by crossing your eyes or by using prisms) a diamond is perceived hovering over a random background (From Julesz, 1971)

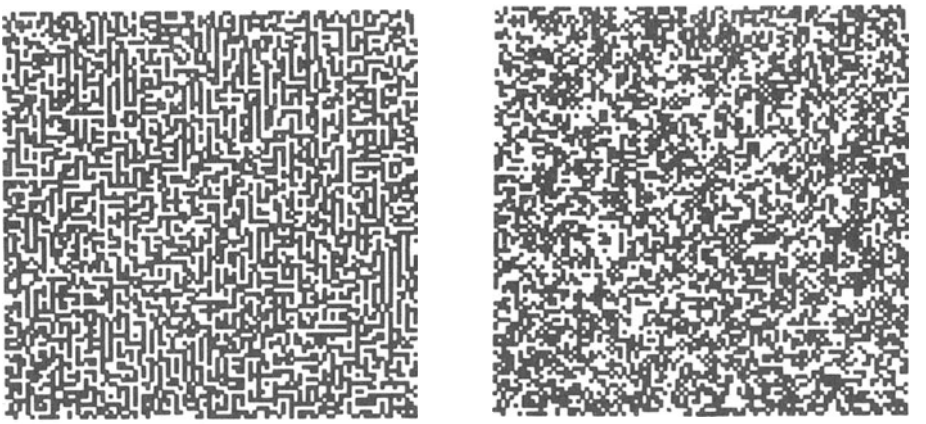


Fig. 3 Random-dot stereogram in which the diagonal connectivity in the left image is broken. 84% of the dots remain identical in the two fields and stereopsis is easy to obtain

are complemented. Whenever three dots along a diagonal are identically black (or white) the center dot is changed to the opposite color. With this manipulation no diagonal connectivity appears in the left image and no 2 x 2 dots or larger micro-pattern array is seen that would be identical in the two images. Yet after binocular fusion, the original hidden form, a center square is seen in depth. Because of the binocular rivalry, owing to the complemented dots, the percept of this square is not so clear as that of the diamond in Fig. 2 but still sharp enough. Thus the hidden binocular correlation is computed by the CNS without the help of form recognition either on macropattern or micropattern level.

In these random-dot stereograms there are millions of ways the dots could be localized. Already by chance alone half of the left and the right dots are in alignment at zero disparity. Similarly at one unit disparity (in the nasal or temporalward direction), again owing to chance, half of the stimulus points are in registration. Why does the visual system ignore these solutions at small disparity values and searches for the diamond at a large disparity value? The answer is simple, the stereopsis mechanism searches for the d e n s e s t possible solution. Instead of stopping at a 50% localization it searches for a disparity shift such that near 100% of the corresponding points in an area should become aligned. If by various horizontal shifts each area of the left and right arrays can be brought in near perfect alignment, the visual system regards this as the solution of the binocular localization problem.

Present neurophysiology has found only the first stage of this mechanism, the l o c a l disparity extractors (that are predominantly in Area 18 of the visual cortex in the monkey (Hubel and Wiesel, 1970)). However, the next required stage, which evaluates all the outputs of these local units and selects that g l o b a l solution which has the densest cluster of firing units with the same disparity, has not been found yet. Before a model of this global stereopsis will be formulated, let me review a few more psychological findings.

For instance, a 15% expansion of one image in Fig. 2 still retains stereopsis. Furthermore, rotation by a few degrees also preserves stereopsis. For instance, in Fig. 4 a random-line stereogram is shown in which the corresponding line segments in the two arrays are rotated by ± 15 degrees. While some of these local elements are at the threshold of fusion (sometimes are seen

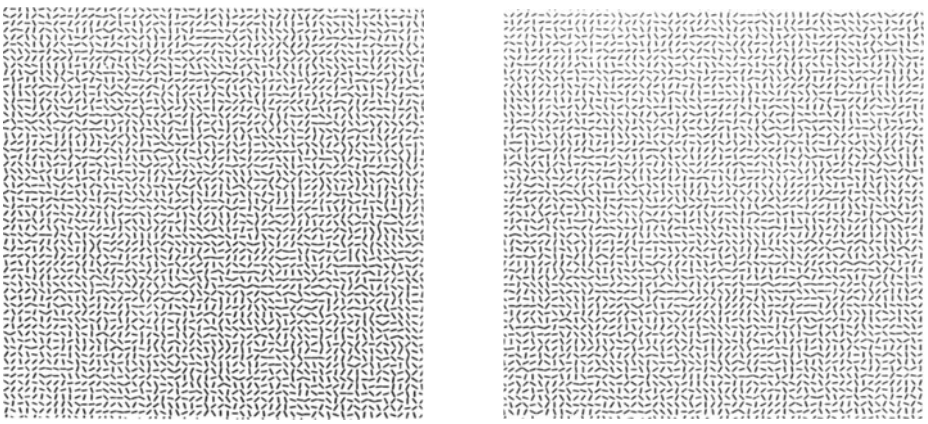


Fig. 4 Random-line stereogram such as used by Marlowe; corresponding line segments in the two fields differ by ± 15°. Global stereopsis can still be obtained

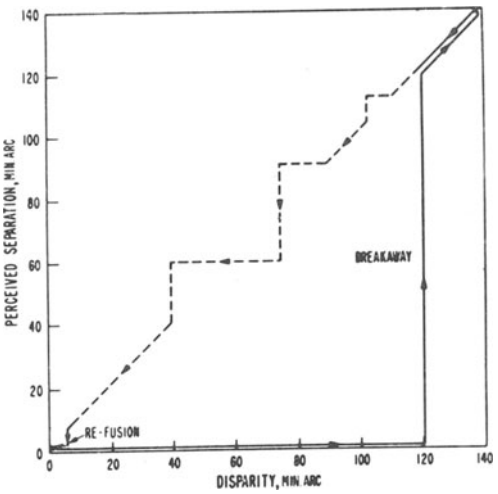


Fig. 5
Hysteresis phenomenon of random-dot stereograms under binocular retinal stabilization (From Fender and Julesz, 1967)

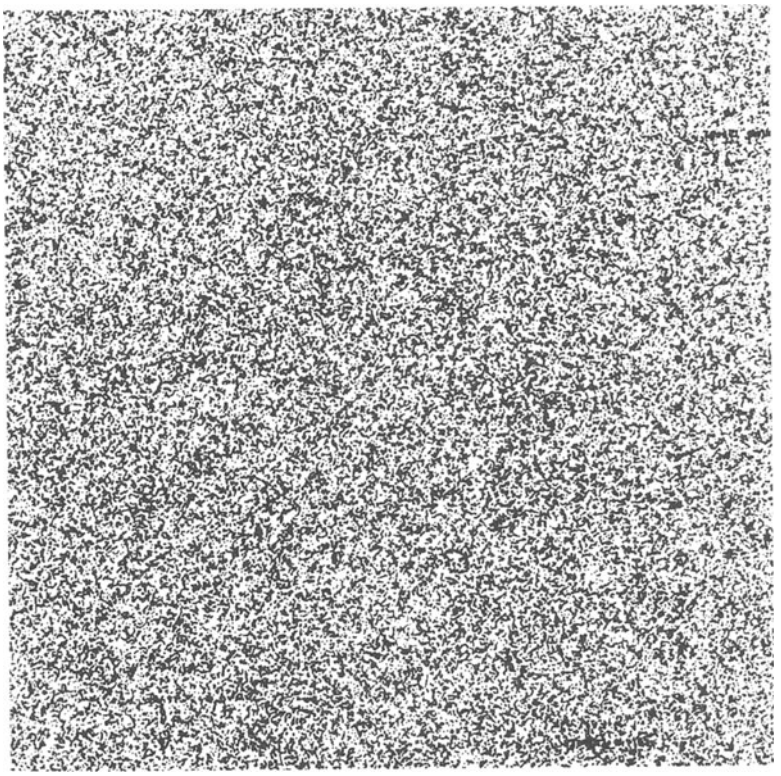
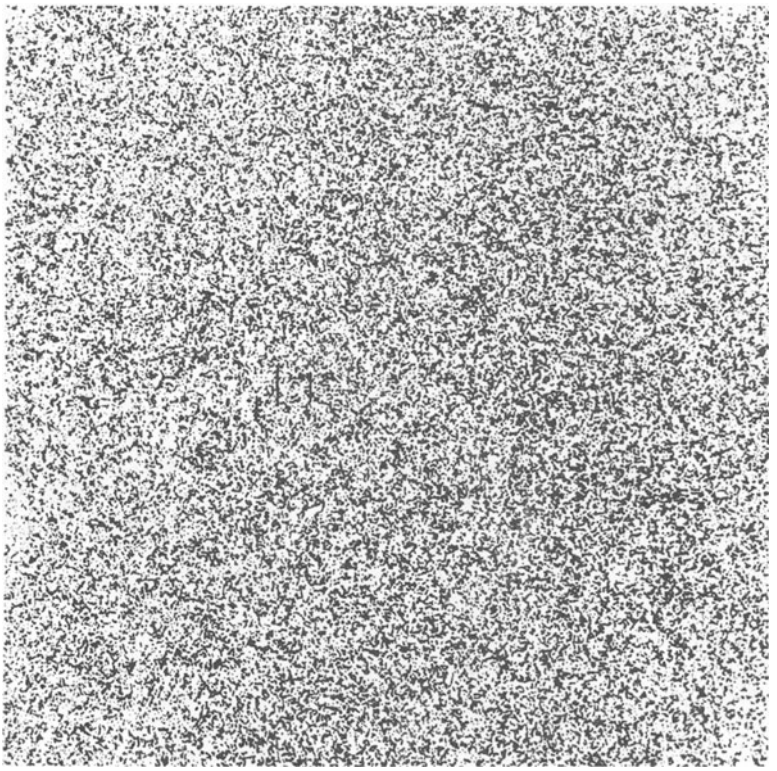


Fig. 6 Random-dot stereogram portraying a complex surface

as double, yet in depth); global stereopsis (i. e. the percept of a large center square above the background) is readily experienced (Julesz, 1971).

Another important finding was obtained under binocular retinal stabilization (Fender and Julesz, 1967). Before this work, it was assumed that Panum's fusional limit was a rigid 6 min arc disparity value in the fovea, above which the fused image would break apart and appear as double. Indeed, under binocular retinal stabilization (using the usual contact lenses and mirrors but for both eyes) the subject cannot fuse the images, which move together with the retinae, until they are actually shifted on the retinae within 6 min arc alignment. Then they suddenly coalesce. However, after fusion the images can be slowly pulled apart by large amounts in the horizontal direction without losing fusion. The limit of pulling depends on the stimulus. For classical targets (a vertical line) the limit is about 40 min arc, whereas for random-dot stereograms it is in excess of 120 min arc. After this limit is reached the fused images suddenly break apart and have to be brought back again within 6 min arc to fuse again. For random-dot stereograms this hysteresis phenomenon of fusion is shown in Fig. 5.

The importance of this finding is that stereopsis is clearly a cooperative phenomenon. Cooperative phenomena in physics exhibit hysteresis (a lag between cause and effect – a primitive type of memory) and the amount of this hysteresis depends on the number of



elements that participate. The finding that a simple line fails to stimulate the entire stereopsis mechanism while complex stereograms do, has many consequences. Moreover any advanced model of stereopsis must also account for whether two points are "corresponding points" or not according to their past history.

Another fact that a model of stereopsis has to explain is the long "learning" required for some stereograms that portray complex surface (Fig. 6). While Fig. 2 can be perceived in depth almost immediately, Fig. 3 initially requires several seconds. It often takes a minute or more till the percept stabilizes and reaches its final extent. But once such a complex stereogram has been fused, at a second trial it can often be fused very rapidly.

A third finding is of interest too. We have seen how the stereopsis mechanism seeks for a solution that gives the densest surface. What would happen if stereograms could be generated that contained more than one dense surface? In case of such an ambiguity what factors might influence the final percept? That ambiguous stereoscopic depth percepts exist is demonstrated by the classically known "wallpaper-effect". When one views a periodic structure (e. g., an old radiator, or bathroom tiles) it is possible to fuse the images at several disparity values, (integral multiples of the periodicity). The result is that a plane can be perceived at different depth levels. Would it be possible to generalize the wallpaper-effect and instead of multiple planes have two (or more) surfaces of any shape?

This problem was solved as illustrated by Fig. 7 [Julesz and Johnson, 1968 (a, b)]. In the case of ordinary random-dot stereograms, the surfaces can be speckled with random dots without any restriction; for ambiguous stereograms this is not true. When we select the color of a dot on surface "A" as P_A, this selection constrains P_B on surface "B" to take the same color.

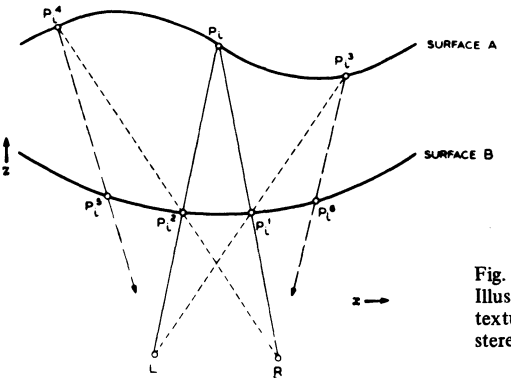


Fig. 7
Illustration of the holistic constraints imposed on the textures of ambiguously perceivable random-dot stereograms (From Julesz and Johnson, 1968b)

However, since P_B can be viewed by the other eye as well, the line of sight through P_B will intersect surface "A" in P_A which has to be identical to P_B and in turn to P_A. This reasoning must continue and a single dot will force its color on a set of points. Theoretically this set has infinite points, but with finite resolution, the set has finite numbers of elements. The result is a "holistic" organization where each element is coupled to the rest. Surface "A" can be the front surface of an object, while surface "B" can be the back (hidden) surface. With such a stereogram one can view the surface from the front or from behind at will. Instead of physically moving around the object (as is the case with holograms) the viewer can sit still and let his mind do the wandering.

Space does not permit me to reprint such ambiguous stereograms. The interested reader may view the anaglyphs shown in Julesz and Johnson, 1968 (2) or Julesz (1971) with the color spectacles supplied.

The essence of these demonstrations is that if one obtains a possible solution, this organization tends to prevail. In order to switch organizations, one must first destroy the prevailing one (by closing his eyes or looking at some unambiguous area at a certain depth). Also, owing to adaptation, after prolonged viewing the prevailing organization weakens, the old percept wanes, and the new one becomes dominant. A satisfactory model must also explain these observations.

3. A model of stereopsis

Now we are in a position to formulate the model. This model was first described in great detail in a book (Julesz, 1971). Here I shall only briefly review it.

This model of stereopsis uses only such simple elements as magnets and springs. A model should not only be isomorphic to the real phenomenon (or an important aspect of it) but should also be better matched to human thought processes than the real kind, and though coupled oscillators with frequency shifts may have provided a more modern version, I prefer the mechanical-magnetic model. I also wish to refrain from neurophysiological models, because we do not possess adequate physiological evidence for a global model of binocular depth perception.

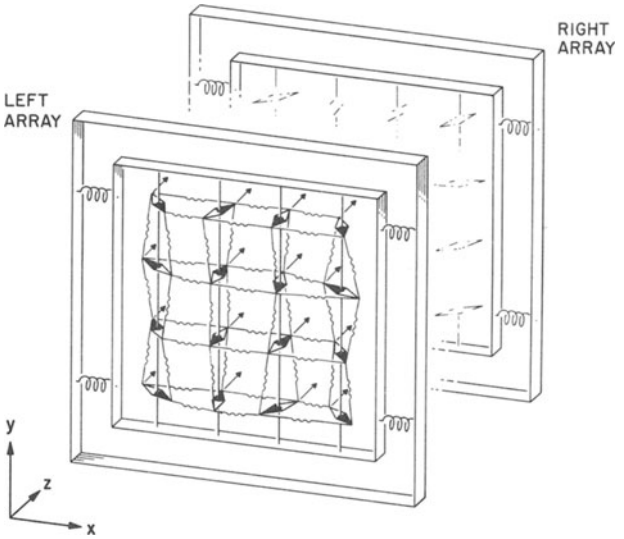


Fig. 8
Spring-coupled dipole model of stereopsis (From Julesz, 1971)

The model is shown in Fig. 8. A two-dimensional array of magnetic dipoles is mounted in ball-joint bearings such that the dipoles can rotate out of the plane of the array. A black or white dot cast on the retina would correspond to a south (S) or north (N) pole turned towards the reader. One such oriented dipole array corresponds to the cortical state that results from stimulating one

retina. The other array corresponds to the cortical state elicited by stimulating the other retina. The two arrays should be imagined in close vicinity sliding over each other. If the patterns on the retina are identical and the two arrays overlap then the S and N poles of corresponding dipoles interlock. Identical arrays of interlocked dipoles in one dimension are shown in Fig. 9a.

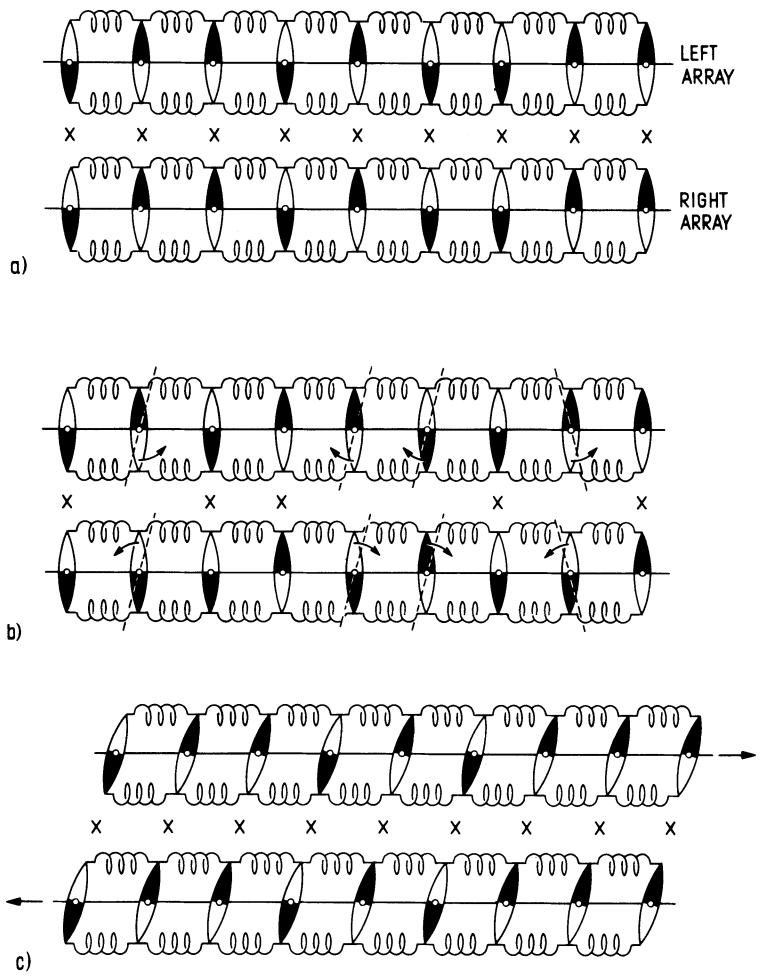
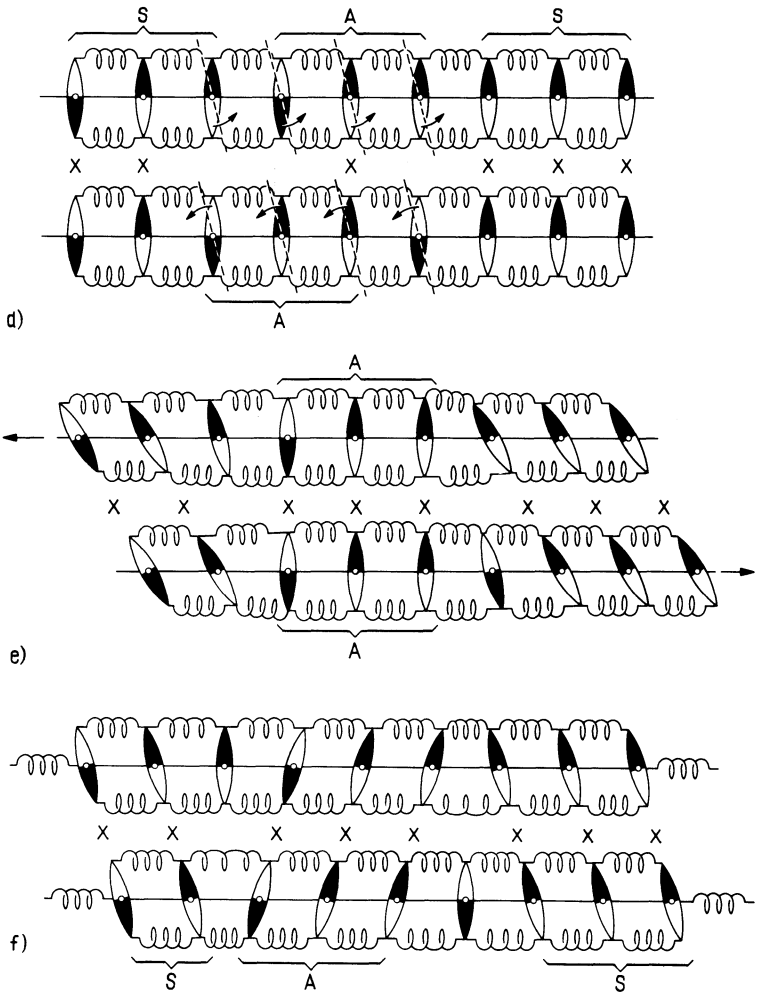


Fig. 9 One-dimensional illustration of the spring-coupled dipole model under various conditions (From Julesz, 1971)

The essence of the model, however, is the spring-coupling of each dipole to its adjacent neighbors. This coupling of all dipoles through their neighbors is essential for hysteresis. (Such coupled dipoles are used in Ising's model of ferromagnetism and were used in brain modeling in some other connotation; Cragg and Temperley, 1954.) The importance of this coupling becomes apparent when we try to explain the fusion of Fig. 1. Before the arrays are fused, by

chance 50% of the dipoles become interlocked. Because of the spring-coupling the other 50% of dipoles are forced to face the corresponding dipoles in the other array with opposite polarities. This is shown in Fig. 9b. Thus 50% of the dipoles attract each other, and 50% repel each other; thus between the arrays the global attraction force is zero. Because of no attraction



the arrays can easily be slid into any other position. However, in a totally interlocked array the pulling of the arrays causes the dipoles to turn, as shown in Fig. 9c. In order to fuse the arrays of Fig. 1 we first align, say, the background areas. This shift which is larger than Panum's fusional limit, is achieved by the convergence-divergence movements of the eye; within Panum's limit the shifts are neurally executed. In the model the arrays are physically moved with respect to each other in the horizontal direction. Thus, when the two arrays are exactly aligned, the corresponding dipoles in the background area (with zero binocular disparity) will interlock.

Once interlocked they exert a large force of attraction between each other. The center area, however, that has a diamond shape (with non-zero disparity) contains dipoles that partly attract and partly repel each other as shown in Fig. 9d. When, as the second step, we shift the center areas into alignment, the magnets of the center area will interlock, however, – and this is the crucial aspect of the model – the already interlocked dipoles of the diamond r e m a i n interlocked but turn (Fig. 9e). If finally, we assume that the two dipole arrays are suspended in a spring loaded frame (as shown in Fig. 9f), after the two areas are interlocked they will be in an equilibrium position.

We now postulate that sensing the degrees of rotation in the horizontal direction corresponds to local stereopsis. The dipoles, of course, can also turn in the vertical direction but this rotation is not sensed. The limit of horizontal rotation that still gives rise to stereopsis is Panum's fusional limit.

We are now in a position to understand most of the phenomena of stereopsis. For instance, if we expand one array by 15% the dipoles across their ball-joints will turn and try to face their corresponding dipoles. It also explains the hysteresis-effect of Fender and Julesz. After the dipole array becomes interlocked it exerts a great force which will cause the springs in Fig. 9f to give way (at least for 120 min arc displacement), but only if the pulling is slow. When the pulling of the stimuli proceeds at a fast rate, the inertia of the springs will cause the arrays to move away from each other, and after a critical distance is reached the dipoles become unlocked. With proper masses, spring constants and frictions this model can be built and can be regarded as an analog computer.

Space does not permit me to go into all the details of this model. The interested reader is referred to my book (Julesz, 1971). However, let me note how the model can clarify the problem of perceptual learning of fusion. When the stereograms are simple, such as Fig. 1, there are only two steps required for fusion: first aligning the background, and then trying a nasal or temporal shift in order to find the other depth plane. If by chance the center area is fused first, a quick cortical shift will fuse the background as well. The only ambiguity is due to the dichotomy of performing a nasal or temporal shift.

However, for stereograms that contain complex surfaces (having many depth planes) this dichotomous decision has to be made for each depth plane. In case of a wrong decision a false localization can be obtained, and often one has to backtrack several decisions and start it from scratch. Nevertheless, if this decision-tree (of nasal-temporal forks) is learned, refusion becomes greatly facilitated.

Finally, the model explains the problems of ambiguities, too. If a solution is reached, that is all dipoles became interlocked, it is impossible to obtain another global solution until the interlocked state is destroyed by some drastic way. The only way to surely find another solution is to unlock the local interlocked dipoles. This can be achieved by using electromagnets with short time constants rather than permanent magnets. After the local elements become unlocked the global fusional process has another chance to search for a different global solution. This explains why in the central nervous system adaptation to stimuli occurs at very early stages! Were adaptation a highly central phenomenon we would never be able to "shake" an obtained global solution. On a higher level, we would never be able to enjoy a pun or a good joke which is based on ambiguous meanings. I am confident that binocular depth perception involves more than localizing objects in depth.

This model explains most of the known findings of stereopsis except for a class of experiments. These experiments were intended to discriminate between a parallel or a serial model (Julesz, 1964). It turned out that in case of ambiguous stereograms in which the two possible dense surfaces had 100% or 96% correlated points, subjects always perceived the 100% correlated surface. In our serial model, however, if by chance a nasal (or temporal) shift had first interlocked the 96% correlated surface, the other solution had not been found. One could incorporate in the model a crude first stage that computes cross correlation such that the largest cross-correlation value determines the amount of the first cortical shift. One could also imagine several identical spring-coupled dipole models that start their search with different initial shifts. Of this parallel array of dipole models that one would dominate whose interlocked domains have the largest density of interlocked dipoles.

4. Problems of random search

This model could easily be generalized. The two arrays that are the cortical representations of the two retinal stimulus patterns could correspond to any two correlated events. Furthermore, the two arrays may not be two-dimensional but could be n-dimensional. For instance, one array could be the n-dimensional feature space of objects that have been stored in memory, while the other array could be the n-dimensional feature space of objects that are presented as input stimuli to an organism. This model then could function as a feature-recognition automation. Unfortunately, in n-dimensional space it would take a very long time to find a random shift that were to result in the interlocking of a large array. You have seen (when fusing Fig. 6) how much time it took in the 2-dimensional array space of stereopsis to localize a complex surface. In a high-dimensional space such a random search might last too long to be tolerated in some practical situation. (Even in stereopsis there is perceptual learning that modifies the random shifting process at repeated trials.)

I have chosen deliberately a generalization of this model in the direction of higher mental processes. Here some of the cooperative features of the model are useful assets but not adequate for a workable process. Random search does not suffice! It seems that an important criterion of intelligent behavior (whether in living organisms or machines) is the built-in heuristics that substitute random search with some more efficient way. At present these enigmatic, goal-oriented heuristic algorithms are beyond our grasp. For instance, in chess, reportedly (Hearst, 1967) grandmasters do not examine a single move further ahead than lower-ranked masters (usually a maximum of six to seven moves ahead), nor do they examine more than two to four first moves per position. The grandmaster is somehow able to "see" the core of the problem immediately. In our terminology, he knows more or less the right direction and amount of the shift around which the corresponding patterns somehow interlock. How this preconscious, automatic selection process operates is still a mystery and prevents progress in the field of artificial intelligence (Dreyfus, 1972).

One nice aspect of the stereopsis model is the simplicity of the search. The cortical shift is a single, scalar quantity within Panum's area (which for foveal targets is ± 6 min arc). Stereo acuity (the finest perceivable disparity change) is a few seconds of arc. Thus there are about ± 100 binocular disparity units which can easily be searched through in a random fashion. This simplicity permits the study of how perceptual learning may affect the random shifting. This is one of the problems I am studying at present.

5. Ising-models versus pattern-matching models

In the introduction it was mentioned that the model of stereopsis is simpler than Ising-models. Now we can undertake this comparison. In the Ising-models the long-term order was obtained as the result of the workings of the model. In our model the long-term order is imposed on the system from outside. The left and right patterns are forced on the system, while the changing states of the system correspond to the extent these patterns become m a t c h e d by the system. This pattern-matching process exhibits disorder-order transitions, hysteresis, etc. and the amount of these effects strongly depends on the number and density of the stimulus elements. This, and similar pattern-matching models thus exhibit all the characteristics of cooperative phenomena yet are conceptually much simpler than the Ising-models. This is the only reason why I accepted the invitation to describe problems of stereopsis to physicists. Yet, beyond this didactic purpose, there might be some phenomena of physics in which two ordered patterns of fields, impurities or crystal lattices are matched. It would be an interesting coincidence, if this model of stereopsis could describe some phenomena of physics.

References

Cragg, B. G., and Temperley, H. N. V.: The organization of neurones: A cooperative analogy, Electroencephalog. Clin. Neurophysiol. **6**, 85–92 (1954).

Dove, H. W.: Über Stereoskopie, Ann. Phys. series 2, **110**, 494–98 (1841).

Dreyfus, H. L.: What computers can't do. New York 1972.

Fender, D. H., and Julesz, B.: Extension of Panum's fusional area in binocularly stabilized vision. J. Opt. Soc. Am. **57**, 819–30 (1967).

Hearst, E.: Psychology across the chessboard. Psychol. Today **1**, 35 (1967).

Hubel, D. H., and Wiesel, T. N.: Stereoscopic vision in macaque monkey, Nature **225**, 41–42 (1970).

Julesz, B.: Binocular depth perception of computer-generated patterns, Bell System Tech. J. **39**, 1125–62 (1960a).

Julesz, B.: Binocular depth perception and pattern recognition, Information Theory, Fourth London Symposium, 1960 (E. C. Cherry). London 1961. (1960b).

Julesz, B.: Binocular depth perception without familiarity cues. Science **145**, 356–62.

Julesz, B.: Foundations of Cyclopean Perception. Chicago 1971.

Julesz, B., and Johnson, S. C.: Stereograms portraying ambiguously perceivable surfaces. Proc. Natl. Acad. Sci. **61**, 437–41 (1968a).

Julesz, B., and Johnson, S. C.: Mental holography: Stereograms portraying ambiguously perceivable surfaces. Bell System Tech. J. **49**: 2075–83 (1968b).

Wheatstone, C.: On some remarkable, and hitherto unobserved, phenomena of binocular vision. Philosophical Trans., Royal Soc. (London) 128: 371–94. Reprinted in: W. Dember, ed., Visual perception: The nineteenth century. New York 1964, 371–77, 386–87 (1838).

Cooperative Phenomena in a Homogeneous Cortical Tissue Model[1] [2]

Hugh R. Wilson

1. Introduction

1. 1. It is readily deducible from two basic characteristics of single nerve cells that functionally significant neural activity in the cerebral cortex must be cooperative in nature. In the first place, the activity impinging upon a neuron must exceed a threshold value in order to produce a response in that cell. It has been estimated that a cortical neuron must receive impulses from at least ten other cells to reach threshold (Eccles, 1964). Secondly, the postsynaptic potential generated within a cell due to impinging activity has a finite decay time, with typical decay constants being on the order of 10 msec. (Eccles, 1964). Thus a cell will reach threshold only if it receives a sufficient number of impulses from other cells within a circumscribed time interval. As the interconnections among neurons have a rather precise spatial organization, it follows that only large scale neural activity which is both spatially and temporally coherent can lead to the activation of a sufficient number of other neurons to permit further processing. Conversely, the activity of a like number of neurons in either a spatially or temporally random fashion will fail to generate further activity. Such random activity will thus be treated by the nervous system as noise and will be effectively filtered out.

1. 2. A further factor contributing to the cooperative nature of functional neural activity is the existence of local redundancy in cell function. This redundancy has two aspects. In the first place, physiological studies have shown that most cells located in a column radially beneath a small element of the cortical surface receive virtually identical signals from the appropriate sense organs (Mountcastle, 1957; Hubel and Wiesel, 1963, 1965). Secondly, anatomical studies suggest that there is a large degree of radial interconnection among cells located at varying depths throughout the thickness of the cortex (Colonnier, 1965; Szentagothai, 1967). In the language of physics, it may be said that the anatomically determined c o r r e l a t i o n l e n g t h in the radial direction is equal to the thickness of the tissue. This radial redundancy suggests that it is not the activity of any single cell, but rather the aggregate activity of those cells located within a radial column beneath an element of the cortical surface which is of functional significance. Such redundancy is, of course, of value in providing further insensitivity to noise and to the daily death of some thousands of neurons.

---

[1]) To Dr. Peter A. Leermakers in Memoriam.

[2]) This research was supported in part by the Alfred P. Sloan Foundation and the Otho S. A. Sprague Memorial Institute.

1. 3. The foregoing considerations suggest several important simplifications in the development of a model of cortical tissue. Due to the presence of local redundancy and to the inherently cooperative nature of cortical activity, it is appropriate to choose the net activity generated within spatially localized neural aggregates as the appropriate level of description of cortical phenomena. Furthermore, due to the largely radial nature of cortical redundancy, these localized neural aggregates may be assumed to extend throughout the thickness of the cortical gray matter. Accordingly, the cortex may be treated as a two-dimensional surface or sheet, with appropriate interactions occurring between aggregates of cells laterally displaced relative to one another along the surface. The point is that the density of radial interconnections is great enough so that radial columns of cells may be expected to function in an almost entirely cooperative fashion. It follows that the functionally important spatial aspect of neural activity resides in its two-dimensional surface distribution. An interesting footnote here is that the increase in volume of the cerebral cortex in more highly evolved vertebrates is mainly due to an increase in surface area rather than in thickness. Thus evolution has favored a sheet-like development of the cortex (Wilson and Cowan, 1972b).

1. 4. The view of cortex as comprising laterally interacting, radially redundant aggregates of neurons is subject to one important modification. It is well known that a variety of cell types are present in the cortex and that these different types play distinct functional roles. A major distinction is that between excitatory and inhibitory neurons: a single neuron will either tend to excite all others with which it synapses, or else it will tend to inhibit them. It is therefore necessary to distinguish between excitatory and inhibitory aggregates and to employ a separate variable for the net activity of each type. Cortical cells may of course be further subdivided into a variety of excitatory and inhibitory types, although this will not be done here.

1. 4. 1. Excitatory and inhibitory cells are assumed to be interspersed in such a manner that they may be represented by the continuous and uniform surface densities ρ_e and ρ_i. Therefore, let us define the variables E (x, t) and I (x, t) such that ρ_e E (x, t) $\delta x\, \delta t$ and ρ_i I (x, t) $\delta x\, \delta t$ give the numbers of excitatory and inhibitory cells becoming active during the interval (t, t + δt) within a region δx around point x on the tissue sheet. E (x, t) and I (x, t) will be referred to as the excitatory and inhibitory neural activities.

1. 5. As there are two types of cells in the model, there will be four types of interactions: excitatory-excitatory, excitatory-inhibitory, inhibitory-excitatory, and inhibitory-inhibitory. These interactions may occur both locally, i. e., among cells within a given aggregate, and laterally, i. e., between aggregates situated at differing points on the tissue sheet. In order to describe these spatial interactions, the connectivity functions β_{ee}, β_{ei}, β_{ie}, β_{ii} will be introduced. These functions indicate the average postsynaptic potential generated in cells at one point by activity in an aggregate at another point. It will be assumed here that the connectivities are distance functions only, thus making the cortical tissue model homogeneous and isotropic. In addition, the connectivity will be assumed to be monotonically decreasing with distance. This is not unreasonable in light of anatomical evidence, (Sholl, 1956) although there are other possibllities which await investigation.

1. 6. Implicit in the use of aggregate activity variables and connectivity functions specifying the average effect of cells in one aggregate upon those in another is the assumption that neural

tissue at this level of description may be treated as a deterministic dynamical system. E (x, t) and I (x, t) thus represent the mean values of statistical distributions of activity. The validity of this approach is based upon the assumption that the local redundancy and the density of interconnections are sufficient to ensure that the variance of these distributions will be quite small, and this is just what is needed for reliable information processing by cortical tissue. The adoption of such a deterministic approach has been guided by an illuminating contrast between the history of brain research and that of thermal physics. Thermal physics evolved from macroscopic, deterministic thermodynamics to statistical mechanics after the discovery of the atomic and molecular nature of matter. Modern brain research, on the other hand, has been concerned with an analysis of the statistically fluctuating properties of single neurons. What is lacking is a macroscopic, deterministic neural tissue dynamics analogous to thermodynamics. The work reported here is an attempt in that direction.

2. The model

2. 1. The derivation of equations governing the dynamics of neural tissue which will be presented here is not intended to be rigorous. Rather, it should be thought of as an indication of the origin and significance of the various terms appearing in the equations. A much more extended and rigorous derivation may be found elsewhere (Wilson and Cowan, 1972a, 1972b). Also, the equations will be restricted to one spatial dimension, since the various spatial patterns used to stimulate the tissue model will likewise vary in only one dimension. Generalization to two dimensions is obvious.

2. 2. Consider first the excitatory cells at point x. Assuming additivity of postsynaptic effects, the postsynaptic potential (PSP) generated in each of these cells by activity in all other cells in the tissue is given by:

2. 2. 1.

$$\mathrm{PSP}_e(x, t) = \int_{-\infty}^{\infty} \beta_{ee}(|x - x'|)\, E(x', t)\, dx' - \int_{-\infty}^{\infty} \beta_{ie}(|x - x'|)\, I(x', t)\, dx' + P(x, t)$$

or

$$\mathrm{PSP}_e(x, t) = \beta_{ee} \otimes E - \beta_{ie} \otimes I + P(x, t)$$

where P (x, t) is an external input to the tissue from other parts of the brain or from sense organs, and $\otimes$ denotes spatial convolution. Note that inhibitory effects are subtractive.

2. 3. A cell will become active only if its PSP exceeds a threshold value, θ, and if it is sensitive, i. e., if it is not refractory. Assuming that a cell's PSP and its sensitive or refractory status are not correlated, the number which will become active will be given by the product of the proportion which are excited above threshold and the number which are sensitive.

2. 4. Let there be a distribution of thresholds within the excitatory aggregate with the distribution function $D_e(\theta)$. Then the expected number of cells receiving superthreshold excitation will be given by the expression:

2. 4. 1.

$$S_e(PSP_e) = \int_0^{PSP_e} D_e(\theta)\, d\theta$$

It is evident from equation 2. 4. 1 that the response function S (PSP) is monotone increasing and bounded by zero and unity. If in addition D (θ) is unimodal, then S (PSP) will be a sigmoid function such as that shown in Figure 1.

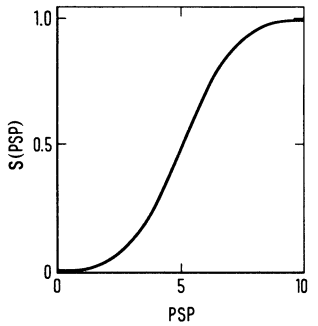


Fig. 1
Plot of a typical sigmoidal function. The particular function shown here is the logistic curve:
S (PSP) = 1/[1 + exp (— a (PSP — θ))] with $\theta = 5$, a = 1.

2. 5. The proportion of excitatory cells at x which are sensitive at time (t + δt) is approximately:

2. 5. 1.

$$1 - rE(x, t)$$

where r is the duration of the refractory period. Combining expressions 2. 2. 1, 2. 4. 1, and 2. 5. 1 results in the equation for the excitatory activity:

2. 5. 2.

$$\tau \frac{\partial E}{\partial t} = -E + (1 - rE)\, S\, [\beta_{ee} \otimes E - \beta_{ie} \otimes I + P(x, t)].$$

Similar considerations yield for the inhibitory activity:

2. 5. 3.

$$\tau \frac{\partial I}{\partial t} = -I + (1 - rI)\, S\, [\beta_{ei} \otimes E - \beta_{ii} \otimes I - Q(x, t)].$$

The time derivatives and exponential decay terms in 2. 5. 2 and 2. 5. 3 arise due to the finite time course of postsynaptic potentials.

2. 6. Before examining the properties of the solutions of equations 2. 5. 2 and 2. 5. 3, it is necessary to choose analytic forms for the functions S (PSP) and $\beta(|x - x'|)$. The actual forms used for calculation were:

2. 6. 1.

$$S\,(PSP) = \frac{1}{1 + \exp\,(-a\,(PSP - \theta))}$$

2. 6. 2.

$$\beta\,(|\,x - x'\,|) = b \exp\,(-|\,x - x'\,|/\sigma)$$

No special significance need be attached to these particular functional forms, other than that S (PSP) is sigmoidal and $\beta\,(|\,x - x'\,|)$ is monotone decreasing with distance. It has been demonstrated elsewhere that the qualitative properties of the solutions to 2. 5. 2 and 2. 5. 3 are independent of the particular analytic form chosen (Wilson and Cowan, 1972a, 1972b). This is a particularly valuable property of the equations, for present anatomical and physiological data is inadequate for a precise specification of these functions. Moreover, the functions will very likely differ from region to region of the cortex. Equation 2. 6. 2 is, however, consistent with the data of Sholl (1956).

2. 7. The appropriate range of parameter values for equations 2. 5. 2 and 2. 5. 3 may be estimated from some elementary data on visual perception. It is known that the highest frequency at which a flickering light source can be distinguished from a uniform source of the same average intensity is about 60 cycles/second (Brindley, 1970). Therefore, functionally significant visual events can be inferred to take place on a time scale not shorter than 10 msec. It is further known that humans cannot perceive spatial gratings at frequencies above about 60 cycles/degree (Campbell and Green, 1965). As one degree of visual angle is represented by about 6 mm of visual cortex in monkeys (Daniel and Whitteridge, 1961), it is apparent that a spatial scale factor of $5 \cdot 10^{-4}$ cm to $10 \cdot 10^{-4}$ cm is appropriate. The parameter values used in this study were in accord with these estimates.

3. Properties of Solutions

3. 1. Equations 2. 5. 2 and 2. 5. 3 were solved numerically subject to several physiologically reasonable constraints on the choice of parameter values. The most important of these are that there must be no spatially homogeneous stable steady states other than the resting state ($E = 0; I = 0$), and that the excitatory-inhibitory connectivity must be of longer range than the excitatory-excitatory connectivity. The implication of these constraints is that short range positive feedback will be balanced by longer range negative feedback. It is this property that is responsible for the self-organizing and cooperative nature of the neural activity.

3. 2. Depending upon the choice of parameter values (subject to the constraints in section 3. 1) three distinct dynamical modes have been obtained. Since the parameters determining the different modes are essentially anatomical in nature, it is apparent that the dynamical modes represent three distinct anatomical specializations of the tissue model. As will be indicated, the dynamics of these differing anatomies appear to be well correlated with physiological properties of differing regions of the brain.

3. 3. Before discussing the dynamical modes in turn, certain properties common to all three modes should be mentioned. In response to stimulation of the tissue via various forms of P (x, t) (see equation 2. 5. 2) threshold effects involving both spatial and temporal integration are manifested. Thus, stimuli of a given intensity will fail to generate a characteristic dynamic response unless they are of sufficient duration. Similarly, stimuli of a particular intensity may excite or fail to excite the tissue depending upon their width. Finally, there is an absolute intensity threshold in that stimuli which are too weak will fail to generate the response characteristic of the mode regardless of their width or duration. All of these threshold properties are important forms of noise insensitivity and are instances of the essentially cooperative nature of cortical function as discussed in section 1. 1.

3. 3. 1. All three modes also exhibit edge enhancement effects at points of sharp transition in stimulus intensity. Such effects are common in the visual system (Ratcliff, 1965) and probably elsewhere as will be discussed later.

3. 4. The first dynamical mode is characterized by the existence of stable, spatially i n h o m o g e n e o u s steady states to which the tissue may be excited by superthreshold stimuli of brief duration. Once attained, these states are maintained by self-excitation in the absence of any further stimulation. Two examples of these steady states and the stimuli which gave rise to them are shown in Figure 2. Narrow stimuli result in a single peak of maintained excitatory activity, whereas broader stimuli lead to peaks of maintained activity marking the boundaries of the region stimulated. It is also apparent that the excitatory activity remains localized because of the longer range inhibitory activity which it generates.

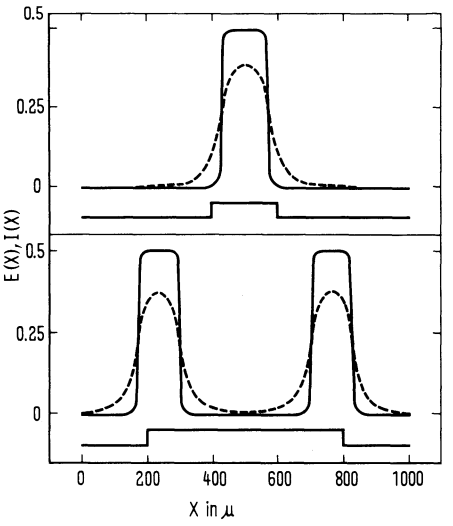


Fig. 2
Spatially inhomogeneous stable steady states of neural activity generated in response to two different stimuli. E (x) is represented by a solid line and I (x) by a dashed line. The rectangular inset below each graph shows the spatial configuration of the stimulus used to excite the tissue to each of the steady state. Once established by brief stimulation, the neural activity is selfmaintained.

3. 4. 1. The possible functional significance of spatially inhomogeneous stable steady state neural activity becomes manifest when it is realized that in two dimensions such steady states will reproduce and retain the contours of a previous stimulus configuration. Thus, it is suggested that such states may play a role in active, short term memory. In support of this suggestion

may be mentioned the work of Fuster and Alexander (1971) who observed a sustained increase in the activity of single neurons in the prefrontal cortex of monkeys during tasks requiring active retention of previous cues.

3. 4. 2. Preliminary studies using a two-dimensional extension of equations 2. 5. 2 and 2. 5. 3 suggest interesting generalization and classification properties of the inhomogeneous steady state solutions. In particular, the solutions appear to exhibit certain preferred angles at corners and preferred radii of curvature for arcs. Therefore, there will be classes of topologically related stimuli which will generate identical steady state responses, i. e., the stimuli will be grouped into invariance classes. These preliminary studies on the two-dimensional tissue model suggest that new and interesting forms of global cooperativity may be present even in an isotropic and homogeneous idealization of the cortex.

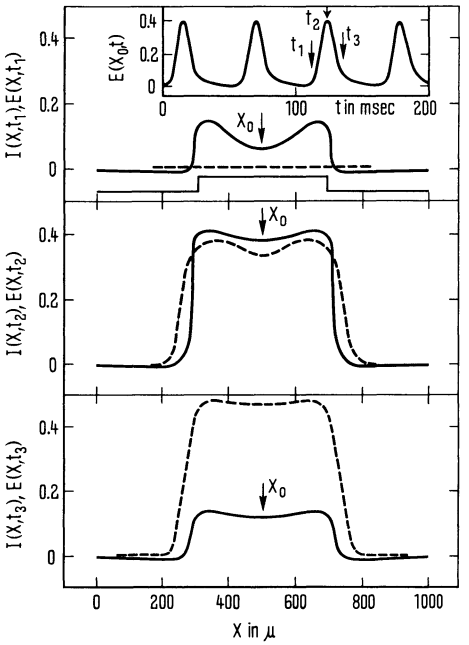


Fig. 3 Three phases in a spatially localized limit cycle oscillation in response to a maintained stimulus with the rectangular profile shown at the bottom of the first diagram. Solid line is E (x); dashed line is I (x). Inset at the top shows temporal details of the oscillation at the point marked x_0 in the other diagrams. The times t_1, t_2 and t_3 show the points on the cycle at which the spatial profiles are plotted.

3. 5. The second dynamical mode is characterized by the existence of limit cycle type oscillations in response to maintained stimulation. Typically these oscillations do not spread, but remain spatially localized as shown by the sequence of spatial profiles in Figure 3. As is indicated in the figure, edge enhancement is obtained for sufficiently broad stimuli.

3. 5. 1. Stimuli which are too weak fall below threshold and fail to generate limit cycles, whereas extremely intense stimuli lead to saturation effects which extinguish the oscillations. Over a wide range of intermediate intensities, however, it has been found that the frequency of oscillation is a monotone increasing function of stimulus intensity and that it is virtually independent of stimulus width. This suggests a mechanism whereby intensity may be frequency coded while width is represented by the distance between oscillations of maximum amplitude. Evidence for frequency coding of stimulus intensity has been obtained in recordings from thalamic nuclei by Poggio and Viernstein (1964).

3. 5. 2. That broadly stimulated regions should oscillate coherently in phase and with the same frequency independently of width is apparently the result of a cooperative frequency entrainment effect. Thus, due to the distributed excitatory and inhibitory feedback, each point in the tissue receives a net input of somewhat different intensity, yet this feedback coupling leads to synchronization and entrainment.

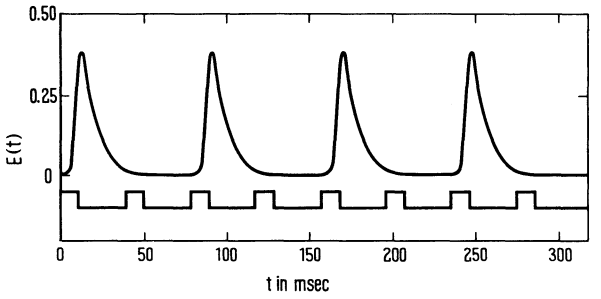


Fig. 4
Frequency demultiplication in the neural response to a stimulus pulse train. Each stimulus pulse had a narrow, rectangular spatial profile.

3. 5. 3. A further characteristic property of thalamic tissue exhibited by the model in the oscillatory mode is frequency demultiplication in response to pulse trains. Figure 4 shows an example of a neural response at one-half the stimulus frequency. In general, demultiplication by factors of three, four, etc., is obtainable in response to appropriate periodic stimulation. Such frequency demultiplication has been observed in the thalamus by Purpura (1970).

3. 5. 4. It is known that most cortical and thalamic regions revive not only a fairly precise topographically organized input, but also a diffuse, nonspecific input affecting large regions relatively uniformly (Chow and Leiman, 1970). It is presumed that this non-specific input serves a threshold setting or biasing function. The effect of such biasing has been studied by providing a uniform and continuous inhibitory input $-Q$ to the inhibitory cells (see equation 2. 5. 3). For sufficiently strong biasing the response to a spatially localized excitatory input $P(x)$ results in the generation of a pair of traveling waves of neural activity during each cycle of the oscillation. These waves travel outward from the stimulated region with maintained shape and amplitude.

3. 5. 5. In recent unpublished work on the spatio-temporal properties of evoked potentials in the olfactory bulb and cortex Dr. Walter J. Freeman has observed both sustained, spatially localized oscillations and propagation of traveling waves depending upon biasing conditions

at the time of stimulation. Furthermore, it is known that the onset of the thalamic alpha rhythm is linked with a transition from localized neural activity to wave propagation. It thus appears that the model anatomy supporting oscillatory dynamics may be associated with that of the thalamus and olfactory systems.

3. 6. The third dynamical mode has been termed the active transient mode; for brief but superthreshold stimuli give rise to active, self-generated transient responses which peak long after the end of stimulation and thereafter decay back to the resting state. Due to the effects of long range inhibitory feedback, active transients of neural activity remain spatially localized. The active transient response may be thought of as qualitatively like action potential generation in the Hodgkin-Huxley equations (Hodgkin and Huxley, 1952), but with an added mechanism to prevent its propagation. Various other simple properties of active transient dynamics are discussed in more detail elsewhere (Wilson and Cowan, 1972b).

3. 6. 1. The simple properties of active transient dynamics alluded to in the preceeding paragraph have been linked with a number of aspects of visual perception (Wilson and Cowan, 1972b). It was therefore felt that the anatomy underlying active transient dynamics might be related to the characteristics of the primary visual areas of the cortex. As a first step in exploring this possibility, the spatial modulation transfer function for the tissue model was obtained. The maximum peak-to-trough amplitude of the active transient response to stimulation of the form: $P(x) = 1 + \cos(2N\pi x/L)$ is plotted as a function of the spatial frequency N in Figure 5. The observed maximal sensitivity to intermediate frequencies of stimulation is characteristic of the psychophysically determined modulation transfer function of the visual system (Campbell and Green, 1965).

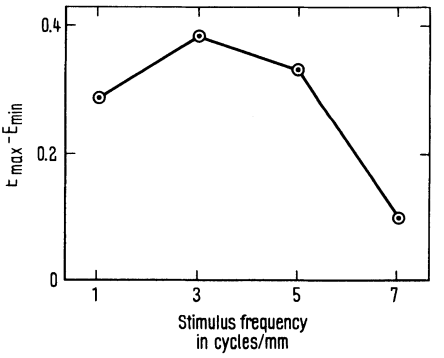


Fig. 5
Spatial modulation transfer function of the tissue model. Stimulation was brief and of the form: $P(x) = 1 + \cos\left(\frac{2N\pi x}{L}\right)$. $E_{max} - E_{min}$ gives the maximum peak to trough amplitude of the excitatory activity.

3. 6. 2. The success of active transient dynamics in reproducing the qualitative features of the visual modulation transfer function prompted an attempt to reproduce the features of two more complex visual experiments. These were the metacontrast or sequential blanking experiments of Mayzner and his colleagues (Mayzner et al., 1966) and the binocular hysteresis effect discovered by Fender and Julesz (1967). Before discussing this work in detail, however, it must be emphasized that the parameter values used in obtaining these results were identical to those used to generate the modulation transfer function in Figure 5. Thus, the results represent the differing responses of the same model anatomy to differing patterns of stimulation.

3. 6. 3. Mayzner et al. (1966) flashed the letters of a word such as CHART on a screen one at a time and in sequence. Each letter appeared at the appropriate position on the screen relative to the other letters in the complete word. With appropriate delays between presentations of the letters, subjects reported that they perceived only the first, third, and fifth letters: C A T. Simulation of this experiment using rectangular spatial pulses in place of letters yielded the results shown in Figure 6. Here the maximal excitatory activity at each position is plotted independently of the time at which it occurred. Evidently, only the first, third, and fifth pulses yield appreciable responses. This may be thought of as a spatio-temporal generalization of the frequency demultiplication discussed in section 3. 5. 3.

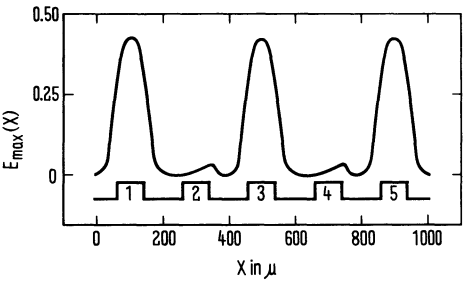


Fig. 6
Simulation of sequential blanking experiment. The line of numbered rectangular pulses shows the location and order in which the stimuli were sequentially presented. E_{max} (x) is the maximum value of the excitatory response at x independent of the time at which it occurred.

3. 6. 4. Fender and Julesz (1967) discovered an hysteresis effect which plays a fundamental role in binocular single vision (Julesz, 1971; see also the article by Julesz in this volume). Using binocular stabilization techniques they were able to independently control the positions of identical vertical bars projected upon the two retinas. With the images projected onto corresponding points on the two retinas the subject perceives a single, fused image. The images were then pulled slowly and symmetrically apart until a critical disparity was reached at which the subject reported loss of fusion and the appearance of a double image. The retinal images were then slowly brought back together until fusion was again established. The striking observation was that the disparity at which fusion was reestablished was significantly smaller than the disparity at which fusion was lost.

3. 6. 5. As anatomical and physiological evidence point to the visual cortex as the site of this binocular hysteresis phenomenon (Julesz, 1971), an attempt was made to simulate the result using the neural tissue model. The binocular stimulus was represented by a pair of identical, sharply peaked Gaussians which were initially superimposed. The Gaussians were then symmetrically translated outward slowly enough so that the neural response was maintained essentially in equilibrium. The excitatory activity in the model showed a single peak until a critical disparity was reached, when the response shifted rapidly to a pair of peaks. Moving the Gaussians slowly together again did not reestablish a single peak of excitatory activity until they were much nearer one another. The results, plotted as distance between maxima of the excitatory activity E (x, t) versus distance between maxima of the two Gaussian stimuli, are shown in Figure 7. This figure may be compared with Figure 3 in the original paper by Fender and Julesz (1967).

3. 6. 6. The success of the model in simulating these varying perceptual phenomena without any alteration in parameter values is gratifying. Indications are that a number of additional

visual effects may be similarly accounted for. Therefore, it is planned to use the model in an attempt to predict new visual phenomena which may be subjected to experimental test.

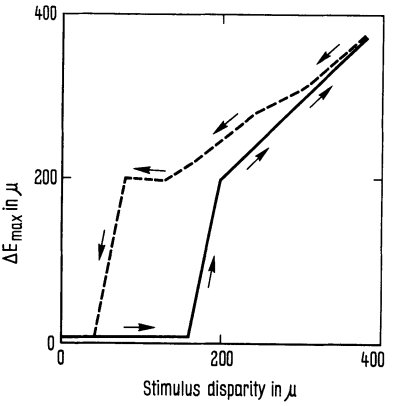


Fig. 7
Simulation of the Fender-Julesz experiment on hysteresis in binocular vision. Stimulus disparity is the distance between maxima of the two displaced Gaussian stimuli while ΔE_{max} is the separation between maxima in the neural response. Arrows show the hysteresis loop generated by first pulling the stimuli apart and then bringing them back into register.

4. Discussion

4. 1. It was pointed out in section 1. 1. that neural cooperativity is a necessary consequence of the existence of neuron thresholds plus the finite decay time of postsynaptic effects. It should be apparent from the properties of the tissue model that this is only a part of the story of neural cooperativity, however. Beyond simple threshold effects for self-excitation, cooperative neural tissue dynamics depends critically upon the interplay between excitatory and inhibitory influences.

4. 2. The excitatory feedback in the model is, of course, responsible for the active, self-generative properties of the dynamics. Excitatory effects are not enough, however; for once threshold was reached at any point in a purely excitatory tissue, activity would spread throughout the tissue until a state of uniformly saturated excitatory activity was reached[1]). Such a purely excitatory tissue could thus function only as a binary device signaling the presence or absence of previous superthreshold stimuli.

4. 3. It is only with addition of inhibitory influences that functionally significant neural cooperativity arises. The longer range inhibitory feedback evoked by superthreshold excitatory activity serves to sculpture that activity by localizing it and enhancing the response to the information-rich contours of the stimulus. Thus, the interaction between excitatory and inhibitory processes is responsible for response localization, edge enhancement, inhomogeneous stable steady states, limit cycle activity, sequential blanking, and the binocular hysteresis effect.

[1]) Beurle (1956) obtained traveling waves in a purely excitatory tissue model, but only through assuming that each cell could fire only once during passage of a wave of activity. Such an assumption is physiologically unreasonable. The wave propagation discussed in section 3. 5. 4 results from excitatory-inhibitory interactions and is not subject to this criticism.

4. 4. The tissue model embodied in equations 2. 5. 2. and 2. 5. 3. may be easily expanded to deal with interactions among N distinct spatially distributed neural aggregates. Such a model would involve N simultaneous partial differential equations coupled by N^2 spatial connectivity functions. In this way it is possible to incorporate many more details of real neural tissue and to further tailor the model to match the anatomy of particular cortical regions. However, the simple form of the model discussed in this paper is actually a good approximation to the anatomy of the olfactory bulb and cortex (Allison, 1953) and to that of the medial lemniscus (Eccles, 1965). As the olfactory system is one of the evolutionarily most primitive parts of the brain, it is felt that the neural tissue model presented here is a useful start towards understanding more highly developed cortical structures.

References

Allison, A. C. (1953): The morphology of the olfactory system in vertebrates. Biol. Rev. (Cambridge) **28**, 195.

Beurle, R. L. (1956): Properties of a mass of cells capable of regenerating pulses. Phil. Trans. Roy. Soc. **B, 240**, 55.

Brindley, G. S. (1970): Physiology of the Retina and Visual Pathway. London

Campbell, F. W. and Green, D. G. (1956): Optical and Retinal Factors Affecting Visual Resolution, J. Physiol. **181**, 576.

Chow, K. L. and Leiman, A. L. (1970): The structural and functional organization of the neocortex. Neurosciences Research Program Bulletin **8**, 153.

Colonnier, M. L. (1965): The structural design of the neocortex. In: Brain and Conscious Experience (Eccles, J. C., ed). Berlin–Heidelberg–New York, 1–23.

Daniel, P. M. and Whitteridge, D. (1961): The representation of the visual field on the cerebral cortex in monkeys. J. Physiol. **159**, 203.

Eccles, J. C. (1964): The Physiology of Synapses. New York–London.

Eccles, J. C. (1965): Inhibition in thalamic and cortical neurones and its role in phasing neuronal discharges. Epilepsia **6**, 89.

Fender, D. and Julesz, B. (1967): Extension of Panum's fusional area in binocularly stabilized vision. J. Opt. Soc. Am. **57**, 819.

Fuster, J. M. and Alexander, G. E. (1971): Neuron activity related to short term memory. Science **173**, 652.

Hodgkin, A. L. and Huxley, A. F. (1952): A quantitative description of membrane current and its application to conduction and excitation in nerve. J. Physiol. **117**, 500.

Hubel, D. H. and Wiesel, T. N. (1963): Shape and arrangement of columns in cat's striate cortex. J. Physiol. **165**, 559.

Hubel, D. H. and Wiesel, T. N. (1965): Receptive fields and functional architecture in two nonstriate visual areas (18 and 19) of the cat. J. Neurophysiol. **28**, 229.

Julesz, B. (1971): Foundations of Cyclopean Perception. Chicago.

Mayzner, M. S., Tresselt, M. E., and Cohen, A. (1966): Preliminary findings on some effects of very fast sequential input rates on perception. Psychonomic Science **6**, 513.

Mountcastle, V. B. (1957): Modality and topographic properties of single neurons of cat's somatic sensory cortex. J. Neurophysiol. **20**, 408.

Poggio, G. F. and Viernstein, L. J. (1964): Time series analysis of impulse sequences of thalamic somatic sensory neurons. J. Neurophysiol. **27**, 517.

Purpura, D. P. (1970): Operations and processes in thalamic and synaptically related neural subsystems. In: The Neurosciences: Second Study Program (Schmitt, F. O., ed.). New York.

Ratcliff, F. (1965): Mach Bands. London

Sholl, D. A. (1956): The Organization of the Cerebral Cortex. London.

Szentagothai, J. (1967): The anatomy of complex integrative units in the nervous system. In: Recent Developments of Neurobiology in Hungary (Lissak, K., ed.). **1**, 9.

Wilson, H. R. and Cowan, J. D. (1972a): Excitatory and inhibitory interactions in localized populations of model neurons. Biophysics J. **12**, 1.

Wilson, H. R. and Cowan, J. D. (1972b): A mathematical theory of the functional dynamics of cortical and thalamic nervous tissue. Submitted to: Proc. Roy. Soc. Lond. **B**.

Cooperative Transitions in Biological Membranes

Gerold Adam

1. Introduction

Many biological regulation processes are mediated by the plasmamembrane of the cell. Since cooperative enzyme kinetics in solution has been established as a regulative device in physiological chemistry [1], it was natural to look for cooperative effects as essential feature of regulation processes involving the plasmamembrane. In this respect, many cases suggestive of cooperative membrane transitions come to attention.

For instance, dramatic effects on processes like mitotic stimulation [2, 3], cell fusion [4], etc. have been observed after changes within narrow ranges of the extracellular calcium concentration, hormone concentration, or concentration of cyclic adenosinmonophosphate.

Another example of this kind is the electric surface charge of trypsinized 3T3-mousefibroblasts, which can be assessed by determining their electrophoretic mobility. We have measured this quantity in dependence on the extracellular calcium concentration and observed a sharp decrease between 4 and 6 mM Ca^{++}, as can be seen on Fig. 1. Cells fixed in formaldehyde show a very similar behavior (Fig. 1), indicating that probably the membrane phospholipids rather than conformational changes of the membrane proteins are involved in this transition.

In any case, the sharpness of the transition in this or other [2-4] examples precludes any simple explanation, for instance by non-cooperative Ca^{++}-binding in the membrane.

In another field of biological phenomena, that of electrical membrane excitation, the idea of postulating a cooperative transition of proteolipid subunits was advanced already about ten years ago [5] and thereafter formulated quantitatively by several authors [6-13].

Any quantitative cooperative model of the membrane exhibits essentially two aspects:

(a) formulation of the basic mechanism of interaction of the external stimulus with the elementary subunit of the membrane
(b) formulation of the cooperative interaction between subunits.

In the nerve membrane for instance, the external stimulus might be a change of the membrane potential, the elementary subunit a proteolipid complex of the membrane, and the mechanism of interaction between both an ion exchange within a specific site of the subunit [6, 8, 9, 13].

The specific behavior of the squid axon, for one, in addition seems to require the transition of the subunits to be cooperative [6-11, 13].

In each step of this or a similar procedure, a molecular model of the membrane unfortunately cannot be set up on a very firm footing.

In the first place, very little is known about the chemical nature of the excitable substructure of the membrane. One very promising way out of this difficulty is to investigate the properties of well characterized experimental model membrane systems such as the phospholipid bilayer membrane. With this system, conductivity mechanisms [14, 15] as well as cooperative thermal transitions have been studied [16, 17]. One can safely expect that model studies of this kind will greatly contribute to our understanding of the behavior of the native biological membrane.

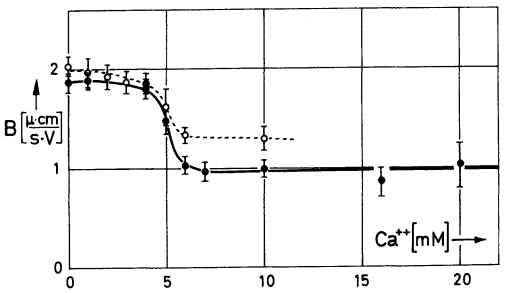


Fig. 1 Plot of electrophoretic mobility of 3T3-mousefibroblasts (full line) and formaldehyde treated 3T3-mousefibroblasts (broken line) versus Ca^{++}-concentration of the suspension medium. Bars indicate mean error of the mean. Before measurement, cells were grown in Dulbecco-Eagle medium with 10% calf serum until confluent, with last change of medium 20-24 h before measurement, suspended by treatment for 5-10 min with 0.25% trypsin solution, washed twice in the suspension medium without Ca^{++} (150 mM Na Cl, 5 m M KCl, 1 mM glucose, 10 mM Triethanolamin buffer); thereafter the Ca^{++}-concentration was adjusted to the values given. After measurement the cells were viable as judged from Trypan blue staining. For formaldehyde treatment, the trypsinized cells were washed twice in Ca^{++}-free suspension medium and kept for 30 days in this medium supplemented with 1.5% formaldehyde. Thereafter the cells were washed twice in Ca^{++}-free suspension medium, the Ca^{++}-concentration adjusted as required, and measured.

In any case, even after one is fixed to a certain molecular model including interaction between subunits, there remains the second difficulty of describing the cooperative aspect of the model. For instance, attempts have been made to describe the excitable axon membrane by an extended two-dimensional Ising-lattice of interacting subunits [6-11,13]. In equilibrium, the two-dimensional Ising-model has been solved exactly only for a special case [18]. Even less is known about the exact time course of this system after a change of an external variable. As a first step in the description of the kinetics, the author has used an approximation equivalent to the molecular field formulation in equilibrium statistical mechanics [19, 20] and applied to the description of the time course of the K^+-conductivity of the squid axon [13]. Although the theory describes well many aspects of this conductance behavior, there are certain deficiencies. In particular, the induction behavior observed experimentally at small times is hardly reproduced by the theory [13]. There is the possibility, that this deficiency stems from the rough nature of the molecular-field-approximation.

Therefore, to provide for a more realistic treatment of the kinetics of the two-dimensional Ising-model, we have solved its master-equation in a second approximation, which in equilibrium corresponds to the quasichemical or Bethe-approximation. Some of the results of these calcula-

tions will be given in the following; a more detailed account is intended for publication in Zeitschrift für Physikalische Chemie N.F.

In the following, we first set up the basic differential equations of the model in this approximation and solve them for the case of large deviations from equilibrium, as corresponds to a typical voltage clamp experiment on the squid axon.

Secondly, we derive the kinetic equations for small deviations from equilibrium in the same approximation and apply the results to temperature jump studies [17] on lipid model systems.

2. Basic equations

We use the approach put forward by Volkenstein and coworkers [21], who set up general kinetic equations describing the transitions of an Ising-model. In this formulation, the subunits can only change their states singly, whereby their transition probabilities depend only on the states of the nearest neighboring subunits. To solve this master-equation, a method of successive approximations using a hierarchy of partial distribution functions was applied [21, 22]. In this approach, the partial distribution functions are required to be spatially homogeneous, which is most restrictive for the lower order partial distribution functions.

For concreteness, we now use the model of a cooperative particle exchange on a two dimensional surface lattice described in detail in our earlier kinetic treatment in the molecular field approximation [19]. In this system the subunits are lattice sites, which bind two types of particles from a reservoir. If a particle of type 0 is bound on a site, then the subunit is in a state 0, similarly for particle type 1 conferring state 1 to a subunit. Empty as well as multiply occupied sites are excluded. The transition probability of a specified site from one state to the other depends on the states of the next neighbors, the number a of which may be in state 1, whereas (z-a) are in state 0. Then, we can write for the probabilities W_{01} and W_{10} of the transitions $0 \rightarrow 1$ and $1 \rightarrow 0$, respectively [19]:

$$W_{01}(z, a) = k_0' \exp\{w_{00}(z - a)/kT\} \exp\{w_{01} a / kT\} \tag{1}$$

$$W_{10}(z, a) = k_1' \exp\{w_{11} a/kT\} \exp\{w_{01}(z - a)/kT\} \tag{2}$$

Here, k_0' and k_1' may contain the interaction energies between a particle and its site, the activities of particles 0 and 1 in the reservoir facing the surface lattice, and activation factors independent of a. The quantities w_{00}, w_{01}, w_{11} are contributions to the binding energies of particle 0 or 1 on a site arising from interactions 00, 01, and 11 pairs of nearest neighbors, respectively.

In the approximation used in the following, the partial distribution functions $g_i(z, a)$ for a site in state i (i = 0, 1) and its z nearest neighbors, a of which are in state 1 can be written [21, 22]:

$$g_i(z, a) = g_{i0}^{(z-a)} \cdot g_{i1}^{a} / g_i^{z-1}; \quad i = 0, 1 \tag{3}$$

where g_{00}, g_{01}, g_{11} are the pair distribution functions; g_0 and g_1 are the single distribution functions, the indices indicating the states.

There are the following obvious relations between these partial distribution functions

$$g_{00} + g_{01} = g_0; \quad g_{01} + g_{11} = g_1; \quad g_0 + g_1 = 1 \tag{4}$$

Now we can enumerate all elementary transitions leading to or from a particular state of the chosen site and obtain:

$$\frac{dg_1}{dt} = \sum_{a=0}^{z} \binom{z}{a} W_{01}(z, a) g_0(z, a) - \sum_{a=0}^{z} \binom{z}{a} W_{10}(z, a) g_1(z, a) \tag{5}$$

Similarly, we evaluate the balance for the pair distribution function g_{01} pertaining to a chosen pair of sites and obtain

$$\frac{dg_{01}}{dt} = g_{00} \sum_{a=0}^{z-1} \binom{z-1}{a} W_{01}(z, a) g_0(z-1, a) - g_{01} \sum_{a=0}^{z-1} \binom{z-1}{a} W_{01}(z, a+1) \times$$

$$\times g_0(z-1, a) + g_{11} \sum_{a=0}^{z-1} \binom{z-1}{a} W_{10}(z, a+1) g_1(z-1, a) -$$

$$- g_{01} \sum_{a=0}^{z-1} \binom{z-1}{a} W_{10}(z, a) g_1(z-1, a) \tag{6}$$

After combining Eqs. (1) through (6) and summing up the binomials, one obtains:

$$\frac{dn}{dt} = k_0 \frac{[p(1-n) + (q-p)m]^z}{(1-n)^{z-1}} - k_1 \frac{[p^{-1}n + (q-p^{-1})m]^z}{n^{z-1}} \tag{7}$$

$$\frac{dm}{dt} = k_0 \frac{[p(1-n) - (q-p)m]\,[p(1-n) + (q-p)m]^{z-1}}{(1-n)^{z-1}} -$$

$$- k_1 \frac{[-p^{-1}n + (q+p^{-1})m]\,[p^{-1}n + (q-p^{-1})m]^{z-1}}{n^{z-1}} \tag{8}$$

where for abbreviation of notation was used:

$$n = g_1; \quad m = g_{01} \tag{9}$$

$$p = \exp\left\{-\frac{(w_{11} - w_{00})}{2kT}\right\}; \; q = \exp\left\{-\frac{w}{2kT}\right\}; \quad w = w_{00} + w_{11} - 2w_{01} \tag{10}$$

$$k_0/k_0' = k_1/k_1' = \exp\left\{\frac{(w_{00} + w_{11})z}{2kT}\right\} \tag{11}$$

Thus, the cooperative system in this approximation can be described by a system of two non-linear differential equations, given by Eqs. (7) and (8). These relations are nearly the same as Eqns. (23) and (24) of reference [22], where a printing error is to be corrected.

In order to asses the nature of the approximation, we use the steady state relations $dn/dt = 0$ and $dm/dt = 0$, yielding the equilibrium values $\bar{n}$ and $\bar{m}$:

$$y = \frac{k_0}{k_1} p^z = \frac{\bar{n}}{1-\bar{n}} \left[\frac{\beta - 1 + 2\bar{n}}{\beta + 1 - 2\bar{n}}\right]^{z/2} \left[\frac{1-\bar{n}}{\bar{n}}\right]^{z/2} \tag{12}$$

and

$$\bar{m} = \frac{\beta - 1}{2(q^2 - 1)} \tag{13}$$

where

$$\beta = [1 + 4\bar{n}(1-\bar{n})(q^2 - 1)]^{1/2} \tag{14}$$

These relations are identical with those of the quasichemical or Bethe-approximation of equilibrium statistical mechanics [23]. For illustration, the equilibrium values $\bar{n}$ and $\bar{m}$ are plotted on Fig. 2 in dependence on log y, which in our specific model is proportional to the difference in chemical potential of particles 1 and 0 in the reservoir. The dotted parts of the curves are thermodynamically unstable.

3. Kinetics of large deviations from equilibrium

We wish to apply the Eqs. (7) and (8) to the following kinetic problem, involving large deviations from equilibrium. At time t = 0 the system may be in the equilibrium state $\bar{n} = 0.05 \ll 1$, corresponding to some $y < 1$ (see Fig. 2). At times $t > 0$, the external variable y is switched to another

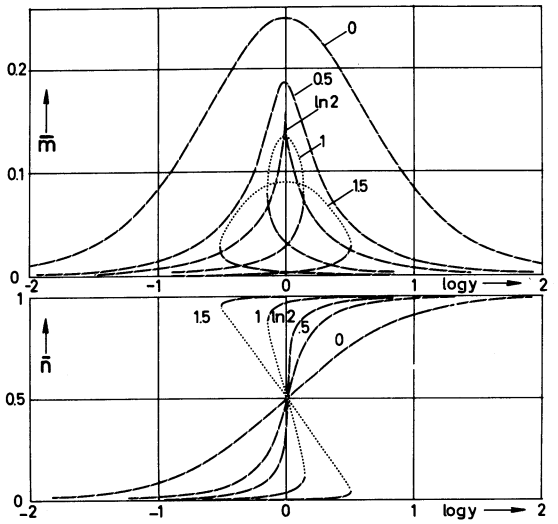


Fig. 2

Plot of equilibrium values $\bar{n}$ and $\bar{m}$ versus log y for the Ising-model in the quasichemical approximation as described in the text. For these curves, z = 4 and – w/2kT, as given as parameter to the curves, have been used. The broken parts of the curves represent thermodynamically unstable regions.

value $y > 1$, and the time course of the system is followed to the new equilibrium values of n and m. This sudden change of y might be implemented by a sudden increase of the concentration of particles 1 in the reservoir. In the membrane process of axon excitation, two reservoirs are implied; then y is a function of the membrane potential [9, 13]. Eqs. (7) and (8) have been integrated numerically on a Hewlett-Packard HP9100B-computer, applying a Runge-Kutta procedure. For these computations we have used z = 4 and $w_{00} = w_{11}$, i.e. p = 1. The results for different values of y and w/2kT = 0,-ln 2, and -1 are plotted on Figs. 3, 4, and 5, respectively. The time scales are arbitrary by the factor

$$\rho = k_1 \exp\{w_{11} z/kT\} \tag{15}$$

As expected, w = 0 gives exponential time courses for n (see Fig. 3). The case $w/2kT = -\ln 2$ corresponds to the critical isotherm (Fig. 4).

In the case $w/2kT = -1$, the system goes through a phase transition of first order (see Fig. 2). In its time course, this system shows an induction behavior at small ρt (see for instance the curves for log y = 1.517 and 2.744 in Fig. 5). For comparison we have plotted in Fig. 5 the time course of the potassium conductivity on arbitrary ordinate and abscissa scales, as derived from the

empirical Hodgkin-Huxley formulation [24] for depolarization of 78 mV of the squid axon. Although the description of the induction period is improved compared to the molecular field treatment [13], the description of the complete time course is still relatively crude. A more

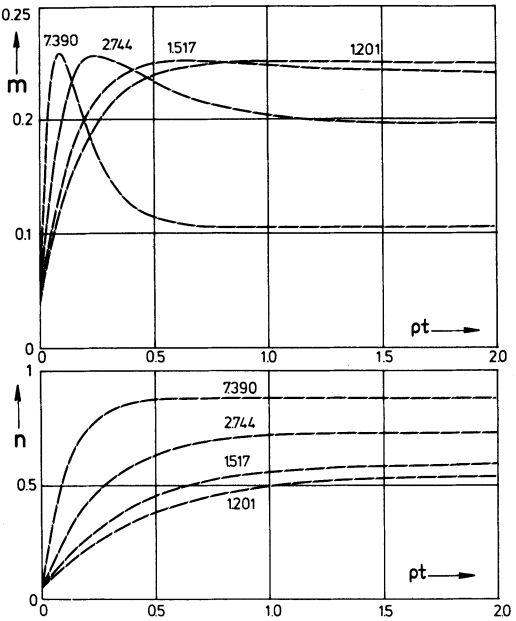


Fig. 3 Time course of n and m for the non-cooperative Ising-model after change from equilibrium state at $y = 0.05$ to y as given by the curve parameters. Theoretical parameters used: $w/2kT = 0$; $z = 4$; $p = 1$, i.e. $w_{00} = w_{11}$. Time scale is arbitrary by factor ρ (see eq. (15)). Note different ordinate scales for n and m.

detailed comparison of the theoretical description with the potassium current of the squid axon will be given elsewhere.

It will be noticed, that for $\log y = 1.201$ the system described in Fig. 5 remains at small values of n and m. Thus, the approximation used cannot describe transitions from the metastable loops

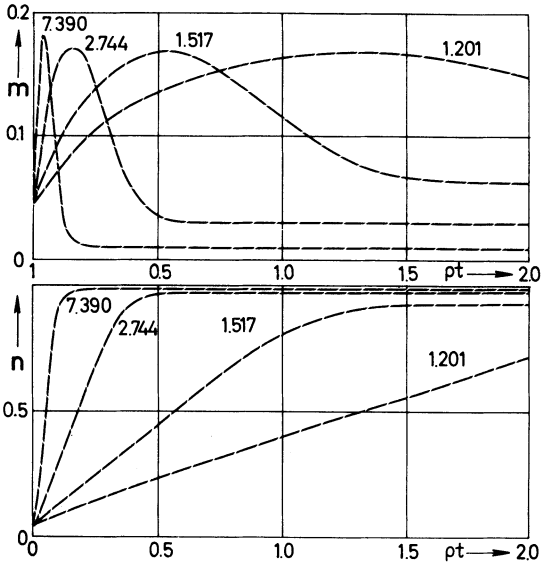


Fig. 4 Time course of n and m for the Ising-model in the critical case after change from equilibrium state at $y = 0.51$ to y as given by the curve parameters. Theoretical parameters used: $w/2kT = -\ln 2$; $z = 4$; $w_{11} = w_{00}$. Time scale is arbitrary by factor ρ (see Eq. (15)). Note different ordinate scales for n and m

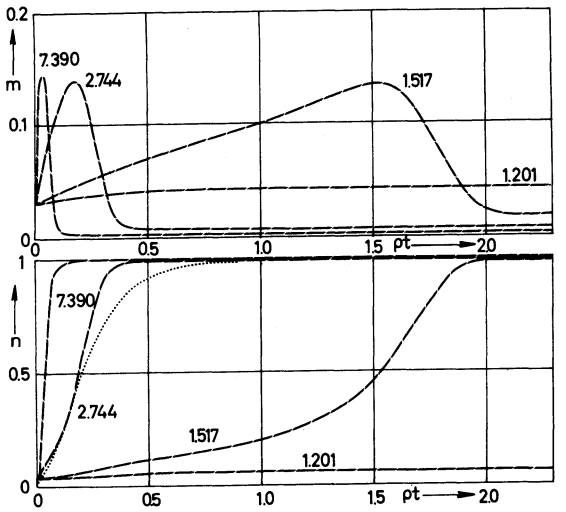


Fig. 5
Time course of n and m for the Ising-model exhibiting a phase transition of first order after change from y = 1 to y as given by the curve parameters. Theoretical parameters: $w/2kT = -1$; $z = 4$; $w_{11} = w_{00}$. Time scale is arbitrary by factor ρ (see Eq. (15)). Note different ordinate scales for n and m.

$\bar{n} < \frac{1}{2}$ to the stable branches $\bar{n} > \frac{1}{2}$ for supercritical values of w/2kT. In this respect, this kinetic theory is similar to the kinetic description in the molecular field approximation [19, 20], the only improvement being a reduction in the size of the loops.

4. Kinetic equations for small deviations from equilibrium

Introducing $n = \bar{n} + \Delta n$ and $m = \bar{m} + \Delta m$ into Eqs. (7) and (8), expanding the right hand sides in powers of Δn and Δm, and neglecting all terms with higher powers than Δn and Δm, yields:

$$\frac{d\Delta\eta}{d\rho' t} = \alpha\Delta n + \beta\Delta m \tag{16}$$

$$\frac{dm}{d\rho' t} = \gamma\Delta n + \delta\Delta m \tag{17}$$

where
$$\alpha = Ar_0\left[\frac{zp^{-1}}{r_0} + \frac{zp}{r_1} - \frac{z-1}{\bar{n}(1-\bar{n})}\right]; \beta = Ar_0 z\left[\frac{q-p^{-1}}{r_0} - \frac{q-p}{r_1}\right] \tag{18}$$

$$\gamma = Ar_2\left[(z-1)\left(\frac{p^{-1}}{r_0} + \frac{p}{r_1} - \frac{1}{\bar{n}(1-\bar{n})}\right) - \frac{p^{-1}}{r_2} + \frac{p}{r_3}\right] \tag{19}$$

$$\delta = Ar_2\left[(z-1)\left(\frac{q-p^{-1}}{r_0} - \frac{q-p}{r_1}\right) + \frac{q+p^{-1}}{r_2} + \frac{q+p}{r_3}\right] \tag{20}$$

with
$$A = -(r_0/\bar{n})^{z-1}\cdot p^z; \quad \rho' = k_1 p^{-z} \tag{21}$$

$$r_0 = p^{-1}\bar{n} + (q-p^{-1})\bar{m}; \quad r_1 = p(1-\bar{n}) + (q-p)\bar{m} \tag{22}$$

$$r_2 = -p^{-1}\bar{n} + (q+p^{-1})\bar{m}; \quad r_3 = p(1-\bar{n}) - (q+p)\bar{m} \tag{23}$$

Thus, the deviations Δn and Δm from equilibrium both relax with two relaxation times τ_+ and τ_-:

$$\tau_\pm = \frac{\alpha+\delta}{2} \pm \left[\left(\frac{\alpha+\delta}{2}\right)^2 - (\alpha\delta - \beta\gamma)\right]^{1/2} \tag{24}$$

Using Eqs. (13), (14) and (16) through (24), the relaxation times can be expressed by the equilibrium quantity $\bar{n}$ of the system.

In Fig. 6, we have plotted the logarithm of both these relaxation times versus $\bar{n}$. For this computation we have used $z = 4$ and $w_{00} = w_{11}$, i.e. $p = 1$. As can be seen, the second relaxation time τ_- is always much smaller than τ_+. In the molecular field approximation only one relaxation time occurs [20] which corresponds to τ_+.

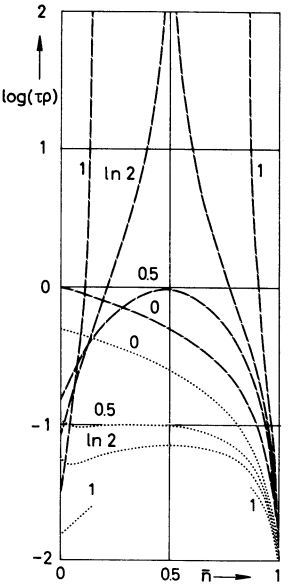


Fig. 6 Relaxation times $\tau+$ (full line), and τ_- (broken lines) for the Ising-model in small deviations from equilibrium, according to Eq. (24) of the text, plotted versus equilibrium value $\bar{n}$. Curve parameters indicate – w/2kT. Parameters used: $z = 4$; $w_{00} = w_{11}$, i.e. $p = 1$.

5. Application of the theory to relaxation studies on lipid systems

An application of the kinetic theory for small deviations from equilibrium is possible to temperature jump experiments done with lamellar phospholipid systems. Träuble [17] has studied the equilibrium and the kinetic behavior of these model systems of membranes by a very elegant optic technique. He has used for instance the absorption of bromthymolblue as indicator for a transition occurring in dipalmitoylphosphatidylcholin-water systems at about 40°C (see Fig. 7). This transition is relatively steep, occurring within a temperature range of about 7°C. Similar transitions have been observed in this and in comparable systems by calorimetric methods [16]. From nuclear magnetic resonance and X-ray studies on such systems further evidence has been obtained by which the transition could be interpreted as arising mainly from a change with temperature in structural order and mobility of the hydrocarbon chains of the phospholipid molecules [25]. In fact, Träuble [17, 26] has interpreted this transition by the sudden appear-

ance with raising temperature of "kinks" in the hydrocarbon chains. It is this "kink" model, we wish to describe in a simplified way by the two-dimensional Ising-model as follows. A different molecular interpretation centering on the polar head-groups of the phospholipids is being attempted by Kramer [30], using the molecular-field formulation of the kinetic problem [20].

It is well known from X-ray studies that the hydrocarbon chains of phospholipid monolayer, bilayer, or lamellar systems pack in a hexagonal lattice with an average chain distance of about 4.8 Å [27, 28]. A "kink" is defined as an interruption of the all-trans configuration of the methylene groups of a hydrocarbon chain by a sequence of internal rotations about the C—C bonds, that allows for a parallel continuation of the hydrocarbon chain and thus for a minimal distortion of the chain packing [29]. There are several possible kink types, the simplest being the 2 g 1 – kink, which consists of rotations about two consecutive methylene C—C bonds by + 120° and – 120°, respectively [29]. Thus, in our description a subunit in state 0 is defined by an all-trans hydrocarbon chain, whereas a subunit in state 1 is a chain with one kink. In this crude

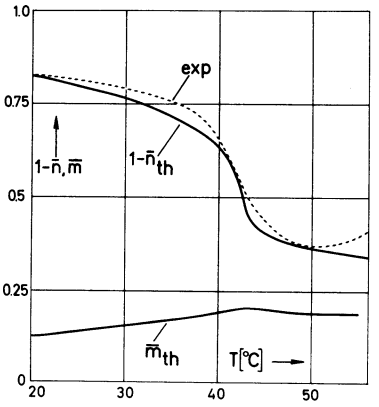


Fig. 7 Plot of equilibrium values $\bar{n}$ and $\bar{m}$ according to theory (full line) and experiments [17] for dipalmitoyllecithin-water measured by absorption changes of bromthymolblue (broken curve) in dependence of temperature. Theoretical parameters: $w = -0.505$ Kcal/mol; $w_{11} - w_{00} = 0.2$ Kcal/mol; $z = 6$; $T_u = 42.5°C$.

treatment we neglect the existence of different types of kinks, as well as the possibility of more than one kink in the same hydrocarbon chain. The latter assumption seems reasonable in view of volumetric data [26]. In a first approximation, we may further neglect the fact that the hydrocarbon chains have restraints by being coupled in pairs by the glycerol-residue in the polar region of the phospholipid molecules. The interaction of the polar groups of the phospholipid molecules probably contributes appreciably to the general cohesion of the bilayer-packing. However, we feel that configurational changes in this region are not basic to the transition discussed here. Estimates of inter- and intramolecular interactions in these regions and comparison to those worked out for the hydrocarbon regions [29, 31] could give more information on this aspect.

Thus, in our simplified description, the interaction energies w_{00}, w_{01}, and w_{11} are between two all-trans chains, between one all-trans and one with a kink, and between two chains with a kink each, respectively. The coordination number in the hexagonal lattice of chains is $z = 6$. Since the chain packing in the neighborhood of a kink is not distorted appreciably [29], the same z can be used at these places too. For description of the system in equilibrium, we use Eqs. (10), (12), (13), and (14). Defining the transition temperature T_u by $\bar{n} = 0.5$, one obtains from Eqs. (10). (12), and (14):

$$k_0/k_1 = \exp\{(w_{11} - w_{00})z/2kT_u\}$$

Taking $T_u = 42.5°C$ from Träuble's data on dipalmitoyllecithin [17], w_{11}-$w_{00} = 0$, 2 Kcal/mol, and $w = -0.505$ Kcal/Mol, the latter being very near to the critical value, the theoretical curve in Fig. 7 was calculated. The experimental results in Fig. 3 of Ref. [17] being given only between 20°C and 56°C and on a relative scale, we have plotted the experimental curve for dipalmitoyllecithin as obtained by bromthymolblue absorption [17] so that it coincides with the theoretical one at 20°C and 50°C (see Fig. 7).

With the same set of parameters given above, we have computed from Eqs. (16) through (24) the relaxation times of the system. Of these τ_- is very small and nearly independent of $\bar{n}$ and, thus, of T. The more interesting relaxation time τ_+ is plotted on Fig. 8 versus $\theta = (\bar{n} - \bar{n}_{50°C})/(\bar{n}_{20°C} - \bar{n}_{50°C})$, the latter quantity being used in analogy to the procedure of Ref. [17]. The experimental points in Fig. 8 are taken from data on pure dipalmitoyllecithin in water, as measured with bromthymolblue as absorption indicator by temperature jump experiments [17].

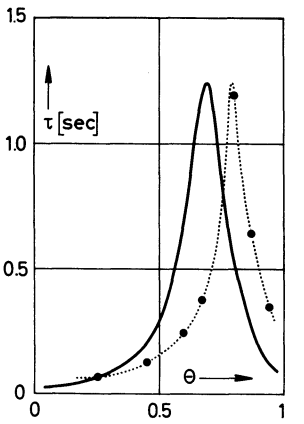


Fig. 8 Plot of relaxation times according to experiment [17] for dipalmitoyllecithin-water as measured by absorption changes of bromthymolblue (points and broken curve) and theoretical curve for τ_+ according to Eq. (24).
Abscissa $\theta = (\bar{n} - \bar{n}_{50°C})/(\bar{n}_{20°C} - \bar{n}_{50°C})$.
Theoretical parameters as in Fig. 7.

As can be seen from Figs. 7 and 8, the proposed theory describes fairly well both equilibrium and kinetic measurements on this system. The agreement could have improved by adjusting the parameters more delicately. However, in view of other uncertainties of the theoretical model and the comparison with the experiments, this does not seem to give much more information. By larger figures of w_{11}-w_{00}, one can shift the maximum of τ_+ to lower θ. For instance, with w_{11}-$w_{00} \approx 1$ kcal/mol it is at about $\theta = \frac{1}{2}$. By lowering of $-w$, one can decrease τ_+ drastically. These effects might contribute to the understanding of similar changes observed on systems including calcium and/or cholesterol in addition to the phospholipid [17]. However, it is not clear at present, how maxima of the relaxation time observed at $\theta < \frac{1}{2}$ can be explained by the theory.

The energy parameters $w_{11} - w_{00} = 0.2$ and $w = -0.505$ kcal/mol are almost by a magnitude smaller as those discussed for kinks in paraffin chains [26, 29]. However, in view of even smaller energy parameters needed in a similar application of the molecular field approximation, an improvement of this aspect can be expected in a higher approximation of the kinetic treatment. The relative magnitude of the two parameters is reasonable, if one considers the improvement in chain packing encountered with two neighboring kinks in the configuration of a so called "kink-block" [29]. In any case, the proposed theory seems to open a quantitative way for a mechanistic analysis of equilibrium and kinetic studies on such lamellar lipid model systems.

6. Conclusion

The kinetic theory in this second approximation of the general kinetic formulation of the Ising-model has been applied above only to a very limited amount of experimental data. In particular, only two-dimensional problems have been treated. Nevertheless, the reasonable description of the basic features of the transitions in lamellar phospholipid systems gives rise to the expectation that other two-dimensional systems too can be treated profitably by this approximation. Since this second approximation is not limited to two-dimensional systems, there is the hope that by the same formulation also three-dimensional cooperative transitions and their kinetics can be described successfully.

An improvement of the kinetic description by use of higher order partial distribution functions seems straighforward; in reference [21] is given a system of differential equations pertaining to the z + 1 subunits of a cooperative region and the nearest neighbors to it. Because of the rapid convergence, of the approximations of successive orders of the partial distribution functions, as discussed for the one-dimensional Ising model in references [21] and [32], there is the hope of arriving at a very good if not exact formulation of the two- and three-dimensional cases by apparently limited but quite a bit more analytic effort, than that spent in the present study.

Acknowledgement: The author wishes to thank Dr. G. Adam for her collaboration in the experiments, and Drs. H. Träuble and L. Kramer for stimulating exchange on the theoretical aspects of the present work. Part of this work was supported by a grant of the Deutsche Forschungsgemeinschaft.

References

[1] Koshland, D. E. Jr.: Conformational Aspects of Enzyme Regulation, Current Topics in Cellular Regulation **1**, 1 (1969).

[2] Whitfield, J. F., Rixon, R. H., Perris, A. D., and Youdale, T.: Stimmulation by Calcium of the Entry of Thymic Lymphocytes into the Deoxyribonucleic Acid-Synthetic (S) Phase of the Cell Cycle, Experimental Cell Research **57**, 8 (1969).

[3] Whitfield, J. F., MacManus, J. P., Youdale, T., and Franks, D. J.: The Roles of Calcium and Cyclic AMP in the Stimulatory Action of Parathyroid Hormone on Thymic Lymphocyte Proliferation, J. Cellular Physiology **78**, 355 (1971).

[4] Shainberg, A., Yagil, G., and Yaffe, D.: Control of Myogenesis in Vitro by Ca^{2+}-Concentration in Nutritional Medium, Experimental Cell Research **58**, 163 (1969).

[5] Tasaki, I.: Permeability of Squid Axon Membrane to Various Ion, J. General Physiology **46**, 755 (1963).

[6] Adam, G.: Nervenerregung als kooperativer Kationenaustausch in einem zweidimensionalen Gitter, Berichte der Bunsengesellschaft **71**, 829 (1967).

[7] Hill, T. L.: Electric Fields and the Cooperativity of Biological Membranes, Proc. Natl. Acad. Sci. US. **58**, 111 (1967).

[8] Adam, G.: Ionenstrom nach einem depolarisierenden Sprung im Membranpotential, Z. Naturforschung **23b**, 181 (1968).

[9] Adam, G.: Theory of Nerve Excitation as a Cooperative Cationexchange in a Two-dimensional Lattice, in: (Snell, F., Wolken, J., Iverson, G. J., and Lam, J., eds.) Physical Principles of Biological Membranes, New York 1970, p. 35 ff.

[10] Blumenthal, R., Changeux, J. P., and Lefever, R.: Membrane Excitability and Dissipative Instabilities, J. Membrane Biology **2**, 351 (1970).

[11] Hill, T. L. and Chen, Y.: Cooperative Effects in Models of Steady State Transport across Membranes, I: Proc. Natl. Acad. Sci. US. **65**, 1069 (1970), II: ibid. **66**, 189 (1970), III: ibid. **66**, 607 (1970), IV: Biophysical Journal **11**, 685 (1971).

[12] Hill, T. L. and Chen, Y.: On the Theory of Ion Transport across the Nerve Membrane, Potassium Ion Kinetics and Cooperativity II: Proc. Natl. Acad. Sci. US. **68**, 1711 (1971), III: ibid. **68**, 2488 (1971).

[13] Adam, G.: A Molecular Mechanism for Regulation of the K^+-Current through the Axon Membrane, in: Broda, E., Locker, A., and Springer-Lederer, H.: Proceedings First European Biophysics Congress, Wien 1971, Vol. V, p. 193 ff.

[14] Mueller, P. and Rudin, D. O.: Resting and Action Potentials in Experimental Bimolecular Lipid Membranes, J. Theoretical Biology **18**, 222 (1968).

[15] Stark, G., Ketterer, B., Benz, R., and Läuger, P.: The Rate Constants of Valinomycin-Mediated Ion Transport Through Thin Lipid Membranes, Biophysical Journal **11**, 981 (1971).

[16] Ladbrooke, B. D. and Chapman, D.: Thermal Analysis of Lipids, Proteins and Biological Membranes, Chemistry and Physics of Lipids **3**, 304 (1969).

[17] Träuble, H.: Phasenumwandlungen in Lipiden, Mögliche Schaltprozesse in biologischen Membranen, Die Naturwissenschaften **58**, 277 (1971).

[18] Onsager, L.: Crystal statistics I. A Two-Dimensional Model with an Order-Disorder Transition, Phys. Rev. **65**, 117 (1944).

[19] Adam, G.: Vergleich zwischen molekularkinetischer und quasithermodynamischer Theorie der Kinetik einer Phasenumwandlung, Z. Physikalische Chemie, N.F. **68**, 113 (1969).

[20] Adam, G.: Kinetik des zweidimensionalen Ising-Modelles in der Bragg-Williams-Näherung: kleine Auslenkungen aus dem Gleichgewicht, Z. Physikalische Chemie N.F. **74**, 78 (1971).

[21] Volkenstein, M. V., Gotlib, Y. Y., and Ptitsyn, O. B.: The Kinetics of Cooperative Processes, Soviet Physics-Solid State **3**, 306 (1962).

[22] Gotlib, Y. Y.: The Kinetics of Cooperative Processes. The Temperature Dependence of Relaxation Properties of the Simplest Cooperative Systems (The Ising Model) Soviet Physics-Solid State **3**, 1574 (1962).

[23] Hill, T. L.: Introduction to Statistical Thermodynamics, Reading-London 1960, p. 254.

[24] Hodgkin, A. L. and Huxley, A. F.: A Quantitative Description of Membrane Current and its application to Conduction and Excitation in Nerve, J. Physiology (London) **117**, 500 (1952).

[25] Chapman, D. and Wallach, D. F. H.: Recent Physical Studies of Phospholipids and Natural Membranes in: (Chapman, D., ed.) Biological Membranes, London and New York (1968) p. 125 ff.

[26] Träuble, H. and Haynes, D. H.: The Volume Change in Lipid Bilayer Lamellae at the Crystalline-Liquid Crystalline Phase Transition, Chemistry and Physics of Lipids **7**, 324 (1971).

[27] Engelman, D. M.: Lipid Bilayer Structure in the Membrane of Mycoplasma Laidlawii, J. Molecular Biology **58**, 153 (1971).

[28] Luzzatti, V.: X-ray Diffraction Studies of Lipid-Water Systems, in: (Chapman, D., ed.) Biological Membranes, London and New York 1968, p. 71 ff.

[29] Pechhold, W.: Molekülbewegung in Polymeren, Kolloid-Z. und Z.f. Polymere **228**, 1 (1968).

[30] Kramer, L.: private communication.

[31] Salem, L.: The Role of Long-Range Forces in the Cohesion of Lipoproteins, Canadien J. Biochemistry and Physiology **40**, 1287 (1962).

[32] Schwarz, G.: General Co-operative Kinetics on a Linear Ising Lattice, Berichte der Bunsengesellschaft **75**, 40 (1971).

A Model for Muscle Contraction

Kazuhisa Tomita

1. Introduction

It has been shown repeatedly that, beyond a threshold of a certain parameter, cooperativity leads to a coherent behaviour of a macroscopic system. In other words, the coherent behaviour of a few degrees of freedom predominates the overall situation.

In a closed system near equilibrium this property is recognized very often as a replicatory coherence, i.e. as a structural regularity either in an ordinary or in a momentum space and its cooperative onset is called "phase transition".

Quite similar situations emerge in the steady state of an open system which is far from equilibrium; however, in this case coherence appears not only in the replication behaviour, but also in its regulatory and metabolic behaviours. Data processing and collective response or perception are the examples of the regulatory coherence, and mechano-chemical conversion is an example of the metabolic coherence. It is interesting to note that all three kinds of coherence seem to be required for a biological system to be in a "living state". It is a "new state" of a huge agregate of matter, which is induced by an influx of matter, information and energy, "new" in the sense that it is difficult to anticipate such a state while one is looking at the neighbourhood of an equilibrium.

It is widely known that life is impossible without mechanical motion, i.e. without movement in space of the organism and its functional parts. Among various topics which fall into this category, muscular contraction is a classical problem in biophysics, and is probably the best studied process of mechanical motion in living nature [1-4].

The available experimental information on the behaviour and mechanism of muscle contraction may be classified roughly into three classes according to the means of observation, i.e. (a) structural data based on X-ray and electron microscope investigation [5-10], (b) data obtained by thermal and mechanical measurements [13-15] and (c) biochemical data indicating the role of individual molecular species as they are involved in the process of contraction.

Structural data based on X-ray scattering [5-7] and electron microscopy [8-10] on frog sartorious muscles indicate that main components of the sarcomere of a cross-striated muscle are thick myosin filaments and thin actin filaments which may move relative to each other. From the myosin rod there project out heavy-meromyosin heads towards actin threads, which are double helices themselves, ready to form cross-bridges.

It has been recognized that the main differences between various states of muscles, e.g. rest, rigor, and contraction, can be located in the behaviour of myosin heads, or rather in that of

cross-bridges. In particular, it was established by X-ray analysis that a "pronounced azimuthal and possibly also radial movement of cross-bridges occurs during contraction". These movements were interpreted as repetitive cyclic interaction of myosin heads with the active sites of actin filaments, which results in the development of steady sliding force between the two different kinds of filaments. This is the sliding model of Huxley (A. F.) and Huxley (H. E.). [11].

Huxley's mathematical representation [11-12] aimed at explaining the physiological observations of A. V. Hill [13-15]. These observations included both mechanical and thermal measurements. An important difference between muscle contraction and an ordinary elastic deformation lies in the fact that the external stress P determines the "time rate of the strain" in the former, and not the "strain" itself as in the latter. Hill [13] summarized his observations by giving an empirical formula for the rate v of the isotonic contraction in terms of the externally applied stress, i.e.

$$v = \frac{b(P_0 - P)}{a + P} \tag{A}$$

where P_0 is the maximum possible stress to be generated, being called isometric stress. It was later shown that P_0 is proportional to the length of the region Λ in which the myosin rods and actin threads overlap [16-17].

$$P_0 \propto \Lambda. \tag{B}$$

Hill [13] also measured the heat release associated with the state of contraction. It was established that there exists a finite amount of heat release even in an isometric contraction, which is called "maintenance heat". When there appears a shortening motion, an extra amount of heat is released, which is proportional to the shortening rate, thus it was called "shortening heat". The total extra energy release $\Delta\dot{E}$ which is the sum of work done against the external stress and the heat of shortening, was found to be given by

$$\Delta\dot{E} = Pv + \alpha_s v = b'(P_0 - P) \tag{C}$$

where α_s is the parameter which characterizes the heat of shortening. It was later found by the same author [15] that α_s actually depends on the external stress P, and was given by the formula

$$\alpha_s = 0.16\, P_0 + 0.18\, P \tag{D}$$

The behaviour of muscle on stretch, or lengthening, is a problem yet to be elucidated; however, it was reported by Hill [13] and was later confirmed by Abbot, Aubert and Hill [18], that the "heat of lengthening" is actually negative (E). Further it was shown by Hill [19] that it is possible to dispense with a certain amount of chemical reaction by way of the mechanical work done by the external stress during the stretch (F). This last point is connected with an essential feature of the energy conversion involved in muscle contraction. Namely, the chemical energy seems to be converted directly into mechanical energy, without essentially passing through the stage of thermal energy as in the case of heat engine. This kind of process is called "mechano-chemical conversion" and is quite characteristic of the muscle contraction.

The chemical processes underlying muscle contraction have long been the subject of biochemical analysis. It has been established that adenosine triphosphate (ATP) plays a central role [2, 20-21] in the mechanism and energetics of the interaction between myosin and actin, i.e. the effect of activating and energizing the cross bridge comes from ATP (about 10 Kcal per mole) on its liberation of terminal phosphate ion. In other words, a strong enzymatic activity was attri-

buted to the combined acto-myosin bridge for this hydrolytic reaction. This property was considered to be connected with the fact that the super-precipitation, i.e. the acto-myosin complex, disperses away on increasing the concentration of ATP. It certainly has an effect of breaking the acto-myosin bonds.

On the other hand it was made clear that Ca^{++} ions are needed for the formation of acto-myosin bonds. In fact the behaviour of the contractile system has been shown entirely controlled by the presence of Ca^{++} ions [3], i.e. they control the interaction of actin and myosin through the aid of troponin, the Ca^{++} receptor, and trapomyosin, the mediator of the signal from troponin to actin. In the absence of Ca^{++} troponin represses the interaction between myosin and actin, and Ca^{++} ions act as de-repressors, giving rise to a burst of interaction.

Based on biochemical information and on the idea of sliding model a number of conjectures have been put forth for the consecutive steps through which the relative sliding of action thread against myosin rod emerges [22-29]. Although the details have to be confirmed by experiments, the essential behaviour of the system may be described in a broad kinetic language, which is not restricted by strict molecular details of particular mechanism.

Since the original proposal of the sliding theory, there have been put forth several theories which try to understand the underlying dynamics: however, as to what extent the cooperativity plays a role the opinions diverge.

In this paper a simple kinetic theory is presented for muscle contraction which is capable of describing most of the observed characteristics, and in particular covers the following points, i.e.

(1) a kinetic representation of the mechano-chemical conversion,
(2) an explicit inclusion of adenosine-triphosphate (ATP) as a metabolic agent,
(3) an explicit inclusion of Ca^{++} as a regulatory agent,
(4) the cooperative onset of sliding motion.

2. Transconfirmation cycle (metabolic cycle)

Let us focus our attention on the state of myosin bridges, or projections, and denote the number density of the closed bridges in the normal state by α. In an activated state, i.e. in the state of "contraction", there appears an excited conformation of the myosin projection of which the concentration is indicated by β. This process is accompanied by the consumption of ATP, of which the concentration is indicated by T. One may write down a simple reaction formula for this process, i.e.

$$\alpha + T \rightleftharpoons \beta. \tag{I}$$

The forward rate constant is denoted by k_t. A small backward rate k_t' is assumed, which occurs probably through a different channel.

The excited conformation on the other hand, is known to be responsible for the generation of the stress which causes actin filaments to slide against myosin rods. Denoting the density of the generated stress by P, one may write down a reaction formula for the second process as

$$\beta \longrightarrow \alpha + P, \tag{II}$$

where k_s is the forward rate constant.

When an applied stress counter-balances the generated stress exactly, there appears no motion, although the reaction goes on in a cyclic manner, and the muscle is said to be in the state of

"isometric contraction". When, however, the external stress fails to counter-balance the generated stress, there must appear a relative sliding motion between actin and myosin. In this case the reaction (II) cannot exhaust the backward channel.

The discrepancy in the stress may be compensated only by an extra counter stress due to the relative motion. In other words, if the channel (II) is reserved for the generation of stress which is numerically equal to the applied stress, there has to be a surplus concentration in β, which must come back to α via another channel producing relative motion. Denoting the concentration of the generated momentum by ψ, and noting that the state $\psi = 0$ is self-supporting, one can write down a third formula, i.e.

$$\beta + \psi \longrightarrow \alpha + 2\psi, \tag{III}$$

where k_β is the forward rate constant, and the feed-back is taken into account in its simplest form.

When the motion starts there appear two additional processes which tend to restrict the motion. First, the system performs a reversible mechanical work against the external stress. This process is symbolically described by

$$P + \psi \longrightarrow C, \tag{IV}$$

Where k_p is the forward rate constant, and C corresponds to the mechanical work. Secondly, there appears an irreversible rapture and formation of the cross-bridges forced by the sliding motion, which is different from the reversible transconformation by nature. The forced rapture is described by

$$\alpha + \psi \longrightarrow A \tag{V}$$

with the forward rate constant k_α, and the formation of cross-bridges is described by

$$A + \psi \longrightarrow \alpha \tag{VI}$$

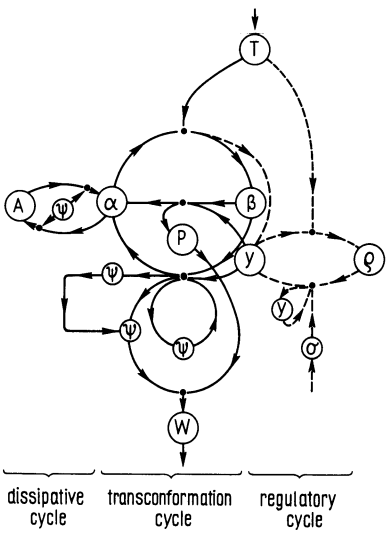


Fig. 1
Schematic representation of the set of kinetic equations with regulatory mechanism. The set consists of three cycles which correspond to transconformation, dissipation, and control by Ca^{++} ions.

with the forward rate constant k_A. One might call the last two processes as "dissipative cycle", which is not very different from ordinary friction.

The symbolic formulae (I) to (VI) completes the transconformation cycle, or the metabolic cycle, as is shown in Fig. 1, and one is now ready to write down the kinetic equations.

Assuming that the stress P and the concentration T of ATP are controlled externally, equations for the time rate of change of the quantities A, α, β, and ψ are written down as follows.

$$\frac{dA}{dt} = k_\alpha \alpha\psi - k_A A\psi, \tag{1}$$

$$\frac{d\alpha}{dt} = -\frac{dA}{dt} - \frac{d\beta}{dt}, \tag{2}$$

$$\frac{d\beta}{dt} = k_t T\alpha - k_t'\beta - k_s\beta - k_\beta\beta\psi, \tag{3}$$

$$\frac{d\psi}{dt} = k_\beta\beta\psi - k_p P\psi - k_A A\psi - k_\alpha\alpha\psi. \tag{4}$$

It is not difficult to solve equations (1) to (4) for a steady state condition, and discuss the stability of the solutions.

The steady state solutions for ψ and β are found as follows

(1) Isoletric solution:

$$\psi = 0, \tag{5a}$$

$$\beta = \frac{k_p}{k_\beta}(a + P_0), \tag{5b}$$

which is stable only for $P \gtrsim P_0$.

(2) Isotonic solution:

$$\psi = \frac{b(P_0 - P)}{a + P}, \tag{6a}$$

$$\beta = \frac{k_p}{k_\beta}(a + P), \tag{6b}$$

which is stable only for $P < P_0$. It is clear that the situation $P = P_0$ corresponds to the marginal stability.

All the parameters appearing in the above expressions may be expressed in terms of rate constants and the quantities P, T, and A, which are considered constant during the process of contraction, i.e.

$$a = \frac{2k_A}{k_p}A, \tag{7}$$

$$b = \frac{k_s + k_t'}{k_\beta}, \tag{8}$$

and

$$P_0 = \left(\frac{k_\beta k_t T}{k_s + k_t'} - 2k_\alpha\right) \frac{k_A}{k_\alpha k_p} A = \eta a, \tag{9}$$

where

$$\eta = \frac{k_t T k_\beta}{2k_\alpha(k_s + k_t')} - 1. \tag{10}$$

One has only to remember that ψ stands essentially for the velocity V in order to recognize that (6a) has exactly the same form as Hill's experimental formula (A) for the isotonic situation. It is interesting to note that the maximum stress P_0, i.e. the isometric stress, vanishes for a finite value T_r of T, where

$$T_r = \frac{2k_\alpha(k_s + k_t')}{k_\beta k_t} \tag{11}$$

is standing for the critical ratio of cleavage rate vs. clamp rate. For $T < T_r$ the state of contraction cannot exist, and the structure of (11) suggests that the situation corresponds to the state of rigor, or superprecipitation because the clamping dominates the situation. It is obvious that the quantity A is proportional to the length Λ of the overlapping region between two kinds of filaments, so that equation (9) indicates that P_0 is proportional to Λ, which is the result (B) established experimentally.

Energetics of shortening may be discussed on the same basis if one supposes that the total energy influx is given by

$$\dot{E} = -\epsilon\left(\frac{\partial T}{\partial t}\right)_{\text{consumption}} \tag{12}$$

The maintenance energy for the steady state is given by

$$\dot{E}_m = \epsilon \frac{k_A k_s k_t T}{k_\alpha(k_s + k_t')} A, \tag{13}$$

and the total extra energy release is given by

$$\Delta\dot{E} \equiv \dot{E} - \dot{E}_m = b'(P_0 - P), \tag{14}$$

where

$$b' = \xi b, \tag{15}$$

and

$$\xi = \frac{\epsilon k_p k_t'}{k_s + k_t'}. \tag{16}$$

The heat of shortening q is obtained as the difference between the extra energy release and the work Pv done by the system, and it is found that

$$q = \alpha_s v, \tag{17}$$

where

$$\alpha_s = (\xi\eta^{-1})P_0 + (\xi - 1)P. \tag{18}$$

One can recognize that equation (14) is identical with Hill's result (C) and the relation (18) is of the same form with (D) observed by Hill.

3. Regulatory cycle

It is not difficult to introduce a regulatory mechanism due to Ca^{++} ions [3], and examine the coupling with the foregoing mechanism operating in the contractile system. It is known that the Ca^{++} binding capacity of the sarcoplasmic reticulum is much greater than that of contractile system, and that this property is maintained by the consumption of ATP. This process may be represented by

$$\gamma + T \longrightarrow \rho, \tag{VII}$$

where γ stands for the concentration of liberated Ca^{++} ions, which may easily be bound to troponin sites on actin filaments, ρ stands for that of captured Ca^{++} in the reticulum, and k_γ is the forward rate constant.

When a muscle cell is stimulated, the action potential propagates over the whole cell through sarcolemma, the depolarization of the membrane being conducted into the muscle cell through the transverse tubule (T-system). The membrane of the sarcoplasmic reticulum, which is facing the T-system, is then depolarized, and this triggers the release of Ca^{++} ions from the reticulum. The Ca^{++} ions, thus released, induce the interaction between thick and thin filaments, which has been repressed in their absence, and this leads to the onset of the sliding motion in the contractile system. In addition it is recognized experimentally that Ca^{++} release occurs in a regenerative way. Introducing the intensity of depolarization σ one may summarize the above process in the following form

$$\rho + \sigma + \gamma \rightleftharpoons 2\gamma, \tag{VIII}$$

where k_ρ is the forward rate constant, and a small backward rate k'_ρ is assumed. (VII) and (VIII) form the Ca^{++} cycle; however, in order for $\gamma(\neq 0)$ to trigger the interaction of thick and thin filaments, the foregoing Eqs. (1) to (4) have to be modified. In the simplest case, however, β may be considered always conjugate with γ, as shown in Fig. 1. In the kinetic equation this means that β is replaced by $\beta\gamma$ everywhere.

In this case it is clear that most of the results obtained in the previous section are unaltered, except the value of β which is now clearly dependent on the value of γ. On the other hand, the steady state solutions for the regulatory cycle yields the following results.

1 Rest solution:

$$\gamma = 0, \tag{19}$$

which is stable only for $\sigma \lesssim \sigma_c$.

2 Contraction solution:

$$\gamma = \frac{k_\rho \rho \sigma - k_\gamma T}{k_\rho} = \frac{k_\rho \rho}{k'_\rho}(\sigma - \sigma_c), \tag{20}$$

which is stable only for $\sigma > \sigma_c$. In these equations σ_c is given by

$$\sigma_c = \frac{k_\gamma T}{k_\rho \rho}, \tag{21}$$

and it is clear that the situation $\sigma = \sigma_c$ corresponds to the marginal stability, which indicates that there exists a finite threshold for the excitation. For the fixed value of σ the marginal stability is represented by T_R, which is defined by

$$T_R = \frac{k_\rho \rho \sigma}{k_\gamma}. \tag{22}$$

This indicates that for $T \gtrless T_R$ there is expected a state of rest, or relaxation. This seems to correspond to the disappearance of super-precipitation on increasing the concentration of ATP.

In so far as the steady state motion is independent of the value of γ, Ca^{++} cycle acts as an ideal switch for the contraction. In practice, however, it is evident that there exist morphological upper bounds for the number of myosin bridges as well as that of active sites on actin filaments. These facts modify the behaviour of the system for small values of γ, although the details are not discussed here.

4. Discussion

In the same way one may treat the case of lengthening, again with coupling with Ca^{++} control. Although the details are not given here, the following results may be mentioned.

(1) Corresponding to formula (A) one finds

$$v_+ = \frac{b_+(P_{0+} - P)}{a - P}, \tag{23}$$

where $$P_{0+} > P_0. \tag{24}$$

(2) It is shown that by doing work to the system from the outside, one may actually dispense with the chemical reaction which is responsible for the metabolism (cf (F)).

(3) The lengthening heat q_ϱ is expressed as $\alpha_\varrho v_+$, and

$$\alpha_\varrho = (\xi \eta_+^{-1}) P_{0+} - (\xi - 1) P, \tag{25}$$

which may be negative, provided the value for the case of shortening may be used for the case of lengthening (cf. (E)).

Under the assumption of conjugate relation between β and γ, one may write down a kind of "phase" diagram covering all the situations. For example, suppose the value of T is fixed at an appropriate value, and increase the external stress P, then there appear various "phases" in the following order: isotonic shortening → isometric situation → isotonic lengthening → forced relaxation. This seems to correspond roughly to what is happening in the real muscle.

Although the time independent characteristics of a spatially uniform system correspond satisfactorily to the behaviour of muscle, the model is yet to be tested by looking into the transient behaviour and the non-uniform situation. In addition there is a certain amount of evidence that the existence of an ordinary elastic component cannot be neglected in trying to understand the transient behaviour of real muscle [27-28]. The problem of morphological restriction, mentioned in the previous section, should also be important in treating the real muscle.

The author wishes to thank Prof. H. Shimizu and Prof. S. Ebashi for their kind introduction to this field of research.

References

[1] Katz, B.: Nerve, Muscle, and Synapse. New York 1966.

[2] Bendall, J. R.: Muscles, Molecules, and Movement. New York 1969.

[3] Ebashi, S. and Endo, M.: Progr. Biophys. and Mol. Biol. **18**, 123 (1968).
Ebashi, S., Endo, M., and Ohtsuki, I.: Quarterly Review of Biophysics, **2**, 351 (1969).

[4] Vol'kenstein, M. V.: Soviet Physics Uspehi **13**, 269 (1970).

[5] Huxley, H. E. and Brown, W.: J. Mol. Biol. **30**, 383 (1967).

[6] Hanson, J.: Quart. Rev. Biophys. **1**, 177 (1968).

[7] Elliott, G., Lowy, J., and Millman, B.: J. Mol. Biol. **25**, 31 (1967).

[8] Hanson, J. and Huxley, H. E.: Nature **172**, 530 (1953).
Huxley, H. E.: Biochim. Biophys. Acta **12**, 387 (1953).

[9] Huxley, H. E.: J. Mol. Biol. **7**, 281 (1963).

[10] Huxley, H. E. and Hanson, J.: in Structure and Function of Muscle (ed. G. H. Bourne). Vol. 1, p. 183. New York 1960.
Huxley, H. E.: Sci. American **213**, 18 (1965).

[11] Huxley, H. E. and Hanson, J.: Nature **173**, 973 (1954).
Huxley, A. F. and Niedergerke, R.: Nature **173**, 971 (1954).

[12] Huxley, A. F.: in Progress in Biophysics and Biophysical Chemistry (eds. J. A. V. Butler and B. Katz) Vol. 7, p. 257, London 1957.

[13] Hill, A. V.: Proc. Roy. Soc. **B126**, 136 (1938).

[14] Hill, A. V.: Proc. Roy. Soc. **B136**, 195, 399, 420 (1949); **B137**, 268, 320 (1950); **B139**, 464 (1952); **B141**, 161 (1953); **B148**, 397 (1958).

[15] Hill, A. V.: Proc. Roy. Soc. **B159**, 297 (1964).
Hill, A. V.: Proc. Roy. Soc. **B159**, 319, 589, 596 (1964).

[16] Edman, K. A. P.: J. Physiol. **183**, 407 (1966).

[17] Gordon, A. M., Huxley, A. F., and Julian, F. J.: J. Physiol. **184**, 170 (1966).

[18] Abbott, B. D., Aubert, X., and Hill, A. V.: Proc. Roy, Soc. **B139**, 86 (1951).

[19] cf. Hill, A. V.: Science, **131**, 897 (1960).

[20] Engelhardt, W. A. and Ljubimova, M. N.: Nature **144**, 669 (1939); Biokhimiya **4**, 716 (1939); **7**, 668 (1942).

[21] Lymn, R. W. and Taylor, E. W.: Biochemistry **10**, 4617 (1971).

[22] Davies, R. E.: Nature **199**, 1068 (1963).

[23] Oosawa, F., Asabura, S., and Ooi, T.: Progr. Theor. Phys., Suppl. No. 17 (1961).

[24] Hill, T. L.: Proc. Nat. Acad. Sci. U.S.A. **57**, 227 (1967); Hill, T. L. and White, G. M.: Proc. Nat. Acad. Sci. U.S.A. **58**, 454 (1967).
Hill, T. L.: Proc. Nat. Acad. Sci. U.S.A. **61**, 98 (1968).
Hill, T. L. and White, G. M.: Proc. Nat. Acad. Sci. U.S.A. **61**, 514, 889 (1968).

[25] Deshcherevskiĭ, V. I.: Biofizika **13**, 928 (1968).

[26] Vol'kenshteĭn, M. V.: Molek. Biologiya, **3**, No. 6 (1969); Biochim. Biophys. Acta **180**, 562 (1969).

[27] Huxley, H. E.: Science **164**, 1356 (1969).

[28] Huxley, A. F. and Simmons, R. M.: Nature **233**, 533 (1971).

[29] Shimizu, H.: (to be published).

Organisation and Long Range Selective Interaction in Biological and other Pumped Systems

H. Fröhlich

The present century has witnessed a growing understanding of the structure and the dynamical behaviour of atoms and molecules. From this, it is often thought, follows uniquely the behaviour of larger systems composed of atoms (and molecules) such that great surprises in their properties cannot be expected. Biological systems would then differ from systems usually considered in physics by their much greater complexity but could be understood without the introduction of new concepts.

That this is a fallacy follows already from the statement: We have excellent knowledge of the atomic structure and behaviour of metals; consequently we can derive from this the properties of all machines built of metals (even those not yet invented) without the introduction of new concepts. More general, macrophysics uses concepts which are quite different from those used in microphysics. Deriving a complete set of microscopic solutions and then selecting from them those which are macroscopically of interest, even if it were possible, is quite useless. For this would present a problem even more formidable than finding these solutions in the first place. Derivation of macro laws from microphysics in fact requires sometimes considerable subtlety: The appropriate macro concept must first be expressed in terms of micro concepts; the macroscopic dynamical behaviour must then be extracted from the exact micro laws.

One might have been tempted to suggest that all important macro concepts have been found quite long ago. This is disproved by the recent discovery of macro wave functions which dominate the dynamical behaviour of superconductors and of superfluid helium. As a consequence these systems which from the point of view of the spatial arrangement of the electrons (in superconductors) or of the He atoms show the same disorder as other liquids exhibit a very subtle "order of motion".

Some time ago I have proposed that biological systems when they are active i. e. when energy is pumped through them should also exhibit an "order of motion" in terms of electric vibrations (polarisation waves) of the materials [1]. While superfluids and superconductors exhibit this type of order at very low temperatures in equilibrium it is proposed that biological systems should exhibit it at room temperature through biological pumping. It should be mentioned that lasers, too, exhibit this type of order, maintained by optical pumping.

This implies that one or a few modes of vibration are very strongly, coherently, excited. It might be thought that maintenance of coherent excitation requires an exceptionally high rate of biological pumping. Once excited, however, coherent vibrations lead to various kinds of deformation [2] and displacements which tend to stabilize the coherent excitation.

Though the possibility of this type of excitation has been derived previously [1] I should like to present here a more elegant derivation. Consider a system with polarisation waves with frequencies in a narrow range between ω_1 and ω_z which is placed into a heat bath of temperature T. Let energy be supplied to all modes ω_k of the polarisation waves at a rate of s quanta independent of k. Then if n_k is the number of quanta in ω_k and if we replace $\hbar\omega_k/kT$ by ω_k then the kinetic equation for n_k is

$$\dot{n}_k = s - \phi\,(n_k e^{\omega_k} - (n_k + 1)) - \chi \sum_j (n_k\,(1 + n_j)\,e^{\omega_k} - n_j\,(1 + n_k)\,e^{\omega_j}) \quad (1)$$

This assumes interchange of single quanta with the heat bath described by a transition probability ϕ, and of processes involving absorption and emission of a quantum (second order processes) with transition probability χ. Furthermore detailed balance for each type has been assumed. More general, of course, ϕ could depend on k, and χ on k and j. Both ϕ and χ are temperature dependent.

Consider the stationary case, $\dot{n}_k = 0$. Summation of (1) over all z modes ω_k yields

$$sz = \phi \sum_{k=1}^{z} (n_k e^{\omega_k} - (n_k + 1)) \quad (2)$$

The χ terms do not contribute. Using

$$N = \sum n_k, \quad (3)$$

and

$$\omega_1 \leqslant \omega_k \leqslant \omega_z \quad (4)$$

we find from (2) that

$$\frac{s/\phi + 1}{e^{\omega_z} - 1} < \frac{N}{z} < \frac{s/\phi + 1}{e^{\omega_1} - 1} \quad (5)$$

which shows that for sufficiently large energy supply N increases proportional to s.

On the other hand equation (1) with $\dot{n}_k = 0$ yields immediately

$$n_k = \left(1 + \frac{\phi}{z\chi}(1 - e^{-\mu})\right) \frac{1}{e^{\omega_k - \mu} - 1} \quad (6)$$

where

$$e^{-\mu} = \frac{\phi + \chi \sum (1 + n_j)}{\phi + \chi \sum n_j e^{\omega_j}} \quad (7)$$

and (2) has been used. In view of (7) equation (6) represents a rather complicated implicit equation for n_k.

It is nevertheless possible to draw conclusions on the general behaviour of n_k if we notice that (2), (7) together with the requirement $n_k \geqslant 0$ yields

$$\omega_1 > \mu \geqslant 0 \quad (8)$$

For then from (6),

$$N = \sum n_k < \left(1 + \frac{\phi}{z\chi}(1 - e^{-\omega_1})\right) \sum \frac{1}{e^{\omega_k - \mu} - 1} \quad (9)$$

It will be noticed that s has been eliminated from expression (6) for n_k through (2). Similarly the upper limit (9) for N does not contain s explicitly. Now from (5) it follows that sufficiently large s require $N \propto s$. For large s this can be achieved only if μ comes very close to ω_1, so that n_1 is very much larger than all the other n_k. The system thus exhibits a phenomenon very well known for massive particles in thermal equilibrium: Einstein condensation of a Bose gas.

It should be emphasised that this result rests on the importance of the χ terms which describe a non linear interaction with the heat bath. We might have added non linear direct interaction terms between polarisation waves. Such terms would counteract the influence of the heat bath; if large compared with both ϕ and χ terms they would tend to establish for n_k a Planck distribution ($\mu = 0$) at a temperature higher than T. Coherent excitation (Bose condensation) of ω_1 thus implies that the interaction with the heat bath is predominant.

Strong excitation of the single mode ω_1 may have far reaching secondary consequences. Thus, for instance, through non linear effects it may lead to deformations of the system and stabilisation of the coherent mode.

As an example [3] consider two pumped systems, a and b separated by a mean distance

$$R \gtrsim L \tag{10}$$

where L is the linear dimension of each of the systems. In view of (10) these systems interact principally through the polarisation waves with zero wave vector, $q = 0$ i. e. through their giant dipole oscillations. In the system consisting of the combination of a and b all polarisation waves $q \neq 0$ can be treated as independent but $q = 0$ gives rise to combined vibrations modified by the interaction. Let ω_a and ω_b, respectively, be the lowest $q = 0$ frequencies of the separated systems (assumed non-degenerate) then the frequencies $\omega_\pm$ of the combined systems are given by

$$\omega_\pm^2 = \frac{\omega_a^2 + \omega_b^2}{2} \pm (\Delta^2 + \frac{\beta^4}{\epsilon_\pm^2})^{1/2} \tag{11}$$

where if $\omega_a \geqslant \omega_b$, and $A \sim 1$,

$$\Delta = \frac{\omega_a^2 - \omega_b^2}{z}; \quad \beta^2 = Ae^2 \sqrt{z_a z_b} / MR^3. \tag{12}$$

z_a and z_b respectively are the numbers of individual ions constituting the two systems; all are assumed of equal mass M; $\epsilon_\pm$ is the dielectric constant of the intermediate medium at frequency $\omega_\pm$.

Clearly for large distance R the frequency ω_- need not be the lowest frequency of the system. As R is considered to decrease, however, β^2 increases so that ω_- decreases and may become the lowest frequency, in particular in the case of resonance when $\omega_a \simeq \omega_b$. In this case, ω_- is strongly excited by pumping so that the systems a and b attract each other strongly and hence approach; hence R is further reduced, etc. Thus through a (mildly) non linear effect, ω_- is moved below the band of polarisation waves $q \neq 0$, and the oscillations are presumably stabilized (I have not yet considered this in detail).

In the case of resonance, $\omega_a \simeq \omega_b$, the interaction is particularly strong and of long range $\propto 1/R^3$. B. Holland [4] has proposed to denote this as dynamic specificity and invoked it to interpret the specific attraction which in meiosis brings a homologous pair of chromosomes together – an interaction which according to Watson [5] has till now remained "a great mystery".

The vibrations, if realised, would have many other consequences on biological dynamics and in fact could dominate many such processes. Thus they might be relevant for the action of enzymes [6, 7] where according to Koshland [8] ". . . the essential mystery of their enormous catalytic power remains . . ."

Another consequence refers to the storage and transport of energy. Thus Szent György [7] has emphasized that whenever chemical energy is released at one place but used in another then it must travel – most likely as an electric wave. Paul Fatt has suggested that in the eye when the energy of an absorbed light quantum travels through a layer of about thousand membranes, this might take place in the form of a coherent wave as treated above. Experiments in this respect are in progress.

One of the main questions to be answered before experimental evidence can be obtained refers to the frequency of the vibrations. If they are connected with oscillations in the dipolar field of a membrane, then my estimate is $10^{11} - 10^{12}$ sec^{-1}. Quite different frequencies, however, due to other mechanisms might be possible.

The ideas described here rest on the excitation by pumping of a few degrees of freedom very much beyond their excitation in thermal equilibrium. The rest of the system is assumed to stay in thermal equilibrium. Now all thermodynamic functions such as entropy have magnitudes proportional to the total number of atoms. Their change through excitation of a few degrees of freedom may, therefore, be expected to be minute – yet this must have a dominating effect on the macro behaviour of the system. This is a very general feature; a cup for instance (kept at constant temperature) is out of thermal equilibrium by the few degrees of freedom determining its shape which in thermal equilibrium would be a solid sphere. Yet these few degrees of freedom dominate the macro behaviour (the usefulness in this case). It is my impression that we lack here a concept which might be described as "organisation" and which should give a quantitative measure for it.

Finally in view of the sociological problems to be discussed in the last section of this book it may be of interest to give a sociological interpretation to the above calculation. Consider a large population (heat bath) in which a fraction is interested in rather exclusive activities ω_k (in a narrow range (4)). In spite of this the exclusive fraction retains a strong tendency to be seen to conform at their level (strong interaction with the heat bath). They are subjected to external influences (s) which tend to increase their number, retaining the possibility of small variations ($\omega_1 \leqslant \omega_k \leqslant \omega_z$). When the external influences become too strong then the tendency to conform drives most of them into e x a c t l y the same activity (condensation into ω_1; safety in numbers). Ultimately in spite of the tendency to conform they will transform the whole of society (non linear processes).

Note added on May 12th. The first confirmation of the prediction of the coherent excitation of vibrational modes by biological pumping has just come to my notice [9] though the authors were not aware of the prediction. With the help of laser techniques they find a very strong very sharp line – shifted by 170 Hz from the primary one – in a suspension of ATP containing erythrocyte ghosts (red blood cells which have their haemoglobin removed). There is evidence that this line arises from the $N_a^+ - K^+$ "pump" as it disappears when this pumping action is inhibited.

The very low frequency might arise from plasma oscillations of the ghosts though more subtle interpretations referring to properties of single ghosts are feasable.

References

[1] Fröhlich, H.: Long-Range Coherence and Energy Storage in Biological Systems. Intern. J. Quantum Chem. **2**, 641 (1968).
[2] Fröhlich, H.: J. of Collective Phenomena (to be published).
[3] Fröhlich, H.: Selective Long Range Dispersion Forces between Large Systems, Phys. Letters 39A, 153 (1972).
[4] Holland, B.: Dynamic Specificity and the Pairing of Homologous Chromosomes in Meiosis. J. Theor. Biology (in print).
[5] Watson, J. D.: Molecular Biology of the Gene, New York, p. 187 (1970).
[6] Fröhlich, H.: Long Range Coherence and the Action of Enzymes. Nature **228**, 1093 (1970).
[7] Conference report. From Theoretical Physics to Biology (in print).
[8] Koshland, D. E. and Neet, K. E.: Am. Rev. Biochem. **37**, 359, 380 (1968).
[9] Bargeron, C. B., McCally, R. L., Tatham, P. E. R., Cannon, S. M., and Hart, R. W.: Phys. Rev. Letters **28**, 1105 (1972).

Nonlinear Transport as a Possible Key to Physical Understanding in Biology

M. Wagner

1. Introduction

For a long time a majority of biologists have denied the principal possibility that the essential characteristics of living matter can be understood on the basis of the laws of physics. Rather, there was the belief that "understanding" in biology was something quite different from "understanding" in physics. This attitude has changed in the last decades.

On the other hand, since the fundamental work of Lotka, Volterra and others, there has been a growing branch of mathematical biology. In this branch dynamical equations for model-control-systems are established, which reproduce the behaviour of living matter or dynamical sub-units thereof [1]. These mathematical methods are very similar to those in the theory of electrical and mechanical control-systems, in the theory of non-linear oscillations and chemical reactions [2]. Therefore, a rich variety of results and models can be taken over from these branches of science.

Naturally, the mechanical as well as the electrical engineer is able to construct devices, which in their real behaviour follow the mathematical description based on the mechanical or electrodynamical principles of physics. However, in biology in most cases the mathematical description is only a phenomenological or empirical way of reproducing in the mathematical results the biological dynamics, without relying on basic principles. Therefore there is the urgent need of finding theories which on the one hand are based on the principles of physics and on the other hand display the behaviour of control systems reproducing the dynamics of living units.

At present already a considerable number of physicists are involved in this field. The overall agreement is that the theories have to belong to the huge field of non-equilibrium statistical mechanics. But within this scope a large variety of approaches is possible.

In my talk I should like to present first some of the predominant features of macroscopic control systems and non-linear differential equations and some of the consequences for biological units. In a second part the connection to physical transport theory in the molecular regime is made. My choice of the formalism will be a very direct one, from which I hope that it will be understandable also to those not working in the field. In the final part I will comment on a few special items of biology in the spirit of transport theory, in particular about the problem of dynamical energy storage and about the possibility of a modified dynamical kind of mutation theory. From this some conclusions can be made concerning the "philosophy of evolution".

2. Control-systems

Everybody in his all-day life is surrounded by all kinds of mechanical and electrical control devices. The automobile and all machines are also control-systems. For illustration we discuss briefly a very simple species of a control-system: the one, which regulates the room temperature. In this system there is a heating, which pumps energy into the room, there are losses through walls and windows and there is a steering element, the thermostat.

Mathematically, if we do not bother about some simplifications, this control system is governed by the following couple of differential equations:

$$\frac{d\tau}{dt} = -a\tau + bT \tag{1}$$

$$\frac{dT}{dt} = -cT + d \cdot \theta(\bar{\tau} - \tau) \tag{2}$$

where τ and T are the room and heating temperatures respectively. The first terms on the r.s. are the heat losses and the second ones the heat gains. $\theta(\bar{\tau} - \tau)$ is the step function and represents the thermostat; it is this term which makes the system highly non-linear.

The solution of Eqs. (1, 2) is represented in Fig. 1. We note that there is no stationary solution, but there is an asymptotically stable stationary fluctuation. It is worth emphasizing three

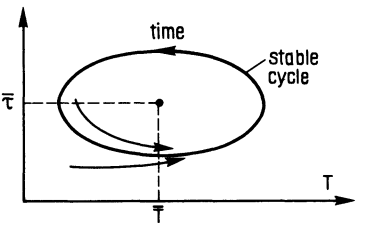


Fig. 1 Schematical diagram for the solution of the room-temperature control-system.

features which are common to a large group of macroscopic control devices. (a) There is a single stable solution, constructed by the engineer. (b) There is transport through the system, i.e. a flow of energy, of matter, of electric charge or of a combination of them. In our case the energy flows from the heating through the room to the outside world. In the automobile both matter and energy flow from the tank through the motor to the street. (c) The system is not in thermal equilibrium with its surrounding. In our case the room temperature is higher than the one outside. In a machine a singular degree of freedom is highly excited (i.e. this macroscopic excitation would correspond to millions of degrees Kelvin).

3. Dynamical system theory in biology [1]

In mathematical biology one considers quite generally kinetic equations of the form

$$\frac{dx_i}{dt} = f_i(x_1, \ldots, x_n), \; i = 1, \ldots, n \tag{3}$$

which are said to be autonomous, if the f_i do not depend on time explicitely, and non-autonomous in the other case. Normally only autonomous systems have been discussed and we

shall restrain to them here too. Since these equations are of particular interest, if f_i is non-linear, a manageable discussion of the solutions requires constraints onto them.

One may either keep to the lowest kind of non-linearity (quadratic). In this case the number of kinetic variables x_i may be high and methods of statistical mechanics may be employed. This has been done in the lectures of Montroll [3] and of Kerner [4] earlier in this conference.

On the other hand one may assume that only a few kinetic variables are coupled strongly and that the couplings to the other variables are taken as "parasitic" parameters, a term intro-

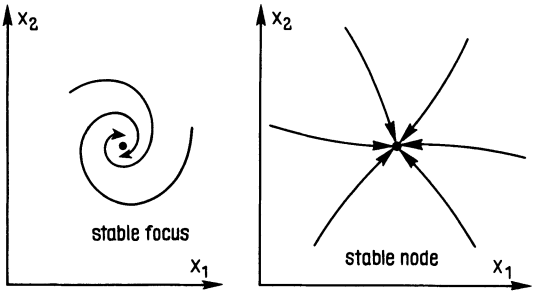


Fig. 2 Stable points of solutions, which are reached asymptotically by all paths of solution in a certain neighbourhood.

duced in the famous book by Andronov et al. [2]. The latter approach will be taken here. Naturally, constraints of this kind have to be discussed carefully in any specific case, but for the purpose of illustration we shall assume here that there are only pairs of strongly coupled variables, whereas each of this pairs is coupled to the others in an average, "parametric" way.

Under this supposition the solutions may be drawn in a plane. We are only interested in the types which have stability, because only these are of biological relevance. There are two of them, and they are depicted in Figs. 2 and 3. These types of solution are characterized by stability in a

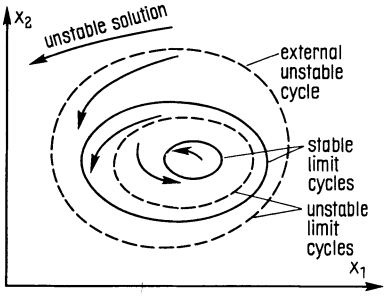


Fig. 3 The sequence of stable and unstable limit cycles, appearing in the manifold of solutions of certain non-linear differential equations of transport type. x_1 and x_2 are functions of time.

bounded region, either in the form of a "static" stability (stable singular points, see Fig. 2) or of a "dynamical" one, as shown in Fig. 3. In mathematical biology the behaviour of a biological unit may be discussed primarily in terms of these stable solutions.

A great number of dynamical properties can be explained in this way. For instance "homeostasis" would mean that the "overall" pair of kinetic variables move on a stable limit cycle. In the case of acquisition and storage of information one may think that a pulse of information (electrical, chemical or optical) throws the respective pair of kinetic variables $x_1(t)$ and $x_2(t)$ across an unstable cycle (see Fig. 3), which separates the regions of two stable cycles, and the system approaches a new stable limit cycle, i.e. the information is stored.

One may consider this as a case of a "dynamical phase transition". Thinking now of the conception that a biological unit is an arrangement of intercorrelated non-linear control systems, such a "dynamical phase transition" in one control-unit may have a steering effect onto a neighbour energy unit, so that energy is set free for reaction onto the original information pulse.

It is easy to imagine that there is an enormous number of possibilities in constructing mathematically complex dynamical control-systems of a great variety of properties. This indeed has been done for the phenomenological description of many characteristics of living matter. Examples of this kind are models for self-replication, biological clocks, self-organization, natural automata, temporal organization in cells, etc.

Although all these phenomenological ways of mathematical description are highly successful, they are not particularly satisfactory to the pure physicist, because most of these models do not even belong to the region of "phenomenological physics", not to speak from a foundation on the first principles. Therefore physicists have started to aim at the results of "Biomathematics" on the basis of physical ideas, both of phenomenological and first principle nature. In this context one must mention the work of Prigogine, Glansdorf, Lefever, Nicolis and their collaborators about dissipative structures [5] in biological systems, a work which is based on a generalization of thermodynamics. Another branch of investigation is that of Montroll [3] and of Kerner [4] about the application of methods of statistical mechanics onto the Volterra models of interacting population. Finally the work of Haken and Graham [6] should be mentioned who succeeded in deriving first principle nonlinear evolution equations for the laser, which they showed to apply also to other kinds of dynamical phase-transitions and which have a close structural similarity to the purely phenomenological kinetic equations of Eigen [7] and his school.

4. Microscopic transport theory

Transport theory up to now has been mainly concerned with regions near thermal equilibrium, which means that the fluxes have been assumed to be infinitesimally small, so that one could linearize the transport equations. This then leads what one now-a-days calls the "thermodynamic branches" of solutions.

If one tries to go far away from thermal equilibrium, at least for some singular degrees of freedom, the difficulties increase enormously and one is bound to make a series of simplifying assumptions. Let me illustrate this in a very simple system: The linear, monatomic harmonic chain, which is coupled anharmonically to a local excitation. For this system the Hamiltonian reads:

$$H = \underbrace{\sum_{\kappa} \omega(\kappa)\, a_\kappa^+ a_\kappa}_{\text{phonon system}} + \underbrace{\omega_s b_s^+ b_s}_{\text{loc.exc.}} + \underbrace{\sum_{\kappa\kappa'} [v_{\kappa\kappa'} b_s^+ a_\kappa a_{\kappa'} + \text{c.c.}]}_{\text{coupling } H_{ps}} \tag{4}$$

This is a quasi-particle operator, but we could have chosen also a system with a real particle Hamiltonian of the same form. To explore transport we introduce a phonon-density operator of the form:

$$\hat{n}(\kappa, x) = \frac{1}{2L} \sum_{\kappa'} [a_\kappa^+ a_{\kappa'}\, e^{-i(\kappa-\kappa')x}\, e^{-\alpha|\kappa-\kappa'|^2} + \text{c.c.}] \tag{5}$$

where L is the length of the chain and α is the localizing factor the magnitude of which can be chosen to our convenience [8]. If all operators are taken in the Heisenberg picture, the equation of motion for $\hat{n}(\kappa, x)$ leads to the transport equation

$$\underbrace{\left[\frac{\partial}{\partial t} + v_\kappa \frac{\partial}{\partial x}\right]}_{\substack{\text{transport} \\ \text{term}}} \langle \hat{n}(\kappa, x) \rangle = \underbrace{\frac{1}{i} \langle [\hat{n}(\kappa, x), H_{ps}] \rangle}_{\text{collision term}} \tag{6}$$

Here already the simplifying assumption is made that either there is small dispersion $d^2\omega(\kappa)/d\kappa^2 \approx 0$ or that the space fluctuations can be neglected (large α). In addition to the transport equation we have for the local excitation $\hat{n}_s = b_s^+ b_s$

$$\frac{\partial}{\partial t} \langle \hat{n}_s \rangle = \frac{1}{i} \langle [\hat{n}_s, H_{ps}] \rangle \tag{7}$$

The system of coupled equations (6, 7) can be simplified strongly, if we introduce the following approximations: (1) the spatial behaviour in the collision term is neglected (i.e. $\alpha \to \infty$ is chosen there); (2) for the time evolution of the occupation numbers $N(\kappa) = \langle a_\kappa^+ a_\kappa \rangle$ the golden-rule of quantum mechanics is applied, which here plays the role of a specific random-phase approximation; (3) the transport term in Eq. (6) is taken as a fixed quantity $T(\kappa)$ given from outside; (4) a factorization procedure of the form $\langle \hat{N}(\kappa) N_s \rangle = N(\kappa) N_s$ is chosen, which is a well-accepted approximation technique for decoupling hierarchical systems. Then the coupled systems looses its space-dependence and reduces to

$$\frac{d}{dt} N(\kappa) = -T(\kappa) + 2\pi \sum_{\kappa'} |v_{\kappa\kappa'}|^2 \{(N(\kappa) + N(\kappa') + 1)N_s - N(\kappa)N(\kappa')\} \times$$

$$\times \delta(\omega(\kappa) + \omega(\kappa') - \omega_s) \tag{8}$$

$$\frac{d}{dt} N_s = 2\pi \sum_{\kappa\kappa'} |v_{\kappa\kappa'}|^2 \{N(\kappa)N(\kappa') - N_s(N(\kappa) + N(\kappa') + 1)\} \times$$

$$\times \delta(\omega(\kappa) + \omega(\kappa') - \omega_s) \tag{9}$$

The important features to note are:

(a) The transport equations take the form of nonlinear evolution equations of the types which also appear for control-systems.

(b) The nonlinearities which appear in the equation for the singular, local and not-transporting degree of freedom s are of higher manifold than those in the equations for the transporting degrees of freedom. Therefore one may think of handling the equations for the transporting variables in a linearized way in contrast to the strong nonlinearity in the non-transporting variable. In this sense one may expect the transport-variables not to deviate strongly from local thermal equilibrium, whereas the local variables may deviate strongly.

(c) The form of the nonlinearities depends essentially on both the form of the coupling term in the Hamiltonian and on the order of perturbation theory where the random-phase approximation is made. (In our example the latter was made in second order perturbation.)

(d) The power or order of the non-linear terms is by no-means restricted to second order terms, and even for low-order interaction-terms in the original Hamiltonian there may be higher order non-linearities in the transport-equations resulting from higher-order random-phase approximations. These may be of importance if they suit for energy-conservation. Thus, in our example several successive and uncorrelated phonon-pulses may kick the local or non-transporting degree of freedom to its excited state.

It is worth going somewhat further in the discussion of our simple example. At a first glance one might think that we have not arrived at a two variable system, but at a multi-variable system. However if the external conditions are not too extreme, the phonon-variables are directly correlated to each other, either by a simple proportionality factor or by a similarly simple direct analytical dependence. Therefore we indeed have practically a two variable system.

Let us consider one of the most interesting types of solutions, drawn again in the form of a diagram in Fig. 4. n_K is the deviation of a typical phonon occupation number from its thermal equilibrium value. It is at the same time a measure for the strength of the transport. If we start imposing transport onto the system from outside, we have a stationary stable solution of the nodal or focal kind, which starts at the thermal equilibrium point. As transport increases, the stable point moves away continuously, until a critical transport is reached, where it becomes unstable. In this situation the system jumps to another stable solution, which may again be of a stationary kind, as drawn in Fig. 4. (It might as well be of a limit cycle type.) We have a transport phase-transition. The new branch of solution is denoted as a non-thermodynamic one in contrast to the thermodynamic one, which starts in thermal equilibrium.

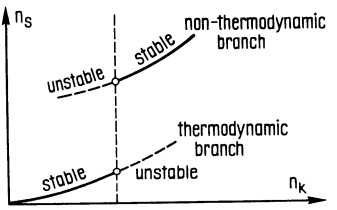


Fig. 4 Transport phase-transition in a one-dimensional chain with anharmonic coupling to the "local" (non-transporting) degree of freedom (simplified illustration!). n_S, n_K = deviation of the occupation number from thermal equilibrium.

This kind of behaviour is highly interesting for biology, since it shows that transport of rather general nature may stabilize singular local degrees of freedom in a state far above thermal equilibrium. This is very similar to the case of the automobile, where also a singular degree of freedom is excited far above thermal equilibrium. In a biological unit the stabilized high excitation may be used to build up chemical compounds of higher complexity, or, to build up order, if you allow me to use this term in such an unqualified manner.

This kind of behaviour indeed seems to have played a decisive role at the beginning of life on earth. Let me just mention the famous experiment of Miller [9] in 1953. In this experiment a gas mixture of methane (CH_4), ammonia (NH_3), hydrogen (H_2) and water vapor (H_2O) is forced to pass an electrical discharge and to circulate continuously. This is a pronounced transport system, specifically one of chemical kinetics, and it is no doubt that the pertinent transport equations are nonlinear. (It is a pure energy transport system, since it is "closed" with respect to matter transport.) After Miller's experiment had run for a week, it was found that there were amino-acids in the condensed water. Amino acids and nucleic acids are the basic chemical compound of living matter.

5. Some speculations about evolution

As everybody knows, mutation is indispensible for the evolution from lower to higher forms of life. On the other hand, for many physicists present day mutation theory is not satisfactory. The first one who gave words of this dissatisfaction was von Neumann.

It is generally assumed today that life started something like 2.6 billion years ago. And in the time until today such a complex being like the "homo sapiens" has been brought about by evolution via mutation and selection. Now, in the mutation theory of Delbrück and Schrödinger, which is still well-accepted, it is assumed that thermal fluctuation is the fundamental process in natural mutation. This is drawn very schematically in Fig. 5. We assume R to be some "configuration parameter" which stands for a spatial or chemical configuration. Now, every physicist engaged in statistical mechanics, knows, that something must be incomplete in this picture, even if one assumes that thermal fluctuation is the fundamental mutational process, because by thermal fluctuation alone higher order structures cannot be created from lower order ones.

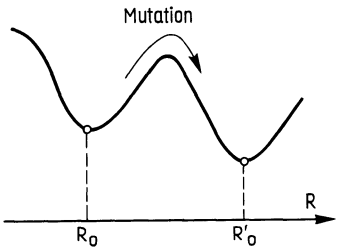


Fig. 5 Schematical and greatly simplified configuration diagram for the illustration of the Delbrück-Schrödinger mutation theory.

On the other hand, one may easily extend this mutation theory on the basis of transport theory. For instance, the introduced configuration variable R may be the singular, non-transporting degree of freedom coupled to a transport system. Then it can be highly excited above thermal equilibrium by transport, which would increase the mutation probability by orders of magnitude. Now two points of view come into strong competition with each other: (a) The new configuration R'_0 may be more stable against "transport excitations" than the old one R_0, but of selective disadvantage. (b) Equally well it may be the other way around. The third and most unique possibility, of course, is, if the mutant configuration has both an advantage in stability and in selection. Only in this case the direction of evolution can be predicted uniquely.

To conclude, let me add two speculative remarks. Firstly I think that any theoretical description of evolution, if phenomenological or more fundamental, should include the competition between stability and selective advantage, which probably would mean an extension of the Neo-Darwinian conception of Eigen. Secondly, although the fundamental process of mutation may be a matter of chance the number of ways of evolution in the competition between selective and stability advantage presumably is countable or even low, whence the question "choice or necessity" raised by Monod [10] has to be answered more in the sense of necessity.

References

[1] An excellent review about these methods is the book by Rosen, R.: Dynamical System Theory in Biology. New York 1970.

[2] We only refer to some of the standard text-books in these fields: (a) Lefshetz, S.: Stability of Nonlinear Control Systems. New York 1965.
(b) Andronov, A. A., Vitt, A. A., and Khaikin, S. E.: Theory of Oscillators. Oxford 1966.
(c) Gavalas, G. R.: Nonlinear Differential Equations of Chemically Reacting Systems. Berlin – Heidelberg – New York 1968.

[3] Montroll, E. W. and West, B. J.: Models of Population Growth, Competition and Rearrangement, this volume, p. 143.

[4] Kerner, E. H.: A Gibbs Ensemble Approach to Biochemical Kinetics, this volume, p. 137.

[5] Prigogine, I. and Lefever, R.: Theory of Dissipative Structures, this volume, p. 124.

[6] Haken, H. and Graham, R.: Synergetik – die Lehre vom Zusammenwirken. Umschau in Wissenschaft und Technik, Heft 6, p. 191, 1971.

[7] Eigen, M.: Self-Organisation of Matter and the Evolution of Macromolecules. Naturwissenschaften Heft 10, October 1971.

[8] Hardy, R. J.: Phys. Rev. **132**, 168 (1963).

[9] Miller, S. L.: A production of amino acids under possible primitive earth conditions. Science **117**, 528 (1953).

[10] Monod, J.: Le hasard et la nécessité. Paris 1970.

Theory of Hierarchical Structures

M. D. Mesarovic

1. Introduction

Concept of a cooperative phenomenon presupposes at least three conditions: The existence and explicit recognition of a set of units, or subsystems; a definite interrelationship between these units so that the performance and behavior of one depends upon the performance of others; an explicitly defined sense in which the units are evaluated as being cooperative or not.

Hierarchical systems satisfy all these three requirements and questions of cooperation can therefore be considered within that framework.

In this presentation we shall be concerned with two topics: first, we shall indicate some of the developments in a general theory of coordination; and second, we shall introduce some basic concepts of hierarchical systems.

2. Coordination theory

Consider a system with the block diagram as given in Fig. 1. There are n processes, $P_1, \ldots, P_n$ each defined simply as the transformation, i.e., P_i is defined on a set M_i, termed the set of (alternative) decisions, and on another set U_i termed the interaction set with the value of P_i in the set Y_i, termed the output set.

The processes are interconnecting so that for any n-tuple $m = (m_1, \ldots, m_n)$ u_i represents the effect of other decisions as applied on the process P_i. It is possible, therefore, to define the overall process P which is the interconnection of $P_1, \ldots, P_n$ and simply maps the n-tuple $m = (m_1, \ldots, m_n)$ into $y = (y_1, \ldots, y_n)$. In addition to the processes, the system also has n + 1 units termed decision units. n of them are in a direct contact with the processes and are termed first level units. A subsystem on the first level is referred to as a decision unit because it is assumed that each of them, say D_i, generates (produces) the output, m_i, by solving a decision problem denoted by Δ_i. The notion of a decision problem will be introduced in a precise manner for a special case, but in view of the fact that a majority of the audience are physicists, it is convenient to view Δ_i as being a "variational problem" as customarily defined in physics and m_i therefore being the solution of such a variational problem.

To be able to define the concept of coordination it is quite essential to introduce the concept of an overall system decision problem Δ. By this we mean that Δ represents in an explicit manner the overall purpose, or the goal, of the system, e.g., again in terms of a variational principle. The precise concept of the overall decision problem for a certain class of systems will be given shortly.

One more rather crucial and quite natural assumption will be made, namely, that each first level unit, say D_i, when solving its own decision problem, is taking into account only its own local process, P_i.

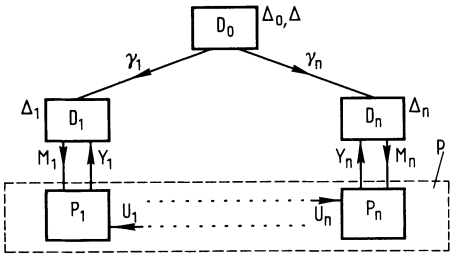


Fig. 1

The remaining unit, D_0, which is also a decision unit, is influencing the first level units through the parameters $\gamma = (\alpha, \beta)$ where $\alpha = (\alpha_1, \ldots, \alpha_n)$, $\beta = (\beta_1, \ldots, \beta_n)$, and (α_i, β_i) are the parameters for D_i. The set of all γ's is parameterizing the set of decision problems on the first level in the sense that for any given (α, β) there is specified a family of decision problems $\Delta_i(\alpha, \beta), \ldots,$ $\Delta_n(\alpha, \beta)$. The task of the top decision unit D_0 is now to select an appropriate set of parameters $(\hat{\alpha}, \hat{\beta})$ so that after the solution of the first level decision problems $\Delta_i(\hat{\alpha}, \hat{\beta}), \ldots, \Delta_n(\hat{\alpha}, \hat{\beta})$ implies the solution of the overall problem Δ, i.e., the first level decision units by solving their own local decision problems also solve the overall decision problem Δ. The first level units are then considered as coordinated by D_0 (with respect to Δ).

We shall now define the coordination problem more precisely for a broad but special class of systems. In doing this we shall use the framework of the mathematical theory of general systems which is based on set-theoretic concepts.

The overall and the local processes are defined by the mappings

$$P: M \to Y \tag{1}$$

and

$$P_i: M_i \times U_i \to Y_i \tag{2}$$

respectively, where $M = M_1 \times \ldots \times M_n$, $U = U_1 \times \ldots \times U_n$, $Y = Y_1 \times \ldots \times Y_n$ and the interactions are generated by the maps

$$K_i: M \to U_i \tag{3}$$

The relationship between (1), (2) and (3) is defined by

$$y = P(m_1, \ldots, m_n), (y_1, \ldots, y_n) = (P_1(m_1, K_1(m)), \ldots, P_n(m_n, K_n(m)))$$

for all $m \in M$.

Turning now to the decision problem we shall first define the overall problem Δ. It is defined in terms of the process $P: M \to Y$ and a function $G: M \times Y \to V$, termed the evaluation function.

The set V is assumed to be linearly ordered; all other sets, it should be noticed, are left completely abstract (i.e., their elements can be defined more specifically in many different ways as the circumstances require: as continuous or discrete time functions, as a string of letters from a given alphabet, as an abstract semi-group etc.). The overall decision problem is now defined by the statement:

Find $\hat{m} \in M$ such that

$$G(\hat{m}, P(\hat{m})) < G(m, P(m)) \quad \text{for all } m \in M.$$

The first level decision problems are defined similarly except that provision for coordination has to be made. $\Delta_i(\alpha, \beta)$ is defined by means of a pair of mappings

$$P_{i\alpha}: M_i \times U_i^{\alpha} \to Y_i$$

$$G_{i\beta}: M_i \times Y_i \times (\beta) \to V$$

where $U_i^{\alpha} \subset U_i$ and $G_{i\beta}$ is the local performance function. α and β reflect the coordination activities. It is assumed that the second level unit selects a parameter β which specifies the local performance function and at the same time instructs the first level unit to restrict its attention only to the subset U_i^{α}. The first level decision problems are then specified except for the uncertainties regarding interactions. Depending on how the first level is instructed to handle interactions several cases can be distinguished. We shall consider here only one of them, in which the second level unit is using the so-called interaction balance principle as the basis for the coordination strategy. In this approach the first level units are instructed to consider the entire interaction set, i.e., $U_i^{\alpha} = U_i$ and furthermore, they are also requested to treat the interactions as additional decision variables, i.e., each first level unit is requested to specify the element from the interaction set it would like to have as an input. With this, the first level decision problem can be fully specified; for the i-th unit it is given by the statement:

Find $(\hat{m}_i(\beta), \hat{u}_i(\beta)) \in M_i \times U_i$ so that

$$G_i(\hat{m}_i(\beta), \hat{u}_i(\beta)) < G_i(m_i, u_i)$$

for all $(m_i, u_i) \in M_i \times U_i$.

It should be noticed that after $(\hat{m}_i(\beta), \hat{u}_i(\beta))$ have been selected only $\hat{m}_i(\beta)$ can be implemented and the resulting interactions, denoted by $u_i(\beta)$, are in general different than the desired, $u_i(\hat{\beta})$, i.e.,

$$\hat{u}_i(\beta) \neq u_i(\beta) = K_i(\hat{m}_i(\beta))$$

It is this difference, now, which serves as the basis for defining the decision problem for the second level unit. Namely, if the characteristics of the system are such that the following holds

$$[(m_i, u_i) = m_i(\beta), u_i(\beta) \text{ and } u_i = K_i(m)] \to m_i = \hat{m}_i \tag{4}$$

for all i, i.e., if whenever the pair m_i, u_i is locally optimal and the equality $u = K(m)$ is satisfied, the same decision m represents the overall optimum and the decision problem of the second level unit Δ can simply be defined to be:

Find β such that implication (4) holds.

In summary, the system's behavior is now the following: First, the coordination unit D_0 selects β such that the balance $u_i = K_i(m)$ holds and subsequently, the first level unit selects m_i by solving the local problem and implement it on the process. By virtue of the implication (4) the overall objective is then satisfied. In connection with this coordination procedure the following should be noted:

(i) The statement (4), which embodies the interaction balance coordination principle, specifies only a defining property of the optimal coordination parameter but does not give a procedure to find it. Two such procedures are of particular interest. The optimal β can be found by an iteration between the levels. The second level specifies β, upon which the first level units derive $(\hat{m}_i(\beta), \hat{u}_i(\beta))$. The second level compares $u_i(\beta) = K_i(\hat{m}(\beta))$ and $\hat{u}_i(\beta)$ and if $u_i(\beta) \neq \hat{u}_i(\beta)$ a new β is specified. The process proceeds until the equality $u_i(\beta) = \hat{u}_i(\beta)$ is achieved upon which the associated control $\hat{m}_i(\beta)$ is implemented. There are a number of ways the second level can select successive β's; e.g., by the steepest descent on the overall performance. The second procedure consists of the establishment of continuous feedback. The structure of the system is shown in Fig. 1. The second level measures the interactions continuously, compares them with the desired interactions generated by the first level units, and, using this difference as an error signal makes the necessary corrections of the coordination parameter[1]).

3. Hierarchical systems and structures

As the second topic, we shall briefly present some basic concepts of hierarchical systems.

The concept of a hierarchy is used in many different contexts and when identifying precisely what is common to all of them one arrives at a rather abstract notion. We shall first present such a general notion of a hierarchical system and then discuss more specifically the basic classes of hierarchical systems.

In the subsequent discussion we shall use the concept of a system (or subsystem) simply as a transformation (or mapping) of stimuli (or inputs) into responses (or outputs).

A generic type of hierarchical system is shown in Fig. 2.

For a system to belong to this class of systems it is necessary at least that:

(i) The system consists of two or more (explicitly recognized) interacting subsystems.

(ii) The subsystems are arranged vertically, i.e., there exists a way to identify the subsystems as being on different levels; this is accomplished when the relationship between the subsystems is such that the behavior of the subsystems on the lower levels is conditioned (determined, specified) by the inputs from the higher levels while the performance (success) of the subsystems on the higher level depends upon the behavior and performance of the lower levels.

The inputs from the higher to the lower levels are referred to as interventions while the inputs in the other direction are referred to as the performance feedbacks. Implicit in the concept of intervention is the priority of actions or goals in the sense that the change of the variable and its implementation is solely determined by the higher level units.

---

[1]) More information on the theory and application of the interaction balance principle can be found in the recent book: Mesarovic, M. D., Macko, D., and Takahara, Y.: Theory of Hierarchical Multilevel Systems. New York 1970.

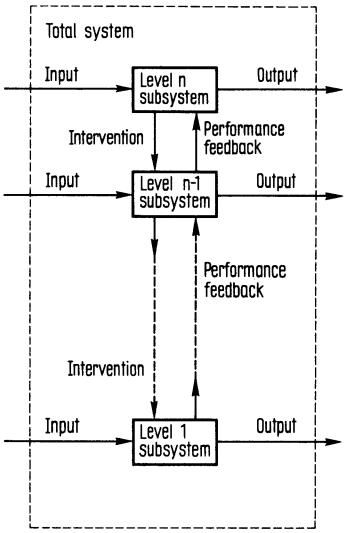


Fig. 2

3.1. Strata-levels of description or abstraction. Truly complex systems almost by definition evade complete and detailed descriptions. The dilemma in description is basically between simplicity, one of the prerequisites for understanding, and the need to take into account numerous behavioral aspects. A resolution of this dilemma is sought through a hierarchical description. One describes the system by a family of models each concerned with the behavior of the system as viewed from a different level of abstraction. For each level there is a set of relevant features, variables, laws and principles in terms of which the system's behavior is described. For such a hierarchy to be effective it is necessary that the description on any level be considered independent of the description on other levels. To distinguish this level-independent description concept of hierarchy from others we use the term stratified system or stratified description. The levels of abstraction involved in a stratified description will be referred to as strata.

Examples of stratified descriptions in the natural sciences are abundant. One can study living organisms on molecular, cellular, organ and total organism levels. Although a complete decoupling of levels cannot be fully justified, the assumed decoupling of levels enables the study of system behavior on any stratum to be performed in considerable detail. Neglecting the cross-strata interdependence leads to an incomplete understanding of the whole system behavior. Indeed, a restriction to biological-type inquires represents in itself an isolation since apparently the system under consideration can also be described on the stratum of concern to chemistry or physics on one hand and ecology and perhaps social science on another.

One can take an electronic computer as an example of a stratified description of a man-made system. Its functioning is customarily described at least on two strata: in terms of physical laws describing the functioning and interconnection of the constituent parts and in terms of processing abstract non-physical entities, such as digits or information sequences. On the stratum of information-processing one is concerned with such problems as computation, programming, etc., and the underlying physical basis of operation is not explicitly considered. Of course, a description of the system or some of its subsystems on other strata may also be of interest; the stratum

of atomic physics for some component design or the so-called systems stratum for problems such as time sharing. Another example of stratification relevant for industrial automation is shown in Fig. 3. On the first stratum the system is described in terms of the laws of physics and chemistry. On the second stratum one is dealing with signals and information and the principles of decision making (control) and information processing are used. On the third stratum the systems operation is described in economic terms.

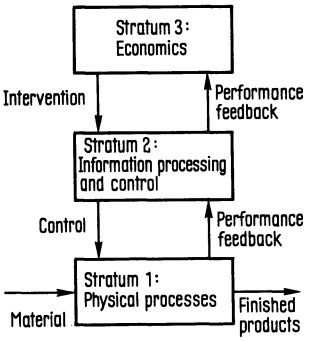


Fig. 3

3.2. Layers – levels of decision complexity. Another concept of hierarchy appears in the context of a complex decision-making process. There are two quite obvious, very important, but conflicting, features of many real-life decision-making situations: on one hand, to avoid decision by default, there is a need to act without delay while, on another, there is an equally great need to take time to understand the situation better. In complex decision-making situations, the resolution of this dilemma is sought in a hierarchical approach. Essentially one defines a family of decision problems whose solution is attempted in a sequential manner. The solution of any problem in the sequence determines and fixes some parameters in the next problem so that the latter is completely specified and its solution can be attempted. In such a way the solution of a complex decision problem is substituted for by the solution of a family of sequentially arranged simpler subproblems, so that the solution of all the subproblems in the family implies the solution of the original problem. Such a hierarchy is termed a hierarchy of decision layers, and the entire decision-making system is referred to as a multi-layer (decision) system.

It is easy to give examples from everyday life of complex decision situations which are approached in a multi-layer fashion. Indeed, personal goals or objectives are as a rule vague and have to be translated into what might be called operational objectives which then provide a basis for the selection of a concrete course of action. For example, the goal of a young man might be to achieve happiness, but that vague goal has to be translated into objectives which lead to some specific actions: an objective has to be selected (say, to attend college, go into business, get married, etc.) which in turn leads to subgoals (say, to select a college, a major field of study, etc.). Very often only after a subgoal is achieved is one in the position to evaluate whether the original goal is approached.

As an example of multilayer hierarchies consider the functional hierarchy in decision problems under true uncertainties. This hierarchy emerges naturally in reference to three essential aspects of decision problems under uncertainty: (i) the selection of strategies to be used in the solution process; (ii) the reduction or elimination of uncertainties; and (iii) the search for a preferable

or acceptable course of action under prespecified conditions. The functional hierarchy is shown in Fig. 4. It contains three layers:

(1) Search layer: The task of this layer is to select the course of action m.

(2) Learning or adaptation layer: The task of this layer is to specify the uncertainty set U used by the search layer.

(3) Self-organizing layer: This layer selects the structure, functions, and strategies which are used on the lower layers so that an overall goal (usually defined in terms which cannot easily be made operational) is being pursued as closely as possible.

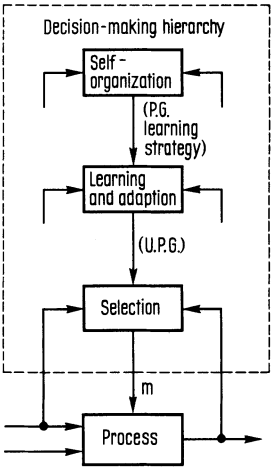


Fig. 4

3.3. **Organizational hierarchies: multi-level multi-goal systems.** For this concept of hierarchy it is necessary that (i) the system consist of a family of interacting subsystems which are recognized explicitly, (ii) some of the subsystems are defined as decision (making) units and (iii) the decision units are arranged hierarchically in the sense that some of them are influenced or controlled by

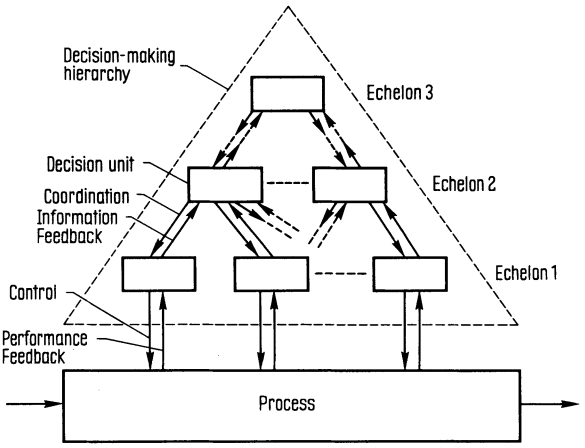


Fig. 5

other decision units (Fig. 5). This relationship between subsystems can be formalized and leads to a partial ordering relationship between subsystems.

These systems are denoted as multi-level multi-goal because, in general, various decision units comprising the system have conflicting goals. This conflict not only appears as the result of the evolution and composition of the system but also can be shown to be necessary (to a degree and in a given sense) for the efficient operation of the overall system.

4. Conclusion

In conclusion, the importance of the hierarchical systems for the cooperative phenomena is two-fold: First, they provide a framework for defining in a precise manner the concepts associated with various phenomena considered as cooperative; this is quite essential particularly when phenomena from different fields (e.g. physics, biology, socio-political sciences, etc.) are compared. Second, various structures derived by theoretical considerations can be used as the models in experimental investigations of the new types of cooperative phenomena.

Metropolitan Models and Public Policy

Benjamin Chinitz, Graham Crampton, Leo P. Kadanoff,
Susan C. Schwartz, John R. Tucker, Herbert Weinblatt

Abstract

The Brown University Urban Analysis Group is currently developing and calibrating simulation models of urban development. These models are designed for studying the direct and indirect effects of current and proposed public policies, both in our local Providence metropolitan area and in the nation as a whole.

Introduction

The Brown University Urban Analysis Group is engaged in an extensive effort aimed at the construction, calibration and use of comprehensive urban simulation models. Our models attempt to describe the linked growth of housing, industry and population in each of several districts within a metropolitan area.

Our long-range goal is to produce realistic, though necessarily simplified, representations of the major socio-economic processes which are generally believed to affect the growth and development of the metropolitan areas of the nation. Such models could then be used as tools for simulating proposed public policies, with the expectation that the results would provide us with insights into some of the direct and indirect effects which might be produced by the programs. These insights, in turn, can then be used in evaluating the policies, as well as in formulating new ones.

The models

Our models are based, in large measures, on Jay Forrester's pioneering "Urban Dynamics" model [1]; however, they go beyond Forrester's in several respects. Most notably, we have attempted to represent the metropolitan areas of the nation in their entirety rather than just a single fully-developed urban area, and we are making a determined effort to relate the models both to real-world data and to generally accepted concepts of urban social theory.

Like Forrester's, our models are each in the form of a purely recursive set of nonlinear difference equations. The recursive property allows the models to be operated inexpensively[1]), thus making it possible to test them frequently during all phases of their development. Such

[1]) The present model is implemented in APL on Brown University's IBM 360/67 and requires about 1.5 seconds of computation time (about 15 ¢) per iteration. (We are currently using a

testing enables us to examine repeatedly the implications of the behavioral assumptions on which our models are based. In this way we hope to avoid the fate of many previous large-scale modeling projects which discovered, after several man-years of effort, that they have created a model which is based on a series of incompatible assumptions and which cannot be modified so that it will produce useful results.

The basic unit of our models is a district within a metropolitan area. The metropolitan areas of the nation are then represented by several such units, with each unit either typifying a set of metropolitan areas or representing a specific metropolitan area, as for example Providence. The relative influence which each unit exerts on the others is specified by a weight which indicates the number and size of the metropolitan areas which it represents. The models also contain a very rough representation of the non-metropolitan areas of the nation so that migration between metropolitan and non-metropolitan areas can be determined.

Each district of a metropolitan area is characterized by certain exogenously specified parameters and variables, as well as by several state variables which (except for initial values) are endogenously generated. The basic state variables describing each district are:

a) The resident labor force in each of three skill categories: managerial-professional (MP), skilled labor (L) and unskilled (U).
b) The amounts of industriy in each of three categories, distinguished by the extent of their market areas: export industry, serving a national market; metropolitan industry, serving the whole of the relevant metropolitan area, and local industry, serving primarily the local district.
c) The amounts of housing in each of six categories.
d) The amount of land occupied by industry and housing.
e) The number of persons working in the district, distinguished by both their skill category and district of residence.

The most important exogenous specifications of each district describe its area and the quality of its "transportation connectivities" to the rest of the nation and to each district in the metropolitan area. The transportation connectivities play an important role both in the metropolitan and national industrial markets as well as in the allocation of the labor force among jobs. Intra-metropolitan transportation connectivities are specified separately for each of the three worker-skill categories.

The present model contains a very simple representation of the fiscal aspects of local government. We are now in the process of both improving this representation and adding the federal and state governments. These changes will make it possible for us to represent realistically the costs of proposed public programs.

Application

Our work has several related purposes. Our first efforts [2, 3, 4, 5] were intended to evaluate and criticize the techniques whereby public policy conclusions are reached with the aid of urban growth models. These critical exercises were carried out using uncalibrated models, and so

---

six-month iteration interval.) We expect to re-implement the model in a production language, such as Fortran, in the near future; such an implementation should greatly reduce the model's operating costs, though it will increase development costs.

represented an attempt to use quantitative models in logical but qualitative argumentation. An example of this kind of argumentation is given in the next section.

The limitations of such analysis have led us to construct a model containing an explicit representation of the Providence metropolitan area. Data from the 1950 and 1960 Census, together with information from surveys and traffic studies are being employed to set the initial values and the parameters within the model. The 1970 Census will be used to furnish data suitable for assessment of the model's operation.

Table 1 Providence Area Population (in thousands)

Date	Providence		Pawtucket & Central Falls		Inner Suburbs		Outer Suburbs	
	Census	Model	Census	Model	Census	Model	Census	Model
1950	248.7		105.0		179.7		52.0	
1960	207.5	209.4	100.9	99.4	234.0	238.8	66.2	68.0
1970	179.2	185.9	95.7	93.4	275.6	274.3	85.2	86.9
1980		170.6		88.3		299.3		116.2

Table 1 shows the result of a primitive run of this procedure. The model was initialized to 1950 populations for four districts within the Providence metropolitan area. Parameters within the model were set so that the correct 1960 populations were approximately generated by a 10-year run. Then the 1970 populations obtained from the Census could be compared with the model's output following a further 10-year run. Of course, the actual calibration will eventually utilize thousands of data elements to set hundreds of parameters.

Once the model's representation of the Providence metropolitan area has been calibrated satisfactorily, we shall use the model to explore certain policy issues of local interest. Among these are urban renewal and transportation planning.

During the past decade, Providence has undertaken several urban renewal projects. We hope to be able to use our model to help us to differentiate better between the changes in the metropolitan area which occurred partly or wholly because of these projects and those which probably would have occurred naturally during this period.

In the area of transportation planning, we expect to investigate various alternatives for alleviating a traffic bottleneck which is building up at the interchange of two interstate highways in downtown Providence. We intend to use our model in conjunction with more conventional transportation analysis techniques to gain new insights into the likely effects of various courses of action on future metropolitan development. These include:

a) Doing nothing.
b) Building a bridge across Narragansett Bay to serve as a partial bypass of the bottleneck.
c) Increasing the capacity of existing roads.
d) Instituting park-and-ride bus and/or rail service.

In the course of this work we benefit from the active cooperation of planners from the State Department of Transportation, the Statewide Planning Agency and the cities of Providence and Warwick.

For the purposes of studying these Providence-area policies, our model need only contain two metropolitan "units": one representing Providence itself and another representing the rest of

the nation. In order to examine national urban policies, however, our model will have to contain several metropolitan units, each representing a different "type" of metropolitan area. Furthermore, each of these units will have to be calibrated. In this task we hope to benefit from our experiences with calibrating the two-unit model.

The resulting "national metropolitan model" can then be used to simulate certain national public programs which affect urban spatial structure and the quality of urban life. Particularly promising subjects of research include various versions of:

a) New towns programs,
b) Revenue sharing, with various degrees of control by the higher levels of government,
c) School tax equalizations, as required by recent court decisions,
d) Urban renewal,
e) Highway and mass transit subsidies,
f) Job training programs,
g) Expansion of employment in the public sector.

An illustrative example

To indicate how models of this type may be used in considering the effectiveness of urban policy, we shall discuss a crude representation of an urban-renewal program with the aid of one of our earliest models.

The program consists of exogenous demolition of 5% of that part of the housing stock which is occupied by the lowest social/skill category. This is more a representation of the practice than of the intent of urban renewal.

Each of the metropolitan areas is divided into a central city and a suburban region. The demolition is assumed to take place only in the central city.

The rationale for the demolition of lower-class housing is to clear central-city land, making it available for new industrial development. Employment is expanded in the city, and economic opportunity and upward social mobility is increased. The major impact should be for the unskilled, where high central city unemployment rates constitute a crucial element of the "urban crisis".

Such reasoning is commonly used to justify demolition programs and urban renewal as a whole. Precisely the same qualitative argument can be made with the aid of a model. Indeed an uncalibrated model can only furnish qualitative arguments. It should be emphasized that the following discussion is not to be viewed as an addition to knowledge about urban renewal, but rather as indicative of the kinds of arguments which might be more convincingly made in the future, when we have better calibrated models available.

The conventional argument for this type of urban renewal is indicated in Fig. 1. The curves represent average metropolitan-area unemployment rates for unskilled workers, derived from different assumptions concerning the extent of the program. One run of our model includes no demolition anywhere; another assumes that demolition occurs in only one city of the nation, and shows results for that city. If we were to use these unemployment rates as an index of policy effectiveness[1]), it would appear that such a program is a rational course of

[1]) Of course, other indexes should be employed [2, 3, 5], but for simplicity and brevity we will restrict ourselves to this evaluation index.

action for any given city. Over the simulated run from 1950 to 1980, a metropolitan area with an ongoing central-city housing-demolition program attains substantially lower unemployment rates for the unskilled than it would without the program.

However, this run says nothing about the desirability of all cities implementing this form of urban renewal. The program is national in origin and financing; it would be reasonable to expect a nationwide application to be effective, but the results in Fig. 1 indicate otherwise.

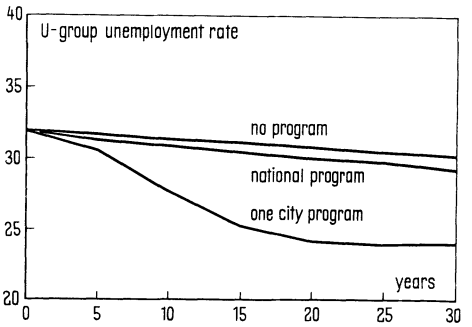


Fig. 1

When the demolition program is run on a nationwide basis, it is much less effective in reducing each area's unskilled unemployment rate than when applied in only one city. In the model, demolition tends to attract industry to the vacated land in renewal areas. Expansion in the total amount of industry, however, is governed by the level of demand for goods and services generated by workers and existing industry, which is only marginally affected by the demolition tends to attract industry to the vacated land in renewal areas. Expansion in the total amount of industry, however, is governed by the level of demand for goods and services national industrial investment aggregate is influenced not only by business expectations, crudely accounted for in the model's demand sector, but by the state of the money market as well. Nevertheless, the qualitative conclusions of the model appear reasonable. While some, possible beneficial, reallocations between central city and suburb do occur in the national program, average metropolitan unemployment rates are little changed, the principal gain arising from increased activity in the construction industry. Clearly, a demolition program implemented in only one city may stimulate investment there at the expense of the rest of the nation, and so appear extremely effective for that particular area.

To emphasize the offsetting nature of the effects of the demolition program as applied to one metropolitan area, we present in Fig. 2 the change in the absolute number of employed unskilled workers caused by the program in the single project area, and in the remainder of the nation. The gain in employment in the area with the program is offset by a corresponding decline in the non-program areas.

Finally in Fig. 3 we plot the results of running a demolition program of this type in exactly half the cities of the nation. The reduction in unemployment rate in the program areas is accompanied by a rise in the other metropolitan areas, so that the total unemployment rate remained nearly unaffected by the program.

The foregoing argument is a somewhat crude example of the use of urban growth models for the qualitative discussion of an important aspect of a national program. We are now working

toward developing more realistic, calibrated, and hopefully more quantitatively accurate urban models. The focus of our policy analysis will be two-fold. On a national level, we hope to provide increasingly sophisticated appraisals of impact for federal urban programs of the

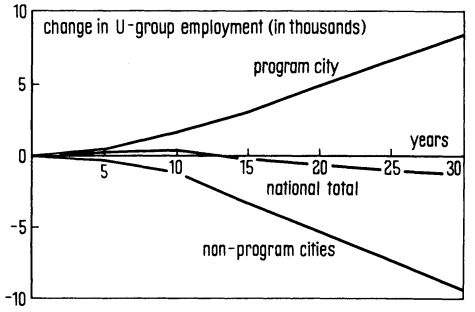


Fig. 2

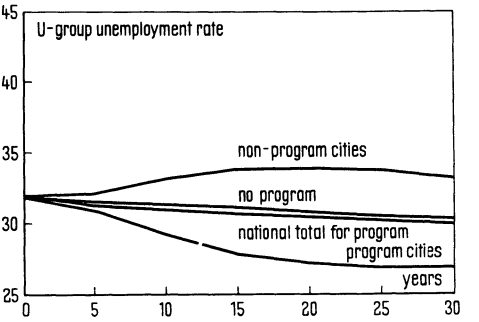


Fig. 3

general type illustrated here. On the local level, we shall be involved in intensive study of the detailed consequences arising from urban renewal, transportation, and other major programs and proposals within the Providence area. These two efforts are mutually reinforcing, and will hopefully provide insights helpful for urban policy planning.

References

[1] Forrester, J.: Urban Dynamics. Cambridge, Mass.–London 1969.

[2] Kadanoff, L. P.: From Simulation Model to Public Policy: An Examination of Forrester's Urban Dynamics. Simulation, June 1971, pp. 261–268. (To be reprinted in: The Awards Papers in Computer Science – 1972, Princeton, N. J.)

[3] Kadanoff, L. P.: From Simulation Model to Public Policy. The American Scientist, January 1972, pp. 74–79.

[4] Kadanoff, L. P.: A Modified Forrester Model of the United States as a Group of Metropolitan Areas in: Urban Dynamics: Extensions and Reflections. San Francisco, 1972.

[5] Kadanoff, L. P. and Weinblatt, H. B.: Public Policy Conclusions from Urban Growth Models. IEEE Transactions on Systems, Man and Cybernetics, January 1972; also in: Urban Dynamics: Extensions and Reflections. San Francisco 1972.

Fokker-Planck Equation Treatment of Interacting Social Groups

W. Weidlich

1. Introduction and general outline

If a physicist attempts to use his methods in other fields of science, some justifying remarks are necessary, because at first sight there seems to be no analogy between physics and, for example, sociology. But, while the fundamental laws governing different fields of science like physics, biology, sociology are very different and no easy direct connection between them can be established at the moment, there exists a unifying level of consideration for all these sciences.

This level is the statistical behaviour of systems with a large number of interacting units. The probabilistic laws governing the statistical dynamics of such systems namely lead to many common features independently of the derivation of these laws from different fundamental levels. Thus for instance usually "macroscopic observables" describing the collective behaviour of such systems can be found. The time development of these macroscopic observables finds its mathematical form in some stochastic equations of various degrees of complexity. The simplest kind of such equations corresponding to Markoff-processes is of the Fokker-Planck equation type in physics and sociology as well. In physics some of the parameters characterizing the collective state of a manybody system, in particular in the stationary and equilibrium case, are wellknown. We mention the thermodynamic potentials, temperature, pressure etc. Also the "critical" behaviour of physical systems at phase transition points of temperature or other relevant parameters can be described with concepts valid for a wide variety of systems.

In sociology, the treatment of the problems by statistical models and the introduction of appropriate collective variables in a similar way should lead to an at least semiquantitative definition of concepts which are normally only used in a qualitative manner. Concepts of such kind describing collective structures in society are for instance: The political (e.g. liberal or totalitarian) "climate"; the "mobility" of a society; the critical character of a situation at the crossroads to different developments etc.

The use of statistical methods in physics, sociology and other fields is, of course, not only possible, but even necessary, due to the high complications of a "microscopic" detailed description. In sociology e.g. idealizations like the subdivision of society into quasi homogeneous groups and the classification of the continuum of possible behaviours into a discrete set necessarily lead to taking averages and therefore to a probabilistic description.

After stressing analogies of different sciences on the level of statistical considerations, some general remarks are due about the specific features of the statistical process of development in society. A coarse classification leads to the distinction between direct and indirect mutual

interaction between individuals with respect to their probabilistic behaviour. The direct interaction consists in the fact, that members of the society can be influenced immediately by the simultaneous actions and attitudes of other individuals. Indirect interaction between individuals arises, if in a society economical, social, ideological, religious "superstructures" are created which thereupon influence the same or other individuals simultaneously or later. The typical form of indirect interaction is the perennial long-range after effect of "traditions".

A complete statistical description of developments of "historical dimension" therefore would have to take into account the extremely non-Markoffian structure of these retarded interactions. Though valid in principle, such a description would not only be very complicated but also at least partially useless, because it takes probabilistic account of all possible developments with many branching points, while only one or a few samples of this probability-distribution are realized. Therefore it is more appropriate to restrict statistical considerations in sociology to short-range developments with well defined initial and boundary conditions. In this case it is again possible to describe the development of the systems with a quasimarkoff-approximation: The influence of the indirect interaction, i.e. the after effect of traditions, of ideologies etc. is taken into account as quasi constant effect on the probabilistic further development; it is defined by the initial and boundary conditions of the momentary situation. The resulting stochastic equation then is of the Fokker-Planck type, as we will discuss it in the next section.

2. A general Fokker-Planck equation for the statistical behaviour of interacting social groups

Let us assume that we have Z homogeneous groups with n_j members in group j (j = 1 . . . Z) and that for each member of each group Ω disjunctive "attitudes"[1]) are possible. With $n_{j,\alpha}$ = number of members of group j with attitude $\alpha(\alpha = 1 \ldots \Omega)$, we have $\sum_{\alpha=1}^{\Omega} n_{j,\alpha} = n_j$.

The fundamental quantity for the further considerations is the transition probability $p^{(i)}_{\alpha\to\beta}(\{n_{j,\gamma}\})$ for a member of group i to change from attitude α to β. This transition probability has to be found by a previous sociological analysis taking into account the initial and boundary conditions of the situation. It can also depend on the configuration $\{n_{j,\gamma}\}$. The equation of motion for the probability distribution $f(\{n_{j,\gamma}\}, t)$

$$\frac{\partial f(\{n_{j,\gamma}\}, t)}{\partial t} = \sum_{\alpha\neq\beta}^{\Omega} \sum_{i=1}^{z} [n'_{i,\alpha} \cdot p^{(i)}_{\alpha\to\beta}(\{n_{j,\gamma}\}'_{i,\alpha\beta}) f(\{n_{j,\gamma}\}'_{i,\alpha,\beta}, t) -$$

$$- n_{i,\alpha} \cdot p^{(i)}_{\alpha\to\beta}(\{n_{j,\gamma}\}) f(\{n_{j,\gamma}\}, t)] \qquad (2.1)$$

then follows from a simple probability balance consideration. In (2.1), $\{n_{j,\gamma}\}'_{i,\alpha,\beta}$ means the group configuration arising from $\{n_{j,\gamma}\}$ by the substitution $n_{i,\alpha} = n'_{i,\alpha} + 1$, and $n_{i\beta} \to n'_{i\beta} = n_{i\beta} - 1$.

Introducing the quasi continuous variables

$$x_{j,\gamma} = n_{j,\gamma}/n_j; \quad \sum_{\alpha=1}^{\Omega} x_{j,\alpha} = 1; 0 \leqslant x_{j,\gamma} \leqslant 1; \Delta x_{j,\gamma} = \frac{1}{n_j} \qquad (2.2)$$

---

[1]) By "attitude" we can understand e.g. political opinions or certain kinds of activities or behaviours.

Eq. (2.1) leads to the approximate Fokker-Planck equation

$$\frac{\partial f(\{x_{j,\alpha}\}, t)}{\partial t} = \sum_{\alpha \neq \beta}^{\Omega} \sum_{i=1}^{z} \left[\left(\frac{\partial}{\partial x_{i,\alpha}} - \frac{\partial}{\partial x_{i,\beta}} \right) + \frac{1}{2n_i} \left(\frac{\partial}{\partial x_{i,\alpha}} - \frac{\partial}{\partial x_{i,\beta}} \right)^2 \right] \times$$

$$\times \left(x_{i,\alpha} p^{(i)}_{\alpha \to \beta}(\{x_{j,\gamma}\}) f(\{x_{j,\gamma}\}; t) \right). \tag{2.3}$$

The meanvalues $\bar{x}_{k,\epsilon}(t)$ of $x_{k,\epsilon}$ are defined by

$$\bar{x}_{k,\epsilon}(t) = \int f(\{x_{j,\gamma}\}, t) x_{k,\epsilon} \, d\tau; \quad d\tau = \prod_{\gamma,i} dx_{i,\gamma} \tag{2.4}$$

Their equation of motion follows from the F.P. equation (2.3)

$$\frac{d\bar{x}_{k,\epsilon}(t)}{dt} = \sum_{\alpha \neq \epsilon}^{\Omega} \int \left(x_{k,\alpha} p^{(k)}_{\alpha \to \epsilon}(\{x_{j,\gamma}\}) - x_{k,\epsilon} p^{(k)}_{\epsilon \to \alpha}(\{x_{j,\gamma}\}) \right) f(\{x_{j,\gamma}\}, t) \, d\tau \tag{2.5}$$

Only, if $f(\{x_{j,\gamma}\}, t)$ is sharply peaked around the mean values, we obtain from (2.5) the following closed set of equations for the $\bar{x}_{k,\epsilon}(t)$:

$$\frac{d\bar{x}_{k,\epsilon}(t)}{dt} = \sum_{\alpha \neq \epsilon}^{\Omega} \left(\bar{x}_{k,\alpha} p^{(k)}_{\alpha \to \epsilon}(\{\bar{x}_{j,\gamma}\}) - \bar{x}_{k,\epsilon} p^{(k)}_{\epsilon \to \alpha}(\{\bar{x}_{j,\gamma}\}) \right) \tag{2.6}$$

Eq. (2.6) represents a system of coupled nonlinear first order differential equations. It is not possible to give general solutions of (2.3) or of (2.6). Only for special form of the coefficients in (2.3), (2.6) methods of solution are known [2, 3, 4]. In the next section we discuss two examples of (2.3), where exact solutions are possible.

3. Applications of the Fokker-Planck equation to simple cases of interacting groups

Example 1. One group with cooperative internal interaction. The Fokker-Planck equation (2.3) reads for the case $Z = 1$; $\Omega = 2$

$$\frac{\partial f(x, t)}{\partial t} = -\frac{\partial j(x, t)}{\partial x} \tag{3.1}$$

$$= -\frac{\partial}{\partial x} [K_1(x) f(x, t)] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [K_2(x) f(x, t)]$$

where we have introduced the variables

$$x = \frac{1}{2}(x_2 - x_1); \quad (x_1 + x_2) = 1. \tag{3.2}$$

and the coefficients

$$K_1(x) = \left\{ \left(\frac{1}{2} - x \right) p_{1 \to 2}(x) - \left(\frac{1}{2} + x \right) p_{2 \to 1}(x) \right\}; \tag{3.3}$$

$$K_2(x) = \frac{1}{n} \left\{ \left(\frac{1}{2} - x \right) p_{1 \to 2}(x) + \left(\frac{1}{2} + x \right) p_{2 \to 1}(x) \right\}.$$

Because of (3.1), the quantity

$$j(x, t) \equiv K_1(x) f(x, t) - \frac{1}{2} \frac{\partial}{\partial x} [K_2(x) f(x, t)] \tag{3.4}$$

may be interpreted as probability current, which therefore has to fulfill the boundary condition

$$j\left(\pm \frac{1}{2}; t\right) = 0. \tag{3.5}$$

The stationary solution is given by

$$f_{st}(x) = c\, K_2^{-1}(x) \exp\left\{2 \int^{x} \frac{K_1(y)}{K_2(y)}\, dy\right\} \tag{3.6}$$

with normalization to

$$\int_{-1/2}^{+1/2} f_{st}(x)\, dx = 1 \tag{3.7}$$

The meanvalue equation and its solution according to (2.6) reads:

$$\frac{d\bar{x}(t)}{dt} = K_1(\bar{x}); \tag{3.8}$$

$$(t - t_0) = \int_{\bar{x}_0}^{\bar{x}} \frac{dy}{K_1(y)},$$

We now make an explicit assumption about the transition probabilities $p_{1\to 2}(x)$; $p_{2\to 1}(x)$:

$$p_{2\to 1}(x) = \nu \exp(-kx - h); \quad p_{1\to 2}(x) = \nu \exp(kx + h) \tag{3.9}$$

This ansatz describes a mutual interaction between the members of the group such that the decision behaviour of the individual is influenced by the attitudes of all other members. k and h are open parameters which because of their meaning in (3.9) may be called k = "adaptation parameter"; h = "preference parameter".

The form (3.9) is analogous to the flipping probability of spins $\sigma_i = \pm 1$ at site i in the Ising model of ferromagnetism [6] which is given by

$$p(\sigma_i \to -\sigma_i) = \nu \exp\left\{ - \frac{H\sigma_i + \sum_j I_{ij}\sigma_j}{kT} \right\}; \quad \sigma_i = \pm 1. \tag{3.10}$$

where H = external magnetic field, I_{ij} = spin-spin interaction constant; T = temperature.

The explicit solution $f_{st}(x; k, h)$ which is found with (3.6) by using (3.9) and (3.3) now changes its structure fundamentally at a value k_0 of k defined by

$$\left.\frac{\partial K_1(k_0, x)}{\partial x}\right|_{x=0} = 0 = \nu(k_0 - 2) \curvearrowright k_0 = 2. \tag{3.11}$$

in the case of h = 0 (compare Figs. 1, 2, 3 and 5). We may therefore call k_0 the phase transition point with respect to the polarization structure of the groups describable by $f_{st}(x; k, h = 0)$. We may also define – in analogy to physical systems – an order parameter which distinguishes between the two phases of the group system. An appropriate order parameter is given by the square of the stable stationary meanvalue following from the stationary solution of (3.8):

$$\begin{aligned} &\bar{x}^2(\infty) = 0 \quad \text{for} \quad k < k_0 \\ &\bar{x}^2(\infty) = x_1^2 > 0 \quad \text{for} \quad k > k_0; \quad K_1(k, x_1) = 0. \end{aligned} \tag{3.12}$$

In Fig. 1 to 5 we show some results of the calculations: Fig. 1 to Fig. 4 show the stationary distribution function f(x), the time development of the meanvalues $\bar{x}(\tau)$ in the dimensionless time coordinate $\tau = \nu t$ and the drift coefficient $K_1(x)$ for different values of adaptationparameter k and preferenceparameter h.

Figure 1, k = 0 means independent decision behaviour of members of the groups; h = 0 means no preference between the attitudes 1 and 2. The most probable groups are those with $n_1 = n_2$ or x = 0. The meanvalues $\bar{x}(\tau)$ quickly relax to $\bar{x}(\infty) = 0$ independently of the initial value.

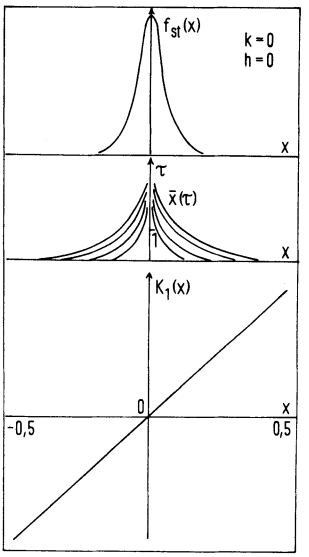


Fig. 1

Figure 2 (k = k_0 = 2; h = 0) gives the group statistics at phase transition point of the adaptation parameter k. The broad distribution function indicates the large critical fluctuations of the single groups around the stationary meanvalue $\bar{x}(\infty) = 0$. The $\bar{x}(\tau)$ show the critical slowing down of the relaxation to $\bar{x}(\infty) = 0$ due to the fact, that the drift coefficient $K_1(x)$ vanishes in higher order at x = 0.

Figure 3 (k = 2, 5; h = 0) shows the behaviour of groups beyond phase transition for strong mutual adaptation of the members of a group. The most probable groups now have $n_1 \gg n_2$ or $n_2 \gg n_1$ (polarization structure). The meanvalues $\bar{x}(\tau)$ now tend to $\pm x_1$ with $K_1(k, x_1) = 0$ dependent on their initial values.

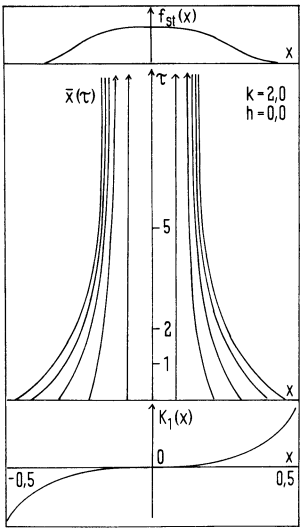


Fig. 2

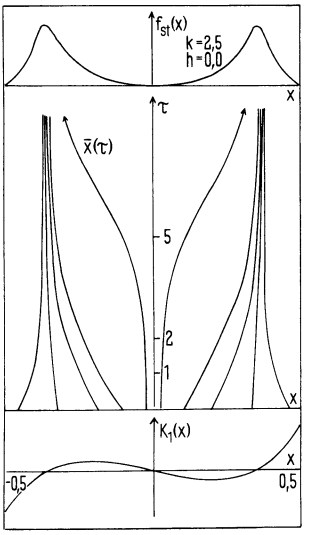


Fig. 3

Figure 4 (k = 2, 5; h = 0, 01) shows the same case with a small preference parameter leading to an asymmetric distribution function f(x, k, h) and an asymmetric relaxation of the $\bar{x}(\tau)$.

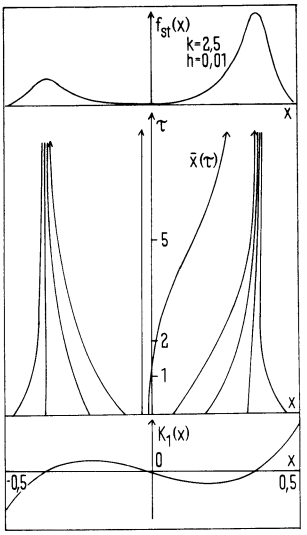


Fig. 4

Figure 5 shows the long range change of parameter k from k = 0 to k = 2, 5, the corresponding change of the distribution function and the development of a typical sample group with critical fluctuations at $k \approx k_0$.

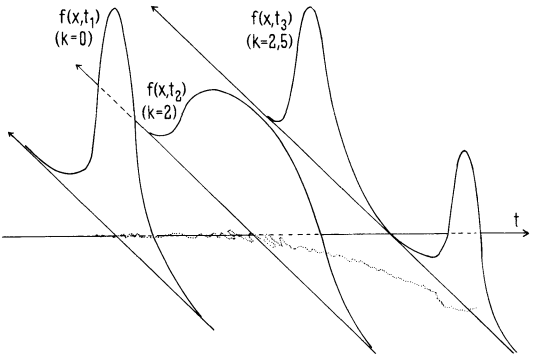


Fig. 5 Development of distribution function and of sample group for slow change of parameter k from k = 0 to k = 2.5.

Example 2. Interaction between two groups. We now consider the case $Z = 2$; $\Omega = 2$; $n_1 = n$, $n_2 = N$. In the coordinates

$$\begin{aligned} x &= (x_2 - x_1); \quad (x_1 + x_2) = 1 && \text{(group 1)} \\ y &= (y_2 - y_1); \quad (y_1 + y_2) = 1 && \text{(group 2)} \end{aligned} \tag{3.13}$$

the Fokker-Planck equation now reads:

$$\frac{\partial f(x, y; t)}{\partial t} = -\frac{\partial}{\partial x}(df) - \frac{\partial}{\partial y}(Df) + \\ + \frac{1}{2}\frac{\partial^2}{\partial x^2}(qf) + \frac{1}{2}\frac{\partial^2}{\partial y^2}(Qf). \tag{3.14}$$

We now assume, that the transition probabilities p_{21}, p_{12} and P_{21}, P_{12} of group 1 and 2, respectively, consist of "internal" parts and of terms describing the influence of the other group. (On the other hand for simplicity we neglect here any dependence of $p_{ij}^{(0)}$, p; $P_{ij}^{(0)}$, P on the coordinates.)

$$\begin{aligned} p_{21} &= p_{21}^{(0)} + py_1; \quad p_{12} = p_{12}^{(0)} + py_2 && \text{(group 1)} \\ P_{21} &= P_{21}^{(0)} + Px_1; \quad P_{12} = P_{12}^{(0)} + Px_2 && \text{(group 2)} \end{aligned} \tag{3.15}$$

The drift coefficients α, D and fluctuation coefficients q, Q then have the form

$$\begin{aligned} d &= 2(x_1 p_{12} - x_2 p_{21}) = r_0 - xp_0 - p(x - y) \\ q &= \frac{4}{n}(x_1 p_{12} + x_2 p_{21}) = \frac{2}{n}(p_0 + p - r_0 x - pxy) \end{aligned} \tag{3.16}$$

with $p_0 = (p_{12}^{(0)} + p_{21}^{(0)})$; $r_0 = (p_{12}^{(0)} - p_{21}^{(0)})$. (group 1)

$$\begin{aligned} D &= 2(y_1 P_{12} - y_2 P_{21}) = R_0 - yP_0 - P(y - x). \\ Q &= \frac{4}{N}(y_1 P_{12} + y_2 P_{21}) = \frac{2}{N}(P_0 + P - R_0 y - Pyx) \end{aligned}$$

with $P_0 = (P_{12}^{(0)} + P_{21}^{(0)}); R_0 = (P_{12}^{(0)} - P_{21}^{(0)})$. (group 2)

The stationary meanvalues

$$\bar{x} = \frac{r_0(P_0 + P) + pR_0}{(p_0P_0 + pP_0 + p_0P)} \qquad \text{(group 1)}$$

$$\bar{y} = \frac{R_0(p_0 + p) + Pr_0}{(p_0P_0 + pP_0 + p_0P)} \qquad \text{(group 2)}$$

are defined by the equations $d(\bar{x};\bar{y}) = 0; D(\bar{x}, \bar{y}) = 0$.

Introducing coordinates

$$u_\varrho = \alpha_{\varrho i}\xi_i, \text{ with } \xi_1 = (x - \bar{x}); \xi_2 = (y - \bar{y}) \tag{3.18}$$

where the $\alpha_{\varrho i}$ satisfy (for $a_{11} = -(p_0 + p); a_{12} = p; a_{21} = P; a_{22} = -(P_0 + P)$)

$$\alpha_{\varrho i}(a_{ik} - \delta_{ik}\lambda_\varrho) = 0 \text{ with } |\, a_{ik} - \delta_{ik}\lambda_\varrho \,| = 0 \tag{3.19}$$

the F.P. equation (3.14) takes the normal form

$$\frac{\partial f(u_j, t)}{\partial t} = -\sum_{i=1}^{2} \lambda_i \frac{\partial}{\partial u_i}(u_i f) + \frac{1}{2}\sum_{i,j=1}^{2} \tilde{Q}_{ij}\frac{\partial^2 f}{\partial u_i \partial u_j} \tag{3.20}$$

where the coordinates u_ϱ the relaxation constants λ_i and the new fluctuation coefficients $\tilde{Q}_{ij}$ are explicitly given by

$$u_1 = \xi_1 + \frac{\alpha}{p}\xi_2; \quad u_2 = -\frac{\alpha}{p}\xi_1 + \xi_2 \tag{3.21}$$

with $\alpha = (\lambda_1 + p_0 + p) = -(\lambda_2 + P_0 + P)$.

$$\lambda_{1/2} = -\frac{1}{2}(P_0 + P + p_0 + p) \pm \tfrac{1}{2}\sqrt{(P_0 + P - p)^2 + 4P \cdot p} < 0 \tag{3.22}$$

$$\tilde{Q}_{11} = \left[q + Q\frac{\alpha^2}{P^2}\right]; \quad \tilde{Q}_{22} = \left[Q + q\frac{\alpha^2}{p^2}\right]. \tag{3.23}$$

$$\tilde{Q}_{12} = \tilde{Q}_{21} = \left[Q\frac{\alpha}{P} - q\frac{\alpha}{p}\right].$$

The wellknown stationary solution of (3.20) reads

$$f_{st}[u_1, u_2] = f_{st}(\xi_1, \xi_2) = \tilde{C}\exp\left\{-\sum_{i,j=1}^{2} \tilde{\delta}_{ij}u_i u_j\right\} =$$

$$= C\exp\left\{-\sum_{i,j=1}^{2} \delta_{ij}\xi_i\xi_j\right\} \tag{3.24}$$

with $((\tilde{\delta}_{ij})) = \left(\left(\frac{2\tilde{Q}_{ij}}{|\lambda_i + \lambda_j|}\right)\right)^{-1}$; $\delta_{mn} = \tilde{\delta}_{ij}\alpha_{im}\alpha_{jn}$ (3.25)

The curves of equal probability in the x/y plane follow from (3.24) and are ellipses defined by

$$(\delta_{11}\xi_1^2 + 2\delta_{12}\xi_1\xi_2 + \delta_{22}\xi_2^2) = \text{const.} = c^2 \tag{3.26}$$

with axes a, b, given by

$$a^2; b^2 = \frac{c^2}{2}\left\{\frac{(\delta_{11}+\delta_{22})}{|\delta|} \pm \frac{\sqrt{(\delta_{11}-\delta_{22})^2+4\delta_{12}^2}}{|\delta|}\right\} \tag{3.27}$$

and angle φ against coordinate axes following from

$$\mathrm{tg}(2\varphi) = \frac{2\delta_{12}}{(\delta_{11}-\delta_{22})} \tag{3.28}$$

Let us now consider the results after some further simplifying assumptions for the transition probabilities (3.15) in two interesting cases.

Case a. Onesided interaction between a "leading" group and a "people" group. The onesided influence of group 1 on group 2 without feedback is described by

$$P = \mu P_0; \quad p = 0; \quad 0 \leqslant \mu = \text{"influence parameter"} \tag{3.29}$$

We further assume, in order to have a well defined situation

$$\begin{aligned} &P_0 \equiv (P_{21}^{(0)} + P_{12}^{(0)}) = p_0 \equiv (p_{21}^{(0)} + p_{12}^{(0)}); \quad n \ll N, \epsilon = \frac{n}{N} \ll 1. \\ &p_{12}^{(0)} = 3p_{21}^{(0)} \curvearrowright r_0 = \frac{p_0}{2}; \quad P_{21}^{(0)} = 3P_{12}^{(0)} \curvearrowright R_0 = -\frac{P_0}{2}. \end{aligned} \tag{3.30}$$

Then the meanvalues according to (3.17) are

$$\bar{x} = \frac{1}{2}; \quad \bar{y} = \frac{1}{2}\frac{(\mu-1)}{(\mu+1)}. \tag{3.31}$$

and the explicit form of the matrix $((\delta_{ij}))$ is

$$\begin{Bmatrix} \frac{2n}{3}\frac{(2+\mu)[\epsilon(1+\mu)(2+\mu)+\mu^2]}{[\epsilon(1+\mu)(2+\mu)^2+\mu^2]}; & -\frac{2n}{3}\frac{\mu(1+\mu)(2+\mu)}{[\epsilon(1+\mu)(2+\mu)^2+\mu^2]} - \\ -\frac{2n}{3}\frac{\mu(1+\mu)(2+\mu)}{[\epsilon(1+\mu)(2+\mu)^2+\mu^2]}; & \frac{2n}{3}\frac{(1+\mu)(2+\mu)^2}{[\epsilon(1+\mu)(2+\mu)^2+\mu^2]} \end{Bmatrix} \tag{3.32}$$

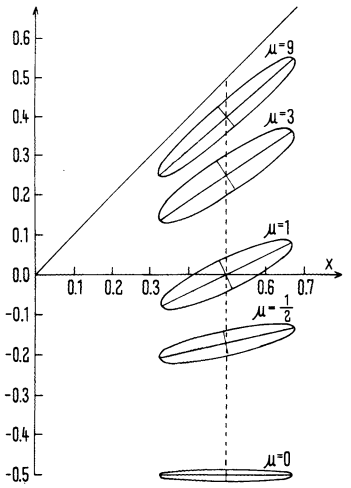


Fig. 6 Equiprobability ellipses for some coupling parameters μ in the case of onesided interaction between two groups for n = 12; N = 1200; ϵ = n/N = 1/100

Figure 6 shows some equiprobability ellipses with the same c^2 derived from (3.27), (3.28), (3.31) and (3.32) for the values 0; 1; 3; 9; of the "influence parameter" μ. For $\mu = 0$ we have the independent fluctuations of values x, y around the meanvalues $\bar{x} = \frac{1}{2}; \bar{y} = -\frac{1}{2}$. With increasing μ, the meanvalue $\bar{x}$ and the fluctuations around $\bar{x}$ of "attitudes" of group 1 remain the same, because there is no feedback influence of group 2 on group 1. On the other hand, with increasing μ, i.e. increasing effect of group 1 on group 2 the meanvalue $\bar{y}$ approaches $\bar{x} = \frac{1}{2}$ and also the fluctuations in group 2 around $\bar{y}$ are now influenced by the fluctuations in group 1, namely for moderate μ in the sense of a higher "mobility" in group 2 (increasing axis b), and for very high μ in the sense of a strong adaptation of any fluctuation around $\bar{y}$ in group 2 to the fluctuation of x around $\bar{x}$ in group 1. In other words: If the influence of the "leading group" on the probabilistic decision behaviour of the "people group" is very strong, then not only the meanvalue of the attitudes of the latter group approaches that of the leading group, also any deviation from the meanvalue in the leading group is closely followed by a corresponding deviation in the "people group".

Case b. Mutual interaction between "leading" group and "people group". The symmetrical influence between group 1 and group 2 can be described by

$$P = p = \mu P_0 = \mu p_0; \quad 0 \leqslant \mu = \text{"influence parameter"} \tag{3.33}$$

The further assumptions agree with (3.30).

The meanvalues according to (3.17) are now

$$\bar{x} = \frac{1}{2} \cdot \frac{1}{(1+2\mu)}; \quad \bar{y} = -\frac{1}{2} \cdot \frac{1}{(1+2\mu)}; \quad \bar{y} = -\bar{x}. \tag{3.34}$$

and the explicit form of $((\delta_{ij}))$ is in this case

$$\begin{pmatrix} \frac{2n}{3} \frac{[2\mu^2(1+\epsilon)+4\epsilon(1+2\mu)]}{[\mu^2(1+\epsilon)^2+4\epsilon(1+2\mu)]}; & -\frac{2n}{3} \frac{2\mu(1+\epsilon)(1+\mu)}{[\mu^2(1+\epsilon)^2+4\epsilon(1+2\mu)]} \\ -\frac{2n}{3} \frac{2\mu(1+\epsilon)(1+\mu)}{[\mu^2(1+\epsilon)^2+4\epsilon(1+2\mu)]}; & \frac{2n}{3} \frac{[2\mu^2(1+\epsilon)+4(1+2\mu)]}{[\mu^2(1+\epsilon)^2+4\epsilon(1+2\mu)]} \end{pmatrix} \tag{3.25}$$

Figure 7 again shows some equiprobability ellipses with the same c^2 for values of the "mutual influence parameter" μ.

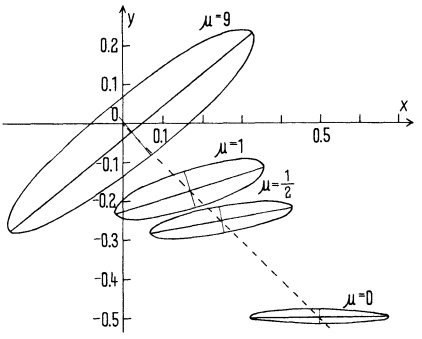


Fig. 7 Equiprobability ellipses for some coupling parameters in the case of mutual interaction between two groups for n = 12; N = 1200; ϵ = n/N = 1/100

Starting with the same case of independent fluctuations around $\bar{x} = \frac{1}{2}$, $\bar{y} = -\frac{1}{2}$ for $\mu = 0$ as in Fig. 6, both groups now develop in a different way for increasing μ. Because of the mutual influence between the groups the meanvalues $\bar{x}$, $\bar{y}$ of the attitude-distribution for both groups approach each other for increasing μ and meet in the middle, i.e. at $\bar{x} = \bar{y} = 0$ for $\mu \to \infty$.

The influence of the mutual adaptation on the fluctuations in both groups is more remarkable: With increasing μ the fluctuations in both groups grow steadily, i.e. higher and higher deviations from the meanvalues are possible and in this sense the "mobility" of the system of both groups increases. The reason is, that the internal probabilities $p_{12}^{(0)}$, $p_{21}^{(0)}$; $P_{12}^{(0)}$ $P_{21}^{(0)}$ leading to the restoration of some definite values $\bar{x}$, $\bar{y}$ now play a minor role against the mutual adaptation probabilities $p = \mu p_0$; $P = \mu P_0$ for which no point in the x/y plane is preferred and which therefore increase the fluctuations.

Concluding remarks

The two simple applications of the Fokker-Planck equation given here intended to show that the inclusion of fluctuations in the treatment of sociological models is not only necessary because of the probabilistic structure of the underlying information, but also leads to additional insights not contained in the consideration of meanvalues only and linking different fields of science by their analogous structure on the level of statistics.

References

[1] comp. Weidlich, W.: The Use of Statistical Models in Sociology, Collective Phenomena 1972, Vol. 1, pp. 51–59.

[2] e.g. Wang, M. Ch. Uhlenbeck, G. E.: Selected Papers in Noise and Stochastic Processes (Wax), New York 1954.

[3] Stratonovich, R. L.: Topics in the Theory of Random Noise Vols. 1, 2. New York and London: Gordon and Breach.

[4] Graham, R. and Haken, H.: Z. Physik **234**, 289 (1971).

[5] Weidlich, W.: Br. J. math. statist. Psychol. **24**, 251 (1971).

[6] Ogita, N. et al.: Suppl. Jap. phys. Soc. **26**, 145 (1969).